

Seismoelectric reflection and transmission at a fluid/porous-medium interface

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The dispersion relation for seismoelectric wave propagation in poroelastic media is formulated in terms of effective densities comprising all viscous and electrokinetic coupling effects. Using Helmholtz decomposition, two seismoelectric conversion coefficients are derived, for an incident *P*-wave upon an interface between a compressible fluid and a poroelastic medium. These coefficients relate the incident *P*-wave to a reflected electromagnetic wave in the fluid, and a transmitted electromagnetic wave in the porous medium. The dependency on angle of incidence and frequency is computed. Using orthodox and interference fluxes, it is shown that energy conservation is satisfied. A sensitivity analysis indicates that electrolyte concentration, viscosity, and permeability highly influence seismoelectric conversion.

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I. INTRODUCTION

When the grain surfaces of soils and rocks are in contact with a fluid electrolyte, they typically acquire a chemically bound surface charge that is balanced by mobile counterions in a thin fluid layer surrounding the grains. The bound charge is immobile, whereas the counterions can move. The distribution of mobile ions is determined by a balance between electrostatic forces and thermal diffusivity. At the interface between the immobile and counterions the so-called zeta-potential is defined. The potential varies exponentially when one moves away from the interface. The corresponding characteristic length scale is called the Debye length, which is on the order of some tens of nanometers for typical grain-electrolyte combinations (Pride, 1994). The ensemble of bound and mobile charge layers is referred to as the electrochemical double layer.

A compressional wave in fluid-saturated porous media will create pressure peaks and troughs on the scale of the wavelength. The resulting hydraulic flow will transport the counterions relative to the immobile, bound charge. In this way, counterions accumulate in pressure troughs and bound charge becomes exposed in pressure peaks, creating an electric coseismic field at the scale of the wavelength (Haines *et al.*, 2007). This electric field drives a conduction current that exactly balances the hydraulic current flow. Thus there is no net electric current for a propagating compressional wave within a homogeneous material. For interfaces, however, this is no longer the case. When a compressional or shear wave traverses an interface with a contrast in electrical or mechanical properties, an electric current imbalance is produced resulting in electromagnetic (EM)-waves that can propagate outside the support of the seismic waves (Haartsen and

Pride, 1997). The generation of these waves at interfaces can thus be associated with new, electrokinetic, reflection, and transmission coefficients.

Several studies measured the conversion of seismic to electromagnetic energy in the field (Thompson and Gist, 1993; Butler *et al.*, 1996; Mikhailov *et al.*, 1997; Garambois and Dietrich, 2001; Haines *et al.*, 2007). The inverse electroseismic conversion was also measured (Thompson *et al.*, 2007; Zhu *et al.*, 2008). So-called electroseismograms in layered media were presented by several authors (e.g., Haartsen and Pride, 1997; Han and Wang, 2001; White and Zhou, 2006; Haines and Pride, 2006). In order to obtain the electrokinetic reflection and transmission coefficients at a fluid/porous-medium interface Block and Harris (2006) approximated Pride's (1994) electrokinetic theory by solving the Biot reflection problem only, from which the relative pore fluid flow is obtained. This flow subsequently acts as a source term in the EM boundary conditions of Pride's (1994) theory. In this paper, we solve the complete boundary value problem in a procedure similar to Denneman *et al.* (2002), where an acoustic wave from a compressible fluid impinges on a poroelastic medium. Two new parameters are defined, describing the conversion from seismic to EM-waves. The incident fluid wave is transmitted into fast and slow compressional waves, a shear wave, and into an EM-wave. Part of the incident wave is reflected as a fluid wave and a fluid EM-wave. In Sec. II we introduce the governing equations and we present the electrokinetic dispersion relations. The dependency of the seismoelectric reflection and transmission coefficients on angle of incidence and frequency is derived in Sec. III. In Sec. IV we recast the reflection and transmission coefficients in vertical energy flux coefficients and perform a sensitivity analysis.

II. GOVERNING EQUATIONS

We consider a poroelastic matrix saturated by an electrolyte and adopt an $\exp[i\omega t]$ convention for time varying

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fields. The governing linearized equations in an isotropic, homogeneous poroelastic medium were given by [Pride and Haartsen \(1996\)](#). We rewrite the equations here in a somewhat different form because we aim to derive the dispersion relations in terms of effective densities. We will confirm full consistency with the original Pride equations in the forthcoming. The governing equations read as

$$-\nabla \cdot \underline{\underline{\hat{\sigma}}} - (1 - \phi) \nabla \hat{p} = -\omega^2[\rho_{11}(\omega)\hat{\mathbf{u}} + \rho_{12}(\omega)\hat{\mathbf{U}}] + \frac{\eta\phi L(\omega)}{k(\omega)}\hat{\mathbf{E}}, \quad (1)$$

$$-\phi \nabla \hat{p} = -\omega^2[\rho_{12}(\omega)\hat{\mathbf{u}} + \rho_{22}(\omega)\hat{\mathbf{U}}] - \frac{\eta\phi L(\omega)}{k(\omega)}\hat{\mathbf{E}}, \quad (2)$$

$$\underline{\underline{\hat{\sigma}}} = -G[\nabla\hat{\mathbf{u}} + (\nabla\hat{\mathbf{u}})^T] - \left[\left(A - Q\frac{1-\phi}{\phi} \right) \nabla \cdot \hat{\mathbf{u}} + \left(Q - R\frac{1-\phi}{\phi} \right) \nabla \cdot \hat{\mathbf{U}} \right] \mathbf{I}, \quad (3)$$

$$\hat{p} = -\frac{1}{\phi}(Q \nabla \cdot \hat{\mathbf{u}} + R \nabla \cdot \hat{\mathbf{U}}), \quad (4)$$

$$\hat{\mathbf{J}} = \sigma(\omega)\hat{\mathbf{E}} + L(\omega)(-\nabla\hat{p} + \omega^2\rho_f\hat{\mathbf{u}}), \quad (5)$$

$$\nabla \times \hat{\mathbf{H}} = i\omega\varepsilon\hat{\mathbf{E}} + \hat{\mathbf{J}}, \quad (6)$$

$$\nabla \times \hat{\mathbf{E}} = -i\omega\mu\hat{\mathbf{H}}. \quad (7)$$

Hats over field variables indicate small frequency-domain quantities, ω is the angular frequency, η is the pore fluid viscosity, ϕ is the porosity, ρ_f is the pore fluid density, μ is the magnetic permeability, ε is the bulk electrical permittivity given by $\varepsilon = \varepsilon_0[\phi(\epsilon_f - \epsilon_s)/\alpha_\infty + \epsilon_s]$, with solid and pore fluid relative permittivities ϵ_s and ϵ_f , and vacuum permittivity ε_0 . $\mathbf{I}$ denotes the identity matrix. The field variables are solid and (pore) fluid displacements $\mathbf{u}$ and $\mathbf{U}$, intergranular stress $\underline{\underline{\sigma}}$, pore pressure p , electric current density $\mathbf{J}$, and the electric and magnetic fields $\mathbf{E}$ and $\mathbf{H}$. A , Q , and R are the generalized elastic coefficients that are related to the bulk modulus of the skeleton grains K_s , the shear modulus G , the bulk modulus of the pore fluid K_f , and the bulk modulus of the framework of grains K_b as follows ([Biot and Willis, 1957](#)):

$$A = \frac{(1 - \phi)^2 K_s K_f - (1 - \phi) K_b K_f + \phi K_s K_b}{K_f \left(1 - \phi - \frac{K_b}{K_s} \right) + \phi K_s} - \frac{2}{3} G, \quad (8)$$

$$Q = \frac{\phi[K_s(1 - \phi) - K_b]K_f}{K_f \left(1 - \phi - \frac{K_b}{K_s} \right) + \phi K_s}, \quad (9)$$

$$R = \frac{\phi^2 K_s K_f}{K_f \left(1 - \phi - \frac{K_b}{K_s} \right) + \phi K_s}. \quad (10)$$

The dynamic permeability $k(\omega)$ describes the transition from viscosity toward inertia-dominated flow. The electrokinetic coupling coefficient $L(\omega)$ couples mechanical and EM-wave behavior. Note that for $L(\omega) = 0$, mechanical and EM-fields are decoupled in Eqs. (1)–(7). Equations (1)–(4) then transform into the original Biot equations ([Biot, 1956a, 1956b](#)) and Eqs. (5)–(7) into Ohm's law and Maxwell's equations. The conductivity of the pore fluid is denoted as $\sigma(\omega)$. Expressions for $k(\omega)$, $L(\omega)$, and $\sigma(\omega)$ are given in Appendix A. As the dynamic permeability and the fluid and solid density terms always appear together, it is convenient to define effective densities $\rho_{11}(\omega)$, $\rho_{12}(\omega)$, and $\rho_{22}(\omega)$ as follows:

$$\rho_{11}(\omega) = (1 - \phi)\rho_s - \rho_{12}(\omega), \quad (11)$$

$$\rho_{12}(\omega) = \phi\rho_f \left[1 + i\frac{\phi\eta}{\omega\rho_f k(\omega)} \right], \quad (12)$$

$$\rho_{22}(\omega) = \phi\rho_f - \rho_{12}(\omega), \quad (13)$$

where ρ_s is the solid density. Adding Eqs. (1) and (2), we obtain the original dynamic equation (3) by [Pride and Haartsen \(1996\)](#) (no source term). With the definitions for $\rho_{12}(\omega)$ and $\rho_{22}(\omega)$, Eq. (2) is written as

$$i\omega\phi(\hat{\mathbf{U}} - \hat{\mathbf{u}}) = L(\omega)\hat{\mathbf{E}} + \frac{k(\omega)}{\eta}(-\nabla\hat{p} + \omega^2\rho_f\hat{\mathbf{u}}), \quad (14)$$

which is Eq. (5) from [Pride and Haartsen \(1996\)](#) (no source term). By adding Eqs. (3) and (4), we obtain an expression for total stress $\underline{\underline{\tau}} = -\underline{\underline{\sigma}} - p\mathbf{I}$, which corresponds to Eq. (6) by [Pride and Haartsen \(1996\)](#). Full consistency with the original Pride equations is thus confirmed.

Substituting Eqs. (3) and (4) into Eqs. (1) and (2) yields

$$G\nabla^2\hat{\mathbf{u}} + (A + G) \nabla \nabla \cdot \hat{\mathbf{u}} + Q \nabla \nabla \cdot \hat{\mathbf{U}} = -\omega^2[\rho_{11}(\omega)\hat{\mathbf{u}} + \rho_{12}(\omega)\hat{\mathbf{U}}] + \frac{\eta\phi L(\omega)}{k(\omega)}\hat{\mathbf{E}}, \quad (15)$$

$$Q \nabla \nabla \cdot \hat{\mathbf{u}} + R \nabla \nabla \cdot \hat{\mathbf{U}} = -\omega^2[\rho_{12}(\omega)\hat{\mathbf{u}} + \rho_{22}(\omega)\hat{\mathbf{U}}] - \frac{\eta\phi L(\omega)}{k(\omega)}\hat{\mathbf{E}}. \quad (16)$$

Eliminating $(-\nabla\hat{p} + \omega^2\rho_f\hat{\mathbf{u}})$ from Eqs. (5) and (14) we obtain

$$i\omega\phi(\hat{\mathbf{U}} - \hat{\mathbf{u}}) = L(\omega)\hat{\mathbf{E}} + \frac{k(\omega)}{\eta} \left(\frac{\hat{\mathbf{J}} - \sigma(\omega)\hat{\mathbf{E}}}{L(\omega)} \right). \quad (17)$$

Substituting Eq. (6) into Eq. (17) we obtain

$$i\omega\phi(\hat{\mathbf{U}} - \hat{\mathbf{u}}) = L(\omega)\hat{\mathbf{E}} + \frac{k(\omega)}{\eta L(\omega)} (\nabla \times \hat{\mathbf{H}} - [i\omega\varepsilon + \sigma(\omega)]\hat{\mathbf{E}}). \quad (18)$$

Finally, substituting Eq. (7) into Eq. (18) we obtain

TABLE I. Parameters of the poroelastic medium and fluid. Parameters of poroelastic medium from [Denneman et al. \(2002\)](#).

Parameters poroelastic medium	
Bulk modulus framework of grains, K_b	5.8×10^9 Pa
Shear modulus framework of grains, G	3.4×10^9 Pa
Bulk modulus skeleton grains, K_s	40×10^9 Pa
Bulk modulus pore fluid, K_f	2.22×10^9 Pa
Pore fluid viscosity, η	0.001 Pa s
Pore fluid density, ρ_f	1000 kg/m ³
Solid density, ρ_s	2760 kg/m ³
Tortuosity of the porous medium, α_∞	2.3
Porosity of the porous medium, ϕ	0.24
Permeability, k_0	0.390×10^{-12} m ²
Temperature, T	295 K
Hydrogen exponent, pH	7
Relative permittivity of the pore fluid, ϵ_f	80
Relative permittivity of the solid, ϵ_s	4
Magnetic permeability ($=\mu_0$), μ	$4\pi \times 10^{-7}$ H/m
Valence of species-1 ion, z_1	1
Valence of species-2 ion, z_2	-1
Mobility of species-1 ion, b_1^a	3.246×10^{11} m/(N s)
Mobility of species-2 ion, b_2^a	4.931×10^{11} m/(N s)
Zeta-potential, ζ^b	-0.065 V
Similarity parameter, M	1
Concentration of the electrolyte, C	0.001 mol/l
Parameters fluid	
Bulk modulus fluid, K_{fl}	2.22×10^9 Pa
Fluid density, ρ_{fl}	1000 kg/m ³
Fluid relative permittivity, ϵ_{fl}	80
Fluid magnetic permeability ($=\mu_0$), μ_{fl}	$4\pi \times 10^{-7}$ H/m
Fluid conductivity, σ_{fl}	5.0×10^{-3} S/m

^aNabighian and Corbett, 1988.

^bFrom Eq. (A5).

$$\omega^2 \mu \bar{\epsilon}(\omega) \hat{\mathbf{E}} + \omega^2 \mu \frac{\eta \phi L(\omega)}{k(\omega)} (\hat{\mathbf{U}} - \hat{\mathbf{u}}) = \nabla \nabla \cdot \hat{\mathbf{E}} - \nabla^2 \hat{\mathbf{E}}, \quad (19)$$

where $\bar{\epsilon}(\omega)$ is the newly defined effective electrical permittivity

$$\bar{\epsilon}(\omega) = \epsilon - i \frac{\sigma(\omega)}{\omega} + i \frac{\eta L^2(\omega)}{\omega k(\omega)}. \quad (20)$$

Equations (15), (16), and (19) form a closed set of equations for the fields $\mathbf{u}$, $\mathbf{U}$, and $\mathbf{E}$. A summary of electrokinetic symbols is given in Table I.

A. Electrokinetic wave velocities

Electrokinetic theory in isotropic, homogeneous media predicts the existence of uncoupled longitudinal and transversal modes. There are two longitudinal waves, the fast P -wave and the slow P -wave, and two transversal waves: an electromagnetic wave and a seismic shear wave ([Pride and Haartsen, 1996](#)). We derive complex electrokinetic wave velocities in a procedure similar to that of [Allard \(1993\)](#). First, we define the following Helmholtz decompositions:

$$\hat{\mathbf{u}} = \nabla \hat{\phi}_s + \nabla \times \hat{\Psi}_s, \quad (21)$$

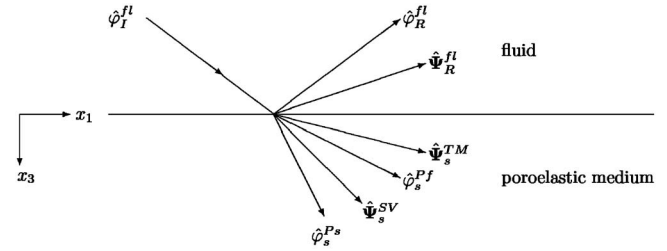


FIG. 1. Seismoelectric reflection and transmission potentials due to an incident fluid wave.

$$\hat{\mathbf{U}} = \nabla \hat{\phi}_f + \nabla \times \hat{\Psi}_f, \quad (22)$$

$$\hat{\mathbf{E}} = \nabla \hat{\phi}_E + \nabla \times \hat{\Psi}_E. \quad (23)$$

These are substituted in Eqs. (15), (16), and (19). This yields two separate sets: one for the longitudinal waves and one for the transversal waves. Subsequently, plane wave propagation is introduced in the $(x_1, 0, x_3)$ plane (see Fig. 1), e.g., $\hat{\phi}_s = \tilde{\phi}_s \exp[-i\mathbf{k} \cdot \mathbf{x}]$ and $\hat{\Psi}_s = (0, \tilde{\psi}_s \exp[-i\mathbf{k} \cdot \mathbf{x}], 0)^T$, where $\mathbf{k}$ is the wavenumber vector and $\mathbf{x}$ is the position vector in that plane. Through elimination of $\tilde{\phi}_E$ in the longitudinal wave equations a system of equations for $\tilde{\phi}_s$ and $\tilde{\phi}_f$ is obtained. The nontrivial solution yields the squared complex slowness (i.e., inverse complex-valued wave velocities) of the longitudinal waves $s_l^2(\omega)$

$$s_l^2(\omega) = \frac{-d_1(\omega)}{2d_2} \mp \frac{1}{2} \sqrt{\left[\frac{d_1(\omega)}{d_2} \right]^2 - 4 \frac{d_0(\omega)}{d_2}}, \quad (24)$$

where $l=Pf, Ps$ refers to the fast and slow longitudinal waves, respectively, and

$$d_0(\omega) = \bar{\rho}_{11}(\omega) \bar{\rho}_{22}(\omega) - [\bar{\rho}_{12}(\omega)]^2,$$

$$d_1(\omega) = -[P \bar{\rho}_{22}(\omega) + R \bar{\rho}_{11}(\omega) - 2Q \bar{\rho}_{12}(\omega)],$$

$$d_2 = PR - Q^2, \quad (25)$$

where $P=A+2G$. As the effective densities and the electrokinetic coupling factor always appear together, it is convenient to define new effective densities $\bar{\rho}_{11}(\omega)$, $\bar{\rho}_{12}(\omega)$, and $\bar{\rho}_{22}(\omega)$ as follows:

$$\bar{\rho}_{11}(\omega) = \rho_{11}(\omega) - E_K(\omega), \quad (26)$$

$$\bar{\rho}_{12}(\omega) = \rho_{12}(\omega) + E_K(\omega), \quad (27)$$

$$\bar{\rho}_{22}(\omega) = \rho_{22}(\omega) - E_K(\omega), \quad (28)$$

$$E_K(\omega) = \frac{\eta^2 \phi^2 L^2(\omega)}{k^2(\omega) \bar{\epsilon}(\omega) \omega^2}. \quad (29)$$

We note that both viscous and electrokinetic coupling effects are now elegantly comprised in effective frequency-dependent densities.

A similar approach is used for the transversal waves. Through elimination of $\tilde{\Psi}_E$ in the transversal wave equations, a system of equations for $\tilde{\Psi}_s$ and $\tilde{\Psi}_f$ is obtained. The nontrivial solution yields the squared complex slowness of

the transversal waves $s_l^2(\omega)$ according to Eq. (24), where now $l=EM$, S refers to the EM and seismic shear waves, respectively, and

$$d_0(\omega) = \mu \bar{\epsilon}(\omega) \frac{\bar{\rho}_{11}(\omega) \bar{\rho}_{22}(\omega) - [\bar{\rho}_{12}(\omega)]^2}{G},$$

$$d_1(\omega) = -\mu \bar{\epsilon}(\omega) \bar{\rho}_{22}(\omega) - \frac{\rho_{11}(\omega) \rho_{22}(\omega) - [\rho_{12}(\omega)]^2}{G},$$

$$d_2(\omega) = \rho_{22}(\omega). \quad (30)$$

From the complex-valued slowness, phase velocities and attenuation coefficients are readily obtained.

III. REFLECTION AND TRANSMISSION AT A PLANE FLUID/POROUS-MEDIUM INTERFACE

A compressible fluid overlays an isotropic, homogeneous porous medium. The horizontal interface is at $x_3=0$ (see Fig. 1). We adopt a right-hand (x_1, x_2, x_3) coordinate system with the x_3 -axis pointing downward. A plane fluid wave (scalar potential $\hat{\phi}_f^{fl}$) impinges on the interface where it reflects as a P-wave (scalar potential $\hat{\phi}_R^{fl}$) and a transverse magnetic (TM) EM-wave (vector potential $\hat{\Psi}_R^{fl}$). It transmits as a TM-wave (vector potential $\hat{\Psi}_s^{TM}$), a fast longitudinal wave (scalar potential $\hat{\phi}_s^{Pf}$), a slow longitudinal wave (scalar potential $\hat{\phi}_s^{Ps}$), and a vertical shear (SV)-wave (vector potential $\hat{\Psi}_s^{SV}$). The reflection and transmission coefficients are obtained from the (open-pore) boundary conditions, which are continuity of fluid volume displacement, fluid pressure, the horizontal magnetic and electric fields, and vanishing vertical and horizontal intergranular stresses (Deresiewicz and Skalak, 1963; Pride and Haartsen, 1996)

$$(1 - \phi) \hat{u}_3 + \phi \hat{U}_3 = \hat{U}_3^{fl}, \quad (31)$$

$$\hat{p} = \hat{p}^{fl}, \quad (32)$$

$$\hat{\sigma}_{13} = \hat{\sigma}_{33} = 0, \quad (33)$$

$$\hat{H}_2 = \hat{H}_2^{fl}, \quad (34)$$

$$\hat{E}_1 = \hat{E}_1^{fl}. \quad (35)$$

The pressure in the upper fluid layer is denoted by p^{fl} . Magnetic and electric fields in the upper fluid layer are indicated by $\mathbf{H}^{fl}$ and $\mathbf{E}^{fl}$, respectively. We relate p^{fl} to upper fluid layer displacement $\mathbf{U}^{fl}$ by $p^{fl} = -K_{fl} \nabla \cdot \mathbf{U}^{fl}$, where K_{fl} is the bulk modulus of the upper fluid layer. The upper fluid layer wave speed c is related to the upper layer fluid density ρ_{fl} by $c = \sqrt{K_{fl}/\rho_{fl}}$. Viscosity does not play a role in the overlying fluid because the ratio of viscous and inertia terms is $\eta\omega/c^2\rho_{fl}$, which is on the order $10^{-11} - 10^{-6}$ for the frequencies considered in this paper.

Scalar potentials $\hat{\phi}_s$, $\hat{\phi}_f$, and $\hat{\phi}_E$, see Eqs. (21)–(23), are composed of contributions of the longitudinal waves, e.g., $\hat{\phi}_s = \hat{\phi}_s^{Pf} + \hat{\phi}_s^{Ps}$, and, similarly, vector potentials $\hat{\Psi}_s$, $\hat{\Psi}_f$, and $\hat{\Psi}_E$ are composed of contributions of the transversal waves,

e.g., $\hat{\Psi}_s = \hat{\Psi}_s^{TM} + \hat{\Psi}_s^{SV} = (0, \hat{\psi}_s^{TM}, 0)^T + (0, \hat{\psi}_s^{SV}, 0)^T$. The dispersion relations given in Eq. (24) not only predict the phase velocities and attenuation coefficients, but also the fluid-solid amplitude ratios β^m and β^n and the amplitude ratios α^m and α^n (V/m^2) that describe the strength of the electric field with respect to the solid displacement amplitude, for each wave

$$\beta^m(\omega) = \frac{\hat{\phi}_f^m}{\hat{\phi}_s^m} = \frac{\bar{\rho}_{11}(\omega) - P s_m^2(\omega)}{Q s_m^2(\omega) - \bar{\rho}_{12}(\omega)}, \quad (36)$$

$$\beta^n(\omega) = \frac{\hat{\psi}_f^n}{\hat{\psi}_s^n} = \frac{G s_n^2(\omega) - (1 - \phi) \rho_s}{\phi \rho_f}, \quad (37)$$

$$\alpha^m(\omega) = \frac{\hat{\phi}_E^m}{\hat{\phi}_s^m} = \frac{\eta \phi L(\omega)}{k(\omega) \bar{\epsilon}(\omega)} [1 - \beta^m(\omega)], \quad (38)$$

$$\alpha^n(\omega) = \frac{\hat{\psi}_E^n}{\hat{\psi}_s^n} = \frac{\mu \eta \phi L(\omega)}{k(\omega) [\mu \bar{\epsilon}(\omega) - s_n^2(\omega)]} [1 - \beta^n(\omega)], \quad (39)$$

for $m=Pf$ or Ps , and $n=TM$ or SV . We now specify the following plane wave propagation expressions:

$$\hat{\phi}_f^{fl} = \tilde{\phi}_f^{fl} \exp[-i(k_1 x_1 + k_3^{fl} x_3)], \quad (40)$$

$$\hat{\phi}_R^{fl} = \tilde{\phi}_R^{fl} \exp[-i(k_1 x_1 - k_3^{fl} x_3)], \quad (41)$$

$$\hat{\Psi}_R^{fl} = (0, \tilde{\psi}_R^{fl} \exp[-i(k_1 x_1 - k_3^E x_3)], 0)^T, \quad (42)$$

$$\hat{\phi}_s^m = \tilde{\phi}_s^m \exp[-i(k_1 x_1 + k_3^m x_3)], \quad (43)$$

$$\hat{\Psi}_s^n = (0, \tilde{\psi}_s^n \exp[-i(k_1 x_1 + k_3^n x_3)], 0)^T, \quad (44)$$

where k_3^{fl} and k_3^E are the vertical wavenumbers of the incident fluid wave and reflected EM-wave, respectively. The reflection and transmission coefficients are

$$R^E = \frac{\tilde{\psi}_R^{fl}}{\tilde{\phi}_f^{fl}}, \quad R^M = \frac{\tilde{\phi}_R^{fl}}{\tilde{\phi}_f^{fl}}, \quad T^{Pf} = \frac{\tilde{\phi}_s^{Pf}}{\tilde{\phi}_f^{fl}},$$

$$T^{Ps} = \frac{\tilde{\phi}_s^{Ps}}{\tilde{\phi}_f^{fl}}, \quad T^{TM} = \frac{\tilde{\psi}_s^{TM}}{\tilde{\phi}_f^{fl}}, \quad T^{SV} = \frac{\tilde{\psi}_s^{SV}}{\tilde{\phi}_f^{fl}}, \quad (45)$$

where R^M indicates the mechanical reflection coefficient and R^E indicates the seismoelectric reflection coefficient. Applying boundary conditions (31)–(35) yields a linear system of six equations and six unknowns

$$\underline{\underline{A}} \cdot (R^E, R^M, T^{Pf}, T^{Ps}, T^{TM}, T^{SV})^T = (k_3^{fl}, \phi \rho_{fl}, 0, 0, 0, 0)^T, \quad (46)$$

where elements of matrix $\underline{\underline{A}}$ are given in Appendix B. We solve Eq. (46) by Cramer's rule and show the results for R^E and T^{TM} as a function of incidence angle θ at a fixed frequency of 10 Hz in Fig. 2. The parameter values, representative for a water/shallow-sandstone interface, are given in Table I. We use these values in all computations of the reflection and transmission coefficients. For $\sin \theta$ between ~ 0 and ~ 0.6 , $|R^E|$ is approximately constant. Note that for

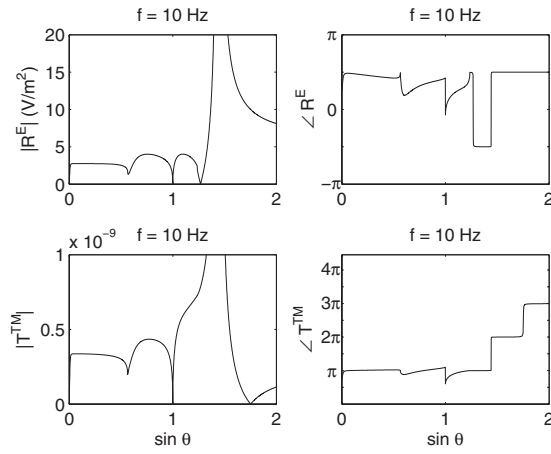


FIG. 2. Amplitudes (left) and phase values (right) of seismoelectric reflection (R^E) and transmission (T^{TM}) coefficients at a fluid/porous-medium boundary.

sin $\theta=0$, no reflected EM-wave is generated due to the prescribed boundary conditions. For the parameters of Table I, $|R^E|$ is approximately 3 V/m², which is equivalent to $\sim 8 \times 10^{-7}$ V/Pa at 10 Hz. At sin $\theta \approx 0.57$, $|R^E|$ has a local minimum associated with critical reflection of the fast P -wave. Obviously, sin $\theta=1$ corresponds to grazing angle of incidence and there is no seismoelectric conversion ($|R^E|=0$). We associate sin $\theta \approx 1.24$ with critical reflection of the SV-wave and we associate the peak at sin $\theta \approx 1.44$ with the Stoneley wave. The above mentioned behavior is also visible in the phase behavior of R^E , and the behavior of T^{TM} . At sin $\theta=0$ and sin $\theta=1$, R^E and T^{TM} are identically zero. We note that generally $|T^{TM}|$ is on the order 10^{-10} .

Next we show in Fig. 3 the frequency-dependence of $|R^E|$ and $|T^{TM}|$ for 45° angle of incidence. Both seismoelectric coefficients increase strongly with frequency, which corresponds to increasing EM-wave velocities in the diffusive regime for both the fluid and poroelastic medium. We note that the increase in $|T^{TM}|$ sharply diminishes at $\omega = \omega_t$, i.e., for the frequency that separates viscosity-dominated flow from inertia-dominated flow.

IV. SENSITIVITY ANALYSIS

A natural way to perform a sensitivity analysis is to analyze the ratio of reflected/transmitted wave energy fluxes to the incident flux; see, e.g., Dutta and Odé (1983) and Santos *et al.* (1992).

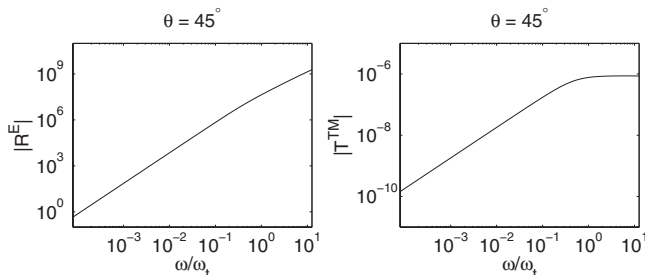


FIG. 3. Magnitude of seismoelectric coefficients R^E and T^{TM} as a function of normalized frequency at $\theta=45^\circ$ angle of incidence.

A. Vertical electrokinetic energy fluxes

The electrokinetic vertical energy flux $Y_3(\mathbf{x}, t)$ of the porous medium, defined by the dot product of the Poynting vector $\mathbf{Y}$ and the unit normal to the interface $\mathbf{n}=(0,0,1)^T$, is given by (Haartsen, 1995)

$$\begin{aligned} Y_3(\mathbf{x}, t) &= [\mathbf{E}(\mathbf{x}, t) \times \mathbf{H}(\mathbf{x}, t) + \underline{\tau}(\mathbf{x}, t) \cdot \dot{\mathbf{u}}(\mathbf{x}, t) \\ &\quad - \phi p(\mathbf{x}, t) (\dot{\mathbf{U}}(\mathbf{x}, t) - \dot{\mathbf{u}}(\mathbf{x}, t))] \cdot \mathbf{n} \\ &= E_1(\mathbf{x}, t) H_2(\mathbf{x}, t) + \tau_{31}(\mathbf{x}, t) \dot{u}_1(\mathbf{x}, t) \\ &\quad + \tau_{33}(\mathbf{x}, t) \dot{u}_3(\mathbf{x}, t) - \phi p(\mathbf{x}, t) \\ &\quad \times (\dot{U}_3(\mathbf{x}, t) - \dot{u}_3(\mathbf{x}, t)). \end{aligned} \quad (47)$$

Similar expressions to Eq. (47) were provided by Pride and Haartsen (1996), Dutta and Odé (1983), and Santos *et al.*, (1992). We note that in our geometry (see Fig. 1) $H_1(\mathbf{x}, t) = E_2(\mathbf{x}, t) = \tau_{32}(\mathbf{x}, t) = u_2(\mathbf{x}, t) = 0$. We now write real parts of all frequency-domain variables and calculate a time-average over 1 cycle in time so that $\langle Y_3(\mathbf{x}, t) \rangle$ becomes

$$\begin{aligned} \langle Y_3(\mathbf{x}, t) \rangle &= \frac{1}{2\pi} \int_0^{2\pi} (\text{Re}[\hat{E}_1 \exp[i\omega t]] \text{Re}[\hat{H}_2 \exp[i\omega t]] \\ &\quad + \text{Re}[\hat{\tau}_{31} \exp[i\omega t]] \text{Re}[i\omega \hat{u}_1 \exp[i\omega t]] \\ &\quad + \text{Re}[\hat{\tau}_{33} \exp[i\omega t]] \text{Re}[i\omega \hat{u}_3 \exp[i\omega t]] \\ &\quad - \text{Re}[\hat{p} \exp[i\omega t]] \text{Re}[i\omega \phi (\hat{U}_3 - \hat{u}_3) \\ &\quad \times \exp[i\omega t]]) d(\omega t) \\ &= \frac{1}{2} [\hat{E}_1' \hat{H}_2' + \hat{E}_1'' \hat{H}_2''] + \frac{\omega}{2} [\hat{\tau}_{31}' \hat{u}_1' - \hat{\tau}_{31}'' \hat{u}_1''] \\ &\quad + \frac{\omega}{2} [\hat{\tau}_{33}' \hat{u}_3' - \hat{\tau}_{33}'' \hat{u}_3''] - \frac{\omega}{2} [\phi \hat{p}'' (\hat{U}_3' - \hat{u}_3') \\ &\quad - \phi \hat{p}' (\hat{U}_3'' - \hat{u}_3'')], \end{aligned} \quad (48)$$

where single and double primes denote real and imaginary parts of complex quantities, respectively. Denoting complex conjugation by an asterisk, we find that

$$\begin{aligned} \langle Y_3(\mathbf{x}, t) \rangle &= \frac{1}{4} \{ (\hat{E}_1 \hat{H}_2^* + \hat{E}_1^* \hat{H}_2) - i\omega [(\hat{\tau}_{31} \hat{u}_1^* - \hat{\tau}_{31}^* \hat{u}_1) \\ &\quad + (\hat{\tau}_{33} \hat{u}_3^* - \hat{\tau}_{33}^* \hat{u}_3) - (\phi \hat{p} (\hat{U}_3^* - \hat{u}_3^*) \\ &\quad - \phi \hat{p}^* (\hat{U}_3 - \hat{u}_3))] \}. \end{aligned} \quad (49)$$

Each of the components in Eq. (49) is composed of contributions from the four electrokinetic waves, e.g., $\hat{E}_1 = \hat{E}_1^{Pf} + \hat{E}_1^{Ps} + \hat{E}_1^{TM} + \hat{E}_1^{SV}$. Writing all terms in Eq. (49) in separate contributions from each of the waves, we will have four orthodox fluxes, resulting from wave amplitudes multiplied by themselves, and six interference fluxes, resulting from cross-terms (Dutta and Odé, 1983; Santos *et al.*, 1992). In the non-dissipative fluid layer there is only an orthodox reflected EM-flux $\langle Y_3^f(\mathbf{x}, t) \rangle^{E,E}$, and incident and reflected orthodox acoustic fluxes $\langle Y_3^f(\mathbf{x}, t) \rangle^{Pm,Pm}$ and $\langle Y_3^f(\mathbf{x}, t) \rangle^{Pr,Pr}$. Vertical energy flux conservation requires

TABLE II. Seismoelectric reflection and transmission coefficients for $\theta = 30^\circ$ at $f=10$ Hz and $f=500$ kHz.

	$f=10$ Hz	$f=500$ kHz
$T_E^{Pf,Pf}$	4.6700×10^{-1}	4.7334×10^{-1}
$T_E^{Ps,Ps}$	1.8083×10^{-3}	1.3026×10^{-1}
$T_E^{TM,TM}$	-2.6500×10^{-10}	-4.9895×10^{-7}
$T_E^{SV,SV}$	1.3084×10^{-1}	7.0295×10^{-2}
$T_E^{Pf,Ps}$	-2.0028×10^{-10}	-8.0868×10^{-7}
$T_E^{Pf,TM}$	-2.7470×10^{-12}	-2.5059×10^{-9}
$T_E^{Pf,SV}$	7.4887×10^{-11}	3.0585×10^{-7}
$T_E^{Ps,TM}$	6.7059×10^{-10}	5.2570×10^{-7}
$T_E^{Ps,SV}$	-1.0420×10^{-8}	-3.7939×10^{-5}
$T_E^{TM,SV}$	4.3970×10^{-14}	1.0814×10^{-7}
$R_E^{E,E}$	7.4497×10^{-11}	1.1871×10^{-7}
$R_E^{Pr,Pr}$	-4.0035×10^{-1}	-3.2615×10^{-1}
$T_E - R_E^{E,E} - R_E^{Pr,Pr}$	1.0000×10^0	1.0000×10^0

$$\begin{aligned} \langle Y_3(\mathbf{x}, t) \rangle - \langle Y_3^{fl}(\mathbf{x}, t) \rangle^{E,E} - \langle Y_3^{fl}(\mathbf{x}, t) \rangle^{Pr,Pr} \\ = \langle Y_3^{fl}(\mathbf{x}, t) \rangle^{Pin,Pin}. \end{aligned} \quad (50)$$

Introducing energy transmission coefficients, e.g., $T_E = \langle Y_3(\mathbf{x}, t) \rangle / \langle Y_3^{fl}(\mathbf{x}, t) \rangle^{Pin,Pin}$, we write

$$T_E - R_E^{E,E} - R_E^{Pr,Pr} = 1. \quad (51)$$

We note, again, that T_E consists of four orthodox flux transmission coefficients and six interference flux transmission coefficients. Numerical examples are given in Tables II and III. In Table II, the angle of incidence θ is fixed at 30° and frequencies f of 10 Hz (left column) and 500 kHz (right column) are chosen. Higher frequencies favor seismoelectric conversion. In Table III, the angular frequency is fixed at 10^6 rad/s, and angles of incidence $\theta=30^\circ$ (left column) and $\theta=45^\circ$ (right column) are chosen. Equation (51) is satisfied indeed. $T_E^{Pf,Ps}$ and $T_E^{Ps,SV}$ are negative, because of the out-of-phase behavior of the slow P -wave. $R_E^{Pr,Pr}$ is negative, because the reflected acoustic pressure and fluid displacement have opposite phase. $T_E^{Pf,Pf}$ approaches zero for $\theta=45^\circ$ because this angle is beyond the critical angle of incidence of the fast P -wave.

TABLE III. Seismoelectric reflection and transmission coefficients for $\omega = 10^6$ rad/s at $\theta=30^\circ$ and $\theta=45^\circ$.

	$\theta=30^\circ$	$\theta=45^\circ$
$T_E^{Pf,Pf}$	4.7389×10^{-1}	1.6454×10^{-7}
$T_E^{Ps,Ps}$	1.2478×10^{-1}	2.7971×10^{-1}
$T_E^{TM,TM}$	-5.0880×10^{-7}	-1.6581×10^{-6}
$T_E^{SV,SV}$	7.2674×10^{-2}	6.6128×10^{-1}
$T_E^{Pf,Ps}$	-2.6036×10^{-6}	3.2114×10^{-5}
$T_E^{Pf,TM}$	-3.3667×10^{-9}	4.4496×10^{-9}
$T_E^{Pf,SV}$	9.8734×10^{-7}	5.2354×10^{-5}
$T_E^{Ps,TM}$	7.1950×10^{-7}	2.4385×10^{-6}
$T_E^{Ps,SV}$	-1.2373×10^{-4}	1.9413×10^{-3}
$T_E^{TM,SV}$	1.3113×10^{-7}	8.4593×10^{-7}
$R_E^{E,E}$	1.3503×10^{-7}	5.4312×10^{-7}
$R_E^{Pr,Pr}$	-3.2877×10^{-1}	-5.6982×10^{-2}
$T_E - R_E^{E,E} - R_E^{Pr,Pr}$	1.0000×10^0	1.0000×10^0

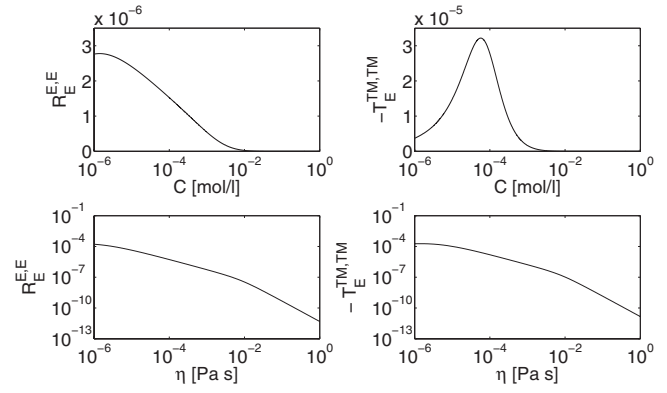


FIG. 4. Variation of $R_E^{E,E}$ and $T_E^{TM,TM}$ as a function of electrolyte concentration (C) and pore fluid viscosity (η) at $\omega=10^6$ rad/s and $\theta=45^\circ$.

B. Parameter variations

Seismoelectric conversions are generated at contrasts in (i) elastic properties (e.g., elastic moduli), (ii) fluid chemistry (pH and electrolyte concentration), and (iii) transport properties such as permeability (Haartsen and Pride, 1997). Electrolyte concentration (C), pore fluid viscosity (η), and permeability (k_0) are dominant seismoelectric conversion parameters while pH, pore fluid bulk modulus (K_f), porosity (ϕ), and tortuosity (α_∞) influence the seismoelectric conversion to a smaller extent (see also Haartsen and Pride, 1997; Mikhailov *et al.*, 2000; Garambois and Dietrich, 2002; Bordes *et al.*, 2006). We investigate the variation of $R_E^{E,E}$ and $T_E^{TM,TM}$ as a function of these parameters in Figs. 4–7 at fixed angular frequency and angle of incidence: $\omega=10^6$ rad/s and $\theta=45^\circ$.

In Fig. 4, $R_E^{E,E}$ decreases with increasing electrolyte concentration because the absolute value of the zeta-potential decreases, which reduces the electrokinetic coupling, as can be seen from Eq. (A4). $T_E^{TM,TM}$ peaks around $C = 10^{-4}$ mol/l. This behavior is related to the vertical electrokinetic energy flux, which depends on the TM-wave velocity. We found that the corresponding maximum attenuation of this wave is at $C=10^{-4}$ mol/l.

Both coefficients strongly decrease with increasing fluid viscosity. This is intuitively explained because a low viscosity will favor EM-wave generation as fluid circulation is enhanced, as was noted by Garambois and Dietrich (2002).

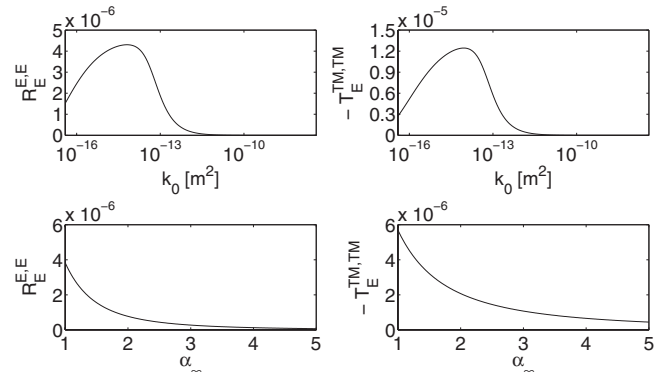


FIG. 5. Variation of $R_E^{E,E}$ and $T_E^{TM,TM}$ as a function of permeability (k_0) and tortuosity (α_∞) at $\omega=10^6$ rad/s and $\theta=45^\circ$.

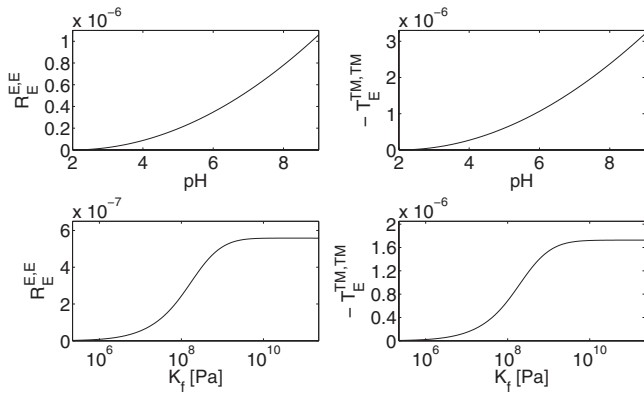


FIG. 6. Variation of $R_E^{E,E}$ and $T_E^{TM,TM}$ as a function of pH and bulk modulus of the pore fluid (K_f) at $\omega = 10^6$ rad/s and $\theta = 45^\circ$.

In the upper two graphs of Fig. 5 we see that seismoelectric coefficients vary over several orders of magnitude as a function of the permeability k_0 . This behavior is attributed to the influence of k_0 on the electrokinetic coupling coefficient $L(\omega)$. From Eq. (A3) we see that by increasing permeability we also increase the pore volume-to-surface ratio Λ , which has two competing effects on $L(\omega)$. Initially the length scale of Λ approaches the characteristic length of the electric double layer d so that $L(\omega)$ increases with increasing Λ . For permeabilities larger than 10^{-14} m² the influence of Λ^2 in the denominator of $L(\omega)$ becomes dominant so that the seismoelectric coefficients decrease. The effect of tortuosity on the seismoelectric coefficients is shown in the lower two graphs of Fig. 5. Tortuosity also influences $L(\omega)$, but its effect is smaller than that of k_0 because of its smaller variability. Again, increasing tortuosity effectively increases Λ . For these parameter value combinations, d/Λ is always negligible so that the seismoelectric coefficients decrease with increasing tortuosity.

$R_E^{E,E}$ and $T_E^{TM,TM}$ increase approximately linearly with pH (upper two graphs of Fig. 6), which is a result of the linear relation used for the zeta-potential [see Eq. (A5)]. Also $L(\omega)$ is directly proportional to ζ [see Eq. (A4)].

In the lower two graphs of Fig. 6, we notice that as the pore fluid becomes stiffer, seismoelectric coefficients $R_E^{E,E}$ and $T_E^{TM,TM}$ increase toward a limiting value for large pore fluid bulk moduli. This increase corresponds with a larger contrast in seismoelectric properties across the interface as the compressional wave velocities increase with increasing incompressibility of the pore fluid.

$R_E^{E,E}$ and $T_E^{TM,TM}$ as a function of porosity ϕ are shown in

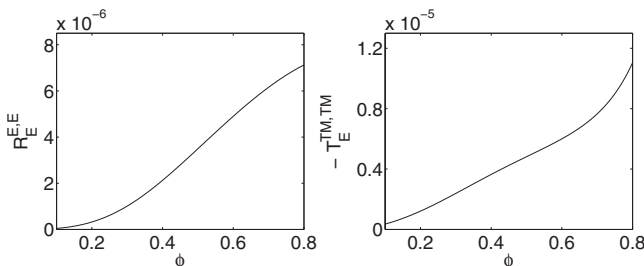


FIG. 7. Variation of $R_E^{E,E}$ and $T_E^{TM,TM}$ as a function of porosity (ϕ) at $\omega = 10^6$ rad/s and $\theta = 45^\circ$.

Fig. 7. The increase with porosity of both seismoelectric coefficients is explained by the fact that $L(\omega)$ depends approximately linearly on ϕ [see Eq. (A4)], implying that an increase in surface area increases electrokinetic coupling.

This sensitivity analysis indicates that for a water-saturated natural sandstone, electrolyte concentration, pore fluid viscosity, and permeability should be sufficiently low ($C \leq 10^{-3}$ mol/l, $\eta \leq 10^{-3}$ Pa s, and $k_0 \leq 10^{-12}$ m²), and the pH sufficiently high (say pH=7 or more) in order to have appreciable seismoelectric conversion.

V. CONCLUSIONS

The dispersion relations for seismoelectric wave propagation in poroelastic media were solved in terms of effective densities. This allows an elegant incorporation of both viscous and electrokinetic coupling in the effective densities. We solved the seismoelectric reflection and transmission coefficients at a fluid/porous-medium boundary. The dependence of the reflected and transmitted EM-waves due to an incident fluid wave on angle of incidence and frequency was shown. Recasting seismoelectric reflection and transmission coefficients to vertical energy fluxes, while accounting for orthodox and interference fluxes, it was asserted that energy conservation is obeyed. For a water/shallow-sandstone interface, seismoelectric conversion at 10 Hz is on the order 10^{-10} and at 500 kHz it is on the order 10^{-7} for energy flux coefficients. A sensitivity analysis was performed. Electrolyte concentration, pore fluid viscosity, and permeability were varied within realistic ranges and it was shown that this influenced seismoelectric conversions over several orders of magnitude. This sensitivity analysis allows the possibility to devise experiments in which the seismoelectric effect can be detected. Moreover, it allows us to appreciate the use of the electrokinetic effect as an exploration technique that is complementary to conventional seismic surveys for exploring hydrocarbon reserves.

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APPENDIX A: DYNAMIC COEFFICIENTS

The dynamic permeability $k(\omega)$ is given by (Johnson *et al.*, 1987)

$$k(\omega) = k_0 \left[\sqrt{1 + i \frac{\omega M}{\omega_t}} + i \frac{\omega}{\omega_t} \right]^{-1}, \quad (\text{A1})$$

where k_0 is permeability. The transition frequency ω_t and the similarity parameter M are given by

$$\omega_t = \frac{\phi \eta}{\alpha_\infty k_0 \rho_f}, \quad (\text{A2})$$

$$M = \frac{8\alpha_\infty k_0}{\phi \Lambda^2}, \quad (\text{A3})$$

where α_∞ is the tortuosity and Λ is a weighted pore volume-to-surface ratio. The electrokinetic coupling coefficient $L(\omega)$ is obtained from [Pride's \(1994\)](#) theory

$$L(\omega) = -\frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \epsilon_f \zeta}{\eta} \left(1 - 2\frac{d}{\Lambda}\right) \times \left[1 + 2i\frac{\omega}{M\omega_f} \left(1 - 2\frac{d}{\Lambda}\right)^2 \left(1 + d\sqrt{\frac{i\omega\rho_f}{\eta}}\right)^2\right]^{-1/2}, \quad (\text{A4})$$

where d is the Debye length (defined below) and ζ is the zeta-potential. A semi-empirical relationship between the zeta-potential, electrolyte concentration (C), and pH is used ([Pride and Garambois, 2005](#)) as follows:

$$\zeta = [0.010 + 0.025 \log_{10}(C)] \frac{\text{pH} - 2}{5}. \quad (\text{A5})$$

The dynamic conductivity $\sigma(\omega)$ is given by

$$\sigma(\omega) = \frac{\phi\sigma_f}{\alpha_\infty} \left[1 + \frac{2[C_{\text{em}} + C_{\text{os}}(\omega)]}{\sigma_f \Lambda}\right], \quad (\text{A6})$$

where σ_f is conductivity of the pore fluid, C_{em} is excess conductance associated with electromigration of excess charge, and $C_{\text{os}}(\omega)$ is excess conductance associated with electroosmosis. Expressions for σ_f , C_{em} , and $C_{\text{os}}(\omega)$ are

$$\sigma_f = \sum_{l=1}^{N_i} (ez_l)^2 b_l N_l, \quad (\text{A7})$$

$$C_{\text{em}} = 2d \sum_{l=1}^{N_i} (ez_l)^2 b_l N_l \left[\exp\left(-\frac{ez_l \zeta}{2k_B T}\right) - 1 \right], \quad (\text{A8})$$

$$C_{\text{os}}(\omega) = \frac{(\epsilon_0 \epsilon_f)^2 \zeta^2}{2d\eta} P \left[1 + \frac{2}{P} d \sqrt{\frac{i\omega\rho_f}{\eta}} \right]^{-1}, \quad (\text{A9})$$

$$P = \frac{8k_B T d^2}{\epsilon_0 \epsilon_f \zeta^2} \sum_{l=1}^{N_i} N_l \left[\exp\left(-\frac{ez_l \zeta}{2k_B T}\right) - 1 \right], \quad (\text{A10})$$

$$d = \sqrt{\left[\sum_{l=1}^{N_i} \frac{(ez_l)^2 N_l}{\epsilon_0 \epsilon_f k_B T} \right]^{-1}}, \quad (\text{A11})$$

where k_B is Boltzmann's constant, e is electron charge, T is temperature, and N_l is the amount of species l ions per volume. In this article we consider a binary symmetric electrolyte ($N_i=2$; $z_1=-z_2=1$; $N_1=N_2$) with mobilities b_1 and b_2 (see Table I).

APPENDIX B: ELEMENTS OF $\underline{\underline{A}}$

Applying the boundary conditions for the seismoelectric reflection and transmission problem yields a linear system

$$\underline{\underline{A}} \cdot (R^E, R^M, T^{Pf}, T^{Ps}, T^{TM}, T^{SV})^T = (k_3^{fl}, \phi\rho_f, 0, 0, 0, 0)^T, \quad (\text{B1})$$

where elements of matrix $\underline{\underline{A}}$ are

$$A(1,1) = 0,$$

$$A(1,2) = k_3^{fl},$$

$$A(1,3) = k_3^{Pf} [1 - \phi + \phi\beta^{Pf}(\omega)],$$

$$A(1,4) = k_3^{Ps} [1 - \phi + \phi\beta^{Ps}(\omega)],$$

$$A(1,5) = k_1 [1 - \phi + \phi\beta^{TM}(\omega)],$$

$$A(1,6) = k_1 [1 - \phi + \phi\beta^{SV}(\omega)], \quad (\text{B2})$$

$$A(2,1) = 0,$$

$$A(2,2) = -\phi\rho_f,$$

$$A(2,3) = [Q + R\beta^{Pf}(\omega)] s_{Pf}^2(\omega),$$

$$A(2,4) = [Q + R\beta^{Ps}(\omega)] s_{Ps}^2(\omega),$$

$$A(2,5) = 0,$$

$$A(2,6) = 0, \quad (\text{B3})$$

$$A(3,1) = 0,$$

$$A(3,2) = 0,$$

$$A(3,3) = k_1 k_3^{Pf},$$

$$A(3,4) = k_1 k_3^{Ps},$$

$$A(3,5) = k_1^2 - \frac{\omega^2}{2} s_{TM}^2(\omega),$$

$$A(3,6) = k_1^2 - \frac{\omega^2}{2} s_{SV}^2(\omega), \quad (\text{B4})$$

$$A(4,1) = 0,$$

$$A(4,2) = 0,$$

$$A(4,3) = k_1^2 - \omega^2 s_{Pf}^2(\omega) \frac{N_1(\omega)}{2G},$$

$$A(4,4) = k_1^2 - \omega^2 s_{Ps}^2(\omega) \frac{N_2(\omega)}{2G},$$

$$A(4,5) = -k_1 k_3^{TM},$$

$$A(4,6) = -k_1 k_3^{SV}, \quad (\text{B5})$$

$$\begin{aligned}
A(5,1) &= -\frac{1}{\mu_{fl}} s_E^2(\omega), \\
A(5,2) &= 0, \\
A(5,3) &= 0, \\
A(5,4) &= 0, \\
A(5,5) &= \frac{\alpha^{TM}(\omega)}{\mu} s_{TM}^2(\omega), \\
A(5,6) &= \frac{\alpha^{SV}(\omega)}{\mu} s_{SV}^2(\omega), \\
A(6,1) &= -k_3^E, \\
A(6,2) &= 0, \\
A(6,3) &= k_1 \alpha^{Pf}(\omega), \\
A(6,4) &= k_1 \alpha^{Ps}(\omega), \\
A(6,5) &= -k_3^{TM} \alpha^{TM}(\omega), \\
A(6,6) &= -k_3^{SV} \alpha^{SV}(\omega),
\end{aligned} \tag{B6}$$

where $N_1(\omega) = P - Q(1 - \phi) / \phi + [Q - R(1 - \phi) / \phi] \beta^{Pf}(\omega)$, $N_2(\omega) = P - Q(1 - \phi) / \phi + [Q - R(1 - \phi) / \phi] \beta^{Ps}(\omega)$, and $s_E(\omega)$ is the slowness of the electromagnetic wave in the fluid, which can be calculated using $s_E^2(\omega) = (\omega^2 \mu_{fl} \epsilon_0 \epsilon_{fl} - i \omega \mu_{fl} \sigma_{fl}) / \omega^2$ (Nabighian and Corbett, 1988) with upper fluid layer permeability, relative permittivity, and conductivity denoted by μ_{fl} , ϵ_{fl} , and σ_{fl} , respectively.

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Laboratory measurements and theoretical modeling of seismoelectric interface response and coseismic wave fields

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A full-waveform seismoelectric numerical model incorporating the directivity pattern of a pressure source is developed. This model provides predictions of coseismic electric fields and the electromagnetic waves that originate from a fluid/porous-medium interface. An experimental setup in which coseismic electric fields and interface responses are measured is constructed. The seismoelectric origin of the signals is confirmed. The numerically predicted polarity reversal of the interfacial signal and seismoelectric effects due to multiple scattering are detected in the measurements. Both the simulated coseismic electric fields and the electromagnetic waves originating from interfaces agree with the measurements in terms of travel times, waveform, polarity, amplitude, and spatial amplitude decay, demonstrating that seismoelectric effects are comprehensively described by theory. © 2011 American Institute of Physics. [doi:10.1063/1.3567945]

I. INTRODUCTION

Grain surfaces in contact with an electrolyte acquire a chemically bound surface charge that is balanced by mobile counter ions in a thin fluid layer surrounding the grains. The distribution of the mobile ions is determined by a balance between the electrostatic forces and thermal diffusivity. At the interface between the immobile and counter ions the so-called zeta-potential is defined. This electric potential decreases exponentially when one moves away from the interface. The characteristic decay length (the Debye length) is on the order of a few nanometers for typical grain-electrolyte combinations.¹ The ensemble of bound and mobile charge layers is referred to as the electric double layer.

When the pore fluid is in motion, the mobile counter ions in the double layer are transported by the flow with respect to the charge bound to the grain surface. Thus a seismic wave that travels through a porous medium generates an electric wave current on the scale of the wavelength. A comprehensive theory for coupled seismic and electromagnetic wave propagation identifying the zeta-potential as the key parameter was derived by Pride.¹ Two seismoelectric effects can be distinguished: (1) a coseismic electric field is confined to seismic waves and therefore propagates with seismic wave velocity² and (2) a seismic wave that traverses a boundary between two distinct porous media produces electromagnetic (EM)-waves that propagate outside the support of the seismic waves with much higher EM-wave speeds.^{2,3}

Following the terminology in Ref. 4 we refer to the latter as the interface response. Both coseismic and interface response effects are visible in full-waveform numerical simulations.^{3–7} Furthermore, a wide range of field and laboratory validations of the coseismic field and the interface response was presented in the literature.^{8–21} Several of these works compare either field measurements^{15,17} or laboratory measurements²⁰ with a seismoelectric wave propagation model, corroborating that the coseismic and interface response fields are predicted by the theory.

In this article we complement the study presented in Ref. 20 by developing a full waveform seismoelectric model for the coseismic and interface response fields based on the Sommerfeld integral, and comparing the numerical predictions with laboratory measurements. We demonstrate the seismoelectric nature of the measured electric potentials in a control experiment. The results are distinct from previous studies in the sense that measured seismoelectric multiples as well as a polarity reversal across an interface are predicted by our model and a typical directivity pattern of the acoustic source wavefield is incorporated in the simulation.

II. MODELING OF COSEISMIC AND INTERFACE RESPONSE FIELDS

We describe our modeling approach in the context of numerical simulations of our laboratory experiments. An isotropic homogeneous fluid-saturated porous medium layer is within a compressible fluid (see Fig. 1). The acoustic source pressure is modeled as a modified spherical wave source²²

$$\hat{p}(\omega, R, \theta) = \frac{A(\omega)D(\theta)}{R} \exp[-ikR], \quad (1)$$

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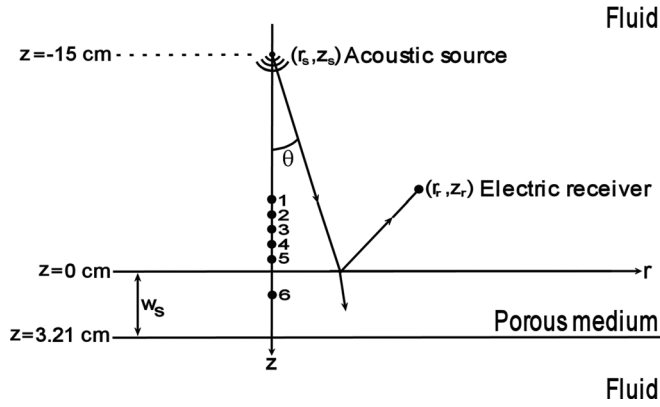


FIG. 1. Geometry of the seismoelectric model and measurements. The total electric potential wavefield is given by an integral over complex-valued θ [Eqs. (5) and (8)]. Positions 1–5 are separated by 0.5 cm with position 1 at $(r, z) = (0, -2.3)$ cm and position 5 at $(r, z) = (0, -0.3)$ cm. Position 6 is at $(r, z) = (0, 1)$ cm.

where ω is the radial frequency, $R = \sqrt{(r - r_s)^2 + (z - z_s)^2}$ is the distance to the source, θ the angle of incidence, $A(\omega)$ is the amplitude spectrum and k is the fluid wavenumber. The directivity function $D(\theta)$, that characterizes the radiation pattern of the source, is given by:

$$D(\theta) = \frac{J_1(ka \sin \theta)}{ka \sin \theta}. \quad (2)$$

Here, J_1 is the Bessel function of the first kind and first order and a is the radius of the transducer. The modeled source pressure wavefield is expanded into conical waves by means of the Sommerfeld integral.^{23,24} The total reflected electric potential in the fluid $\hat{\phi}(\omega, r_r, z_r)$, received at position (r_r, z_r) , can then be expressed as ($r_s = 0$; $z_s, z_r < 0$):

$$\hat{\phi}(\omega, r_r, z_r) = -iA(\omega) \int_0^\infty \frac{k_r}{k_z} D(k_r) J_0(k_r r_r) \exp[ik_z z_s] \times R^E(k_r) \exp[ik_z^E z_r] dk_r, \quad (3)$$

where $k_r = k \sin \theta$ and $k_z = k \cos \theta$ are the radial and vertical components of k , respectively, and k_z^E is the vertical component of the fluid EM wavenumber. The seismoelectric conversion coefficient, $R^E(k_r)$, relates the incident pressure wavefront to an EM signal in the fluid. This conversion coefficient was derived in Ref. 25 adopting full seismoelectric theory. Note that this coefficient is distinct from an ordinary reflection coefficient as it describes the conversion of mechanical energy into EM energy. Also note that the notation in Ref. 25 is slightly different (see Table I). Expressing k_r and k_z in terms of k and θ , Eq. (3) is written as,

TABLE I. Relation between seismoelectric conversion coefficients in this paper and those given in Ref. 25. The fluid density is denoted by ρ_f .

This paper	Reference 25
R^E	$R^E/\rho_f \omega^2$
T_f^{TM}	$\alpha^{TM} T_f^{TM}/\rho_f \omega^2$
αT_f^{Pf}	$\alpha^{Pf} T_f^{Pf}/\rho_f \omega^2$

$$\hat{\phi}(\omega, r_r, z_r) = -iA(\omega) \int_0^{\pi/2+i\infty} D(\theta) k \sin \theta J_0(kr_r \sin \theta) \times \exp[ikz_s \cos \theta] R^E(\theta) \exp[ik_z^E(\theta) z_r] d\theta, \quad (4)$$

where $k_z^E(\theta) = \omega \sqrt{1/c_E^2(\omega) - \sin^2 \theta/c_P^2}$ and $\text{Im}[k_z^E(\theta) < 0]$, with $c_E(\omega)$ and c_P the fluid EM and pressure wave velocities, respectively. The path of integration is along straight lines from 0 to $\pi/2$ and from $\pi/2$ to $\pi/2 + i\infty$ in the complex θ -plane. The second integral over complex θ is simplified further by employing the substitution $\theta = \pi/2 + i \ln[\sqrt{\gamma^2 + 1} + \gamma]$:

$$\hat{\phi}(\omega, r_r, z_r) = -\frac{iA(\omega)}{a} \int_0^{\pi/2} J_0(kr_r \sin \theta) J_1(ka \sin \theta) \times \exp[ikz_s \cos \theta] R^E(\theta) \exp[ik_z^E(\theta) z_r] d\theta + \frac{A(\omega)}{a} \int_0^\infty J_0(kr_r \sqrt{\gamma^2 + 1}) \frac{J_1(ka \sqrt{\gamma^2 + 1})}{\sqrt{\gamma^2 + 1}} \times \exp[kz_s \gamma] R^E(\gamma) \exp[ik_z^E(\gamma) z_r] d\gamma, \quad (5)$$

where Eq. (2) was used for $D(\theta)$. Equation (5) can readily be numerically evaluated and yields the total reflected EM wavefield in the fluid.

In the porous medium, pressure waves propagate accompanied by their coseismic fields. We take into account the coseismic field belonging to the pressure wave transmitted from the fluid to the porous medium, and the one belonging to the pressure wave that is subsequently reflected from the back of the sample. Only the (fast) pressure (Pf)-wave is considered as it carries virtually all of the acoustic energy. The acoustic waves generate freely propagating transverse magnetic (TM) EM-waves when they encounter interfaces. We take three subsequently generated interface responses into account. We incorporate these coseismic and interface response fields into the Sommerfeld integral. Analogously to Eq. (3) we arrive at the following expression for the electric potential within the porous medium ($r_s = 0$; $z_s < 0$ and $z_r > 0$),

$$\hat{\phi} = -iA \int_0^\infty \frac{k_r}{k_z} D J_0 \exp[ik_z z_s] \left\{ T_f^{TM} \exp[-ik_z^{TM} z_r] + T_f^{Pf} [\alpha \exp(-ik_z^{Pf} z_r) + \exp(-ik_z^{Pf} w_s) S^E] \right\} dk_r, \quad (6)$$

where the functional dependencies are not explicitly written. The width of the porous sample is denoted as w_s . The term $T_f^{TM} \exp[-ik_z^{TM} z_r]$ describes the propagation of the first TM-wave to the receiver at z_r . This wave is generated at $z = 0$ by the incident acoustic wave from the fluid. T_f^{TM} represents the transmission of converted electric potential due to incident acoustic potential at the front of the sample. The term $\alpha T_f^{Pf} \exp[-ik_z^{Pf} z_r]$ describes the coseismic electric field traveling along with the Pf-wave in the porous medium. The amplitude ratio between the electric and the pressure fields is described by α , and T_f^{Pf} is the acoustic transmission coefficient. The Pf-wave travels to the back of the porous sample where it is reflected upward. The associated seismoelectric effects are described by S^E ,

$$S^E = R_b^{TM} \exp[-ik_z^{TM}(w_s - z_r)] + R_b^{Pf} \{ \alpha \exp[-ik_z^{Pf}(w_s - z_r)] + \exp[-ik_z^{Pf}w_s] R_f^{TM} \exp[-ik_z^{TM}z_r] \}. \quad (7)$$

The first two terms in Eq. (7) describe the upwardly propagating second TM and coseismic waves, respectively. The last term describes the third interface response that is generated at $z=0$ by the incident Pf-wave from below and now propagates downward as a TM-wave to the receiver. The seismoelectric conversion coefficients at the back and the front of the sample are denoted as R_b^{TM} and R_f^{TM} , respectively. The reflection coefficient for the Pf-wave at the back of the sample is denoted R_b^{Pf} . These coefficients at a fluid/porous-medium interface due to an incident Pf-wave are derived in a procedure similar to the one described in Ref. 25. The notations are slightly different from those in Ref. 25 (see Table I). In a similar way as for Eq. (5), we find from Eq. (6):

$$\begin{aligned} \hat{\phi} = & -\frac{iA}{a} \int_0^{\pi/2} J_0 J_1 \exp[ikz_s \cos \theta] \\ & \times \left\{ T_f^{TM} \exp[-ik_z^{TM}z_r] + T_f^{Pf} [\alpha \exp(-ik_z^{Pf}z_r) \right. \\ & \left. + \exp(-ik_z^{Pf}w_s) S^E] \right\} d\theta \\ & + \frac{A}{a} \int_0^\infty J_0 \frac{J_1}{\sqrt{\gamma^2 + 1}} \exp[kz_s \gamma] \\ & \times \left\{ T_f^{TM} \exp[-ik_z^{TM}z_r] + T_f^{Pf} [\alpha \exp(-ik_z^{Pf}z_r) \right. \\ & \left. + \exp(-ik_z^{Pf}w_s) S^E] \right\} d\gamma, \end{aligned} \quad (8)$$

TABLE II. Parameters of the porous medium. The temperature is 293.15 K.

Bulk modulus skeleton grains ^a	49.9×10^9	Pa
Bulk modulus (pore) fluid ^b	2.2×10^9	Pa
Pore fluid viscosity ^b	1×10^{-3}	Pa s
(pore) Fluid density ^b	998	kg/m ³
Relative permittivity of the (pore) fluid ^b	80.1	—
Relative permittivity of the solid ^b	4	—
(fluid) Magnetic permeability ($= \mu_0$)	$4\pi \times 10^{-7}$	H/m
Weighted pore volume-to-surface ratio ^c	9.43×10^{-6}	m
Bulk modulus framework of grains ^d	0.93×10^9	Pa
Shear modulus framework of grains ^d	0.88×10^9	Pa
Porosity of the porous medium ^d	0.52	—
Solid density ^d	2570	kg/m ³
Permeability ^d	3.4×10^{-12}	m ²
Tortuosity ^d	1.7	—
(pore) Fluid conductivity ^c	4.8×10^{-2}	S/m
Zeta-potential ^f	-4×10^{-2}	V
Sample width ^e	3.21×10^{-2}	m

^aReference 26.

^bReference 27. We take the value of Pyrex 7070 glass for the solid permittivity.

^cReferences 1, 28–30.

^dSee N5B in Ref. 31.

^eMeasured values.

^fSee Ref. 32. We assume that conductivity is due to a NaCl salt solution and pH = 6.

where Eq. (2) was used for $D(\theta)$. Equation (8) describes the electric potential within the porous medium. Equations (5) and (8) are numerically evaluated through the inverse fast Fourier transform. For each frequency we use a recursive adaptive Simpson quadrature algorithm implemented in MATLAB. An experimentally recorded pressure waveform at $(r, z) = (0, 0)$ is used for $A(\omega)$ and a 144–896 kHz numerical band-pass filter is applied. The input parameters of Table II are used. Figure 2 (left) shows the resulting modeled reflected electric potentials at positions 1–6 along the z -axis (Fig. 1).

An electric pulse is computed around 0.10 ms at each position, which corresponds to the travel time of the acoustic wave from the source to the porous medium surface, where it generates an EM signal. Due to the large fluid EM-wave speed ($\sim 1.0 \times 10^7$ m/s), the electric pulses arrive almost simultaneously at each position. Note that the pulse decays in amplitude as the distance from the interface increases (positions 1–5 in Fig. 1). Position 6 is inside the porous medium. The first pulse arriving around 0.10 ms at this position has a reversed polarity with respect to the fluid computations. This observation is in agreement with the description of the interface response modeled as the radiation from an oscillating electric dipole oriented vertically with respect to the interface.^{3,15} The second pulse at position 6 arrives 5.7 μ s later. It travels along with the Pf-wave at

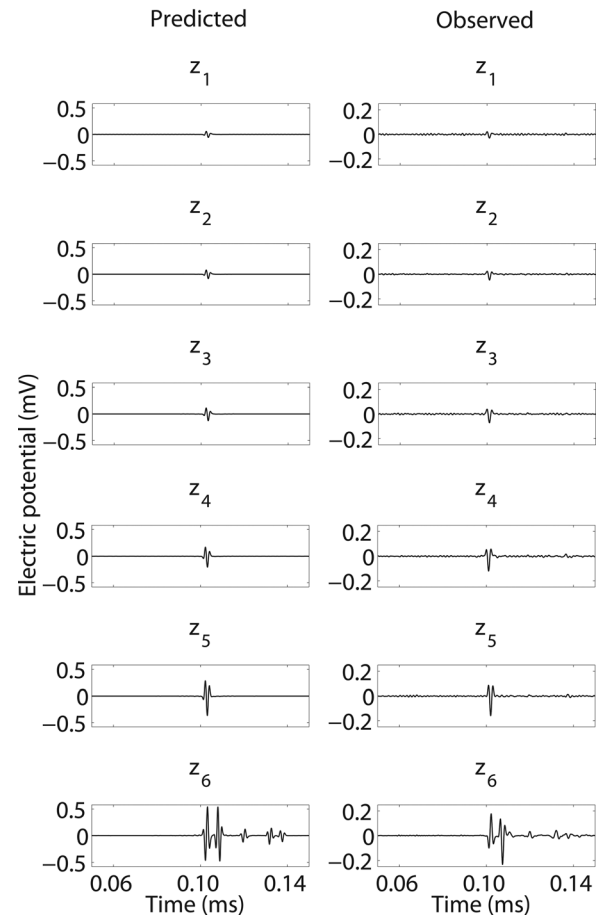


FIG. 2. Numerical electric potential waveform predictions (left) and measurements (right) within the fluid (positions 1–5) and within the porous sample (position 6).

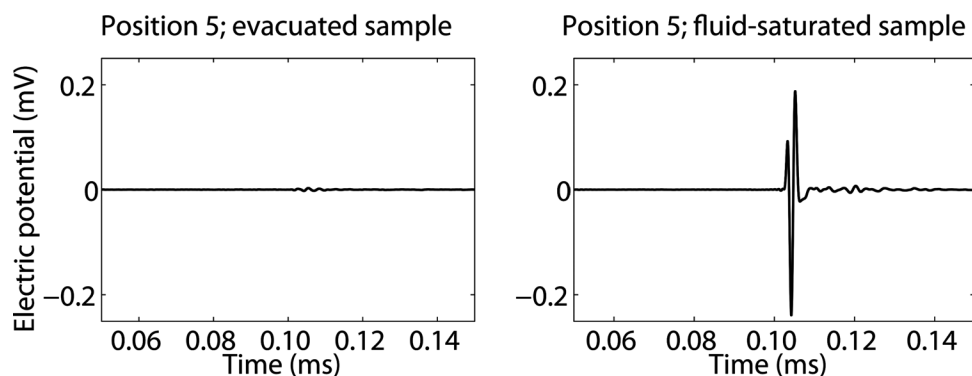


FIG. 3. Electric potential measurement at position 5 on evacuated (left) and fluid-saturated (right) porous sample.

approximately 1900 m/s from the front of the sample to position 6. Therefore it is due to the coseismic electric field supported by the Pf-wave. The pulse arriving around 0.12 ms corresponds with the Pf-wave travel time to the back of the porous sample, where it generates a reflected EM-wave. The fourth and fifth pulses are, respectively, coseismic and interface responses of the compressional wave multiple.

III. SEISMOELECTRIC EXPERIMENTS

A. Seismoelectric experimental setup and measurements

The experimental setup consists of a $58 \times 39 \times 28$ cm water tank in which an acoustic wave source transducer (Panametrics V3638), grounded electrodes (A-M Systems, Inc., Ag/AgCl diameter 0.015 in.), and a porous sample are installed. The geometry of the experiment is depicted in Fig. 1. The sample is made of sintered crushed glass (parameters listed in Table II), carefully saturated, and immersed in tap water. The electrode at position 6 is installed in a 1.4 mm diameter cylindrical hole, which was drilled in the porous sample. A 500 kHz single sine pulse (with a 500 mV peak-to-peak amplitude) from a waveform generator (Agilent Technologies 33220A) is fed into an amplifier (ENI 2100L RF Power Amplifier) set at 50 dB gain. The output signal is coupled into a piezoelectric source transducer. A 144–896 kHz band-pass filter is applied in all measurements. Electric potential measurements were performed at positions 1–6. These on-axis measurements are averaged over 8192 source pulses and shown in Fig. 2 (right). Indeed, the measured seismoelectric pulse patterns are predicted by the model.

The predicted electric potentials (Fig. 2, left) agree with the measurements (Fig. 2, right) in terms of travel times, waveform, polarity, and the spatial amplitude decay. Note that the panels on the right are scaled down by a factor of 2.5. This means that the measured amplitudes are lower than predicted, which can be explained by uncertainties in the parameters given in Table II. Notably, our assumption on the zeta-potential may not be fully appropriate. Identical amplitude scaling is present in all the reflected interface response signals at positions 1–5 and the transmitted signal at position 6 that contains coseismic and interface response effects. This means that, by introducing a scaling factor, all coseismic and interface response effects are correctly described by theory.

The small remaining differences in waveform are probably related to minor geometric misalignment in the experiments (the transducer-porous sample-electrode geometry deviates slightly from the model geometry) and to the ruggedness of the porous sample surface. Also, the predicted polarity reversal across the fluid/porous-medium interface (positions 1–5 versus position 6) is clearly visible in the measurements. In Ref. 13 a polarity reversal was also reported in a borehole penetrating the water table. Faint arrivals after the first interface response signal are observed in the recordings at positions 4–5. These do not appear in the computed signals, because only the first interface response is modeled for the fluid positions.

B. Seismoelectric control experiment

We also performed a control experiment to identify seismoelectric conversion as the single mechanism for the electric potential generation. A sample was dried and jacketed before it was immersed in water. A vacuum pump continuously evacuated the sample. Figure 3 (left) shows the measurement at position 5. No appreciable signal can be detected by the electric receiver. Subsequently, the jacketed sample was carefully saturated with de-ionized water and the seismoelectric experiment was repeated. Figure 3 (right) shows the result. Now a clear pulse arrives around 0.10 ms. This experiment unambiguously identifies seismoelectric conversion as the mechanism for electric signal generation. In Refs. 10 and 20 similar conclusions are drawn from different seismoelectric experiments. It is well-known that solids may generate electric effects when under strain, which is referred to as piezoelectricity. This effect however, cannot explain the observations in Fig. 3, as any piezoelectricity would be screened by the electrolyte.²

IV. CONCLUSIONS

A seismoelectric numerical model incorporating a typical directivity pattern of the acoustic source wavefield provided full-waveform numerical predictions for both the coseismic and interface response effects. The existence of the seismoelectric coupling phenomenon was validated in our experimental setup. Measurements of the coseismic electric field and the fluid/porous-medium interface response were performed. Agreement between the predicted and

measured wavefields was shown in terms of travel times, waveform, polarity, and the spatial amplitude decay. Furthermore, the predicted polarity reversal and Pf-wave multiple scattering effects were validated by the measurements. The unresolved scaling issue is probably primarily caused by variations in pore fluid conductivity and the zeta-potential that are inaccessible to direct independent measurements. Yet, the overall agreement between the measurements and numerical predictions convincingly validates the existing seismoelectric theory. As both the numerical and experimental techniques can be generalized to field situations, this work allows us to appreciate the use of seismoelectric effects as exploration techniques that are complementary to conventional seismic surveys for exploring the subsurface.

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Seismoelectric interface electromagnetic wave characteristics for the finite offset Vertical Seismoelectric Profiling configuration: Theoretical modeling and experiment verification

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Abstract: The seismoelectric interface electromagnetic characteristics have been studied for the finite offset Vertical Seismoelectric Profiling (VSEP) configuration. The approach consists of theoretical modeling and laboratory verification. The results show that the wave variation characteristics along the finite offset measurement line are markedly different from those along the zero-offset line. More interestingly, the wave characteristics for both configurations can be satisfactorily explained by the electric dipole model for the seismoelectric interface wave radiation. Besides, the experiment confirms the modeling result based on the seismoelectric coupling theory and validates the VSEP technique as an effective method for subsurface interface delineation.

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1. Introduction

Subsurface porous rocks contain electric double layers which, when deformed by passing seismic/acoustic waves, will result in relative movement of the charged ions and generate electromagnetic (EM) signals. This phenomenon is known as seismoelectric effect (Pride, 1994; Pride and Haartsen, 1996). In general, the seismoelectric effect induces two kinds of electric signals. One is the coseismic electric field signal that travels along with the seismic wave. The coseismic signal can only provide localized information around the receivers (Zhu and Toksöz, 2003). Another seismoelectric signal is the interface EM wave, which is generated when the seismic wave impinging onto an elastic or electric interface in the porous medium. Examples of such an interface include rock formation boundary, oil/water contact, and fracture, etc. The seismoelectric interface EM wave propagates or radiates away from the interface with a velocity near the speed of light (Zhu and Toksöz, 2003, 2005; Gao *et al.*, 2017). For such interfaces, the seismoelectric interface EM signal can locate them with the seismic resolution and detect the fluid content using the EM property. Thus, this type of EM signal has great potential in various geophysical applications, such as hydrocarbon exploration and production, ground water contamination, and permafrost interface detection, etc. (Hu *et al.*, 2007; Revil and Jardani, 2010).

Because the seismoelectric interface EM wave is a weak signal and has a relatively low signal-to-noise ratio, the Vertical Seismoelectric Profiling (VSEP) configuration has been used both in the field and laboratory measurements (Dupuis *et al.*, 2009; Schakel *et al.*, 2011; Peng *et al.*, 2017). Similar to Vertical Seismic Profiling, in the VSEP measurements the receiving electrodes are close to the interface of the radiation EM signal generation; this configuration minimizes the propagation loss and can thus improve the signal-to-noise ratio relative to the surface measurements. Depending on

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the seismic source position relative to the receiving electrodes, the VSEP can be conducted in zero offset and finite offset configurations. Using the finite offset VSEP, Dupuis *et al.* (2009) have conducted the field measurement and analyzed the seismoelectric EM wave amplitude characteristics. In their measurement, the source and electrodes are located at the same side of the interface. For this configuration, however, the first arrival is the coseismic wave which causes a strong interference with the latter small interface EM wave arrival. The interference would be worse in the VSEP measurement where the electrode is close to the interface. Consequently, for the solid-porous model, we can get measurement results at only one side of the interface. Schakel *et al.* (2011) and Peng *et al.* (2017) have measured the response characteristics of the seismoelectric interface EM wave using the fluid-porous model in the laboratory. Because there is no seismoelectric coupling in water, the fluid-porous model can avoid the interference caused by the large amplitude coseismic waves and allows for measuring the pure seismoelectric interface EM wave. Following the work of these researchers for the zero-offset VSEP, this study conducts research for the finite offset VSEP configuration, because the latter configuration is more relevant for subsurface geophysical measurements.

In what follows, we first model the seismoelectric interface EM wave response for the fluid-porous model using the Pride (1994) theory for seismoelectric coupling. The resulting amplitude characteristics of the signals resemble those of an electric dipole at the interface. To verify the theoretical result, we conduct a controlled laboratory experiment and find that the measurement and the theory are in good agreement. The result of this work not only helps understand the mechanism of the seismoelectric interface phenomenon, but also shows that the finite offset VSEP is an effective way for determining a subsurface interface.

2. Theoretical modeling

2.1 Theory

Pride governing equations (Pride, 1994) are widely used in seismoelectric modeling. The modeling results have been validated in numerous laboratory and field experiments (e.g., Schakel *et al.*, 2011; Dupuis *et al.*, 2009). We use the theory to simulate the seismoelectric interface EM wave response of the fluid-porous sandstone interface due to the incidence of an acoustic wave. It should be noted that, unlike the point acoustic source used by Schakel *et al.* (2011) and Gao *et al.* (2017), we use a finite width pulse acoustic source in order to compare the modeling with experiment. The source can radiate the acoustic beam with a sharp directivity similar to that of the acoustic transducer used in the experiment. This finite width pulse source can be adjusted to generate only the interface EM wave for the analysis, eliminating the interference from the coseismic signals, which would otherwise be generated at the fluid-porous model if a point source is used (Peng *et al.*, 2017).

Figure 1 is the seismoelectric model of the fluid-porous sandstone with a finite width pulse acoustic source. In the Cartesian coordinate system, the acoustic source is located at $(0, z_s)$ in the fluid region. The receiving electrodes are evenly distributed along a measurement line penetrating the interface; the horizontal offset of the receiver line is D , indicating the finite offset configuration. Zhang (2004) analyzed this configuration for the elastic wave problem. Following their formulation, we analyze the seismoelectric EM wave using the following equations:

$$\varphi_r(x, z) = \iint A(k, \omega) R^E(k, \omega) \exp[-i(k_z^i z_s - k_z^{Er} z + kx - \omega t)] dk d\omega, \quad (1)$$

$$\varphi_t(x, z) = \iint A(k, \omega) T^E(k, \omega) \exp[-i(k_z^i z_s + k_z^{Et} z + kx - \omega t)] dk d\omega, \quad (2)$$

where φ_r and φ_t are, respectively, the electrical potentials in the fluid and porous sandstone; x and z are the horizontal and vertical coordinates of the receiver; z_s is the distance of the acoustic source to the interface. $k_1 = \omega/V_f$ is the fluid acoustic wavenumber, and V_f is the acoustic velocity. k and $k_z^i = \sqrt{k_1^2 - k^2}$ are the horizontal and vertical acoustic wavenumbers, respectively. $A(k, \omega) = \exp[-(k - k_0)^2/k_b^2] * S(\omega)$ is the acoustic source frequency-wavenumber spectral function, which controls the source frequency, radiation angle, and radiation beam width; $S(\omega)$ is the acoustic source spectrum and ω is the angular frequency; k_z^{Er} and k_z^{Et} denote the vertical wavenumber of the EM wave in the fluid and the porous rock, respectively; $R^E(k, \omega)$ and $T^E(k, \omega)$ are, respectively, the reflection and transmission coefficients of the seismoelectric interface EM wave, which can be calculated according to Schakel and Smeulders (2010).

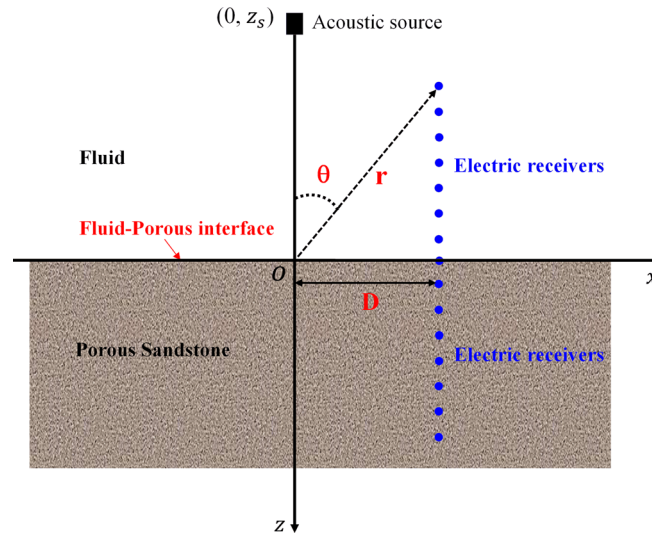


Fig. 1. (Color online) Finite offset VSEP seismoelectric model with a finite width pulse acoustic source.

For a wave incident angle θ , $k_0 = k_1 \sin(\theta)$ determines the radiation direction of the acoustic source; k_b determines the radiation acoustic beam width in space. The role of the exponential function in $A(k, \omega)$ is to insert a two-dimensional bandpass region into the frequency-wavenumber domain of the seismoelectric response function, such that the interface EM signal that falls into this region will be included, whereas other coseismic signals, because their response spectral distribution is outside this region, will be excluded.

2.2 Modeling results and analysis

Using the above-mentioned seismoelectric theory and modeling method, we model the characteristics of the seismoelectric interface EM wave for the finite offset VSEP. The elastic and electrical parameters of the fluid and porous sandstone used in the simulation can be found in [Schakel and Smeulders \(2010\)](#). For the VSEP configuration, the source to interface distance $z_s = 80$ mm and the receiver line offset $D = 45$ mm. The receiving electrodes, as shown in Fig. 1, are evenly spaced along the line from -100 mm to 100 mm, with 1 mm spacing. We set the incident angle $\theta = 0^\circ$ and the radiation width $k_b = 80 \text{ mm}^{-1}$ to make the acoustic beam vertically impinge onto the fluid-porous sandstone. A Ricker wavelet with a center frequency 500 kHz is used for the source function.

Figure 2(a) shows the calculated EM signals along the receiving line. They are colored black, blue, and red when the receiver is above, at, and below the interface, respectively. Three phenomena can be observed from the modeling result. The first is that all signals arrive at a constant time of $T_0 = 52.5 \mu\text{s}$, which is the acoustic travel time from the source to the interface, indicating that they are the seismoelectric interface EM waves generated by the incident acoustic wave at the interface. The second is the polarity reversal for the signals above and below the interface, which is accompanied by the gradual amplitude variation across the interface. The third phenomenon is that the signal amplitude exhibits a non-monotonic variation on each side of the interface.

The first two phenomena of our finite offset VSEP modeling are similar to those of the zero-offset VSEP experiment ([Schakel et al., 2011](#); [Peng et al., 2017](#)). It is the third phenomenon of our modeling that differs greatly from the zero-offset experimental result. (The latter result shows that the signal amplitude monotonically increases toward the interface.) The following analysis is done to reconcile both results.

According to Pride seismoelectric theory ([Haartsen and Pride, 1997](#)), the impinging acoustic wave onto the interface of Fig. 1 will generate a vertical electric dipole at the origin O, thereafter the dipole radiates EM waves outward into the whole space with a dipole radiation directivity given by (see [Jackson, 1999](#))

$$\varphi = f(\mathbf{P}, \varepsilon) \frac{\cos(\theta)}{r^2}, \quad (3)$$

where φ is the electric potential as a function of distance r (measured from origin) and radiation angle θ (measured from the z axis); $\mathbf{P}$ and ε are, respectively, the electric

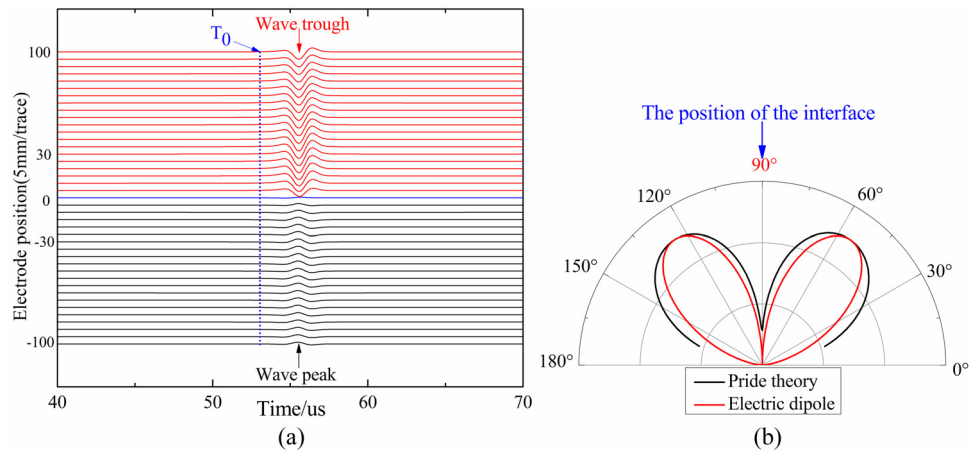


Fig. 2. (Color online) (a) Simulated waveforms of the seismoelectric interface EM wave when the electrodes are located at different positions of the fluid-porous sandstone interface. (b) The modeled (black) electrical potential amplitude variation with the angle θ for the VSEP configuration compared with that of the electric dipole model (red).

dipole moment and the dielectric constant of the medium. For a given medium, $f(\mathbf{P}, \varepsilon)$ is treated as a constant, and the electrical potential radiated from the electric dipole varies with θ and r only, as

$$\varphi \propto \frac{\cos(\theta)}{r^2}. \quad (4)$$

The electric dipole model explains the modeling result of Fig. 2(a) quite well. For the interface located at $\theta = 90^\circ$, the $\cos(\theta)$ function changes signs across the interface. This explains the polarity reversal phenomenon of Fig. 2(a). The modeling data amplitude, after normalization to eliminate the $f(\mathbf{P}, \varepsilon)$ parameter, is compared with that of Eq. (4) in Fig. 2(b), where θ and r for each electrode position can be easily calculated for the configuration in Fig. 1. Figure 2(b) shows that the full modeling result (black) agrees with the dipole model (red) reasonably well. (The slightly fatter pattern of our modeling is presumably caused by the acoustic beam source in the modeling).

The electric dipole model can also explain the result of the zero-offset configuration (Schakel *et al.*, 2011; Peng *et al.*, 2017). For the zero-offset, the source and electrodes are along the same line specified by $\theta = 0^\circ$ [$\cos(\theta) = 1$], such that $\varphi \propto 1/r^2$, resulting in a monotonic increase of signal amplitude toward the interface as $r \rightarrow 0$.

3. Experimental verification

3.1 Experimental apparatus and methods

An experiment was conducted to verify the above theoretical modeling result. The experimental setup is shown in Fig. 3, where a $150 \times 150 \times 110$ mm sandstone block is immersed in a $400 \times 400 \times 400$ mm water tank. The porosity and permeability of the sandstone are 19.7% and 310 mD, respectively. The p -wave velocity of the water-saturated sandstone is 3036 m/s at 500 kHz. The water conductivity is 0.5 mS/cm at

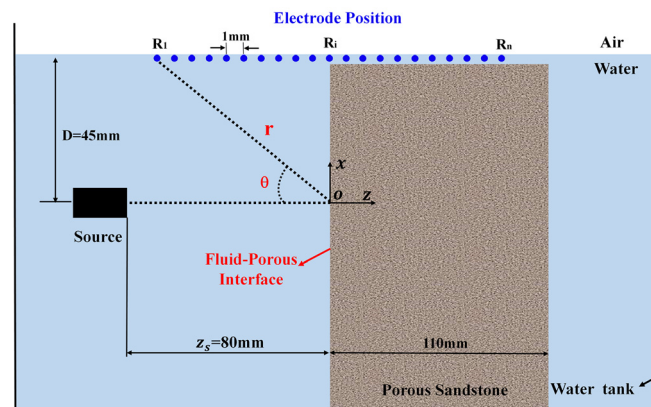


Fig. 3. (Color online) The side view of the experimental measurement setup.

20.2 °C. A 500 kHz center frequency Panametrics, Inc., (Waltham, MA) V391 ultrasonic transducer of 1.125 in. diameter and an A-M system 531 500 electrode (0.008 in. diameter) were used to actuate and receive the seismoelectric signal. The measurement system also includes an Agilent DSO-X-6004A oscilloscope, an Agilent (Santa Clara, CA) 33500B waveform generator, a 50 dB gain Electronic and Innovation (Rochester, NY) 2011L power amplifier, and a 20 dB gain Picotest J2180A preamplifier. In the setup of Fig. 3, the distance between the source and fluid-solid interface is fixed at 80 mm. The distance between the receiving line and the source is $D=45$ mm. A 500 kHz single cycle sine wave with peak to peak amplitude of 300 V is loaded on the source transducer to actuate the incident acoustic wave.

The electrode measurement positions, which are similar to those in Fig. 1, are shown in Fig. 3. The first electrode position R_1 is placed at 70 mm in front of the interface, the last position R_n is placed at 105 mm behind the interface, and R_i is right at the interface position. By varying the measurement position at 1 mm increment, the angle θ , as defined in Fig. 3, varies from $\theta < 90^\circ$ to $\theta > 90^\circ$, with $\theta = 90^\circ$ coinciding with the interface.

3.2 Experimental results and analysis

To verify that the measured electric signal is the seismoelectric EM signal from the interface, three measurements were made. The first was done by removing the sandstone and placing a Panametrics V391 ultrasonic receiver at the origin O facing the source transducer. The measured acoustic signal serves as a reference signal. The other two measurements were made with the electrode at the R_1 position in the presence and absence of the sandstone rock, respectively. The measurement results are shown in Fig. 4(a). It is clear that no electric signal is measured in the absence of the sandstone. In the presence of the rock, the electric (top) and acoustic (bottom) signals occur at the same arrival time of $T_0 = 52.5 \mu\text{s}$, suggesting that the electric signal is generated instantly when the acoustic signal strikes the interface. The former is therefore the seismoelectric EM interface signal radiated from the origin O at the interface at the speed of light in water.

Figure 4(b) shows the measured seismoelectric interface EM signals along the receiving line from left to right (only the section of -70 to 70 mm is shown, at a 5 mm increment). The upper (red) and lower (black) part of the signals were measured for

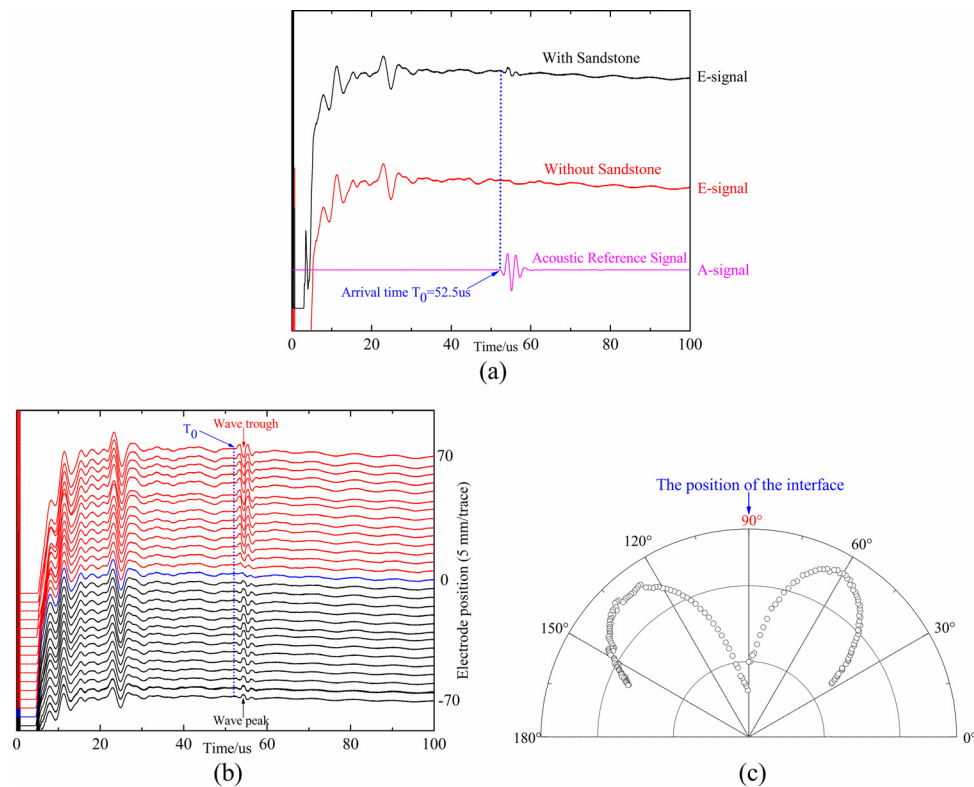


Fig. 4. (Color online) (a) Validation test at receiving position R_1 , where the pink (bottom), red (middle), and black (top) signals are the measured acoustic reference signal, electric signal in the absence, and electric signal in the presence, of the sandstone rock, respectively. (b) Measured seismoelectric interface EM waveforms when the electrode is at the different positions of the measurement line. (c) Normalized signal amplitude of (b) versus the angle θ of the electrode position.

positions on right and left of the interface, respectively. For each measurement position, the angle θ and distance r to the origin can be calculated, and the signal amplitude [normalized for each side of the interface to remove the $f(\mathbf{P}, \varepsilon)$ parameter of Eq. (3)] is plotted in Fig. 4(c). (Note the data point in this figure is plotted for every measurement position at an increment of 1 mm.) The pattern of Fig. 4(c) resembles its theoretical counterpart very well. The experiment result is further discussed below.

First, as shown in Fig. 4(b), all signals occur simultaneously at a constant arrival time of $T_0 = 52.5 \mu\text{s}$. For the same reason used to explain in Fig. 4(a), these are the piezoelectric EM waves radiated from the interface. The phase and amplitude variation, especially the polarity reversal phenomenon across the interface, compare very well with the theoretical modeling results in Fig. 2(a). We now analyze the pattern in Fig. 4(c). The peak amplitude position on the left and right side of the interface occur at $\theta = 60^\circ$ and $\theta = 127.5^\circ$, respectively. In comparison, the theoretical value for these two positions are $\theta = 55^\circ$ and $\theta = 125^\circ$, in good agreement with the experimental results. We can therefore conclude that our experiment has verified the piezoelectric EM interface wave characteristics predicted by the Pride (1994) piezoelectric coupling theory.

4. Conclusion

We have studied the piezoelectric interface EM wave phenomenon using the theoretical modeling and laboratory experiment for the finite offset VSEP configuration. We found that the wave variation characteristics along the finite offset measurement line are markedly different from those of the previous zero-offset measurement results. However, the difference can be satisfactorily explained by the electric dipole model for the piezoelectric interface EM radiation. Our experimental result confirms the theoretical modeling result and suggests that the VSEP is a valid method for subsurface interface delineation.

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Seismoelectric interface response: Experimental results and forward model

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ABSTRACT

Understanding the seismoelectric interface response is important for developing seismoelectric field methods for oil exploration and environmental/engineering geophysics. The existing seismoelectric theory has never been validated systematically by controlled experiments. We have designed and developed an experimental setup in which acoustic-to-electromagnetic wave conversions at interfaces are measured. An acoustic source emits a pressure wave that impinges upon a porous sample. The reflected electric-wave potential is recorded by a wire electrode. We have also developed a full-waveform electrokinetic theoretical model based on the Sommerfeld approach and have compared it with measurements at positions

perpendicular and parallel to the fluid/porous-medium interface. We performed experiments at several salinities. For 10^{-3} and 10^{-2} M sodium chloride (NaCl) solutions, both waveforms and amplitudes agree. For 10^{-4} M NaCl, however, amplitude deviations occur. We found that a single amplitude field scaling factor describes these discrepancies. We also checked the repeatability of experiments. The amplitudes are constant for the duration of an experiment (1–4 hours) but decrease on longer time scales (~ 24 hours). However, the waveforms and spatial amplitude pattern of the electric wavefield are preserved over time. Our results validate electrokinetic theory for the seismic-to-electromagnetic-wave conversion at interfaces for subsurface exploration purposes.

INTRODUCTION

When the grain surfaces of soils and rocks are in contact with a fluid electrolyte, they typically acquire a chemically bound surface charge that is balanced by mobile counterions in a thin fluid layer surrounding the grains. The bound charge is immobile, whereas the counterions can move with the fluid. At the interface between the immobile and counterions, the so-called zeta potential is defined. A comprehensive theory for coupled seismic and electromagnetic (EM) wave propagation with the zeta potential as a key parameter is derived by [Pride \(1994\)](#). From research based on this electrokinetic theory, one can distinguish two seismoelectric effects: (1) a coseismic (electric) field is coupled to seismic waves and therefore propagates with seismic wave velocity (e.g., [Pride and Haartsen, 1996](#)) and (2) a seismic wave that traverses an interface with a contrast in electrical or mechanical properties produces EM waves that propagate outside the support of the seismic waves with much higher EM wavespeeds (e.g.,

[Pride and Haartsen, 1996](#); [Haartsen and Pride, 1997](#)). These effects have been verified experimentally in the laboratory (e.g., [Zhu et al., 1999, 2000](#); [Zhu and Toksöz, 2003](#); [Block and Harris, 2006](#); [Bordes et al., 2006](#)) and in the field (e.g., [Long and Rivers, 1975](#); [Butler et al., 1996](#); [Mikhailov et al., 1997](#); [Beamish, 1999](#); [Garambois and Dietrich, 2001](#); [Dupuis et al., 2007](#); [Haines et al., 2007](#)). Also, the coseismic magnetic fields associated with a Stoneley wave ([Zhu and Toksöz, 2005](#)) and with a shear wave have been measured ([Bordes et al., 2006, 2008](#)).

Electrokinetic theory combines poroelastic and EM theory through coupling in flux/force relations. The so-called dynamic electrokinetic coupling coefficient that occurs in the flux/force relations has been validated experimentally ([Reppert et al., 2001](#)). Seismoelectric exploration can combine seismic resolution and the sensitivity to pore fluids of EM methods ([Haines et al., 2007](#)). [Garambois and Dietrich \(2001\)](#), [Block and Harris \(2006\)](#), and [Dupuis and Butler \(2006\)](#) compare measured and theoretically predicted coseismic electric field amplitudes.

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The numerically predicted amplitude-decay pattern of the seismoelectric conversion at an interface (e.g., Haartsen and Pride, 1997; Garambois and Dietrich, 2002; Haines and Pride, 2006) has been observed by several workers (e.g., Butler et al., 1996; Mikhailov et al., 1997; Dupuis et al., 2007). Mikhailov et al. (1997) compare field measurements of this pattern as a function of horizontal distance with a numerical model and find reasonable agreement, and Charara et al. (2009) model and measure seismoelectric signals at a fluid/porous-medium interface in a laboratory experiment. However, Charara et al. do not consider the spatial variation of reflected waveform and amplitude.

In this article, we test the predictive power of electrokinetic theory for the seismoelectric interface response. Laboratory measurements and modeling on the basis of the Sommerfeld approach and Pride's electrokinetic theory are presented. Measurements are conducted at several positions in a water tank in which a fluid-saturated porous sample is immersed. Excellent agreement between theory and experiments for waveforms and spatial amplitude pattern is found. Measured amplitudes are shown to be dependent on salinity. Only a single amplitude field scaling factor is necessary to find absolute agreement. Our research validates electrokinetic theory for the seismoelectric interface response.

SEISMOELECTRIC EXPERIMENTAL SETUP

The experimental setup consists of a $58 \times 39 \times 28$ cm water tank in which a 1.125 inch diameter P-wave Panametrics source transducer (model V3638, 500-kHz center frequency), a 380 μ m diameter A-M Systems silver/silver chloride (Ag/AgCl)

electrode, and a Technoglas porous sample (P3 glass filter) are mounted (see Figure 1). The sample is made of sintered 40–100 μ m glass particles that are carefully saturated with and immersed in a 10^{-3} M sodium chloride (NaCl) solution. A 500 kHz single sine pulse with a 500 mV peak-to-peak amplitude from an Agilent waveform generator (model 33220A) is fed into an ENI amplifier (model 2100L RF Power Amplifier) set at 50 dB gain. The output signal is coupled into a piezoelectric source transducer. The electrode is clamped in a computer-controlled mounting bracket so it can be positioned in the water layer with high accuracy. An Analogic preamplifier (model D1000) and a Yokogawa oscilloscope (model DL4200) are used for recording. Parameter values are given in Table 1.

The cylindrical coordinate system (r, z) is also indicated in Figure 1. The z -axis connects the centers of the sample and the transducer. At several positions along the z -axis (Figure 1), electric potential measurements are performed. These on-axis recordings, averaged over 8192 source pulses, are shown in Figure 2a. A 144–896 kHz numerical band-pass filter is applied in all measurements. The amplifier is calibrated with respect to amplitude and waveform using typical time signals. An electric pulse arrives at approximately 0.1 ms at each position, corresponding to the traveltime of the acoustic wave from the source to the sample surface. Because of the large EM wavespeed (approximately 1.1×10^7 m/s), the electric pulses arrive almost simultaneously at each position. Note that the pulse decays in

Table 1. Parameters of the porous medium. The values for K_s , K_f , η , ρ_f , Λ , ϵ_f , ϵ_s , and μ are taken from the literature (see below). Values for K_b and G are derived from experiments, and ϕ , ρ_s , k_0 , and α_∞ are measured directly. When applicable, fluid parameters are the same as for the porous medium. The temperature is 293.15 K.

Symbol	Parameter	Value	Units
K_s	Bulk modulus skeleton grains ^a	49.9×10^9	Pa
K_f	Bulk modulus (pore) fluid ^b	2.2×10^9	Pa
η	Pore fluid viscosity ^b	1×10^{-3}	Pa s
ρ_f	(Pore) fluid density ^b	998	kg/m ³
Λ	Weighted pore-volume-to-surface ratio ^c	1.229×10^{-5}	m
ϵ_f	Relative permittivity of the (pore) fluid ^b	80.1	—
ϵ_s	Relative permittivity of the solid ^b	4	—
μ	(Fluid) magnetic permeability ($= \mu_0$)	$4\pi \times 10^{-7}$	H/m
K_b	Bulk modulus framework of grains	6.6×10^9	Pa
G	Shear modulus framework of grains	5.5×10^9	Pa
ϕ	Porosity of the porous medium	0.345	—
ρ_s	Solid density	2.212×10^3	kg/m ³
k_0	Permeability	3.1×10^{-12}	m ²
α_∞	Tortuosity of the porous medium	2.1	—

^aJohnson and Plona (1982).

^bLide (2010).

^cPride (1994); Johnson et al. (1994); Jocker and Smeulders (2009).

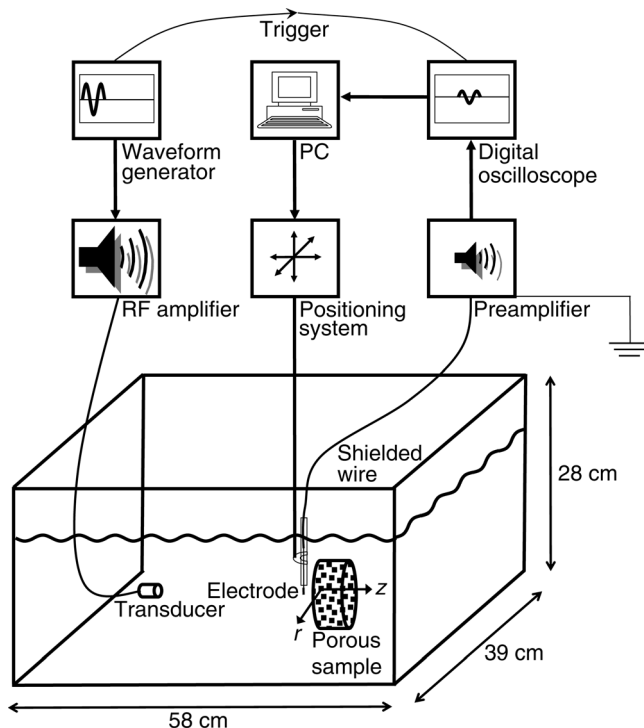


Figure 1. Schematic of the experimental setup. The electrode is clamped in a computer-controlled mounting bracket. Electrode recordings are with respect to ground level. The shielding of the wire electrode is also connected to ground level. The r - and z -axes coincide with those of Figure 3.

amplitude as the distance from the interface increases. The measured (true) peak-to-peak amplitude at $(r, z) = (0, -0.3)$ cm is approximately 0.25 mV. In a separate experiment, acoustic pressure was measured along the z -axis. In Figure 2b, we show the acoustic waveforms measured with a Specialty Engineering Associates needle hydrophone (model PZT-Z44-1000-S/N), calibrated over the frequency range of interest. From these measurements, we can extrapolate that the acoustic pulse will indeed arrive at the interface around 0.1 ms.

FORWARD MODELING OF THE SEISMOELECTRIC INTERFACE RESPONSE

The acoustic source is modeled by (e.g., Hall, 1987)

$$\hat{p}(\omega, R, \theta) = \frac{A(\omega)}{R} e^{-ikR} D(\theta), \quad (1)$$

where ω is the radial frequency, $R = \sqrt{(r - r_0)^2 + (z - z_0)^2}$ is the distance to the source, θ is the angle of incidence (see Figure 3), $A(\omega)$ is the amplitude spectrum, and k is the fluid wavenumber. The directivity function $D(\theta)$ is given by

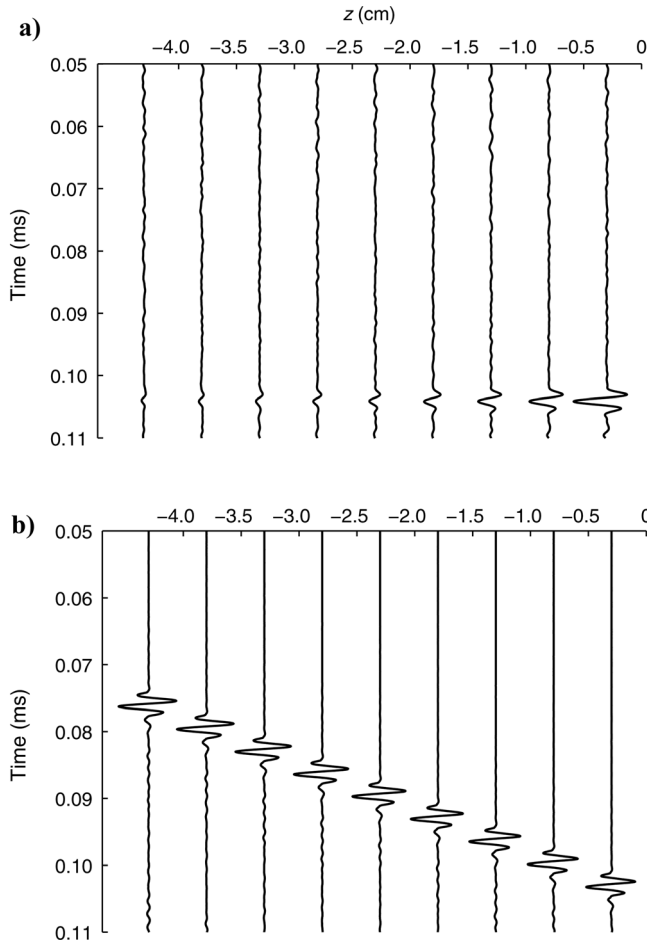


Figure 2. (a) Electric potential and (b) acoustic pressure as a function of vertical distance z at $r = 0$.

$$D(\theta) = \frac{J_1(ka \sin \theta)}{ka \sin \theta}. \quad (2)$$

Here, J_1 is the Bessel function of the first kind and first order, and a is the radius of the transducer. At $(r, z) = (0, 0)$, $A(\omega)$ is determined directly from the measurement (see Figure 4). As an additional test, the source response as a function of r is measured for a 15.3 cm on-axis distance from the transducer. A single sine 500 kHz source pulse with a 500 mV peak-to-peak amplitude is used to excite the source transducer. Results are shown by the asterisks in Figure 5. The dashed line is the theoretical value obtained from equation 1. We notice that the wavefield of the transducer is well described by equations 1 and 2, although the theory predicts interference patterns that are not recorded in the experiment.

The modeled source pressure wavefield is now expanded into conical waves by means of the Sommerfeld integral (see e.g., Brekhovskikh [1960] and Aki and Richards [2002]). The total reflected electric potential $\hat{\phi}(\omega, r_r, z_r)$ can then be expressed in terms of a Sommerfeld integral ($r_0 = 0$; $z_0, z_r < 0$):

$$\hat{\phi}(\omega, r_r, z_r) = -iA(\omega) \int_0^\infty \frac{k_r}{k_z} D(k_r) J_0(k_r r_r) \times e^{ik_z z_0} e^{ik_z^E z_r} R^E(k_r) dk_r, \quad (3)$$

where $k_r = k \sin \theta$ and $k_z = k \cos \theta$ are the radial and vertical components of k , respectively, and k_z^E is the vertical component of the fluid EM wavenumber. The seismoelectric reflection coefficient $R^E(k_r)$ relates the incident pressure wavefront to a reflected EM signal as derived by Schakel and Smeulders (2010), adopting full electrokinetic theory (Pride, 1994). Note that their coefficient is displacement normalized, whereas R^E is pressure normalized (in V/Pa).

Expressing k_r and k_z in terms of k and θ , equation 3 is written as

$$\hat{\phi}(\omega, r_r, z_r) = -iA(\omega) \int_0^{\pi/2 + i\infty} D(\theta) k \sin \theta J_0(k r_r \sin \theta) \times e^{ik z_0 \cos \theta} e^{ik_z^E(\theta) z_r} R^E(\theta) d\theta, \quad (4)$$

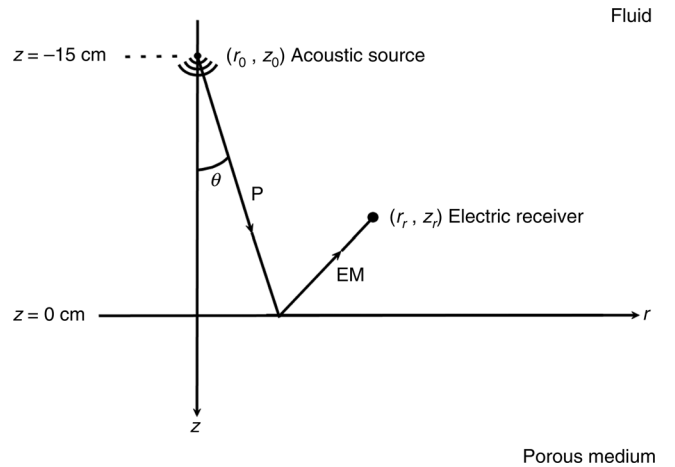


Figure 3. Geometry of seismoelectric model and experiment. A pressure (P-) wave is converted to an EM wave at the interface.

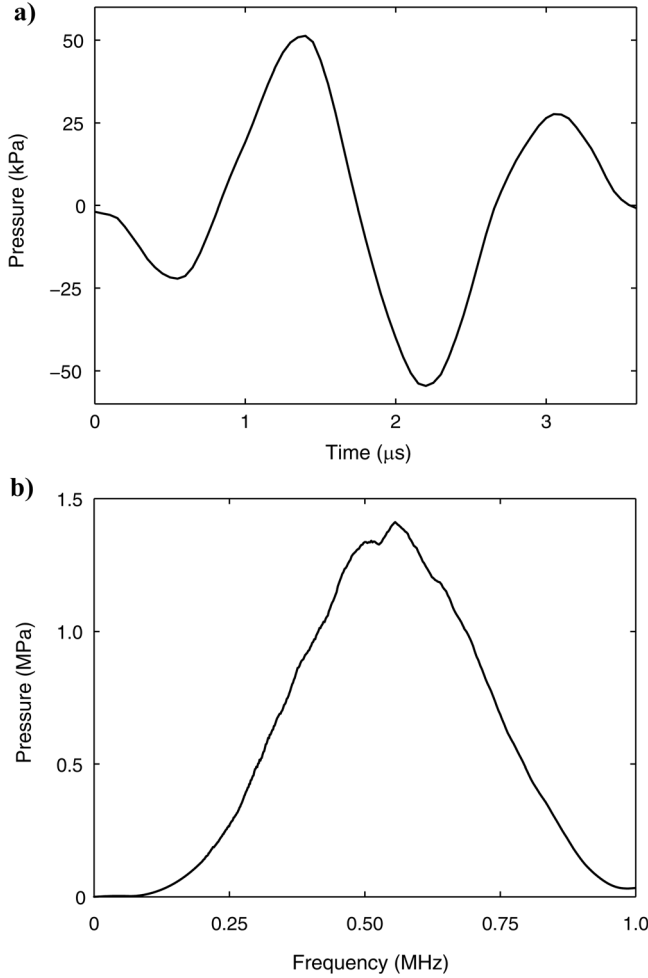


Figure 4. (a) Measured pressure pulse and (b) its frequency spectrum at $(r, z) = (0, 0)$.

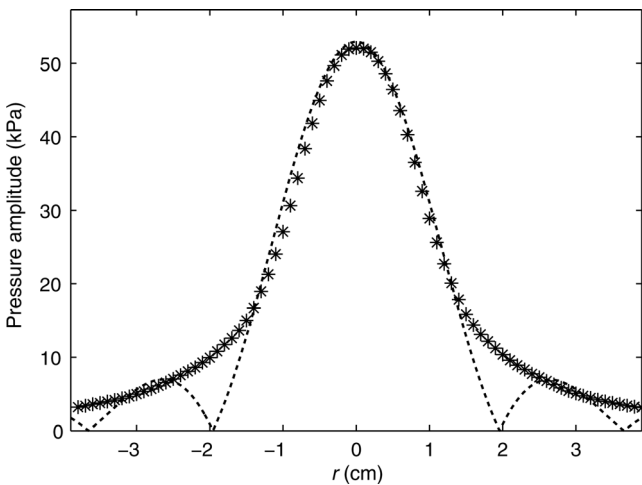


Figure 5. Measured (asterisks) and modeled (dashed line) pressure as a function of r . On-axis distance from the transducer is 15.3 cm ($R = 15.3$ cm when $\theta = 0$).

where $k_z^E(\theta) = \omega \sqrt{(1/c_E^2(\omega)) - (\sin^2 \theta / c_P^2)}$ and $\text{Im}[k_z^E(\theta) < 0]$. Fluid EM and pressure wave velocities are given by, respectively, $c_E(\omega) = \sqrt{(\mu \epsilon_0 \epsilon_f - (i \mu \sigma_f / \omega))^{-1}}$ and $c_P = \sqrt{K_f / \rho_f}$, where ϵ_0 is the vacuum permittivity and σ_f is the fluid conductivity. The path of integration is along straight lines from zero to $\pi/2$ and from $\pi/2$ to $\pi/2 + i\infty$ in the complex θ plane.

The second integral over complex θ is simplified further using the substitution $\theta = \pi/2 + i \ln[\sqrt{\gamma^2 + 1} + \gamma]$:

$$\begin{aligned} \hat{\phi}(\omega, r_r, z_r) = & -\frac{iA(\omega)}{a} \int_0^{\pi/2} J_0(kr_r \sin \theta) J_1(ka \sin \theta) \\ & \times e^{ikz_0 \cos \theta} e^{ik_z^E(\theta) z_r} R^E(\theta) d\theta \\ & + \frac{A(\omega)}{a} \int_0^\infty J_0(kr_r \sqrt{\gamma^2 + 1}) \frac{J_1(ka \sqrt{\gamma^2 + 1})}{\sqrt{\gamma^2 + 1}} \\ & \times e^{kz_0 \gamma} e^{ik_z^E(\gamma) z_r} R^E(\gamma) d\gamma, \end{aligned} \quad (5)$$

where equation 2 is used for $D(\theta)$. Equation 5 can be readily evaluated numerically and yields the total reflected EM wave-field through the inverse fast Fourier transform. The source is located at $(r_0, z_0) = (0, -15)$ cm (Figure 3). For each frequency, we use a recursive adaptive Simpson quadrature algorithm implemented in Matlab. A 144–896 kHz numerical band-pass filter is applied, and the input parameters of Tables 1 and 2 are used. Figure 6 shows the resulting modeled reflected electric potential and its frequency spectrum for a 10^{-3} M NaCl salinity.

COMPARISON BETWEEN EXPERIMENT AND FORWARD MODEL

Figure 7 compares measured waveforms (solid lines) at various positions in the fluid along the negative z -axis (Figure 3) with modeled waveforms (dashed lines). The agreement between measured and modeled waveforms is excellent. The modeled amplitudes are scaled to the measured amplitudes by the amplitude field scaling factor $A_C = 0.19$, which is obtained by comparing the theoretical Fourier spectrum maxima of the entire field to the measured maxima. Thus, $A_C < 1$ means that the theory overpredicts the actual measured electric potentials.

In Figure 8, the scaled theoretical Fourier spectra are compared with the experiments. Note that the measured pulses have

Table 2. Electrochemical parameters and amplitude field scaling factors A_C . Conductivities were measured directly in the fluid surrounding the porous medium, whereas zeta potentials ζ were obtained from Zetasizer Nano ZS measurements on crushed porous sample material suspensions in the corresponding salt solutions.

Salinity (M NaCl)	Conductivity (S/m)	ζ (mV)	A_C
10^{-4}	1.27×10^{-3}	−51.7	0.03
10^{-3}	1.20×10^{-2}	−61.5	0.19
10^{-2}	1.01×10^{-1}	−58.1	0.41

their energy distributed over slightly smaller frequencies than predicted by theory. Figure 9 compares measured waveforms (solid lines) at various positions in the fluid along the r -axis at $z = -1.3$ cm (Figure 3) with modeled waveforms (dashed lines). Again, $A_C = 0.19$ is used. As in the measurements along the z -axis, good agreement in the waveforms as well as in the spatial amplitude pattern is observed. We notice a slight asymmetry in the measured data. This is most likely caused by a small misalignment in the setup. In Figure 10, the corresponding frequency spectra are shown. We recall that the $A_C = 0.19$ value is obtained from Fourier spectra of the entire field shown in Figures 8 and 10.

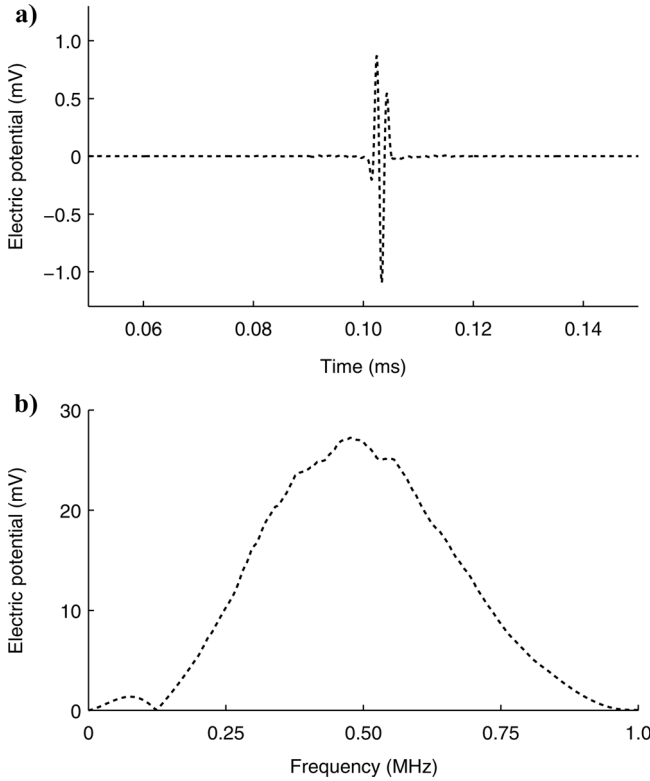


Figure 6. (a) Modeled reflected electric potential for an acoustic source at $(r_0, z_0) = (0, -15)$ cm and (b) its frequency spectrum.

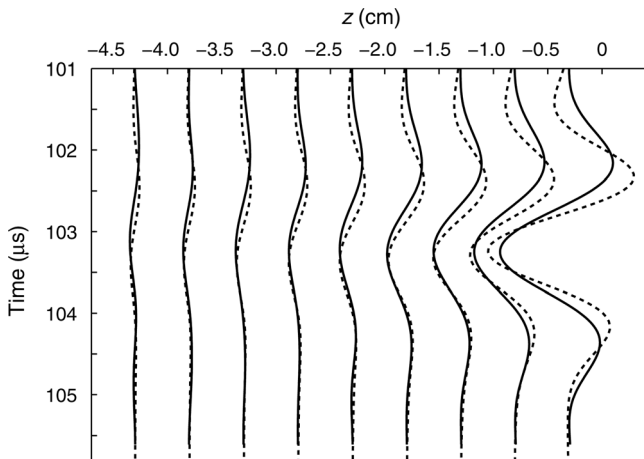


Figure 7. Measured (solid lines) and modeled (dashed lines) electric potential waveforms at various positions in the fluid along the z -axis at $r = 0$ for $C = 10^{-3}$ M NaCl.

quency spectra are shown. We recall that the $A_C = 0.19$ value is obtained from Fourier spectra of the entire field shown in Figures 8 and 10.

To compare predicted amplitudes with measurements more directly, peak-to-peak amplitudes (V_{pp}) are plotted in Figure 11 and compared with our model (dashed lines). Again, the spatial amplitude pattern is excellently predicted by theory for the measurements along the negative z -axis (Figure 11a) and parallel to the r -axis at $z = -1.3$ cm (Figure 11b). We also check the repeatability of the experiment. There is a small amplitude decrease over two days. The waveform is preserved on these longer timescales. In Figure 11 (gray lines), we also plot the vertical electric dipole approximation $A_D z e^{-ik^F d} / (r^2 + z^2)^{3/2}$ (see also, e.g., Dupuis et al. [2007] and Haines et al. [2007]), where d is the distance from the origin and the coefficient A_D is

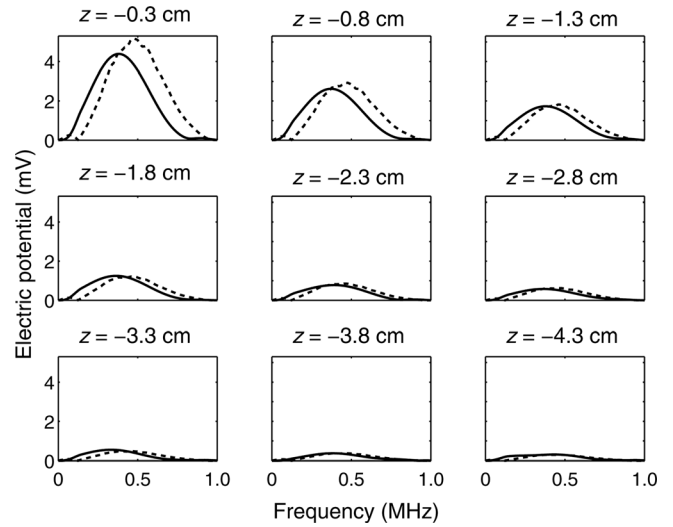


Figure 8. Measured (solid lines) and modeled (dashed lines) electric potential spectra at various positions in the fluid along the z -axis at $r = 0$ for $C = 10^{-3}$ M NaCl.

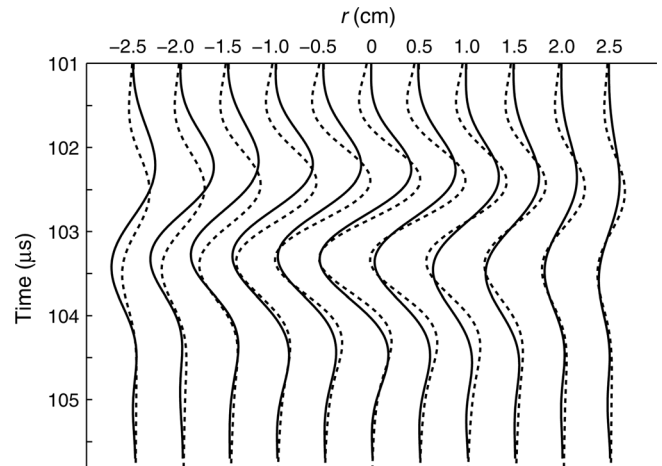


Figure 9. Measured (solid lines) and modeled (dashed lines) electric potential waveforms at various positions in the fluid parallel to the r -axis at $z = -1.3$ cm for $C = 10^{-3}$ M NaCl.

obtained from a best-fit procedure. The vertical electric dipole approximation overestimates the spatial amplitude decay. This means that the full model (equation 5) is needed to explain the amplitude measurements.

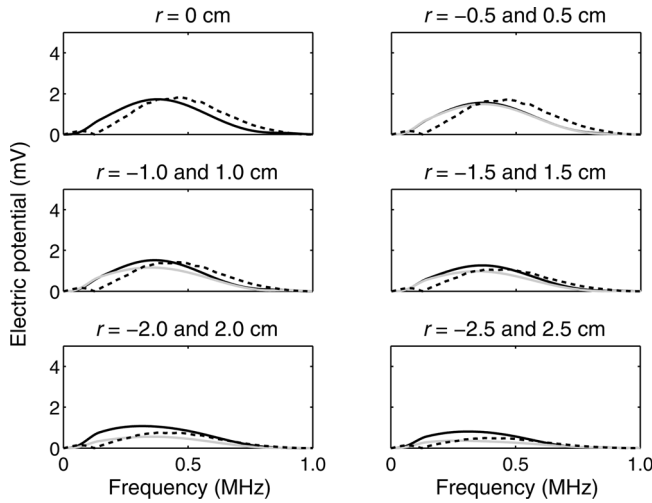


Figure 10. Measured (solid lines) and modeled (dashed lines) electric potential spectra at various positions in the fluid parallel to the r -axis at $z = -1.3$ cm for $C = 10^{-3}$ M NaCl. The black and gray solid lines correspond with measured spectra for negative and positive r -coordinates, respectively.

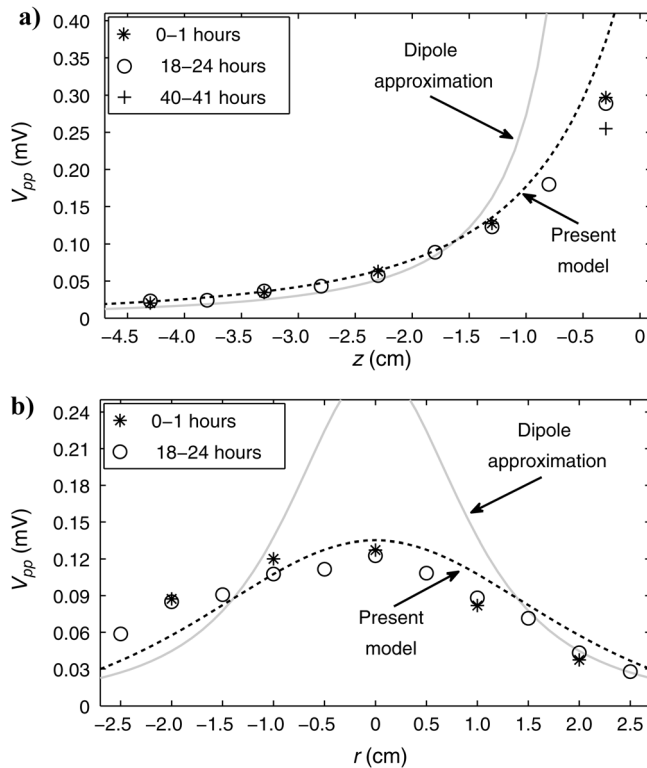


Figure 11. Measured (symbols) and modeled (dashed lines) peak-to-peak amplitudes V_{pp} as a function of (a) z at $r = 0$ and (b) r at $z = -1.3$ cm for $C = 10^{-3}$ M NaCl. Experiments were repeated several times to check repeatability. Here, $A_C = 0.19$. Also, the electric dipole approximation is plotted (gray lines).

Figures 12 and 13 show measured and full-waveform modeled V_{pp} -values where $C = 10^{-4}$ and $C = 10^{-2}$ M NaCl solutions are used, respectively. We determine the A_C -values in the same way as for the 10^{-3} M NaCl solution. Resulting values are given in Table 2. The spatial amplitude pattern is also excellently predicted for these salt solutions.

For higher salinities, the A_C values are closer to (or approach) unity. This observation is in agreement with Block and Harris (2006), who find that the discrepancy between electrokinetic theoretical predictions and measurements decreases with increasing salinity. Block and Harris (2006) also suggest that when low-salinity electrolytes are used to saturate the pores of sandstones, surface conductivity effects other than those predicted by Pride's electrokinetic theory may become dominant. The Pride theory used here incorporates diffuse-layer excess conduction associated with electromigration and electroosmosis. When the pore width is much larger than the Debye length, which is a measure of the diffuse-layer thickness, surface conductivity effects become negligible with increasing salinity. Our sample has a 16–40 μm pore width, whereas the Debye length is on the order of 3–30 nm.

We also observe that for all salinities, the A_C values differ from unity. Several possible reasons can be identified. First, the zeta potentials are obtained from crushed sample material suspensions. A Zetasizer is used for this. Probably, the zeta potential measured on ground-up material has a much higher value than that measured on intact (aged) surface material. This observation alone could be responsible for A_C being different from

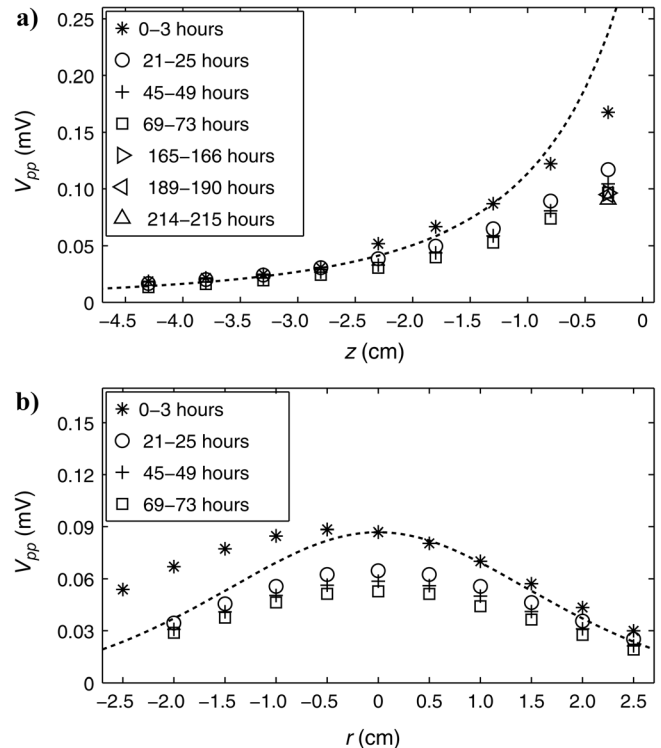


Figure 12. Measured (symbols) and modeled (dashed lines) peak-to-peak amplitudes V_{pp} as a function of (a) z at $r = 0$ and (b) r at $z = -1.3$ cm for $C = 10^{-4}$ M NaCl. Experiments were repeated several times to check repeatability. Here, $A_C = 0.03$.

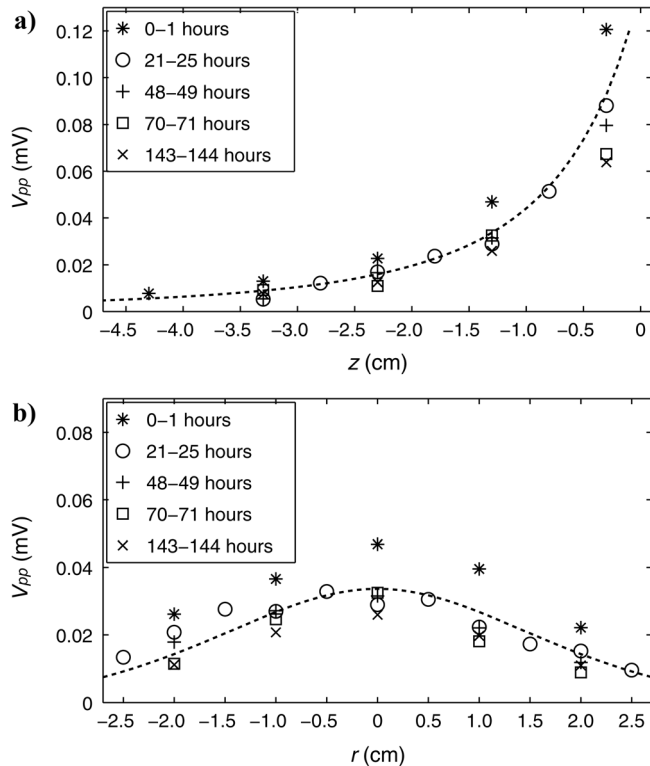


Figure 13. Measured (symbols) and modeled (dashed lines) peak-to-peak amplitudes V_{pp} as a function of (a) z at $r = 0$ and (b) r at $z = -1.3$ cm for $C = 10^{-2}$ M NaCl. Experiments were repeated several times to check repeatability. Here, $A_C = 0.41$.

unity. We also performed zeta potential calibration measurements on 99.8% silicon oxide (SiO_2) powder suspensions. The measured values were different by a factor of approximately two to three from streaming potential measurements reported in literature (e.g., Li and de Bruyn [1966] and Pride and Morgan [1991]), which indicates the uncertainty in our zeta potential determination. Second, the fluid conductivities given in Table 2 are measured in the fluid outside the pores; therefore, the actual pore fluid conductivity may be different. Finally, the seismoelectric interface response amplitudes decrease on timescales that are longer (approximately 24 hours) than the duration of a single measurement series (1–4 hours). For example, the measured amplitudes in Figure 13 from the last measurement series (143–144 hours) are, on average, 1.9 times smaller than those of the first series (0–1 hours). This observation could be related to increasing bulk conductivity (Block and Harris, 2006) and/or incomplete equilibration of the electric double layer. For silica, the electric double layer may take several days to become stable (Morgan et al., 1989), and it may take hours at soil interfaces (Liu et al., 2008). One could also speculate on the electrodes developing a voltage drop at their surface, thus reducing the measured amplitudes over time. However, replacing the electrodes under stable experimental conditions does not change potential readings significantly.

CONCLUSIONS

The electrokinetic conversion of an acoustic wave to an EM wave at an interface was systematically investigated in a labora-

tory setup. We independently determined the mechanical and electrical parameters. It was demonstrated that this effect is well predicted by electrokinetic theory in terms of waveform and spatial amplitude pattern. At $C = 10^{-3}$ and $C = 10^{-2}$ M NaCl, amplitude field scaling factors of 0.19 and 0.41 were used. At $C = 10^{-4}$ M NaCl, however, the value was only 0.03, which may be related to surface conductivity effects other than those present in electrokinetic theory. The magnitude of acoustic-to-EM conversion was shown to depend on the salinity of the saturating fluid, which is promising for applications in the oil and gas industry because the zones of interest are saturated with (resistive) hydrocarbons instead of water. A careful characterization of the source radiation pattern was part of the forward model, indicating that electrokinetic interface response predictions require proper modeling of the source. The results of this study indicate that the acoustic-to-EM wave responses of these rock/fluid combinations can be predicted by a full-waveform model for general subsurface exploration purposes.

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Electroseismic wave properties

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In a porous material saturated by a fluid electrolyte, mechanical and electromagnetic disturbances are coupled. The coupling is due to an excess of electrolyte ions that exist in a fluid layer near the grain surfaces within the material; i.e., the coupling is electrokinetic in nature. The governing equations controlling the coupled electromagnetic-seismic (or “electroseismic”) wave propagation are presented for a general anisotropic and heterogeneous porous material. Uniqueness is derived as well as the statements of energy conservation and reciprocity. Representation integrals for the various wave fields are derived that require, in general, nine different Green’s tensors. For the special case of an isotropic and homogeneous wholespace, both the plane-wave and the point-source responses are obtained. Finally, the boundary conditions that hold at interfaces in the porous material are derived. © 1996 Acoustical Society of America.

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INTRODUCTION

Thompson and Gist (1993), Butler *et al.* (1994), and Mikhailov and Haartsen (1996) have all collected field data demonstrating that when seismic waves propagate through the near-surface sediment layers of the Earth (depths less than, say, 300 m), electromagnetic disturbances are generated that can be recorded at the surface. Thompson and Gist (1993) and Pride (1994) have hypothesized that the most likely coupling mechanism is electrokinetic in nature. As is briefly discussed in Sec. I, piezoelectricity cannot be the coupling mechanism.

Starting from first principles, Pride (1994) has derived equations that control such “electroseismic” phenomena when an electrokinetic-coupling mechanism is assumed. The basic idea is as follows. The grains within sedimentary materials have an excess charge adsorbed to their surface (usually negative) due to the termination of the crystal lattice and certain chemical reactions that occur between the electrolyte ions and terminal sites on the crystal surface. This chemically bound charge is immobile but is balanced by a layer of free counterions in the adjacent fluid. This layer of fluid containing the mobile counterions is called the “diffuse-double layer.” When a seismic wave propagates through the sediments, a relative fluid–solid motion is induced that transports the counterions relative to the fixed charge thus inducing an electrical “streaming” current. This streaming current is the source for the electromagnetic coupling.

For compressional waves in a homogeneous porous material, the streaming current leads to a buildup of counter charge in the rarefactions of the wave and a depletion in the compressions. The resultant electric field drives a conduction current that exactly balances the streaming current thus fixing the amplitude of the charge separation within the wave. In this case, as the compressional wave propagates through the medium, there is a constant electric field fixed to the

wave with no extent outside of the wave. Independently propagating electromagnetic waves are not generated. However, when the compressional wave is incident at an interface, there is a variation in the streaming-current distribution while the wave is traversing the interface leading to a charge separation that varies with the time signature of the wave. This time-varying charge separation does generate an electromagnetic disturbance that will independently propagate to the Earth’s surface where it can be recorded. Somewhat similar comments can be made concerning shear waves (see Sec. II below). These ideas will be elaborated upon and quantified in both this and a forthcoming publication (Haartsen and Pride, 1996).

The overall goal of this paper is to develop the necessary background theory required to solve electroseismic boundary-value problems. The structure of the paper is as follows. In Sec. I, the governing equations for a general anisotropic and heterogeneous porous continuum are presented and discussed. The general statements of energy conservation, uniqueness, and reciprocity are then given as well as the representation integrals for the various wave fields. In Sec. II, we specialize to the case of a homogeneous isotropic whole space. Both the plane-wave and point-source responses are obtained for this special case. Finally, in Sec. III, the boundary conditions that hold at an interface in the porous continuum are derived.

I. GENERAL PROPERTIES

A. Governing equations

Assuming an $e^{-i\omega t}$ time dependence, the equations governing the coupled electromagnetic-acoustic wave propagation in a general anisotropic and heterogeneous porous continuum are

$$\nabla \times \mathbf{E} = i\omega \mathbf{B}, \quad (1)$$

$$\nabla \times \mathbf{H} = -i\omega \mathbf{D} + \mathbf{J} + \mathbf{C}, \quad (2)$$

$$\nabla \cdot \boldsymbol{\tau} = -i\omega(\rho \dot{\mathbf{u}} + \rho_f \dot{\mathbf{w}}) - \mathbf{F}, \quad (3)$$

$$\mathbf{J} = \mathbf{S} \cdot \mathbf{E} + \mathbf{L} \cdot (-\nabla p + i\omega \rho_f \dot{\mathbf{u}} + \mathbf{f}), \quad (4)$$

$$\dot{\mathbf{w}} = \mathbf{L} \cdot \mathbf{E} + \frac{\mathbf{K}}{\eta} \cdot (-\nabla p + i\omega \rho_f \dot{\mathbf{u}} + \mathbf{f}), \quad (5)$$

$$\boldsymbol{\tau} = {}_4\mathcal{L} : \nabla \mathbf{u} + \mathcal{C} \nabla \cdot \mathbf{w}, \quad (6)$$

$$-p = \mathcal{E} : \nabla \mathbf{u} + M \nabla \cdot \mathbf{w}, \quad (7)$$

$$\mathbf{B} = \boldsymbol{\mu} \cdot \mathbf{H}, \quad (8)$$

$$\mathbf{D} = \boldsymbol{\epsilon} \cdot \mathbf{E}. \quad (9)$$

The notation $\dot{\mathbf{u}}(\omega) = -i\omega \mathbf{u}(\omega)$ is being employed. These equations are the natural generalization to the anisotropic case of the isotropic equations obtained by Pride (1994) using volume averaging. The equations naturally group into the four sets following: Eqs. (1)–(3) are the dynamical equations; Eqs. (4)–(5) are the transport equations; Eqs. (6)–(7) are the poroelastic constitutive laws; and Eqs. (8)–(9) are the electromagnetic constitutive laws. These are linear equations so that the wave fields obey superposition. There is a limitation on the maximum frequency that can be treated defined by the condition that wavelengths must be much greater than grain sizes. For sediments in the Earth, this maximum frequency is on the order of 10^6 Hz.

All coupling is in the transport equations and is due to the electrokinetic-coupling tensor $\mathbf{L}$. If $\mathbf{L}$ were set to zero, these equations would separate into Maxwell's and Biot's (1956) equations. Despite the allowance for anisotropy, there is no allowance for piezoelectric coupling. This is justified when the fluid is an electrolyte because fluid cations will accumulate on the negatively polarized portion of a grain surface and anions on the positive portion thus screening the piezoelectric field of each grain. Such ionic accumulations will take place on time scales of ϵ_f/σ_f , where ϵ_f is the fluid's permittivity ($\sim 10^{-9}$ F/m for water-based electrolytes) and σ_f is the fluid's conductivity ($> 10^{-3}$ S/m for electrolytes of interest). Thus the screening time will always be $< 10^{-6}$ s which is faster than wave periods of interest so that the piezoelectric fields of the grains always remain screened.

If $\mathbf{u}_f(\mathbf{r})$ and $\mathbf{u}_s(\mathbf{r})$ are defined, respectively, as the average displacement of the fluid and solid phases within an averaging volume V_A positioned at point $\mathbf{r}$ (V_A being larger than the grains but much smaller than the wavelengths), then the $\mathbf{u}$ and $\mathbf{w}$ in Eqs. (3)–(7) are defined $\mathbf{u} \equiv \mathbf{u}_s$ and $\mathbf{w} \equiv \phi(\mathbf{u}_f - \mathbf{u}_s)$, where ϕ is porosity. The filtration velocity $\dot{\mathbf{w}}(\omega) = -i\omega \mathbf{w}(\omega)$ represents the volume of fluid traversing unit area in unit time. The displacement that a macroscopic sensor is sensitive to, $\mathbf{U}$ say, is the total average within V_A ; i.e., $\mathbf{U} = \phi \mathbf{u}_f + (1 - \phi) \mathbf{u}_s = \mathbf{u} + \mathbf{w}$. The bulk-stress tensor $\boldsymbol{\tau}$ of Eqs. (3) and (6) is defined $\boldsymbol{\tau} = \phi \boldsymbol{\tau}_f + (1 - \phi) \boldsymbol{\tau}_s$, where $\boldsymbol{\tau}_s$ and $\boldsymbol{\tau}_f$ are the average stress tensors in the grains and fluid, respectively. It is easily shown (e.g., Pride, 1994) that over frequencies where grain scattering is unimportant, the fluid supports no averaged shear deformation and the average fluid stress is purely isotropic; i.e., $\boldsymbol{\tau}_f = -p\mathbf{I}$, where $\mathbf{I}$ is the identity tensor and p is the fluid pressure. The electromag-

netic fields $\mathbf{E}$, $\mathbf{H}$, $\mathbf{D}$, and $\mathbf{B}$ represent their average value throughout all of V_A . It can be assumed that $\mathbf{E}$ and $\mathbf{B}$ are the fields that macroscopic antennas record. For the time-harmonic description being employed, all of these wave field variables are complex and frequency dependent.

If a source for mechanical waves exerts an average force density $\mathbf{f}_f$ on the fluid phase and $\mathbf{f}_s$ on the solid phase, then the source terms $\mathbf{F}$ and $\mathbf{f}$ of Eqs. (3)–(5) are defined $\mathbf{F} \equiv \phi \mathbf{f}_f + (1 - \phi) \mathbf{f}_s$ and $\mathbf{f} \equiv \mathbf{f}_f$. For many exploration sources (explosions in particular) it can be assumed that the force applied to the fluid phase is the same as that applied to the solid; i.e., $\mathbf{f}_f = \mathbf{f}_s$ or $\mathbf{F} = \mathbf{f}$. However, in what follows, $\mathbf{F}$ and $\mathbf{f}$ will be treated distinctly. The primary source for the electromagnetic fields is the current-source density $\mathbf{C}$. A magnetic-source term is not included on the right-hand side of Faraday's law [Eq. (1)] because such a term is always due to loops of applied current; i.e., a magnetic source is exactly represented by a particular distribution of applied current $\mathbf{C}$ and is, thus not a fundamental (or necessary) quantity.

The transport tensors $\mathbf{S}$ (electrical conductivity), $\mathbf{L}$ (electrokinetic-coupling coefficient), and $\mathbf{K}$ (fluid-flow permeability) are complex and frequency dependent. In the generalized Darcy's law [Eq. (5)], η is the fluid's shear viscosity. The porous-continuum statement of Onsager reciprocity requires that both $\mathbf{S}$ and $\mathbf{K}$ are symmetric; however, no such general constraints are placed on the symmetry of $\mathbf{L}$. Nonetheless, for the commonly assumed case in which the charge per unit area adsorbed to the grain surfaces is uniform throughout an averaging volume and, simultaneously, the diffuse-double layers are thin, one can show that $\mathbf{L}$ is symmetric and, indeed, possesses the same symmetry properties as $\mathbf{S}$ (this is seen by generalizing the arguments in Pride, 1994, to the anisotropic case). Throughout what follows, it will be assumed that $\mathbf{L}$ is symmetric. In the isotropic case we define $\mathbf{S} = \sigma \mathbf{I}$, $\mathbf{L} = L \mathbf{I}$, and $\mathbf{K} = k \mathbf{I}$. Analytical expressions relating the isotropic coefficients $\sigma(\omega)$, $L(\omega)$, and $k(\omega)$ to frequency, fluid chemistry, and pore geometry are given by Pride (1994) to which the reader is referred for more detail.

The fourth-order stiffness tensor ${}_4\mathcal{L}$ is expressed as ${}_4\mathcal{L} = \mathcal{L}_{ijkl} \hat{\mathbf{x}}_i \hat{\mathbf{x}}_j \hat{\mathbf{x}}_k \hat{\mathbf{x}}_l$ in polyadic notation where the unit base vectors $\hat{\mathbf{x}}_i$ may correspond to any curvilinear coordinate system; however, we assume from the outset that they are orthogonal ($\hat{\mathbf{x}}_i \cdot \hat{\mathbf{x}}_j = 0$; $i \neq j$). Material independent symmetry properties of ${}_4\mathcal{L}$ are obtained, as usual, by the requirements that: (1) rigid-body rotations produce no stress ($\mathcal{L}_{ijkl} = \mathcal{L}_{ijlk}$); (2) no couple stresses are present ($\mathcal{L}_{jikl} = \mathcal{L}_{ijlk}$); and (3) there exists a strain energy function ($\mathcal{L}_{ijkl} = \mathcal{L}_{klij}$). As usual, these symmetries reduce the number of possible independent constants within ${}_4\mathcal{L}$ to 21. Any material symmetries will reduce this number further. The second-order cross-stiffness tensor $\mathcal{C}$ must also be symmetric ($\mathcal{C}_{ij} = \mathcal{C}_{ji}$) because no couple stresses are being allowed for, while the stiffness M is a scalar. In the isotropic case, we define

$${}_4\mathcal{L} = [(H - 2G) \delta_{ij} \delta_{kl} + G(\delta_{il} \delta_{jk} + \delta_{ik} \delta_{jl})] \hat{\mathbf{x}}_i \hat{\mathbf{x}}_j \hat{\mathbf{x}}_k \hat{\mathbf{x}}_l, \quad (10)$$

$$\mathcal{C} = C \mathbf{I}, \quad (11)$$

where expressions that relate H , C , and M to the underlying fluid, solid, and drained-framework-of-grains elastic moduli are given by Pride *et al.* (1992) (and are equivalent to those of Biot and Willis, 1957) for the special case in which the material in an averaging volume has roughly uniform properties (e.g., if there is a mixture of sand and shale bodies within V_A , then the standard definitions for H , C , and M no longer hold—see Berryman and Milton, 1991 and Berryman and Pride, 1996, for a discussion of such cases). The stiffness G is the porous-material shear modulus. All of the above stiffnesses may be assumed complex and frequency dependent. The imaginary parts correspond to anelastic processes not directly related to macroscopic flux. In this work, we use the notation $c=c'+ic''$ to denote the separation into real and imaginary parts. It will be seen that the second law of thermodynamics requires that the principle components of ${}_4\mathcal{L}$ as well as M'' must all be negative when there is an assumed $e^{-i\omega t}$ time dependence.

The symmetric, second-order tensors μ and ϵ are the magnetic permeability and electrical permittivity. For nearly all porous media of interest it can be assumed that $\mu=\mu_0\mathbf{I}$ (iron, nickel, and cobalt are not significantly present in most sedimentary materials). In the isotropic case, we define $\epsilon=\epsilon\mathbf{I}$. In general, ϵ will be complex and frequency dependent. However, as discussed by Pride (1994), if $\epsilon_s/\epsilon_f\ll 1$ (where ϵ_s and ϵ_f are the real permittivities of the grains and fluid) and if electric double layers are thin relative to grain dimensions, then effectively no grain-scale ionic polarization will take place and ϵ can be taken as real and frequency independent. These two conditions are usually met in porous media such as sedimentary rocks. Nonetheless, if there are losses associated with the processes of either electrical or magnetic polarization, it will be seen that the second law requires that the principal components of μ'' and ϵ'' must be positive.

Lastly, the bulk density ρ in Eq. (3) is defined $\rho=\phi\rho_f+(1-\phi)\rho_s$, where ρ_f and ρ_s are the purely real mass densities of the fluid and solid.

In the following sections we will derive a series of related results that are most easily obtained if the transport equation for fluid flux $\dot{\mathbf{w}}$ [Eq. (5)] is dot multiplied by $\eta\mathbf{K}^{-1}$ to give an effective force balance on the fluid in relative motion

$$\eta\mathbf{K}^{-1}\cdot\dot{\mathbf{w}}=-\nabla p+i\omega\rho_f\dot{\mathbf{u}}+\mathbf{f}+\eta(\mathbf{K}^{-1}\cdot\mathbf{L})\cdot\mathbf{E}. \quad (12)$$

The frame of reference for this force balance is fixed on the grains and is accelerating as $-i\omega\dot{\mathbf{u}}(\omega)$ thus giving rise to the apparent body force $i\omega\rho_f\dot{\mathbf{u}}(\omega)$. The inverse of a symmetric tensor is symmetric as is the dot product of two symmetric tensors. Thus $\mathbf{K}^{-1}\cdot\mathbf{L}$ is symmetric.

B. The energy balance

We now obtain the porous-continuum energy balance using the complex wave fields defined above. From Eqs. (1) and (2), the following products are constructed where a star denotes the complex conjugate

$$\mathbf{H}^*\cdot[\nabla\times\mathbf{E}=i\omega\mathbf{B}], \quad (13)$$

$$\mathbf{E}\cdot[\nabla\times\mathbf{H}^*=i\omega\mathbf{D}^*+\mathbf{J}^*+\mathbf{C}^*]. \quad (14)$$

Subtracting Eq. (13) from (14) and using $\nabla\cdot(\mathbf{a}\times\mathbf{b})=\mathbf{b}\cdot\nabla\times\mathbf{a}-\mathbf{a}\cdot\nabla\times\mathbf{b}$ then gives

$$-\nabla\cdot(\mathbf{E}\times\mathbf{H}^*)=i\omega(\mathbf{E}\cdot\mathbf{D}^*-\mathbf{H}^*\cdot\mathbf{B})+\mathbf{E}\cdot\mathbf{J}^*+\mathbf{E}\cdot\mathbf{C}^*, \quad (15)$$

which is the complex-energy balance for the electromagnetic fields.

Similarly, from Eqs. (3) and (12), the following products are constructed

$$\dot{\mathbf{u}}^*\cdot[\nabla\cdot\boldsymbol{\tau}=-i\omega(\rho\dot{\mathbf{u}}+\rho_f\dot{\mathbf{w}})-\mathbf{F}], \quad (16)$$

$$\dot{\mathbf{w}}^*\cdot[-\nabla p=-i\omega\rho_f\dot{\mathbf{u}}-\mathbf{f}+\eta\mathbf{K}^{-1}\cdot\dot{\mathbf{w}}-\eta\mathbf{K}^{-1}\cdot\mathbf{L}\cdot\mathbf{E}]. \quad (17)$$

Combining with $\nabla\cdot(\mathbf{a}\cdot\mathbf{A})=\mathbf{A}:\nabla\mathbf{a}+(\nabla\cdot\mathbf{A}^T)\cdot\mathbf{a}$ and the fact that $\boldsymbol{\tau}=\boldsymbol{\tau}^T$ then gives

$$\begin{aligned} \nabla\cdot(\dot{\mathbf{u}}^*\cdot\boldsymbol{\tau}-\dot{\mathbf{w}}^*p) &= i\omega[\boldsymbol{\tau}:(\nabla\mathbf{u})^*-p(\nabla\cdot\mathbf{w})^*] \\ &\quad -i\omega[\rho|\dot{\mathbf{u}}|^2+\rho_f(\dot{\mathbf{u}}^*\cdot\dot{\mathbf{w}}+\dot{\mathbf{u}}\cdot\dot{\mathbf{w}}^*)] \\ &\quad +\dot{\mathbf{w}}^*\cdot\mathbf{g}-(\dot{\mathbf{u}}^*\cdot\mathbf{F}+\dot{\mathbf{w}}^*\cdot\mathbf{f}), \end{aligned} \quad (18)$$

where $\mathbf{g}$ is the mechanical force driving relative flow and is defined

$$\mathbf{g}=-\nabla p+i\omega\rho_f\dot{\mathbf{u}}+\mathbf{f}. \quad (19)$$

Equation (18) is the complex-energy balance for the mechanical fields.

We next add the electromagnetic and mechanical energy balances and separate into real and imaginary parts. This results in two purely real balances

$$-\nabla\cdot\mathbf{J}'_e=Q'+S'_e, \quad (20)$$

$$-\nabla\cdot\mathbf{J}''_e=Q''+S''_e, \quad (21)$$

where $\mathbf{J}_e$ is the complex-energy flux and is defined

$$\mathbf{J}_e=\mathbf{E}\times\mathbf{H}^*-(\dot{\mathbf{u}}^*\cdot\boldsymbol{\tau}-\dot{\mathbf{w}}^*p), \quad (22)$$

while S_e is the rate density at which the sources create complex energy and is defined

$$S_e=\mathbf{E}\cdot\mathbf{C}^*-(\dot{\mathbf{u}}^*\cdot\mathbf{F}+\dot{\mathbf{w}}^*\cdot\mathbf{f}). \quad (23)$$

The real vector $\mathbf{J}'_e$ represents the total amount of energy traversing unit area of the porous continuum in unit time while S'_e represents the total amount of energy being created by the sources per unit time and volume. The real scalar Q' is the rate at which energy is being irreversibly lost to heat per unit volume while Q'' is the rate at which energy is being reversibly stored per unit volume (Q'' is the reactive energy). Both Q' and Q'' are defined by substituting the constitutive laws into the complex-energy balances to obtain

$$\begin{aligned}
Q' + iQ'' = & i\omega \epsilon^* : \frac{1}{2}(\mathbf{E}\mathbf{E}^* + \mathbf{E}^*\mathbf{E}) - i\omega \mu : \frac{1}{2}(\mathbf{H}\mathbf{H}^* + \mathbf{H}^*\mathbf{H}) \\
& + i\omega {}_4\mathcal{L} :: \frac{1}{2}[(\nabla \mathbf{u})(\nabla \mathbf{u})^* + (\nabla \mathbf{u})^*(\nabla \mathbf{u})] \\
& + i\omega \mathcal{C} : [(\nabla \cdot \mathbf{w})(\nabla \mathbf{u})^* + (\nabla \cdot \mathbf{w})^*(\nabla \mathbf{u})] \\
& + i\omega M(\nabla \cdot \mathbf{w})^*(\nabla \cdot \mathbf{w}) - i\omega \rho |\dot{\mathbf{u}}|^2 \\
& - i\omega \rho_f(\dot{\mathbf{u}} \cdot \dot{\mathbf{w}}^* + \dot{\mathbf{u}}^* \cdot \dot{\mathbf{w}}) + \mathbf{S}^* : \frac{1}{2}(\mathbf{E}\mathbf{E}^* + \mathbf{E}^*\mathbf{E}) \\
& + \frac{\mathbf{K}^*}{\eta} : \frac{1}{2}(\mathbf{g}\mathbf{g}^* + \mathbf{g}^*\mathbf{g}) + \mathbf{L}^* : (\mathbf{E}\mathbf{g}^* + \mathbf{E}^*\mathbf{g}).
\end{aligned} \tag{24}$$

In this expression, the factors involving the wave fields ($\mathbf{E}$, $\mathbf{H}$, $\mathbf{u}$, $\mathbf{w}$, and $\mathbf{g}$) are all purely real. Thus the separation into real and imaginary parts ($Q' + iQ''$) is defined entirely by the real and imaginary parts of the various constitutive coefficients. In writing Eq. (24) in this form, we have used the fact that $\mathbf{a}^* \cdot \mathbf{A} \cdot \mathbf{a} = \mathbf{A} : (\mathbf{a}\mathbf{a}^* + \mathbf{a}^*\mathbf{a})/2$ for a symmetric tensor $\mathbf{A}$.

Equation (20) is the conservation of energy and is the central result of this section. Perhaps the main usefulness of Eq. (21) is that it helps provide physical insight into the meaning of the (real) vector $\mathbf{J}_e''$: away from the sources, any accumulation of the energy transported as $\mathbf{J}_e''$ does so reversibly and is associated with the difference between energy being stored in the fields (the potential energies) and the kinetic energies.

Finally, it is convenient to establish here conditions under which the second law of thermodynamics $Q' > 0$ is satisfied. From Eq. (24), Q' may be written as

$$\begin{aligned}
Q' = & \omega \mathbf{H}^* \cdot \boldsymbol{\mu}'' \cdot \mathbf{H} + [\mathbf{E}^*, \mathbf{g}^*] \cdot \begin{bmatrix} \mathbf{S}' + \omega \boldsymbol{\epsilon}'' & \mathbf{L}' \\ \mathbf{L}' & \mathbf{K}' \end{bmatrix} \cdot \begin{bmatrix} \mathbf{E} \\ \mathbf{g} \end{bmatrix} \\
& + \omega [(\nabla \cdot \mathbf{w})^*, e_1^*, \dots, e_6^*] \\
& \cdot \begin{bmatrix} -M'' & -\mathcal{C}_1'' & \dots & -\mathcal{C}_6'' \\ -\mathcal{C}_1'' & -\mathcal{L}_{11}'' & \dots & -\mathcal{L}_{16}'' \\ \vdots & \vdots & \ddots & \vdots \\ -\mathcal{C}_6'' & -\mathcal{L}_{16}'' & \dots & -\mathcal{L}_{66}'' \end{bmatrix} \cdot \begin{bmatrix} \nabla \cdot \mathbf{w} \\ e_1 \\ \vdots \\ e_6 \end{bmatrix}. \tag{25}
\end{aligned}$$

In the third term, the strains e_α and the moduli $\mathcal{C}_\alpha''$ and $\mathcal{L}_{\alpha\beta}''$ are related to their corresponding tensors by the usual mapping (e.g., Nye, 1985)

$$\begin{aligned}
\frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right) & \rightarrow e_\alpha, \\
\mathcal{C}_{ij}'' & \rightarrow \mathcal{C}_{\alpha}'', \quad \mathcal{L}_{ijkl}'' \rightarrow \mathcal{L}_{\alpha\beta}'',
\end{aligned} \tag{26}$$

where the $ij \rightarrow \alpha$ map is defined $11 \rightarrow 1$, $22 \rightarrow 2$, $33 \rightarrow 3$, $23 \rightarrow 4$, $13 \rightarrow 5$, and $12 \rightarrow 6$.

Each of the three terms comprising Q' are in quadratic form (i.e., of the form $\mathbf{a}^* \cdot \mathbf{A} \cdot \mathbf{a}$). The second law requires that $Q' > 0$ unless all the macroscopic fields involved ($\mathbf{E}$, $\mathbf{H}$, $\mathbf{g}$, $\nabla \cdot \mathbf{w}$, and e_α) are zero in which case $Q' = 0$. By considering the presence of arbitrary combinations of field types, we must have that each of the coefficient matrices in the three terms of Eq. (25) is positive definite [a real matrix $\mathbf{A}$ is called “positive definite” if $\mathbf{a}^* \cdot \mathbf{A} \cdot \mathbf{a} > 0$ for an arbitrary vector $\mathbf{a}$].

A positive-definite matrix has the property that each of its diagonal elements must be positive as well as all of its eigenvalues. More completely, the N necessary and sufficient conditions required for an $N \times N$ matrix A_{ij} to be positive definite can be written (e.g., Noble, 1969)

$$A_{11} > 0, \begin{vmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{vmatrix} > 0, \dots, \begin{vmatrix} A_{11} & \dots & A_{1N} \\ \vdots & \ddots & \vdots \\ A_{N1} & \dots & A_{NN} \end{vmatrix} > 0; \tag{27}$$

i.e., the determinant of the minors of A_{ij} must all be positive. Each of the three coefficient matrices in Eq. (25) must satisfy constraints of this form in order to satisfy the second law. Thus the principal components of ${}_4\mathcal{L}''$ and M'' must all be negative while the principal components of $\mathbf{S}'$, $\mathbf{K}'$, $\boldsymbol{\epsilon}''$, and $\boldsymbol{\mu}''$ must all be positive. The positive-definite nature of these coefficient matrices will now be used in establishing uniqueness.

C. Uniqueness

We next ask the following question: Within some volume V of interest, is it possible to have two sets of fields— $\mathbf{u}_1$, $\mathbf{w}_1$, $\mathbf{E}_1$, $\mathbf{H}_1$ and $\mathbf{u}_2$, $\mathbf{w}_2$, $\mathbf{E}_2$, $\mathbf{H}_2$ —that are both generated by the same sources and that both satisfy the same prescribed boundary conditions on the surface S surrounding V but that are otherwise different functions throughout V ? To answer this, the difference fields

$$\begin{aligned}
\delta \mathbf{u} &= \mathbf{u}_1 - \mathbf{u}_2, \quad \delta \mathbf{w} = \mathbf{w}_1 - \mathbf{w}_2, \quad \delta \mathbf{E} = \mathbf{E}_1 - \mathbf{E}_2, \\
\delta \mathbf{H} &= \mathbf{H}_1 - \mathbf{H}_2,
\end{aligned} \tag{28}$$

are introduced. If the difference fields can be shown to be everywhere zero, the fields are said to be “unique.”

The following products of difference fields may be constructed in analogy to Eqs. (13)–(17)

$$\delta \mathbf{H}^* \cdot [\nabla \times \delta \mathbf{E} = i\omega \boldsymbol{\mu} \cdot \delta \mathbf{H}], \tag{29}$$

$$\delta \mathbf{E} \cdot [\nabla \times \delta \mathbf{H}^* = i\omega \boldsymbol{\epsilon}^* \cdot \delta \mathbf{E}^* + \mathbf{S}^* \cdot \delta \mathbf{E}^* + \mathbf{L}^* \cdot \delta \mathbf{g}^*], \tag{30}$$

$$\delta \dot{\mathbf{u}}^* \cdot [\nabla \cdot \delta \boldsymbol{\tau} = -i\omega(\rho \delta \dot{\mathbf{u}} + \rho_f \delta \dot{\mathbf{w}})], \tag{31}$$

$$\delta \dot{\mathbf{w}}^* \cdot [-\nabla \delta p = -i\omega \rho_f \delta \dot{\mathbf{u}} + \eta \mathbf{K}^{-1} \cdot \delta \dot{\mathbf{w}} - \eta \mathbf{K}^{-1} \cdot \mathbf{L} \cdot \delta \mathbf{E}], \tag{32}$$

where the source terms are now missing since they are common to both sets of fields and where $\delta \mathbf{g} = -\nabla \delta p + i\omega \delta \dot{\mathbf{u}}$. The difference stresses $\delta \boldsymbol{\tau}$ and $-\delta p$ have the obvious definition from Eqs. (6) and (7). Going through the equivalent steps that led to the energy balance and then integrating over the volume V and using the divergence theorem gives

$$\int_S \mathbf{n} \cdot \text{Re}\{\delta \mathbf{E} \times \delta \mathbf{H}^* - \delta \boldsymbol{\tau} \cdot \delta \dot{\mathbf{u}}^* + \delta p \delta \dot{\mathbf{w}}^*\} dS = \int_V \delta Q' dV, \tag{33}$$

where

$$\delta Q' = \omega \delta \mathbf{H}^* \cdot \boldsymbol{\mu}'' \cdot \delta \mathbf{H} + [\delta \mathbf{E}^*, \delta \mathbf{g}^*] \cdot \begin{bmatrix} \mathbf{S}' + \omega \boldsymbol{\epsilon}'' & \mathbf{L}' \\ \mathbf{L}' & \mathbf{K}' \end{bmatrix} \cdot \begin{bmatrix} \delta \mathbf{E} \\ \delta \mathbf{g} \end{bmatrix} + \omega [(\nabla \cdot \delta \mathbf{w})^*, \delta e_1^*, \dots, \delta e_6^*] \cdot \begin{bmatrix} -M'' & -\mathcal{C}_1'' & \dots & -\mathcal{C}_6'' \\ -\mathcal{C}_1'' & -\mathcal{L}_{11}'' & \dots & -\mathcal{L}_{16}'' \\ \vdots & \vdots & \ddots & \vdots \\ -\mathcal{C}_6'' & -\mathcal{L}_{16}'' & \dots & -\mathcal{L}_{66}'' \end{bmatrix} \cdot \begin{bmatrix} \nabla \cdot \delta \mathbf{w} \\ \delta e_1 \\ \vdots \\ \delta e_6 \end{bmatrix}. \quad (34)$$

The difference strains δe_α are defined as in Eq. (26).

If it is now required that each of the following three types of boundary conditions are established on the surface S

- (i) $\mathbf{n} \times \mathbf{H}$ or $\mathbf{n} \times \mathbf{E}$ specified,
- (ii) $\mathbf{n} \cdot \boldsymbol{\tau}$ or $\mathbf{u}$ specified, (35)
- (iii) p or $\mathbf{n} \cdot \mathbf{w}$ specified,

then the surface integral in Eq. (33) vanishes. Because V is arbitrary, we then must have

$$\delta Q' = 0 \quad (36)$$

everywhere within V . Because each of the coefficient matrices in the three quadratic forms comprising $\delta Q'$ are positive definite, we then must have

$$\delta \mathbf{E} = 0, \quad \delta \mathbf{g} = 0, \quad \delta \mathbf{H} = 0, \quad \nabla \cdot \delta \mathbf{w} = 0, \quad \delta e_\alpha = 0. \quad (37)$$

From $\delta \mathbf{E} = 0$ and $\delta \mathbf{g} = 0$, Eq. (32) gives (after multiplying by $\mathbf{K}/\eta$ and using the definition of $\delta \mathbf{g}$) that $\delta \mathbf{w} = 0$. From $\nabla \cdot \delta \mathbf{w} = 0$ and $\delta e_\alpha = 0$, we have that $\delta \boldsymbol{\tau} = 0$ so that Eq. (31) gives $\delta \mathbf{u} = 0$. Thus because all the difference fields are zero, the electroseismic problem possesses unique field solutions when the boundary conditions in Eqs. (35) are satisfied.

Under conditions where $\boldsymbol{\epsilon}''$, $\boldsymbol{\mu}''$, M'' , $\mathcal{C}_\alpha''$, and $\mathcal{L}_{\alpha\beta}''$ are all identically zero (the only loss is due to macroscopic transport), the requirement $\delta Q' = 0$ only gives that $\delta \mathbf{E} = 0$ and $\delta \mathbf{g} = 0$. However, Eq. (29) gives $\delta \mathbf{H} = 0$, while Eq. (32) again gives $\delta \mathbf{w} = 0$. Thus in this case, $\delta \mathbf{u}$ satisfies $\nabla \cdot (\mathcal{L} : \nabla \delta \mathbf{u}) = -\omega^2 \rho \delta \mathbf{u}$ which are the elastodynamic equations. It is well known (e.g., Love, 1944, Sec. 124) that the displacement field is unique in elastodynamics so long as 1) $\mathbf{n} \cdot \boldsymbol{\tau}$ or $\mathbf{u}$ are specified on S , and 2) at some initial time the displacements are known. Without this initial condition, the solid-displacement field is nonunique when the framework of grains is lossless. Thus when the conditions of Eq. (35) are satisfied and when there is some initial knowledge of the solid displacements (e.g., $\mathbf{u} = 0$ at time zero), the electroseismic wave fields are always unique throughout V .

D. Reciprocity

Define $\mathbf{E}_1$, $\mathbf{H}_1$, $\mathbf{u}_1$, and $\mathbf{w}_1$ as the fields generated by the distribution of sources $\mathbf{C}_1$, $\mathbf{F}_1$, and $\mathbf{f}_1$, and the fields $\mathbf{E}_2$, $\mathbf{H}_2$, $\mathbf{u}_2$, and $\mathbf{w}_2$ as the fields generated by $\mathbf{C}_2$, $\mathbf{F}_2$, and $\mathbf{f}_2$. All of these fields and sources exist in the same (possibly heteroge-

neous) porous material. Consider the following products involving the governing equations (which have been slightly rewritten)

$$\mathbf{H}_i \cdot [\nabla \times \mathbf{E}_j = i\omega \boldsymbol{\mu} \cdot \mathbf{H}_j], \quad (38)$$

$$\mathbf{E}_i \cdot [\nabla \times \mathbf{H}_j = -i\omega \mathbf{P} \cdot \mathbf{E}_j - \omega^2 \mathbf{L} \cdot \mathbf{R} \cdot \mathbf{w}_j + \mathbf{C}_j], \quad (39)$$

$$-i\omega \mathbf{u}_i \cdot [\nabla \cdot \boldsymbol{\tau}_j = -\omega^2 (\rho \mathbf{u}_j + \rho_f \mathbf{w}_j) - \mathbf{F}_j], \quad (40)$$

$$-i\omega \mathbf{w}_i \cdot [-\nabla p_j = -\omega^2 (\rho_f \mathbf{u}_j + \mathbf{R} \cdot \mathbf{w}_j) + i\omega \mathbf{R} \cdot \mathbf{L} \cdot \mathbf{E}_j - \mathbf{f}_j], \quad (41)$$

where we have defined an effective complex inertia tensor $\mathbf{R}$ for the fluid in relative motion

$$\mathbf{R} \equiv \frac{i\eta}{\omega} \mathbf{K}^{-1}, \quad (42)$$

and an effective complex electrical-permittivity tensor $\mathbf{P}$ as

$$\mathbf{P} \equiv \boldsymbol{\epsilon} + \frac{i}{\omega} \mathbf{S} - \mathbf{L} \cdot \mathbf{R} \cdot \mathbf{L}. \quad (43)$$

Using $\mathbf{R}$ and $\mathbf{P}$ symbolically emphasizes the wave properties of the equations as opposed to the diffusive properties. Both $\mathbf{R}$ and $\mathbf{P}$ are symmetric (if $\mathbf{L}$ is taken as symmetric) so that $\mathbf{R} \cdot \mathbf{L} = \mathbf{L} \cdot \mathbf{R}$. The source indices i and j in Eqs. (38)–(41) equal 1 or 2 with $i \neq j$. Letting $i=1$ and $j=2$ and then $i=2$ and $j=1$, allows the following to be constructed by subtraction

$$\nabla \cdot (\mathbf{E}_1 \times \mathbf{H}_2 - \mathbf{E}_2 \times \mathbf{H}_1) = \omega^2 (\mathbf{L} \cdot \mathbf{R}) : (\mathbf{E}_1 \mathbf{w}_2 - \mathbf{E}_2 \mathbf{w}_1) - (\mathbf{E}_1 \cdot \mathbf{C}_2 - \mathbf{E}_2 \cdot \mathbf{C}_1), \quad (44)$$

$$\begin{aligned} -i\omega \nabla \cdot (\mathbf{u}_1 \cdot \boldsymbol{\tau}_2 - \mathbf{u}_2 \cdot \boldsymbol{\tau}_1) &= -i\omega \mathcal{C} : [(\nabla \mathbf{u}_1)(\nabla \cdot \mathbf{w}_2) - (\nabla \mathbf{u}_2)(\nabla \cdot \mathbf{w}_1)] \\ &\quad + i\omega^3 \rho_f (\mathbf{u}_1 \cdot \mathbf{w}_2 - \mathbf{u}_2 \cdot \mathbf{w}_1) + i\omega (\mathbf{u}_1 \cdot \mathbf{F}_2 - \mathbf{u}_2 \cdot \mathbf{F}_1), \end{aligned} \quad (45)$$

$$\begin{aligned} i\omega \nabla \cdot (\mathbf{w}_1 p_2 - \mathbf{w}_2 p_1) &= i\omega \mathcal{C} : [(\nabla \mathbf{u}_1)(\nabla \cdot \mathbf{w}_2) - (\nabla \mathbf{u}_2)(\nabla \cdot \mathbf{w}_1)] \\ &\quad - i\omega^3 \rho_f (\mathbf{u}_1 \cdot \mathbf{w}_2 - \mathbf{u}_2 \cdot \mathbf{w}_1) - \omega^2 (\mathbf{L} \cdot \mathbf{R}) : (\mathbf{E}_1 \mathbf{w}_2 - \mathbf{E}_2 \mathbf{w}_1) \\ &\quad + i\omega (\mathbf{w}_1 \cdot \mathbf{f}_2 - \mathbf{w}_2 \cdot \mathbf{f}_1). \end{aligned} \quad (46)$$

Adding these equations and integrating over a volume V of the material then gives

$$\begin{aligned} - \int_S \mathbf{n} \cdot [\mathbf{E}_1 \times \mathbf{H}_2 - \mathbf{E}_2 \times \mathbf{H}_1 - i\omega (\mathbf{u}_1 \cdot \boldsymbol{\tau}_2 - \mathbf{u}_2 \cdot \boldsymbol{\tau}_1 - \mathbf{w}_1 p_2 \\ + \mathbf{w}_2 p_1)] dS = \int_V [\mathbf{E}_1 \cdot \mathbf{C}_2 - \mathbf{E}_2 \cdot \mathbf{C}_1 - i\omega (\mathbf{u}_1 \cdot \mathbf{F}_2 - \mathbf{u}_2 \cdot \mathbf{F}_1 \\ + \mathbf{w}_1 \cdot \mathbf{f}_2 - \mathbf{w}_2 \cdot \mathbf{f}_1)] dV. \end{aligned} \quad (47)$$

This is the general statement of reciprocity. If the sources $(\mathbf{C}_i, \mathbf{F}_i, \mathbf{f}_i)$ all have finite spatial extent, then the fields from these sources will necessarily fall off at distances greater than the source dimensions. Thus, for finite sources, if the volume is expanded so that the fields are completely negligible on S , the following statement of reciprocity is obtained

$$\begin{aligned} & \int_{V_{C2}} \mathbf{E}_1 \cdot \mathbf{C}_2 dV - \int_{V_{C1}} \mathbf{E}_2 \cdot \mathbf{C}_1 dV - i\omega \left[\int_{V_{F2}} \mathbf{u}_1 \cdot \mathbf{F}_2 dV \right. \\ & \quad \left. - \int_{V_{F1}} \mathbf{u}_2 \cdot \mathbf{F}_1 dV \right] - i\omega \left[\int_{V_{f2}} \mathbf{w}_1 \cdot \mathbf{f}_2 dV \right. \\ & \quad \left. - \int_{V_{f1}} \mathbf{w}_2 \cdot \mathbf{f}_1 dV \right] = 0, \end{aligned} \quad (48)$$

where the volumes V_{C2} , V_{F2} , etc. are defined to just enclose their respective source terms.

This statement is very useful especially when the source fields are compact enough to be treated as point sources. As a specific example, consider the following two point-source distributions

$$\begin{aligned} \mathbf{F}_1(\mathbf{r}) &= mNg s_1(\omega) \delta(\mathbf{r} - \mathbf{r}_1) \hat{\mathbf{z}}, \\ \mathbf{f}_1(\mathbf{r}) &= mNg s_1(\omega) \delta(\mathbf{r} - \mathbf{r}_1) \hat{\mathbf{z}}, \quad \mathbf{C}_1 = 0, \\ \mathbf{F}_2(\mathbf{r}) &= 0, \quad \mathbf{f}_2(\mathbf{r}) = 0, \\ \mathbf{C}_2(\mathbf{r}) &= -i\omega d s_2(\omega) \delta(\mathbf{r} - \mathbf{r}_2) \hat{\mathbf{x}}. \end{aligned} \quad (49)$$

The first distribution corresponds to a vertical impact source; i.e., a mass m accelerating at Ng ($g=9.8 \text{ m/s}^2$) impacts at the point $\mathbf{r}_1$ in the vertical direction $\hat{\mathbf{z}}$ with a normalized spectra $s_1(\omega)$ (s_1 is unitless and is of order one in amplitude). The second distribution corresponds to that of an electric dipole of moment d , situated at position $\mathbf{r}_2$, oriented in the horizontal direction $\hat{\mathbf{x}}$, and possessing a normalized spectra $s_2(\omega)$. Reciprocity [Eq. (48)] then gives that

$$\hat{\mathbf{z}} \cdot [\mathbf{u}_2(\mathbf{r}_1) + \mathbf{w}_2(\mathbf{r}_1)] = \frac{s_2(\omega)}{s_1(\omega)} \frac{d}{mNg} \hat{\mathbf{x}} \cdot \mathbf{E}_1(\mathbf{r}_2). \quad (50)$$

If we know, for example, the horizontal component of the electric field recorded at position $\mathbf{r}_2$ due to a point impact at $\mathbf{r}_1$ [i.e., $\hat{\mathbf{x}} \cdot \mathbf{E}_1(\mathbf{r}_2)$], then we know automatically the vertical component of the displacement field recorded at position $\mathbf{r}_1$ due to a horizontal-dipole antenna at position $\mathbf{r}_2$ [i.e., $\hat{\mathbf{z}} \cdot [\mathbf{u}_2(\mathbf{r}_1) + \mathbf{w}_2(\mathbf{r}_1)]$].

Continuing with this example, let's say that the impact source is a hammer weighing 4 kg accelerated at $2g$ onto the surface of the Earth. Seismic waves are generated that then traverse an interface at depth generating an electric field. For an interface at a few tens of meters depth, a field strength of 10^{-6} V/m is easily recorded at the Earth's surface—see Mikhailov and Haartsen (1996) for experimental details involving a hammer source and Thompson and Gist (1993) for details involving an explosion source. Furthermore, if a horizontal-dipole current source on the Earth's surface produces an electromagnetic field that traverses this same interface, we expect seismic waves to be generated. In order to detect a seismic displacement on the order of 10^{-7} m (a typical displacement recorded in exploration surveys), Eq. (50) tells us that the dipole moment must be on the order of 10 Cm. Although somewhat large compared to conventional dipole sources used in exploration geophysics, an antenna with a dipole moment on this order could easily be constructed. Thus, as first suggested by Thompson and Gist (1993), there is the distinct possibility that detectable seismic

waves are generated as electromagnetic waves traverse interfaces at depth. This result is but one demonstration of the great utility of reciprocity. Another is given in the following section.

E. Representation integrals

Following the general idea of Gangi (1970), we now use reciprocity to arrive at the representation integrals for $\mathbf{u}$, $\mathbf{w}$, and $\mathbf{E}$ from which $\boldsymbol{\tau}$, p , and $\mathbf{H}$ can be obtained by differentiation. The primary fields $\mathbf{u}$, $\mathbf{w}$, and $\mathbf{E}$ are to be represented (or determined) within a volume V of material—possibly heterogeneous—which is bounded by a surface S . The fields are either due to sources $\mathbf{F}$, $\mathbf{f}$, and $\mathbf{C}$ distributed within V or else are due to sources external to V in which case all field types are assumed known over S .

Let us begin by defining three specific sets of point sources within V

$$\begin{aligned} F \text{ set: } & \mathbf{F}^F = \mathbf{s}^F \delta(\mathbf{r} - \mathbf{r}'), \quad \mathbf{f}^F = 0, \quad \mathbf{C}^F = 0; \\ f \text{ set: } & \mathbf{F}^f = 0, \quad \mathbf{f}^f = \mathbf{s}^f \delta(\mathbf{r} - \mathbf{r}'), \quad \mathbf{C}^f = 0; \\ C \text{ set: } & \mathbf{F}^C = 0, \quad \mathbf{f}^C = 0, \quad \mathbf{C}^C = \mathbf{s}^C \delta(\mathbf{r} - \mathbf{r}'). \end{aligned} \quad (51)$$

The vectors $\mathbf{s}^F$, $\mathbf{s}^f$, and $\mathbf{s}^C$ are assumed spatially constant but are otherwise arbitrary. Corresponding to each of these point-source sets there are produced the “fundamental” vectors $\mathbf{u}^\xi$, $\mathbf{w}^\xi$, $\mathbf{E}^\xi$, and $\mathbf{H}^\xi$ as well as the fundamental stress tensor $\boldsymbol{\tau}^\xi$ and the fundamental pressure p^ξ where $\xi=F, f$, or C depending on the point-source set.

The reciprocity statement is now used. Identifying our unknown fields $\mathbf{u}$, $\mathbf{w}$, $\mathbf{E}$, etc. with the fields “1” in Eq. (47) and our fundamental quantities $\mathbf{u}^\xi$, $\mathbf{w}^\xi$, $\mathbf{E}^\xi$, etc. with the fields “2,” then gives the first variant of the representation integrals

$$\begin{aligned} \mathbf{s}^\xi \cdot \mathbf{X}_\xi = & \int_V \{ \mathbf{E}^\xi \cdot \mathbf{C} - i\omega [\mathbf{u}^\xi \cdot \mathbf{F} + \mathbf{w}^\xi \cdot \mathbf{f}] \} dV - \int_S \mathbf{n} \cdot \{ \mathbf{E} \times \mathbf{H}^\xi \\ & - \mathbf{E}^\xi \times \mathbf{H} - i\omega [\boldsymbol{\tau}^\xi \cdot \mathbf{u} - \boldsymbol{\tau} \cdot \mathbf{u}^\xi - \mathbf{w} p^\xi + \mathbf{w}^\xi p] \} dS, \end{aligned} \quad (52)$$

where, for compactness of expression, we have temporarily redefined the independent wave fields as $\mathbf{X}_F \equiv -i\omega \mathbf{u}$, $\mathbf{X}_f \equiv -i\omega \mathbf{w}$, and $\mathbf{X}_C \equiv \mathbf{E}$. Next, the Green's tensors are introduced as the linear proportionalities between the point-source directions $\mathbf{s}^\xi$ and the fundamental vectors

$$\mathbf{u}^\xi(\mathbf{r}|\mathbf{r}'; \mathbf{s}^\xi) = \mathbf{G}_u^\xi(\mathbf{r}|\mathbf{r}') \cdot \mathbf{s}^\xi, \quad (53)$$

$$\mathbf{w}^\xi(\mathbf{r}|\mathbf{r}'; \mathbf{s}^\xi) = \mathbf{G}_w^\xi(\mathbf{r}|\mathbf{r}') \cdot \mathbf{s}^\xi, \quad (54)$$

$$\mathbf{E}^\xi(\mathbf{r}|\mathbf{r}'; \mathbf{s}^\xi) = \mathbf{G}_E^\xi(\mathbf{r}|\mathbf{r}') \cdot \mathbf{s}^\xi, \quad (55)$$

where again $\xi=F, f$, or C . Thus there are a total of nine primary Green's tensors involved in electroseismic theory. The Green's tensors do not depend on the arbitrary source vectors $\mathbf{s}^\xi$. The notation $\mathbf{r}|\mathbf{r}'$ is indicating, as usual, the response at point $\mathbf{r}$ due to a point source at $\mathbf{r}'$. Because the $\mathbf{s}^\xi$ are spatially constant, when Eqs. (53)–(55) are introduced into the governing equations, the following tensor equations are obtained

$$\nabla \times \mathbf{G}_E^\xi = i\omega \boldsymbol{\mu} \cdot \mathbf{G}_H^\xi, \quad (56)$$

$$\nabla \times \mathbf{G}_H^\xi = -i\omega \mathbf{P} \cdot \mathbf{G}_E^\xi - \omega^2 (\mathbf{L} \cdot \mathbf{R}) \cdot \mathbf{G}_w^\xi + \delta_{C\xi} \mathbf{I} \delta(\mathbf{r} - \mathbf{r}'), \quad (57)$$

$$\nabla \cdot {}_3\mathbf{T}^\xi = -\omega^2 (\rho \mathbf{G}_u^\xi + \rho_f \mathbf{G}_w^\xi) - \delta_{F\xi} \mathbf{I} \delta(\mathbf{r} - \mathbf{r}'), \quad (58)$$

$$-\nabla \mathbf{p}^\xi = -\omega^2 (\rho_f \mathbf{G}_u^\xi + \mathbf{R} \cdot \mathbf{G}_w^\xi) + i\omega (\mathbf{R} \cdot \mathbf{L}) \cdot \mathbf{G}_E^\xi - \delta_{f\xi} \mathbf{I} \delta(\mathbf{r} - \mathbf{r}'), \quad (59)$$

where the third-order Green's bulk-stress tensor ${}_3\mathbf{T}^\xi$ and the Green's fluid-pressure vector $\mathbf{p}^\xi$ are related to the primary tensors as

$${}_3\mathbf{T}^\xi = {}_4\mathcal{L} : \nabla \mathbf{G}_u^\xi + \mathcal{C} \nabla \cdot \mathbf{G}_w^\xi \quad (60)$$

$$-\mathbf{p}^\xi = \mathcal{C} : \nabla \mathbf{G}_u^\xi + M \nabla \cdot \mathbf{G}_w^\xi. \quad (61)$$

The $\delta_{C\xi}$, $\delta_{F\xi}$, and $\delta_{f\xi}$ are Kronecker delta functions. Equations (56)–(61) are the defining differential equations for the Green's tensors.

We now establish the reciprocal properties of the primary Green's tensors. Using Eq. (48), one can establish relations between the fundamental vectors such as

$$\begin{aligned} & \int_{V_{F2}} [\mathbf{u}^F(\mathbf{r}|\mathbf{r}_1; \mathbf{s}_1^F) \cdot \mathbf{s}_2^F] \delta(\mathbf{r} - \mathbf{r}_2) d^3\mathbf{r} \\ &= \int_{V_{F1}} [\mathbf{u}^F(\mathbf{r}|\mathbf{r}_2; \mathbf{s}_2^F) \cdot \mathbf{s}_1^F] \delta(\mathbf{r} - \mathbf{r}_1) d^3\mathbf{r}, \end{aligned} \quad (62)$$

which gives

$$\mathbf{s}_2^F \cdot \mathbf{u}^F(\mathbf{r}_2|\mathbf{r}_1; \mathbf{s}_1^F) = \mathbf{s}_1^F \cdot \mathbf{u}^F(\mathbf{r}_1|\mathbf{r}_2; \mathbf{s}_2^F). \quad (63)$$

The following identities can now be established where $\hat{\mathbf{x}}_i$ is the unit vector along the i th coordinate axis and where summation over the repeated indices is assumed

$$\begin{aligned} \mathbf{G}_u^F(\mathbf{r}_1|\mathbf{r}_2) &= \hat{\mathbf{x}}_i [\hat{\mathbf{x}}_i \cdot \mathbf{G}_u^F(\mathbf{r}_1|\mathbf{r}_2) \cdot \hat{\mathbf{x}}_j] \hat{\mathbf{x}}_j \\ &= \hat{\mathbf{x}}_i [\hat{\mathbf{x}}_i \cdot \mathbf{u}^F(\mathbf{r}_1|\mathbf{r}_2; \hat{\mathbf{x}}_j)] \hat{\mathbf{x}}_j \\ &= \hat{\mathbf{x}}_i [\hat{\mathbf{x}}_j \cdot \mathbf{u}^F(\mathbf{r}_2|\mathbf{r}_1; \hat{\mathbf{x}}_i)] \hat{\mathbf{x}}_j \\ &= \hat{\mathbf{x}}_i [\hat{\mathbf{x}}_j \cdot \mathbf{G}_u^F(\mathbf{r}_2|\mathbf{r}_1) \cdot \hat{\mathbf{x}}_i] \hat{\mathbf{x}}_j \\ &= [\mathbf{G}_u^F(\mathbf{r}_2|\mathbf{r}_1)]^T. \end{aligned} \quad (64)$$

Equation (63) was used between the second and third lines. Identical manipulations on the other fundamental vectors gives

$$\mathbf{G}_u^f(\mathbf{r}_1|\mathbf{r}_2) = [\mathbf{G}_w^F(\mathbf{r}_2|\mathbf{r}_1)]^T, \quad (65)$$

$$-i\omega \mathbf{G}_u^C(\mathbf{r}_1|\mathbf{r}_2) = [\mathbf{G}_E^F(\mathbf{r}_2|\mathbf{r}_1)]^T, \quad (66)$$

$$\mathbf{G}_w^f(\mathbf{r}_1|\mathbf{r}_2) = [\mathbf{G}_w^f(\mathbf{r}_2|\mathbf{r}_1)]^T, \quad (67)$$

$$-i\omega \mathbf{G}_w^C(\mathbf{r}_1|\mathbf{r}_2) = [\mathbf{G}_E^f(\mathbf{r}_2|\mathbf{r}_1)]^T, \quad (68)$$

$$\mathbf{G}_E^C(\mathbf{r}_1|\mathbf{r}_2) = [\mathbf{G}_E^C(\mathbf{r}_2|\mathbf{r}_1)]^T. \quad (69)$$

These define the general reciprocal (symmetry) properties of the Green's tensors.

Using these results in Eq. (52) then gives the representation integrals in final form

$$\begin{aligned} \mathbf{u}(\mathbf{r}) &= \Sigma_F(\mathbf{r}) + \int_V \{ \mathbf{G}_u^C(\mathbf{r}|\mathbf{r}') \cdot \mathbf{C}(\mathbf{r}') + \mathbf{G}_u^F(\mathbf{r}|\mathbf{r}') \cdot \mathbf{F}(\mathbf{r}') \\ &\quad + \mathbf{G}_u^f(\mathbf{r}|\mathbf{r}') \cdot \mathbf{f}(\mathbf{r}') \} d^3\mathbf{r}', \end{aligned} \quad (70)$$

$$\begin{aligned} \mathbf{w}(\mathbf{r}) &= \Sigma_f(\mathbf{r}) + \int_V \{ \mathbf{G}_w^C(\mathbf{r}|\mathbf{r}') \cdot \mathbf{C}(\mathbf{r}') + \mathbf{G}_w^F(\mathbf{r}|\mathbf{r}') \cdot \mathbf{F}(\mathbf{r}') \\ &\quad + \mathbf{G}_w^f(\mathbf{r}|\mathbf{r}') \cdot \mathbf{f}(\mathbf{r}') \} d^3\mathbf{r}', \end{aligned} \quad (71)$$

$$\begin{aligned} \mathbf{E}(\mathbf{r}) &= -i\omega \Sigma_C(\mathbf{r}) + \int_V \{ \mathbf{G}_E^C(\mathbf{r}|\mathbf{r}') \cdot \mathbf{C}(\mathbf{r}') \\ &\quad + \mathbf{G}_E^F(\mathbf{r}|\mathbf{r}') \cdot \mathbf{F}(\mathbf{r}') + \mathbf{G}_E^f(\mathbf{r}|\mathbf{r}') \cdot \mathbf{f}(\mathbf{r}') \} d^3\mathbf{r}', \end{aligned} \quad (72)$$

where the vectors Σ_ξ ($\xi = F, f$, or C) are the surface-integral contributions and can be written

$$\begin{aligned} \Sigma_\xi(\mathbf{r}) &= \int_S \left\{ \frac{1}{i\omega} [[\mathbf{G}_E^\xi(\mathbf{r}'|\mathbf{r})]^T \times \mathbf{n}] \cdot \mathbf{H}(\mathbf{r}') \right. \\ &\quad - [(\omega^2 \boldsymbol{\mu})^{-1} \cdot [\nabla' \times \mathbf{G}_E^\xi(\mathbf{r}'|\mathbf{r})]^T \times \mathbf{n}] \cdot \mathbf{E}(\mathbf{r}') \\ &\quad - \mathbf{n} \cdot [{}_4\mathcal{L} : \nabla' [\mathbf{G}_u^\xi(\mathbf{r}'|\mathbf{r})]^T] \\ &\quad + \mathcal{C} \nabla' \cdot [\mathbf{G}_w^\xi(\mathbf{r}'|\mathbf{r})]^T \cdot \mathbf{u}(\mathbf{r}') \\ &\quad + [\mathcal{C} : \nabla' \mathbf{G}_u^\xi(\mathbf{r}'|\mathbf{r}) \mathbf{n} + M \nabla' \cdot \mathbf{G}_w^\xi(\mathbf{r}'|\mathbf{r}) \mathbf{n}] \cdot \mathbf{w}(\mathbf{r}') \\ &\quad \left. + [\mathbf{G}_u^\xi(\mathbf{r}'|\mathbf{r}) \cdot \mathbf{n}] \cdot \boldsymbol{\tau}(\mathbf{r}') - [\mathbf{n} \cdot \mathbf{G}_w^\xi(\mathbf{r}'|\mathbf{r})] p(\mathbf{r}') \right\} d^2\mathbf{r}'. \end{aligned} \quad (73)$$

Within these surface integrals, we have that $\mathbf{n}$, $\boldsymbol{\mu}$, ${}_4\mathbf{L}$, $\mathbf{C}$, and M are all possible functions of $\mathbf{r}'$. Using the symmetry relations [Eqs. (64)–(69)], it is possible to express all of the Green's tensors within the surface integrals as functions of $\mathbf{r}|\mathbf{r}'$ (instead of $\mathbf{r}'|\mathbf{r}$ as given). Note that only the nine primary Green's tensors defined as $\mathbf{G}_u^\xi(\mathbf{r}|\mathbf{r}')$, $\mathbf{G}_w^\xi(\mathbf{r}|\mathbf{r}')$, and $\mathbf{G}_E^\xi(\mathbf{r}|\mathbf{r}')$, where $\xi = F, f$, or C are required in the representation integrals. If analytical solutions are sought, then only six of these nine ($\mathbf{G}_u^F$, $\mathbf{G}_w^F$, $\mathbf{G}_E^F$, $\mathbf{G}_u^f$, $\mathbf{G}_w^f$, and $\mathbf{G}_E^C$) need be determined from Eqs. (56)–(61) because $\mathbf{G}_u^f$, $\mathbf{G}_u^C$, and $\mathbf{G}_w^C$ are obtainable from the symmetry relations.

II. WAVES IN AN ISOTROPIC AND HOMOGENEOUS WHOLESAPCE

The equations will now be solved for the simple case of a homogeneous and isotropic wholespace. In this case, the complete set of governing equations can be written

$$\begin{aligned} & [(H - G) \nabla \nabla + (G \nabla^2 + \omega^2 \rho) \mathbf{I}] \cdot \mathbf{u} + [C \nabla \nabla + \omega^2 \rho_f \mathbf{I}] \cdot \mathbf{w} \\ &= -\mathbf{F}, \end{aligned} \quad (74)$$

$$[C \nabla \nabla + \omega^2 \rho_f \mathbf{I}] \cdot \mathbf{u} + [M \nabla \nabla + \omega^2 \tilde{\rho} \mathbf{I}] \cdot \mathbf{w} - i\omega \tilde{\rho} L \mathbf{E} = -\mathbf{f}, \quad (75)$$

$$[\nabla \nabla - (\nabla^2 + \omega^2 \mu \tilde{\epsilon}) \mathbf{I}] \cdot \mathbf{E} + i\omega^3 \mu \tilde{\rho} L \mathbf{w} = i\omega \mu \mathbf{C}, \quad (76)$$

where

$$\tilde{\rho}(\omega) = \frac{i}{\omega} \frac{\eta}{k(\omega)} \quad (77)$$

is the effective density of the fluid in relative motion and where

$$\tilde{\epsilon}(\omega) = \epsilon(\omega) + \frac{i}{\omega} \sigma(\omega) - \tilde{\rho}(\omega) L^2(\omega) \quad (78)$$

is the effective electrical permittivity of the porous continuum. Both the plane-wave and the point-source responses of Eqs. (74)–(76) will be obtained.

A. Plane waves

Setting $\mathbf{C}$, $\mathbf{F}$, and $\mathbf{f}$ to zero and working with the plane-wave representations given by

$$\begin{bmatrix} (H-G)s^2(\hat{\mathbf{k}} \cdot \hat{\mathbf{u}})\hat{\mathbf{k}} - (\rho - Gs^2)\hat{\mathbf{u}} & Cs^2(\hat{\mathbf{k}} \cdot \hat{\mathbf{w}})\hat{\mathbf{k}} - \rho_f \hat{\mathbf{w}} & 0 \\ Cs^2(\hat{\mathbf{k}} \cdot \hat{\mathbf{u}})\hat{\mathbf{k}} - \rho_f \hat{\mathbf{u}} & Ms^2(\hat{\mathbf{k}} \cdot \hat{\mathbf{w}})\hat{\mathbf{k}} - \tilde{\rho} \hat{\mathbf{w}} & \frac{i}{\omega} \tilde{\rho} L \hat{\mathbf{e}} \\ 0 & -i\omega \mu \tilde{\rho} L \hat{\mathbf{w}} & s^2(\hat{\mathbf{k}} \cdot \hat{\mathbf{e}})\hat{\mathbf{k}} + (\tilde{\epsilon} \mu - s^2)\hat{\mathbf{e}} \end{bmatrix} \begin{bmatrix} \mathcal{U} \\ \mathcal{W} \\ \mathcal{E} \end{bmatrix} = 0. \quad (83)$$

We will consider in this section the various transverse and longitudinal wave modes associated with these equations.

By definition, a transverse mode is one in which the material response is in a direction perpendicular (transverse) to the direction of propagation (i.e., $\hat{\mathbf{k}} \cdot \hat{\mathbf{u}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{w}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{e}} = 0$). Similarly, a longitudinal mode is one in which the material response and the wave direction are parallel (i.e., $\hat{\mathbf{k}} \cdot \hat{\mathbf{u}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{w}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{e}} = 1$). It is easily shown from Eq. (83) that wave modes with mixed response (e.g., $\hat{\mathbf{k}} \cdot \hat{\mathbf{u}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{w}} = 1$ but $\hat{\mathbf{k}} \cdot \hat{\mathbf{e}} = 0$, etc.) lead to the trivial solution $\mathcal{U} = \mathcal{W} = \mathcal{E} = 0$ and, thus do not exist. Furthermore, one obtains that for both transverse and longitudinal modes $\hat{\mathbf{u}} = \hat{\mathbf{w}} = \hat{\mathbf{e}}$.

In what follows, to obtain the real phase velocity v and the real attenuation coefficient α (units of inverse length) from the complex slownesses s , one must use the expressions

$$v(\omega) = 1/\text{Re}\{s(\omega)\}, \quad (84)$$

$$\alpha(\omega) = \omega \text{Im}\{s(\omega)\}. \quad (85)$$

1. Transverse modes

Upon putting $\hat{\mathbf{k}} \cdot \hat{\mathbf{u}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{w}} = \hat{\mathbf{k}} \cdot \hat{\mathbf{e}} = 0$ and $\hat{\mathbf{u}} = \hat{\mathbf{w}} = \hat{\mathbf{e}}$ in Eq. (83), the transverse-mode system can be written

$$\begin{bmatrix} \rho - Gs^2 & \rho_f & 0 \\ \rho_f & \tilde{\rho} & -\frac{i}{\omega} \tilde{\rho} L \\ 0 & -i\omega \mu \tilde{\rho} L & \tilde{\epsilon} \mu - s^2 \end{bmatrix} \begin{bmatrix} \mathcal{U} \\ \mathcal{W} \\ \mathcal{E} \end{bmatrix} = 0. \quad (86)$$

Nontrivial solutions for $\mathcal{U}$, $\mathcal{W}$, and $\mathcal{E}$ are obtained only if the determinant of the matrix vanishes. Upon taking the determinant, there are found two distinct complex transverse-wave slownesses

$$\mathbf{u} = \mathcal{U} \exp(i\mathbf{k} \cdot \mathbf{r}) \hat{\mathbf{u}}, \quad (79)$$

$$\mathbf{w} = \mathcal{W} \exp(i\mathbf{k} \cdot \mathbf{r}) \hat{\mathbf{w}}, \quad (80)$$

$$\mathbf{E} = \mathcal{E} \exp(i\mathbf{k} \cdot \mathbf{r}) \hat{\mathbf{e}}, \quad (81)$$

where $\hat{\mathbf{u}}$, $\hat{\mathbf{w}}$, and $\hat{\mathbf{e}}$ are unit vectors and where the wave vector $\mathbf{k}$ may be written

$$\mathbf{k} = \omega s(\omega) \hat{\mathbf{k}}, \quad (82)$$

with $s(\omega)$ the complex slowness (to be determined) and $\hat{\mathbf{k}}$ a unit vector in the direction of propagation, allows the governing equations to be written as

$$2s_{s,em}^2 = \frac{\rho_t}{G} + \mu \tilde{\epsilon} \left(1 + \frac{\tilde{\rho} L^2}{\tilde{\epsilon}} \right) \pm \left\{ \left[\frac{\rho_t}{G} - \mu \tilde{\epsilon} \left(1 + \frac{\tilde{\rho} L^2}{\tilde{\epsilon}} \right) \right]^2 - 4\mu \frac{\rho_f^2 L^2}{G} \right\}^{1/2}, \quad (87)$$

where the $+$ in front of the radical corresponds to the mechanical shear-wave slowness s_s and the $-$ corresponds to the electromagnetic (EM) slowness s_{em} . The complex density ρ_t is defined

$$\rho_t = \rho - \rho_f^2 / \tilde{\rho}. \quad (88)$$

Associating the positive square root with the shear wave (and the negative with the EM wave) comes from considering the case where there is no electrokinetic coupling ($L=0$) in which case s_s and s_{em} take on their usual definitions. The allowance for L makes essentially no difference for these transverse-wave slownesses but we will see that it can have a non-negligible effect on the compressional-wave slownesses.

In terms of the amplitude $\mathcal{U}$, Eq. (86) gives

$$\mathcal{W} = \frac{G}{\rho_f} \left(s^2 - \frac{\rho}{G} \right) \mathcal{U}, \quad (89)$$

$$\mathcal{E} = -i\omega \mu \tilde{\rho} L \frac{G}{\rho_f} \left(\frac{s^2 - \rho/G}{s^2 - \mu \tilde{\epsilon}} \right) \mathcal{U}. \quad (90)$$

When $s = s_s$, these amplitudes correspond to that of a plane shear wave, while when $s = s_{em}$, they correspond to a plane EM wave.

There are always two polarizations possible for plane transverse waves. If the plane wave propagates in the x - z plane of a Cartesian system (x, y, z) , then one polarization corresponds to $\hat{\mathbf{u}} = \hat{\mathbf{w}} = \hat{\mathbf{e}} = \hat{\mathbf{y}}$ and is the SH - TE polarization (horizontally polarized displacement of shear waves “ SH ” and horizontally polarized electric field of EM waves “ TE ”). The other polarization is given by $\hat{\mathbf{u}} = \hat{\mathbf{w}} = \hat{\mathbf{e}} = \hat{\mathbf{y}} \times \hat{\mathbf{k}}$

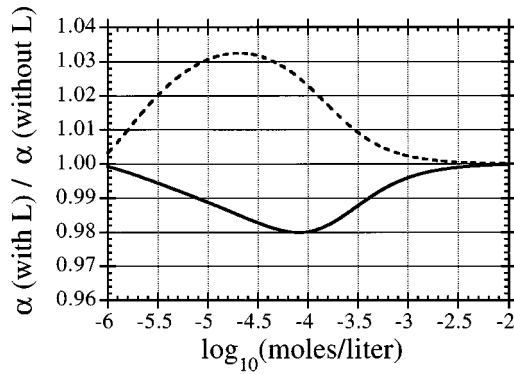


FIG. 1. The effect of the electrokinetic coupling coefficient L on the compressional wave attenuation coefficients as a function of salinity of the saturating electrolyte. The dashed line is the slow-compressional wave while the solid line is the fast-compressional wave.

and is the SV - TM polarization (vertically polarized displacements of shear waves “ SV ” and horizontally polarized magnetic field of EM waves “ TM ”).

2. Longitudinal modes

Putting $\hat{\mathbf{u}}=\hat{\mathbf{w}}=\hat{\mathbf{e}}=\hat{\mathbf{k}}$ in Eq. (83) gives the system

$$\begin{bmatrix} Hs^2 - \rho & Cs^2 - \rho_f & 0 \\ Cs^2 - \rho_f & Ms^2 - \tilde{\rho} & \frac{i}{\omega} \tilde{\rho} L \\ 0 & i\omega\mu\tilde{\rho} L & -\mu\tilde{\epsilon} \end{bmatrix} \begin{bmatrix} \mathcal{U} \\ \mathcal{W} \\ \mathcal{E} \end{bmatrix} = 0. \quad (91)$$

Taking the determinant and setting it to zero gives two distinct complex longitudinal-wave slownesses

$$2s_{pf,ps}^2 = \gamma \mp \sqrt{\gamma^2 - \frac{4\tilde{\rho}\rho}{MH - C^2} \left(\frac{\rho_t}{\rho} + \frac{\tilde{\rho} L^2}{\tilde{\epsilon}} \right)}, \quad (92)$$

where the $-$ corresponds to Biot’s “fast”-compressional-wave slowness s_{pf} while the $+$ corresponds to the “slow”-compressional-wave slowness s_{ps} and where

$$\gamma = \frac{\rho M + \tilde{\rho} H(1 + \tilde{\rho} L^2 / \tilde{\epsilon}) - 2\rho_f C}{HM - C^2}. \quad (93)$$

As $L \rightarrow 0$ (no electrokinetic coupling), these slownesses correspond exactly to those defined by Biot (1956).

In terms of the amplitude $\mathcal{U}$, we then have

$$\mathcal{W} = - \left(\frac{Hs^2 - \rho}{Cs^2 - \rho_f} \right) \mathcal{U}, \quad (94)$$

$$\mathcal{E} = - \frac{i\omega\tilde{\rho} L}{\tilde{\epsilon}} \left(\frac{Hs^2 - \rho}{Cs^2 - \rho_f} \right) \mathcal{U}. \quad (95)$$

When $s = s_{pf}$, these amplitudes correspond to a fast-compressional wave, while when $s = s_{ps}$, they correspond to a slow-compressional wave.

As is seen in Fig. 1, the presence of the coupling coefficient L can have a slight influence on the compressional-attenuation coefficients (on the order of a few percent). The effect on the phase velocity is less than one percent for this example. The material properties used in Fig. 1 correspond to a well-consolidated sandstone with 10-mD permeability

and electrokinetic properties corresponding to quartz. Expressions for the various terms in Eq. (92) are given by Pride (1994) and the electrokinetic properties are consistent with the data of Li and de Bruyn (1966). The wave frequency is 100 Hz. We now interpret Fig. 1.

In the fast wave, the deviations in fluid pressure are generated by the frame compressions associated with the wave—regions of enhanced fluid pressure are generated by a frame contraction while regions of decreased fluid pressure are generated by a frame expansion. The frame compression and fluid compression occur in phase in the fast wave. These half-wavelength scale fluid-pressure gradients generate a fluid flow that carries along the excess counterions of the diffuse-double layer; i.e., the wave generates a streaming current. The counterions accumulate in the low-fluid-pressure regions and are depleted in the high-fluid-pressure regions thus generating an electric field. This electric field drives a conduction current that balances the streaming current so that the final degree of charge separation is controlled by both the electrokinetic and conductivity properties of the material. The generated electric field acts as a body force on the counterions in the diffuse-double layer that opposes the pressure gradient and, thus, reduces the amount of relative flow. Therefore, in the fast wave, losses due to viscous shearing are reduced due to the electrokinetic effects as is seen in Fig. 1.

In the slow wave, the deviations in fluid pressure are generated directly from the fluid compressions associated with this wave. A region of high-fluid pressure generates a frame expansion while a region of low-fluid pressure generates a frame contraction. The frame compression and fluid compression thus occur out of phase in the slow wave resulting in the high fluid-pressure regions having an accumulation of counter charge and the low-fluid-pressure regions having a depletion (opposite from the fast wave). The fluid-pressure gradients will drive a relative flow that tends to reduce these charge separations (and pressure differences); however, the sign of the charge accumulations will not be reversed by this flow. Thus in the slow wave, the electric field acts as a body force that enhances the relative flow driven by the pressure gradient and results in a larger amount of viscous shearing as is seen in Fig. 1.

For both waves, the dependence on salinity is explained as follows. Both the width and the excess-charge content of the diffuse-double layers grow with decreasing salinity while the conductivity decreases. Thus as the salinity drops the amount of charge separation increases. However, when the salinity has decreased to values where surface conduction in the diffuse layer becomes important, the magnitude of the charge separation is decreased and the effect on the relative flow and attenuation is reduced all of which is seen in Fig. 1.

3. The eigen response

We now summarize the above for the case of plane waves in the x - z plane of a Cartesian system (x, y, z) . The plane-wave direction is characterized as

$$\hat{\mathbf{k}} = [\sin \theta, 0, \cos \theta]^T \quad (96)$$

$$= [p/s, 0, q/s]^T, \quad (97)$$

where θ measures the direction of the wave relative to the positive $\hat{\mathbf{z}}$ direction, p is the “horizontal” slowness, q is the “vertical” slowness, s is the appropriate plane-wave slowness, and $p^2 + q^2 = s^2$. In what follows, p will be used to characterize wave direction rather than θ . Note that in general, either p or q (or both) are complex because s is complex.

For the $SH-TE$ transverse waves, the response is

$$\mathbf{u} = \hat{\mathbf{y}} e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{SH-TE}, \quad (98)$$

$$\mathbf{w} = \frac{G}{\rho_f} \left(s^2 - \frac{\rho}{G} \right) \hat{\mathbf{y}} e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{SH-TE}, \quad (99)$$

$$\boldsymbol{\tau} = i\omega s G [\hat{\mathbf{k}} \hat{\mathbf{y}} + \hat{\mathbf{y}} \hat{\mathbf{k}}] e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{SH-TE}, \quad (100)$$

$$p = 0, \quad (101)$$

$$\mathbf{E} = i\omega \mu \frac{\tilde{\rho}}{\rho_f} L G \beta_T \hat{\mathbf{y}} e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{SH-TE} \quad (102)$$

$$\mathbf{H} = i\omega s \frac{\tilde{\rho}}{\rho_f} L G \beta_T \hat{\mathbf{k}} \times \hat{\mathbf{y}} e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{SH-TE}, \quad (103)$$

where

$$\beta_T = -\frac{s^2 - \rho/G}{s^2 - \mu \tilde{\epsilon}}, \quad (104)$$

$$\hat{\mathbf{k}} \hat{\mathbf{y}} + \hat{\mathbf{y}} \hat{\mathbf{k}} = \begin{bmatrix} 0 & p/s & 0 \\ p/s & 0 & q/s \\ 0 & q/s & 0 \end{bmatrix} \quad (105)$$

$$\hat{\mathbf{k}} \times \hat{\mathbf{y}} = [-q/s, 0, p/s]^T, \quad (106)$$

$$\mathbf{k} \cdot \mathbf{r} = \omega(px + qz), \quad (107)$$

and where $s = s_s$ for SH waves and $s = s_{em}$ for TE waves. All responses have been normalized by a scale factor $\mathcal{S}^{SH-TE}$ that we are free to set (i.e., we have put $1/\mathcal{S} = \mathcal{S}^{SH-TE}$). Useful values for such scale factors will be discussed shortly. The plane-wave response is only known to within an arbitrary scale factor because the source amplitude is not fixed (i.e., the conditions for uniqueness, as discussed earlier, are being violated).

Similarly, for the $SV-TM$ transverse waves, the response is

$$\mathbf{u} = (\hat{\mathbf{y}} \times \hat{\mathbf{k}}) e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{SV-TM}, \quad (108)$$

$$\mathbf{w} = \frac{G}{\rho_f} \left(s^2 - \frac{\rho}{G} \right) (\hat{\mathbf{y}} \times \hat{\mathbf{k}}) e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{SV-TM}, \quad (109)$$

$$\boldsymbol{\tau} = i\omega s G [\hat{\mathbf{k}} (\hat{\mathbf{y}} \times \hat{\mathbf{k}}) + (\hat{\mathbf{y}} \times \hat{\mathbf{k}}) \hat{\mathbf{k}}] e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{SV-TM}, \quad (110)$$

$$p = 0, \quad (111)$$

$$\mathbf{E} = i\omega \mu \frac{\tilde{\rho}}{\rho_f} L G \beta_T (\hat{\mathbf{y}} \times \hat{\mathbf{k}}) e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{SV-TM}, \quad (112)$$

$$\mathbf{H} = i\omega s \frac{\tilde{\rho}}{\rho_f} L G \beta_T \hat{\mathbf{y}} e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{SV-TM}, \quad (113)$$

where

$$\hat{\mathbf{k}} (\hat{\mathbf{y}} \times \hat{\mathbf{k}}) + (\hat{\mathbf{y}} \times \hat{\mathbf{k}}) \hat{\mathbf{k}} = \begin{bmatrix} 2qp/s^2 & 0 & (q^2 - p^2)/s^2 \\ 0 & 0 & 0 \\ (q^2 - p^2)/s^2 & 0 & -2qp/s^2 \end{bmatrix} \quad (114)$$

and where $s = s_s$ for SV waves and $s = s_{em}$ for TM waves. Note that in both of these transverse-wave polarizations, there are no induced changes in the fluid pressure, but there is induced relative flow due to grain accelerations. This relative flow results in a streaming current that produces the $\mathbf{H}$ field. Although there is no charge separation produced by the transverse waves, there is a small $\mathbf{E}$ field generated from the $\mathbf{H}$ field by induction. For the shear waves, these $\mathbf{E}$ and $\mathbf{H}$ fields are carried along with the shear wave as part of the material response and have no support outside the extent of the shear wave. In homogeneous media, shear waves do not generate EM waves. Similarly, for the EM waves, the mechanical components are carried along with the EM waves but have no support outside the extent of the EM wave. In homogeneous media, EM waves do not generate mechanical shear waves.

Finally, the longitudinal $Pf-Ps$ response is

$$\mathbf{u} = \hat{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{Pf-Ps}, \quad (115)$$

$$\mathbf{w} = \beta_L \hat{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{Pf-Ps}, \quad (116)$$

$$\boldsymbol{\tau} = i\omega s [(H - 2G + C\beta_L)\mathbf{I} + 2G\hat{\mathbf{k}}\hat{\mathbf{k}}] e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{Pf-Ps}, \quad (117)$$

$$-p = i\omega s (C + M\beta_L) e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{Pf-Ps}, \quad (118)$$

$$\mathbf{E} = i\omega (\tilde{\rho}L/\tilde{\epsilon}) \beta_L \hat{\mathbf{k}} e^{i\mathbf{k} \cdot \mathbf{r}} / \mathcal{S}^{Pf-Ps}, \quad (119)$$

$$\mathbf{H} = 0, \quad (120)$$

where

$$\beta_L = -\frac{Hs^2 - \rho}{Cs^2 - \rho_f}, \quad (121)$$

$$\hat{\mathbf{k}}\hat{\mathbf{k}} = \begin{bmatrix} p^2/s^2 & 0 & pq/s^2 \\ 0 & 0 & 0 \\ pq/s^2 & 0 & q^2/s^2 \end{bmatrix} \quad (122)$$

and where $s = s_{pf}$ for fast waves and $s = s_{ps}$ for slow waves. Note that $\mathbf{H} = 0$; i.e., there is no net electrical current inside of the compressional waves (the streaming current and conduction current exactly balance) as is easily verified using the other components of the response. The electric field is due entirely to the charge separation associated with the accumulation and depletion of the counterions in the compressions and rarefactions of the waves as already discussed. In homogeneous media, the compressional waves carry this $\mathbf{E}$ field along with them as part of the material response but do not generate EM waves.

Finally, we conclude with two suggestions for the real scale factors $\mathcal{S}$. It is usually desirable to normalize the above responses so that each plane-wave type carries a unit amount

of some specified physical quantity when the wave is at the origin ($\mathbf{r}=0$). Note that because s is complex in general, this quantity will vary when the plane wave is no longer at the origin due to attenuation. A first suggestion is to select $\mathcal{S}$ so that the plane-waves have a unit total-displacement amplitude at the origin; i.e.,

$$|\mathbf{u} + \mathbf{w}| = 1 \quad (123)$$

at $\mathbf{r}=0$. The other is to select $\mathcal{S}$ so that the plane waves have a unit of real power flux at the origin; i.e.,

$$\text{Re}\{\hat{\mathbf{k}}\} \cdot \text{Re}\{\mathbf{E} \times \mathbf{H}^* - i\omega(\mathbf{u}^* \cdot \boldsymbol{\tau} - \mathbf{w}^* p)\} = 1 \quad (124)$$

at $\mathbf{r}=0$ [recall Eq. (22)]. We do not pause to give the detailed expressions for these scale factors since they are most conveniently determined on the computer [using the responses given above and Eqs. (123) or (124)] in the course of a numerical calculation.

B. Point-source response

We now consider the solution to Eqs. (74)–(76) when the source terms are given as

$$\mathbf{C} = \mathbf{C}_0 \delta(\mathbf{r} - \mathbf{r}'), \quad (125)$$

$$\mathbf{F} = \mathbf{F}_0 \delta(\mathbf{r} - \mathbf{r}'), \quad (126)$$

$$\mathbf{f} = \mathbf{f}_0 \delta(\mathbf{r} - \mathbf{r}'). \quad (127)$$

The response is represented in terms of Green's tensors as

$$\mathbf{u} = \mathbf{G}_u^F \cdot \mathbf{F}_0 + \mathbf{G}_u^f \cdot \mathbf{f}_0 + \mathbf{G}_u^C \cdot \mathbf{C}_0, \quad (128)$$

$$\mathbf{w} = \mathbf{G}_w^F \cdot \mathbf{F}_0 + \mathbf{G}_w^f \cdot \mathbf{f}_0 + \mathbf{G}_w^C \cdot \mathbf{C}_0, \quad (129)$$

$$\mathbf{E} = \mathbf{G}_E^F \cdot \mathbf{F}_0 + \mathbf{G}_E^f \cdot \mathbf{f}_0 + \mathbf{G}_E^C \cdot \mathbf{C}_0. \quad (130)$$

In isotropic and homogeneous media, the Green's tensors must possess spherical symmetry; i.e., the tensors must depend only on $x = |\mathbf{r} - \mathbf{r}'|$ and they must be symmetric. Thus from our previous symmetry relations [Eqs. (64)–(69)], we have that

$$\mathbf{G}_w^F = \mathbf{G}_u^f, \quad (131)$$

$$\mathbf{G}_E^F = -i\omega \mathbf{G}_u^C, \quad (132)$$

$$\mathbf{G}_E^f = -i\omega \mathbf{G}_w^C, \quad (133)$$

for isotropic and homogeneous media. It is thus only necessary to explicitly solve for the six Green's tensors $\mathbf{G}_u^F$, $\mathbf{G}_w^F$, $\mathbf{G}_E^f$, $\mathbf{G}_w^f$, $\mathbf{G}_E^C$, and $\mathbf{G}_w^C$.

The Green's tensors are found using Fourier transforms where the transform pair is defined

$$\tilde{\mathbf{u}}(\mathbf{k}) = \int \mathbf{u}(\mathbf{r}) e^{-i\mathbf{k} \cdot \mathbf{r}} d^3\mathbf{r}, \quad (134)$$

$$\mathbf{u}(\mathbf{r}) = \frac{1}{(2\pi)^3} \int \tilde{\mathbf{u}}(\mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{r}} d^3\mathbf{k}. \quad (135)$$

Upon Fourier transformation, Eqs. (74)–(76) are written in the matrix form

$$\begin{bmatrix} (k^2 - \omega^2 \mu \tilde{\epsilon}) \mathbf{I} - \mathbf{k}\mathbf{k} & i\omega^3 \mu \tilde{\rho} L \mathbf{I} & \mathbf{0} \\ -i\omega \tilde{\rho} L \mathbf{I} & \omega^2 \tilde{\rho} \mathbf{I} - M \mathbf{k}\mathbf{k} & \omega^2 \rho_f \mathbf{I} - C \mathbf{k}\mathbf{k} \\ \mathbf{0} & \omega^2 \rho_f \mathbf{I} - C \mathbf{k}\mathbf{k} & (-Gk^2 + \omega^2 \rho) \mathbf{I} - (H - G) \mathbf{k}\mathbf{k} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{E}} \\ \tilde{\mathbf{w}} \\ \tilde{\mathbf{u}} \end{bmatrix} = \begin{bmatrix} i\omega \mu \mathbf{C}_0 \\ -\mathbf{f}_0 \\ -\mathbf{F}_0 \end{bmatrix}. \quad (136)$$

To start with, the response $[\tilde{\mathbf{E}}^F, \tilde{\mathbf{w}}^F, \tilde{\mathbf{u}}^F]^T$ associated with $\mathbf{F}_0$ is obtained. Setting $\mathbf{C}_0=0$ and $\mathbf{f}_0=0$, Eq. (136) is put into upper-triangular form and then solved using the identity

$$[\mathbf{I} - \alpha \mathbf{k}\mathbf{k}]^{-1} = \mathbf{I} - \frac{\mathbf{k}\mathbf{k}}{k^2 - 1/\alpha}. \quad (137)$$

After a somewhat lengthy (yet straightforward) algebraic manipulation, one finds

$$\begin{aligned} \tilde{\mathbf{G}}_u^F(\mathbf{k}) &= \frac{k^2 - \omega^2 \mu \tilde{\epsilon} (1 + \tilde{\rho} L^2 / \tilde{\epsilon})}{G(k^2 - \omega^2 s_s^2)(k^2 - \omega^2 s_{em}^2)} \left(\mathbf{I} - \frac{\mathbf{k}\mathbf{k}}{k^2} \right) \\ &+ \left(\frac{M}{MH - C^2} \right) \frac{k^2 - \omega^2 \tilde{\rho} (1 + \tilde{\rho} L^2 / \tilde{\epsilon}) / M}{(k^2 - \omega^2 s_{pf}^2)(k^2 - \omega^2 s_{ps}^2)} \frac{\mathbf{k}\mathbf{k}}{k^2}, \end{aligned} \quad (138)$$

$$\begin{aligned} \tilde{\mathbf{G}}_w^F(\mathbf{k}) &= -\frac{\rho_f}{\tilde{\rho}} \frac{k^2 - \omega^2 \mu \tilde{\epsilon}}{G(k^2 - \omega^2 s_s^2)(k^2 - \omega^2 s_{em}^2)} \left(\mathbf{I} - \frac{\mathbf{k}\mathbf{k}}{k^2} \right) \\ &- \left(\frac{C}{MH - C^2} \right) \frac{k^2 - \omega^2 \rho_f / C}{(k^2 - \omega^2 s_{pf}^2)(k^2 - \omega^2 s_{ps}^2)} \frac{\mathbf{k}\mathbf{k}}{k^2}, \end{aligned} \quad (139)$$

$$\begin{aligned} \tilde{\mathbf{G}}_E^F(\mathbf{k}) &= \frac{i\omega^3 \mu \rho_f L}{G(k^2 - \omega^2 s_s^2)(k^2 - \omega^2 s_{em}^2)} \left(\mathbf{I} - \frac{\mathbf{k}\mathbf{k}}{k^2} \right) \\ &- i\omega \frac{\tilde{\rho} L}{\tilde{\epsilon}} \left(\frac{C}{MH - C^2} \right) \frac{k^2 - \omega^2 \rho_f / C}{(k^2 - \omega^2 s_{pf}^2)(k^2 - \omega^2 s_{ps}^2)} \frac{\mathbf{k}\mathbf{k}}{k^2}. \end{aligned} \quad (140)$$

To find the response associated with $\mathbf{f}_0$, it is advised to interchange rows two and three of Eq. (136) and then interchange columns two and three so that $[\tilde{\mathbf{E}}^f, \tilde{\mathbf{u}}^f, \tilde{\mathbf{w}}^f]^T$ is the unknown wave field vector. Upon setting $\mathbf{C}_0=0$ and $\mathbf{F}_0=0$, manipulating into upper triangular form and solving, one finds

$$\begin{aligned} \tilde{\mathbf{G}}_w^f(\mathbf{k}) &= \frac{(k^2 - \omega^2 \rho / G)(k^2 - \omega^2 \mu \tilde{\epsilon})}{\omega^2 \tilde{\rho} (k^2 - \omega^2 s_s^2)(k^2 - \omega^2 s_{em}^2)} \left(\mathbf{I} - \frac{\mathbf{k}\mathbf{k}}{k^2} \right) \\ &+ \left(\frac{H}{MH - C^2} \right) \frac{k^2 - \omega^2 \rho / H}{(k^2 - \omega^2 s_{pf}^2)(k^2 - \omega^2 s_{ps}^2)} \frac{\mathbf{k}\mathbf{k}}{k^2}, \end{aligned} \quad (141)$$

$$\begin{aligned}\tilde{\mathbf{G}}_E^f(\mathbf{k}) = & i\omega\mu L \frac{k^2 - \omega^2\rho/G}{(k^2 - \omega^2s_s^2)(k^2 - \omega^2s_{em}^2)} \left(\mathbf{I} - \frac{\mathbf{kk}}{k^2} \right) \\ & + i\omega \frac{\tilde{\rho}L}{\tilde{\epsilon}} \left(\frac{H}{MH - C^2} \right) \\ & \times \frac{k^2 - \omega^2\rho/H}{(k^2 - \omega^2s_{pf}^2)(k^2 - \omega^2s_{ps}^2)} \frac{\mathbf{kk}}{k^2}.\end{aligned}\quad (142)$$

Lastly, $\tilde{\mathbf{G}}_E^C$ is obtained as

$$\begin{aligned}\tilde{\mathbf{G}}_E^C(\mathbf{k}) = & i\omega\mu \frac{k^2 - \omega^2\rho_t/G}{(k^2 - \omega^2s_s^2)(k^2 - \omega^2s_{em}^2)} \left(\mathbf{I} - \frac{\mathbf{kk}}{k^2} \right) \\ & - i\omega \left(\frac{\tilde{\rho}L}{\tilde{\epsilon}} \right)^2 \left(\frac{H}{MH - C^2} \right) \\ & \times \frac{k^2 - \omega^2\rho/H}{(k^2 - \omega^2s_{pf}^2)(k^2 - \omega^2s_{ps}^2)} \frac{\mathbf{kk}}{k^2}.\end{aligned}\quad (143)$$

These are the six independent Green's tensors for a homogeneous and isotropic whole space.

The spatial response is now determined by taking the inverse-Fourier transform in spherical $\mathbf{k}$ coordinates. The radial k integral is obtained using the residue theorem. We have verified that there is no contribution from the poles at $k=0$ (thus providing a check on the validity of the above expressions). The final spatial response is separated into transverse, longitudinal, and near-field contributions as

$$\begin{aligned}\mathbf{G}_\eta^\xi(\mathbf{r}|\mathbf{r}') = & \sum_{s=s_s}^{s_{em}} T_{\eta,s}^\xi \frac{e^{i\omega s x}}{4\pi x} (\mathbf{I} - \hat{\mathbf{x}}\hat{\mathbf{x}}) + \sum_{s=s_{pf}}^{s_{ps}} L_{\eta,s}^\xi \frac{e^{i\omega s x}}{4\pi x} \hat{\mathbf{x}}\hat{\mathbf{x}} \\ & + \left[\sum_{s=s_s}^{s_{em}} T_{\eta,s}^\xi \left(\frac{i}{\omega s x} - \frac{1}{(\omega s x)^2} \right) \frac{e^{i\omega s x}}{4\pi x} \right. \\ & \left. - \sum_{s=s_{pf}}^{s_{ps}} L_{\eta,s}^\xi \left(\frac{i}{\omega s x} - \frac{1}{(\omega s x)^2} \right) \frac{e^{i\omega s x}}{4\pi x} \right] (\mathbf{I} - 3\hat{\mathbf{x}}\hat{\mathbf{x}}),\end{aligned}\quad (144)$$

where, in this expression,

$$\hat{\mathbf{x}} = \frac{\mathbf{r} - \mathbf{r}'}{x}, \quad x = |\mathbf{r} - \mathbf{r}'| \quad (145)$$

and where $\xi = F, f$, or C indicates source type and $\eta = u, w$, or E indicates field type. The tensorial polarization $\mathbf{I} - \hat{\mathbf{x}}\hat{\mathbf{x}}$ corresponds to transverse waves while $\hat{\mathbf{x}}\hat{\mathbf{x}}$ corresponds to longitudinal waves. Thus as usual, the near-field polarization $\mathbf{I} - 3\hat{\mathbf{x}}\hat{\mathbf{x}}$ can be thought of as a combination of transverse and longitudinal modes. At distances large compared to wavelengths ($\omega s x \gg 1$), the near-field terms become negligible. We now list the various transverse $T_{\eta,s}^\xi$, and longitudinal $L_{\eta,s}^\xi$ amplitudes:

$$T_{u,s_s}^F = \frac{s_s^2 - \mu\tilde{\epsilon}(1 + \tilde{\rho}L^2/\tilde{\epsilon})}{G(s_s^2 - s_{em}^2)}, \quad (146)$$

$$T_{u,s_{em}}^F = \frac{s_{em}^2 - \mu\tilde{\epsilon}(1 + \tilde{\rho}L^2/\tilde{\epsilon})}{G(s_{em}^2 - s_s^2)}, \quad (147)$$

$$L_{u,s_{pf}}^F = \left(\frac{M}{MH - C^2} \right) \frac{[s_{pf}^2 - \tilde{\rho}(1 + \tilde{\rho}L^2/\tilde{\epsilon})/M]}{(s_{pf}^2 - s_{ps}^2)}, \quad (148)$$

$$L_{u,s_{ps}}^F = \left(\frac{M}{MH - C^2} \right) \frac{[s_{ps}^2 - \tilde{\rho}(1 + \tilde{\rho}L^2/\tilde{\epsilon})/M]}{(s_{ps}^2 - s_{pf}^2)}, \quad (149)$$

$$T_{w,s_s}^F = -\frac{\rho_f}{\tilde{\rho}} \frac{(s_s^2 - \mu\tilde{\epsilon})}{G(s_s^2 - s_{em}^2)}, \quad (150)$$

$$T_{w,s_{em}}^F = -\frac{\rho_f}{\tilde{\rho}} \frac{(s_{em}^2 - \mu\tilde{\epsilon})}{G(s_{em}^2 - s_s^2)}, \quad (151)$$

$$L_{w,s_{pf}}^F = -\left(\frac{C}{MH - C^2} \right) \frac{(s_{pf}^2 - \rho_f/C)}{(s_{pf}^2 - s_{ps}^2)}, \quad (152)$$

$$L_{w,s_{ps}}^F = -\left(\frac{C}{MH - C^2} \right) \frac{(s_{ps}^2 - \rho_f/C)}{(s_{ps}^2 - s_{pf}^2)}, \quad (153)$$

$$T_{E,s_s}^F = \frac{i\omega^3\mu\rho_f L}{G(s_s^2 - s_{em}^2)}, \quad (154)$$

$$T_{E,s_{em}}^F = \frac{i\omega^3\mu\rho_f L}{G(s_{em}^2 - s_s^2)}, \quad (155)$$

$$L_{E,s_{pf}}^F = -i\omega \frac{\tilde{\rho}L}{\tilde{\epsilon}} \left(\frac{C}{MH - C^2} \right) \frac{(s_{pf}^2 - \rho_f/C)}{(s_{pf}^2 - s_{ps}^2)}, \quad (156)$$

$$L_{E,s_{ps}}^F = -i\omega \frac{\tilde{\rho}L}{\tilde{\epsilon}} \left(\frac{C}{MH - C^2} \right) \frac{(s_{ps}^2 - \rho_f/C)}{(s_{ps}^2 - s_{pf}^2)}, \quad (157)$$

$$T_{w,s_s}^f = -\frac{(s_s^2 - \rho/G)(s_s^2 - \mu\tilde{\epsilon})}{\omega^2\tilde{\rho}(s_s^2 - s_{em}^2)}, \quad (158)$$

$$T_{w,s_{em}}^f = -\frac{(s_{em}^2 - \rho/G)(s_{em}^2 - \mu\tilde{\epsilon})}{\omega^2\tilde{\rho}(s_{em}^2 - s_s^2)}, \quad (159)$$

$$L_{w,s_{pf}}^f = \left(\frac{H}{MH - C^2} \right) \frac{(s_{pf}^2 - \rho/H)}{(s_{pf}^2 - s_{ps}^2)}, \quad (160)$$

$$L_{w,s_{ps}}^f = \left(\frac{H}{MH - C^2} \right) \frac{(s_{ps}^2 - \rho/H)}{(s_{ps}^2 - s_{pf}^2)}, \quad (161)$$

$$T_{E,s_s}^f = i\omega\mu L \frac{(s_s^2 - \rho/G)}{s_s^2 - s_{em}^2}, \quad (162)$$

$$T_{E,s_{em}}^f = i\omega\mu L \frac{(s_{em}^2 - \rho/G)}{s_{em}^2 - s_s^2}, \quad (163)$$

$$L_{E,s_{pf}}^f = i\omega \frac{\tilde{\rho}L}{\tilde{\epsilon}} \left(\frac{H}{MH - C^2} \right) \frac{(s_{pf}^2 - \rho/H)}{(s_{pf}^2 - s_{ps}^2)}, \quad (164)$$

$$L_{E,s_{ps}}^f = i\omega \frac{\tilde{\rho}L}{\tilde{\epsilon}} \left(\frac{H}{MH - C^2} \right) \frac{(s_{ps}^2 - \rho/H)}{(s_{ps}^2 - s_{pf}^2)}, \quad (165)$$

$$T_{E,s_s}^C = i\omega\mu \frac{(s_s^2 - \rho_t/G)}{s_s^2 - s_{em}^2}, \quad (166)$$

$$T_{E,s_{em}}^C = i\omega\mu \frac{(s_{em}^2 - \rho_t/G)}{s_{em}^2 - s_s^2}, \quad (167)$$

$$L_{E,s_{pf}}^C = -i\omega \left(\frac{\tilde{\rho} L}{\tilde{\epsilon}} \right)^2 \left(\frac{H}{MH - C^2} \right) \frac{(s_{pf}^2 - \rho/H)}{(s_{pf}^2 - s_{ps}^2)}, \quad (168)$$

$$L_{E,s_{ps}}^C = -i\omega \left(\frac{\tilde{\rho} L}{\tilde{\epsilon}} \right)^2 \left(\frac{H}{MH - C^2} \right) \frac{(s_{ps}^2 - \rho/H)}{(s_{ps}^2 - s_{pf}^2)}. \quad (169)$$

This completes the specification of the Green's tensors.

We have confirmed by direct calculation that Eqs. (131)–(133) are satisfied thus providing another check on the validity of the above expressions. It is seen that from any one of the sources ($\mathbf{F}$, $\mathbf{f}$, or $\mathbf{C}$), all four wave types are generated (shear, EM, fast compressional, and slow compressional). Setting $L=0$ in the above gives the Green's tensors associated with Maxwell's and Biot's equations. Previous work on the point-source response of Biot theory (e.g., Burridge and Vargas, 1979; Norris, 1985) forgot to include the important contributions from $\mathbf{f}$.

III. BOUNDARY CONDITIONS

In this section, the general boundary conditions that must be satisfied at a conformal interface between two dissimilar porous materials are established.

In the section on uniqueness, it was seen that so long as the field components in Eq. (35) are specified on the surface that encloses some region of interest, the wave fields are uniquely determined. However, uniqueness does not help in any way to determine what these specific boundary values should be. The boundary conditions at internal surfaces must come from conservation laws and kinematic constraints as will now be outlined. It will be seen that the boundary conditions so obtained do not violate the uniqueness conditions.

Imagine a porous material 1 conformally overlying another porous material 2. At the interface (denoted by S_I), the grains from the two materials come into contact. In linear wave problems, no distinction need be made between welded and nonwelded grain contacts so long as the overlying layer is more than a few tens of centimeters thick. In this case, the static normal force on nonwelded contacts (due to gravity) is significantly greater than the wave-induced deviations, so that the interface grains remain in contact during wave passage.

This leads to the first boundary condition. It is assumed that the grains of material 1 do not penetrate into or slip relative to material 2. This kinematic constraint is expressed (using the porous-continuum variables) as

$$\mathbf{u}_1 = \mathbf{u}_2 \quad \text{on } S_I. \quad (170)$$

The remaining boundary conditions come from Maxwell's equations and the conservation laws governing momentum flux, fluid-mass flux, and energy flux.

In what follows, we use the standard device of integrating Maxwell's equations and the conservations laws over a thin disk of volume V_D that straddles the interface in the limit as the disk thickness vanishes. Upon integrating Eqs. (1) and (2) (the porous-continuum Maxwell's equations) and using Green's theorem, one finds that

$$\mathbf{n} \times [\mathbf{E}_2 - \mathbf{E}_1] = 0, \quad (171)$$

$$\mathbf{n} \times [\mathbf{H}_2 - \mathbf{H}_1] = 0 \quad \text{on } S_I, \quad (172)$$

where $\mathbf{n}$ is the normal to the interface directed from material 1 toward material 2. The horizontal components of $\mathbf{E}$ and $\mathbf{H}$ are continuous. An exception to this condition arises if it is assumed that the interface carries a free charge per unit area $\mathcal{Q}$ before the disturbances arrive. Such a surface-charge density could arise electrokinetically if there was a background fluid flow traversing the interface. When a seismic wave shakes such a precharged interface, there is an effective surface current $i\omega \mathbf{n} \times \mathbf{n} \times \mathbf{u} \mathcal{Q}$ that must be present on the right-hand side of Eq. (172). Such a discontinuity in the horizontal component of the $\mathbf{H}$ field represents a possible means by which independently propagating EM waves can be generated from a seismic wave at an interface. However, order of magnitude calculations indicate that such a source will typically be quite small compared to the "standard" source involving the mismatch in the seismically induced streaming-current density across an initially uncharged interface.

Next, the conservation of total linear momentum [Eq. (3)] is integrated giving

$$\mathbf{n} \cdot [\boldsymbol{\tau}_2 - \boldsymbol{\tau}_1] = 0 \quad \text{on } S_I, \quad (173)$$

which states that the total-momentum flux into the interface equals the total-momentum flux out. Although it is possible to write Biot's equations as momentum balances involving the momentum-flux rates through the solid $(1-\phi)\boldsymbol{\tau}_s$ and through the fluid $\phi\boldsymbol{\tau}_f$, it is important to note that neither $\mathbf{n} \cdot [(1-\phi)\boldsymbol{\tau}_s]$ nor $\mathbf{n} \cdot [\phi\boldsymbol{\tau}_f]$ are necessarily continuous across an uncharged interface. Only their sum $\mathbf{n} \cdot \boldsymbol{\tau}$ must be continuous. This is because momentum may be transferred from one phase to another across an interface. A possible exception to Eq. (173) again arises if the interface is assumed to carry a free-charge per unit area $\mathcal{Q}$ before the disturbances arrive. In this case, we must replace the right-hand side by the effective body force per unit area $-\mathcal{Q}\mathbf{E}$.

Next, the conservation of fluid mass is integrated. Dividing the fluid compressibility law [Eq. (7)] by M gives the porous-continuum statement of the conservation of fluid mass

$$\nabla \cdot \mathbf{w} = -\frac{1}{M} p - \frac{\mathcal{C}}{M} : \nabla \mathbf{u}. \quad (174)$$

Integrating this over the infinitesimal disk gives

$$\mathbf{n} \cdot [\mathbf{w}_2 - \mathbf{w}_1] = 0 \quad \text{on } S_I; \quad (175)$$

i.e., fluid flux into the interface equals fluid flux out (recall that $\mathbf{n} \cdot \mathbf{w}$ is the Darcy flux in the direction $\mathbf{n}$). There is no way to convert fluid to solid at the interface so that this constraint must always be satisfied. To see that $(\mathcal{C}/M) : \nabla \mathbf{u}$ volume integrates to zero at the interface $z=0$ with $\mathbf{n}=\hat{\mathbf{z}}$, we note that in the limit as the disk thickness vanishes

$$\begin{aligned} & \int_{V_D} \left(\frac{\mathcal{E}}{M} : \nabla \mathbf{u} \right) dV \\ &= \int_{V_D} \left\{ \nabla \cdot \left(\frac{\mathcal{E}}{M} \cdot \mathbf{u} \right) - \left[\nabla \cdot \left(\frac{\mathcal{E}}{M} \right) \right] \cdot \mathbf{u} \right\} dV \\ &= \int_{z=0} \hat{\mathbf{z}} \cdot \left[\left(\frac{\mathcal{E}}{M} \right)_2 \cdot \mathbf{u}_2 - \left(\frac{\mathcal{E}}{M} \right)_1 \cdot \mathbf{u}_1 \right] dS \end{aligned} \quad (176)$$

$$- \int_{V_D} \delta(z) \hat{\mathbf{z}} \cdot \left[\left(\frac{\mathcal{E}}{M} \right)_2 - \left(\frac{\mathcal{E}}{M} \right)_1 \right] \cdot \mathbf{u} dV \quad (177)$$

$$= 0, \quad (178)$$

where $\mathcal{E}/M(\mathbf{r}) = (\mathcal{E}/M)_1 + [(\mathcal{E}/M)_2 - (\mathcal{E}/M)_1] H(z)$ with $H(z)$ the step function was used along with continuity of $\mathbf{u}$.

Finally, the conservation of total complex energy [cf. Eqs. (30)–(32)]

$$\nabla \cdot [\mathbf{E} \times \mathbf{H}^* - i\omega(\boldsymbol{\tau} \cdot \mathbf{u}^* - p\mathbf{w}^*)] = Q \quad (179)$$

is integrated over the infinitesimal disk giving that $\mathbf{n} \cdot [\mathbf{E} \times \mathbf{H}^* - i\omega(\boldsymbol{\tau} \cdot \mathbf{u}^* - p\mathbf{w}^*)]$ is continuous across the interface; i.e., the total-energy flux into the interface equals the total-energy flux out. Noting that $\mathbf{n} \cdot (\mathbf{E} \times \mathbf{H}^*) = (\mathbf{n} \times \mathbf{E}) \cdot \mathbf{H}^*$, where $\mathbf{n} \times \mathbf{E}$ is both continuous across the interface and parallel to it and that the horizontal components of $\mathbf{H}$ are continuous, one has that $\mathbf{n} \cdot (\mathbf{E} \times \mathbf{H}^*)$ is continuous. Since $\mathbf{n} \cdot \boldsymbol{\tau}$, $\mathbf{u}$, and $\mathbf{n} \cdot \mathbf{w}$ have already been shown to be continuous, one finally arrives at

$$p_1 = p_2 \quad \text{on } S_I \quad (180)$$

as the final boundary condition. This is a direct consequence of energy conservation and is always satisfied. If the interface carries an initial surface charge $\mathcal{Q}$, the contribution from the discontinuity in $\mathbf{n} \cdot \boldsymbol{\tau}$ exactly cancels that from $\mathbf{n} \times \mathbf{H}$, and Eq. (180) remains valid. Deresiewicz and Skalak (1963) hypothesize that a possible nonalignment of pores at the interface can result in p_1 being different from p_2 . They argue, essentially, that the resultant mismatch in energy-flux densities is made up for by a dissipation source term due to Darcy flow across the interface. However, the interface, by definition, has no thickness and can produce no dissipation. If the “interface” is better characterized as a thin transition zone, then a proper treatment requires the explicit inclusion of a thin porous layer (or “membrane”) between the two half-spaces. In this case, the boundary conditions at both of the half-space-layer interfaces are as given above. Deresiewicz and Skalak also allow for the possibility that the grains from material 1 exactly plug the pores of material 2 at the interface so that the interface is effectively sealed to Darcy flow. In this case, the boundary condition $p_1 = p_2$ is replaced by $\mathbf{n} \cdot \mathbf{w} = 0$. However, two granular media can never achieve this situation unless the grains are perfect polyhedra and, even then, only for certain very special packing configurations. This situation never arises in naturally occurring porous materials and need not be worried about in general.

To summarize, at a conformal interface separating two arbitrary porous materials and upon which no free charge is accumulated prior to the arrival of the disturbances, the vector

$$\mathbf{b} = [\mathbf{u}, \mathbf{n} \cdot \mathbf{w}, \mathbf{n} \cdot \boldsymbol{\tau}, -p, \mathbf{n} \times \mathbf{E}, \mathbf{n} \times \mathbf{H}]^T \quad (181)$$

must be continuous. This is the primary result of this section. The continuity of $\mathbf{b}$ satisfies the uniqueness requirements but is in no way a result of uniqueness. At a planar interface, a given incoming plane wave (of any type) will generate, in general, four reflected waves and four transmitted waves (these wave types are shear, EM, fast compressional, and slow compressional). Thus a total of eight boundary conditions are required at a porous–porous interface and the continuity of $\mathbf{b}$ [Eq. (181)] provides exactly these eight equations (note that continuity of the normal and horizontal components of $\mathbf{u}$ and $\mathbf{n} \cdot \boldsymbol{\tau}$ represents four equations).

IV. CONCLUSIONS

We now briefly summarize a few of the key ideas coming out of this work. In addition to the experimentally measured effect of seismic waves generating electromagnetic waves at interfaces within the Earth, reciprocity has shown that electromagnetic waves from reasonable dipole antennas can generate detectable seismic waves. It may be that under certain conditions it is more convenient to carry out the EM to seismic experiment than the seismic to EM experiment. Reciprocity shows the information content of the two experiments to be equivalent.

Internal to the compressional waves there exist electric fields that act on the diffuse-double-layer charge and can enhance (for slow waves) or retard (for fast waves) the relative flow between the compressions and rarefactions. This effect can alter the attenuation coefficients of the compressional waves by up to 10% for realistic material properties. In shear waves, the electric field is generated by induction and is too small to significantly alter the relative flow. Although the seismic waves generate electromagnetic fields that move along with them as part of the material response, they do not produce independently propagating electromagnetic waves in a homogeneous material. An interface is required for there to be a conversion of wave type.

Because there are three different types of sources in the electroseismic problem ($\mathbf{F}$, $\mathbf{f}$, and $\mathbf{C}$) and three different primary wave fields ($\mathbf{u}$, $\mathbf{w}$, and $\mathbf{E}$) there are nine different Green’s tensors required. In homogeneous-isotropic media, these Green’s tensors separate into the usual transverse, longitudinal, and near-field polarizations expected in any wave problem. In general, all four wave types (shear, EM, fast compressional, and slow compressional) are generated from each of the three sources ($\mathbf{F}$, $\mathbf{f}$, and $\mathbf{C}$).

Finally, the mechanical boundary conditions at interfaces in porous media are obtained by requiring that: (1) no voids are generated at an interface, (2) the total-momentum flux into an interface equals the total-momentum flux out, (3) the total-fluid-mass flux into an interface equals the total-fluid-mass flux out, and (4) the total-energy flux into the interface equals the total-energy flux out. Details of how pores are aligned at the interface are unimportant.

We conclude by noting that there is no way to properly model the electromagnetic fields generated by seismic waves without using Biot theory to calculate the amount of induced relative flow. In this sense, the electroseismic problem represents one of the most significant applications of Biot theory. Interestingly, the first quantitative work on the theory

of seismic waves in porous media (Frenkel, 1944) had as its goal the explanation of electroseismic phenomena.

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Governing equations for the coupled electromagnetics and acoustics of porous media

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The macroscopic governing equations controlling the coupled electromagnetics and acoustics of porous media are derived here from first principles. The porous material is assumed to consist of a packing of solid grains that is saturated by an electrolyte. A sedimentary rock is an example of such a material. The approach is to volume average the equations known to apply in the fluid and solid phases while allowing for the boundary conditions that exist on the fluid-solid interface. The coupling is due to a layer of excess charge adsorbed to the surfaces of the solid grains that is balanced by mobile ions in the fluid electrolyte; i.e., the coupling is electrokinetic in nature. The derived equations have the form of Maxwell's equations coupled to Biot's equations with coupling occurring in the flux-force (or transport) relations. The frequency-dependent macroscopic-transport coefficients are explicitly obtained and related to each other. Onsager reciprocity is derived and is not simply postulated.

I. INTRODUCTION

This paper is concerned with the macroscopic dynamics of two-phase (fluid and solid) porous media possessing continuously distributed phases. A packing of grains saturated by a fluid is an example of such a material. The term "macroscopic," as used here, means that the wavelengths of the applied disturbances are much greater than the dimensions of the grains. When a macroscopic-mechanical disturbance propagates through such a material, a small amount of relative motion is induced between the fluid and solid phases. This relative flow will carry along the excess ions in the electric double layers near the grain surfaces. Thus, a mechanical wave can act as a current source for macroscopic-electromagnetic disturbances. Similarly, when an electromagnetic disturbance propagates, the electric field will act on the charge excesses of the double layers producing pressure gradients in the fluid and, in principle, macroscopic-mechanical disturbances. Thompson and Gist¹ have made field measurements clearly demonstrating that seismic waves can induce electromagnetic disturbances in saturated sediments in the Earth. It is believed that their observations can be explained by the electrokinetic mechanism mentioned above. In this paper, the equations controlling such behavior will be derived and the coefficients defined.

There has not been much previous work in this direction. In 1944, Frenkel² postulated equations that estimated the amount of relative fluid motion induced by a seismic wave. He attempted to allow for flow-induced electric fields by employing, incorrectly, the Helmholtz-Smoluchowski equation. This equation assumes that the total electric current in the porous material, which is a sum of the mechanically driven streaming current (due to convection of double-layer ions) and an electrically driven conduction current, is everywhere zero. Thus, in Frenkel's model, the generated electric field exists only where there is seismically induced relative flow. Correctly, the total current should be present in Ampère's

law and the full set of Maxwell's equations utilized. This is the more general result obtained here.

Neev and Yeatts³ also postulated a set of equations that attempt to model the interaction between mechanical waves and electric fields due to electrokinetics. They also do not allow for the full set of Maxwell's equations which leads them to the erroneous conclusion that mechanical shear waves do not generate electromagnetic disturbances. They only allow for electric fields generated by charge separation. A mechanical shear wave, however, generates a divergence-free (no induced-charge excess) streaming current that will act as a current source in Ampère's law, thus demonstrating the need to use the complete set of Maxwell's equations. Both Neev and Yeatts and Frenkel completely ignore the frequency dependence of their proposed transport laws.

The central problem with both of these cited works is that their equations are not derived from the underlying constituent properties. In this paper, the continuum equations known to apply in the solid grains and fluid electrolyte are volume averaged to obtain the macroscopic-governing equations. The final equations have the form of Maxwell's equations coupled to Biot's⁴ equations. The coupling arises naturally from just two postulates: (1) The solid grains have a uniform surface-charge density adsorbed to their surface, and (2) prior to the arrival of a disturbance, the net charge in a volume of the porous material is zero. The definition and frequency dependences of the transport coefficients will be given. Onsager reciprocity will be derived and not simply assumed.

A brief summary of the major assumptions and/or limitations of the derived theory is now given. Only linear disturbances are considered (i.e., those that obey superposition). The fluid is assumed to be an ideal electrolyte, thus restricting salt concentrations to be less than 1 mol/liter. Both the solid grains and all the macroscopic-constitutive laws are assumed to be isotropic. All wave-induced diffusion effects (flux due to wave-induced ion-

concentration gradients) are, in the end, ignored. It is demonstrated that two conditions must be fulfilled in order to make this assumption at the pore and grain scale: (1) The dielectric constant of the grains must be much less than that of the electrolyte (say a factor of 10 or more), and (2) the thickness of the double layer should be much less than the radii of curvature of the solid grains. The specific class of porous media that are of primary interest here are water-saturated sediments and sedimentary rocks. For these materials, both of these assumptions are valid. Last, no wave scattering from the individual grains of the porous material will be allowed for. This implies that wavelengths λ are always much greater than grain sizes (e.g., $\lambda \geq 1$ mm). Thus, the largest applied frequency that can be considered (for mechanical waves) is on the order of 10^6 Hz.

The structure of the paper is as follows. In Sec. II, the grain-scale equations and boundary conditions for the fluid and solid phases are stated. In Sec. III, these equations are volume averaged to obtain the macroscopic equations of interest. However, several integrals present in the macroscopic-transport equations require the local electric and fluid-flow fields in the pore space to be related to the macroscopic fields generating them. Thus, in Sec. IV, the boundary-value problems controlling the pore-scale fields are treated and in Sec. V, the remaining integrals evaluated to give the macroscopic-transport coefficients. Last, in Sec. VI, the governing equations are presented in final form and summarized.

II. PORE AND GRAIN-SCALE GOVERNING EQUATIONS

There is assumed to be a layer of electrolyte ions and structured (hydrogen-bonded) water molecules that are chemically and physically adsorbed to the surface of the solid grains. This layer also includes the ionized surface sites present on the grain surfaces. The molecules and ions in this layer are assumed to be immobile. In the lexicon of the electric double layer,⁵ this adsorbed layer corresponds to the “Stern layer” or, equivalently, to both the “inner and outer Helmholtz layers.” If there is a net excess of charge in the adsorbed layer (the usual situation), it is balanced by an opposite amount of excess mobile ions distributed in the adjacent fluid. This region of fluid balancing the charge of the adsorbed layer is called the “diffuse layer.” The adsorbed layer and diffuse layer together constitute the electric double layer. The surface that separates the diffuse layer from the adsorbed layer is called the “shear plane” and will be the surface used to define the fluid-solid interface S_w in what follows (the subscript w stands for the “wall”). The adsorbed layer is so thin (≤ 10 Å) that it cannot be properly treated as a separate phase. Thus, in the continuum description that follows, the physical properties of the adsorbed layer show up as boundary conditions on S_w , namely, as a uniform surface-charge density Q^0 and as the no-slip flow condition.

A. Electromagnetic equations

The solid grains (e.g., quartz) are assumed to be electrically insulating while the fluid electrolyte is assumed to possess L ionic species. Maxwell’s equations for the solid and fluid phases take the form, respectively,

$$\nabla \cdot \mathbf{B}_s = 0, \quad (1)$$

$$\nabla \cdot \mathbf{D}_s = 0, \quad (2)$$

$$\nabla \times \mathbf{E}_s = -\dot{\mathbf{B}}_s, \quad (3)$$

$$\nabla \times \mathbf{H}_s = \dot{\mathbf{D}}_s, \quad (4)$$

and

$$\nabla \cdot \mathbf{B}_f = 0, \quad (5)$$

$$\nabla \cdot \mathbf{D}_f = \sum_{\ell=1}^L ez_{\ell} N_{\ell}, \quad (6)$$

$$\nabla \times \mathbf{E}_f = -\dot{\mathbf{B}}_f, \quad (7)$$

$$\nabla \times \mathbf{H}_f = \dot{\mathbf{D}}_f + \mathbf{J}_f, \quad (8)$$

where $\mathbf{J}_f$ is the ionic-current density and has contributions from the diffusion, electromigration, and convection of ions,

$$\mathbf{J}_f = \sum_{\ell=1}^L ez_{\ell} [-kTb_{\ell} \nabla N_{\ell} + ez_{\ell} b_{\ell} N_{\ell} \mathbf{E}_f + N_{\ell} \dot{\mathbf{u}}_f]. \quad (9)$$

$\dot{\mathbf{u}}_f$ is the instantaneous fluid velocity (the fluid may be accelerating). The ionic properties are the valences z_{ℓ} (ez_{ℓ} represents the net charge and sign on each species- ℓ ion), the number densities N_{ℓ} (the number of species- ℓ ions per unit volume), and the mobilities b_{ℓ} (units of velocity per unit force). The Einstein-Stokes approximation $b_{\ell} = 1/(6\pi\eta R_{\ell})$, where η is the fluid’s shear viscosity and R_{ℓ} is the effective “ion radius,” is often accurate and is quite convenient. The mobilities are related to the ionic diffusivities D_{ℓ} as $D_{\ell} = kTb_{\ell}$. Note that some authors employ an alternative definition of ionic mobility, say, u_{ℓ} , given by $u_{\ell} = |ez_{\ell}|b_{\ell}$.

The above estimate for the current density is strictly valid only when ions move independently from each other (an ideal electrolyte). By considering the ion-separation distant at which electrostatic-interaction energies become significant compared to the thermal energy kT , it can be concluded that as long as ion concentrations are less than, say, 1 mol/liter, the ideal electrolyte assumption is (at least roughly) valid. Although the fluid has a velocity $\dot{\mathbf{u}}_f$, the Lorentz electric field $\dot{\mathbf{u}}_f \times \mathbf{B}_f$ that should be added to $\mathbf{E}_f$ in the current density is completely negligible if $\mathbf{B}_f$ is associated with the same disturbance that produces $\mathbf{E}_f$; i.e., the fluid velocity is dramatically less than the speed of the electromagnetic disturbance. Alternatively, if $\mathbf{B}_f$ is taken to be the maximum possible value of the Earth’s field at the Earth’s surface ($\sim 10^{-4}$ T), one has that $|ez_{\ell} b_{\ell} \dot{\mathbf{u}}_f \times \mathbf{B}_f|/|\dot{\mathbf{u}}_f|$ is of the order 10^{-12} ; i.e., the current generated by the Lorentz force is completely negligible compared to that generated by convection of the free double-layer ions (although, of course, the Lorentz-force current will operate regardless of whether

there are charge excesses in the diffuse double layer). Last, although the grains are assumed to be a material like quartz, no piezoelectric effects will be allowed for. This is because the grains of the material are assumed to be oriented at random to one another so that the average polarization of a group of grains is zero.

The boundary conditions that hold on the surface S_w (obtained by volume integrating Maxwell's equations over an infinitesimally thin disk that straddles S_w) are

$$\mathbf{n} \cdot (\mathbf{B}_s - \mathbf{B}_f) = 0, \quad (10)$$

$$\mathbf{n} \cdot (\mathbf{D}_s - \mathbf{D}_f) = Q, \quad (11)$$

$$\mathbf{n} \times (\mathbf{E}_s - \mathbf{E}_f) = 0, \quad (12)$$

$$\mathbf{n} \times (\mathbf{H}_s - \mathbf{H}_f) = Q\dot{\mathbf{u}}_s, \quad (13)$$

$$\mathbf{n} \cdot \mathbf{J}_f = \dot{Q}, \quad (14)$$

where $\mathbf{n}$ is the normal to S_w directed from fluid to solid, $\dot{\mathbf{u}}_s$ is the instantaneous velocity of the solid ($= \dot{\mathbf{u}}_f$ on S_w), and Q is the free charge per unit area of the adsorbed layer. Equation (14) follows, as usual, by taking the divergence of Ampère's law and the time derivative of Coulomb's law (i.e., it is a consequence of charge conservation). The right-hand side of Eq. (12) follows only if the charge of the adsorbed layer is uniformly distributed along S_w . An additional current term could be added to the right-hand side of Eq. (13) if it was assumed that conduction can occur in the adsorbed layer. Although this possibility is ignored here, hydrogen ions may efficiently conduct through the network of hydrogen-bonded water molecules present in the adsorbed layer. This effect can be included in future work.

Last, the above equations are closed by the constitutive laws

$$\mathbf{B}_\xi = \mu_0 \mathbf{H}_\xi, \quad (15)$$

$$\mathbf{D}_\xi = \epsilon_0 \kappa_\xi \mathbf{E}_\xi, \quad (16)$$

where $\xi = f$ or s and where κ_ξ is the dielectric constant.

All the field variables will now be written in the form

$$Q(t) = Q^0 + \text{Re}\{q(\omega)e^{-i\omega t}\}, \quad (17)$$

$$N_\ell(t) = N_\ell^0 + \text{Re}\{n_\ell(\omega)e^{-i\omega t}\}, \quad (18)$$

$$\mathbf{E}_f(t) = \mathbf{E}_f^0 + \text{Re}\{\mathbf{e}_f(\omega)e^{-i\omega t}\}, \quad (19)$$

and so on. The first term in these expressions represents the static-equilibrium field prior to the arrival of any disturbance, while the second term represents the deviation in the field due to a time-harmonic disturbance. It will be assumed, here, that no free charge q is induced on the surface (note that the free charge q does not include the polarization charge dielectrically induced on the surface). The only way for the free charge of the adsorbed layer to change is if chemical reactions occur and, thus, such chemical reactions are ignored here. It is assumed that no steady currents operate prior to disturbance and that there are no static magnetic fields present. Because Q^0 is assumed to be uniform over the grain surfaces, it can be shown using a simple argument based on Stoke's law

that $\mathbf{E}_s^0$ must be zero (the net circulation of $\mathbf{E}_s^0$ around a closed path that runs along the surface of the grain and then passes arbitrarily through the grain must be zero).

The only nonzero static fields are then $\mathbf{E}_f^0$ and N_ℓ^0 . The equations governing these fields are [from Eqs. (6)–(9) and (17)–(19)]

$$\epsilon_0 \kappa_f \nabla \cdot \mathbf{E}_f^0 = \sum_{\ell=1}^L e z_\ell N_\ell^0, \quad (20)$$

$$\nabla \times \mathbf{E}_f^0 = 0, \quad (21)$$

$$-kT \nabla N_\ell^0 + N_\ell^0 e z_\ell \mathbf{E}_f^0 = 0, \quad (22)$$

with the boundary conditions on S_w ,

$$\epsilon_0 \kappa_f \mathbf{n} \cdot \mathbf{E}_f^0 = Q^0, \quad (23)$$

$$\mathbf{n} \times \mathbf{E}_f^0 = 0. \quad (24)$$

The dielectric constant is assumed constant right up to S_w (Ref. 6 shows that there is at most a 1% deviation in κ_f across the diffuse portion of the double layer). Equations (21) and (22) are satisfied by

$$\mathbf{E}_f^0 = -\nabla \Phi^0, \quad (25)$$

$$N_\ell^0 = \mathcal{N}_\ell \exp\left(-\frac{e z_\ell}{kT} \Phi^0\right), \quad (26)$$

where $\mathcal{N}_\ell$ are the bulk-ionic concentrations. Thus, Coulomb's law leads to the so-called "Poisson-Boltzmann" equation

$$\nabla^2 \Phi^0 = - \sum_{\ell=1}^L \frac{e z_\ell \mathcal{N}_\ell}{\epsilon_0 \kappa_f} \exp\left(-\frac{e z_\ell}{kT} \Phi^0\right). \quad (27)$$

The solution to this problem completely solves the static case.

In Appendix A, solutions to the Poisson-Boltzmann equation are considered. The potential near a plane wall has, approximately, an exponential distribution given by

$$\Phi^0 = \zeta \exp(-\chi/d), \quad (28)$$

where χ is a local coordinate measuring distance normal to the wall surface, d is called the Debye length and is defined as

$$\frac{1}{d^2} = \sum_{\ell=1}^L \frac{(e z_\ell)^2 \mathcal{N}_\ell}{\epsilon_0 \kappa_f kT}, \quad (29)$$

and ζ is called the zeta potential and is the static electric potential at the shear plane (i.e., the surface S_w). Validity of the simple exponential distribution (the Debye approximation) and the relation between Q^0 and ζ are discussed in Appendix A. Here it is emphasized that the Debye length, which is a measure of the thickness of the diffuse double layer, is much smaller than any geometri-

cal length associated with the porous material. For example, a univalent electrolyte at room temperature gives $d = 3 \times 10^{-10} / \sqrt{C}$ (m), where C is the bulk-electrolyte molarity. In practice, it is uncommon ever to work with electrolyte more dilute (in the pores) than 10^{-4} mol/liter which corresponds to $d \leq 3 \times 10^{-8}$ m. In sedimentary materials, typical sand-grain radii are on the order of 10^{-4} m, while clay particles have characteristic lengths of the order of 10^{-6} m. The point is that for porous media of interest, the double layer can be modeled as being planar relative to the grain surfaces. This "thin-double-layer" approximation will be made throughout the entire analysis.

The time-harmonic disturbances are governed by [from Eqs. (1)–(9) and (17)–(19)]

$$\nabla \cdot \mathbf{b}_s = 0, \quad (30)$$

$$\nabla \cdot \mathbf{d}_s = 0, \quad (31)$$

$$\nabla \times \mathbf{e}_s = i\omega \mathbf{b}_s, \quad (32)$$

$$\nabla \times \mathbf{h}_s = -i\omega \mathbf{d}_s, \quad (33)$$

and

$$\nabla \cdot \mathbf{b}_f = 0, \quad (34)$$

$$\nabla \cdot \mathbf{d}_f = \sum_{\ell=1}^L e z_{\ell} n_{\ell}, \quad (35)$$

$$\nabla \times \mathbf{e}_f = i\omega \mathbf{b}_f, \quad (36)$$

$$\nabla \times \mathbf{h}_f = -i\omega \mathbf{d}_f + \mathbf{j}_f, \quad (37)$$

where

$$\mathbf{j}_f = \sum_{\ell=1}^L e z_{\ell} \left[-k T b_{\ell} \nabla n_{\ell} + e z_{\ell} b_{\ell} (N_{\ell}^0 \mathbf{e}_f + n_{\ell} \mathbf{E}_f^0) + N_{\ell}^0 \dot{\mathbf{u}}_f(\omega) \right]. \quad (38)$$

The notation $\dot{\mathbf{u}}_f(t) = \text{Re}\{\dot{\mathbf{u}}_f(\omega)e^{-i\omega t}\}$ is being employed. Products of disturbances have been neglected in $\mathbf{j}_f$. The neglect of these terms linearizes the equations. The boundary conditions on S_w are

$$\mathbf{n} \cdot (\mathbf{b}_s - \mathbf{b}_f) = 0, \quad (39)$$

$$\mathbf{n} \cdot (\mathbf{d}_s - \mathbf{d}_f) = 0, \quad (40)$$

$$\mathbf{n} \times (\mathbf{e}_s - \mathbf{e}_f) = 0, \quad (41)$$

$$\mathbf{n} \times (\mathbf{h}_s - \mathbf{h}_f) = Q^0 \dot{\mathbf{u}}_s(\omega), \quad (42)$$

$$\mathbf{n} \cdot \mathbf{j}_f = 0. \quad (43)$$

B. Mechanical equations

The conservation of linear momentum governs the dynamics of both the fluid and solid phases and is expressed for time-harmonic disturbances as

$$-i\omega \rho_f \dot{\mathbf{u}}_f = \nabla \cdot \boldsymbol{\tau}_f + \sum_{\ell=1}^L e z_{\ell} (N_{\ell}^0 \mathbf{e}_f + n_{\ell} \mathbf{E}_f^0), \quad (44)$$

$$-i\omega \rho_s \dot{\mathbf{u}}_s = \nabla \cdot \boldsymbol{\tau}_s. \quad (45)$$

The summation on the right-hand side of the fluid balance is the electrical body force acting on any excess charge (with products of disturbances again neglected). The Lorentz force $\mathbf{j}_f \times \mathbf{B}_f$ has been neglected in the fluid balance for the same reasons that the Lorentz electric field is negligible. The nonlinear convective accelerations $\dot{\mathbf{u}} \cdot \nabla \dot{\mathbf{u}}$ have also been ignored as justified by Pride, Gangi, and Morgan.⁷ The fluid and solid stress tensors are given by

$$\boldsymbol{\tau}_f = K_f \nabla \cdot \mathbf{u}_f \mathbf{I} - i\omega \eta \left(\nabla \mathbf{u}_f + \nabla \mathbf{u}_f^T - \frac{2}{3} \nabla \cdot \mathbf{u}_f \mathbf{I} \right), \quad (46)$$

$$\boldsymbol{\tau}_s = K_s \nabla \cdot \mathbf{u}_s \mathbf{I} + G \left(\nabla \mathbf{u}_s + \nabla \mathbf{u}_s^T - \frac{2}{3} \nabla \cdot \mathbf{u}_s \mathbf{I} \right), \quad (47)$$

where $-i\omega \mathbf{u}(\omega) = \dot{\mathbf{u}}(\omega)$. Finally, the boundary conditions on S_w are

$$\mathbf{n} \cdot (\boldsymbol{\tau}_s - \boldsymbol{\tau}_f) = -Q^0 \mathbf{e}_s, \quad (48)$$

$$\mathbf{u}_s - \mathbf{u}_f = 0. \quad (49)$$

The right-hand side of Eq. (48) represents the electrical body force (per unit area) acting on the excess charge of the adsorbed layer.

III. AVERAGING THE GOVERNING EQUATIONS

The above microscopic (i.e., pore and grain scale) equations will now be volume averaged to obtain the macroscopic equations of interest. The averaging volume V_A is assumed to be larger than the grains but much smaller than the wavelengths of the applied disturbances. If the volume average for some microscopic field $\mathbf{a}_{\xi}$ associated with the ξ th phase is defined as

$$\langle \mathbf{a}_{\xi} \rangle = \frac{1}{V_A} \int_{V_{\xi}} \mathbf{a}_{\xi} dV, \quad (50)$$

where V_{ξ} represents the volume of the ξ th phase within V_A , then the following theorem due to Slattery⁸ is easily established:⁷

$$\langle \nabla \mathbf{a}_{\xi} \rangle = \nabla \langle \mathbf{a}_{\xi} \rangle + \frac{1}{V_A} \int_{S_w} \mathbf{n}_{\xi} \mathbf{a}_{\xi} dS, \quad (51)$$

where $\mathbf{n}_{\xi}$ is the normal to S_w and is defined as

$$\mathbf{n}_f = \mathbf{n}, \quad (52)$$

$$\mathbf{n}_s = -\mathbf{n}, \quad (53)$$

with $\mathbf{n}$ directed from the fluid to the solid. When applying the averaging theorem to the porous material, it will be assumed that the fluid and solid phases each have homogeneous material properties throughout V_A . Furthermore, it is assumed that macroscopic-material properties

such as porosity ϕ ($\equiv V_f/V_A$) vary slowly as V_A is moved over distances the size of V_A or less. Boundary conditions at singular jumps in the macroscopic-material properties are treated as part of another paper.⁹

It is convenient to define two other averages that are simply related to $\langle \mathbf{a}_\xi \rangle$,

$$\bar{\mathbf{a}}_\xi = \langle \mathbf{a}_\xi \rangle / \varphi_\xi, \quad (54)$$

$$\bar{\mathbf{A}} = \sum_\xi \langle \mathbf{a}_\xi \rangle = \sum_\xi \varphi_\xi \bar{\mathbf{a}}_\xi, \quad (55)$$

where φ_ξ is the volume fraction of the ξ th phase,

$$\varphi_\xi = V_\xi / V_A; \quad (56)$$

i.e., $\varphi_f = \phi$ (porosity) and $\varphi_s = 1 - \phi$. $\bar{\mathbf{a}}_\xi$ may be called the “phase average” because it represents the average value within the ξ th phase, while $\bar{\mathbf{A}}$ may be called the “total average” because it represents the average value of a given field type throughout all of V_A .

A. Averaging the electromagnetic equations

To start with, the static field $\mathbf{E}_f^0 = -\nabla\Phi^0$ is averaged to give

$$\langle \mathbf{E}_f^0 \rangle = -\nabla \langle \Phi^0 \rangle - \frac{1}{V_A} \int_{S_w} \mathbf{n} \Phi^0 dS. \quad (57)$$

Because $\Phi^0 (= \zeta)$ is assumed to be constant over S_w , the surface integral vanishes. Thus, if the material has uniform macroscopic properties so that $\langle \Phi^0 \rangle = \text{const}$, then $\langle \mathbf{E}_f^0 \rangle = 0$; i.e., there are no macroscopic static fields.

The equations governing the disturbances [Eqs. (30)–(38)] are now averaged in the following manner: (1) The volume average is applied; (2) Slattery’s theorem is used; (3) the corresponding solid and fluid equations are added; (4) boundary conditions (39)–(43) are applied; (5) the relative fluid-solid flow vector $\mathbf{v}$ is introduced,

$$\mathbf{v} = \dot{\mathbf{u}}_f - \dot{\mathbf{u}}_s, \quad (58)$$

where $\dot{\mathbf{u}}_s$ is the phase-averaged velocity of the solid phase; and (6) the electroneutrality condition prior to disturbances is applied,

$$\sum_{\ell=1}^L e z_\ell \langle N_\ell^0 \rangle V_A + Q_s^0 S_w = 0. \quad (59)$$

Such a procedure directly results in the following macroscopic form of Maxwell’s equations:

$$\nabla \cdot \bar{\mathbf{B}} = 0, \quad (60)$$

$$\nabla \cdot \bar{\mathbf{D}} = \phi \sum_{\ell=1}^L e z_\ell \bar{n}_\ell, \quad (61)$$

$$\nabla \times \bar{\mathbf{E}} = i\omega \bar{\mathbf{B}}, \quad (62)$$

$$\nabla \times \bar{\mathbf{H}} = -i\omega \bar{\mathbf{D}} + \bar{\mathbf{J}}, \quad (63)$$

where $\bar{\mathbf{B}}$, $\bar{\mathbf{D}}$, $\bar{\mathbf{E}}$, $\bar{\mathbf{H}}$, and $\bar{\mathbf{J}}$ are “total” averages [Eq. (55)] while $\bar{n}_\ell$ is a phase average in the fluid. The macroscopic-current density $\bar{\mathbf{J}}$ has four distinct contributions:

$$\bar{\mathbf{J}} = \phi(\mathbf{J}_d + \mathbf{J}_c + \mathbf{J}_s + \mathbf{J}_n). \quad (64)$$

$\mathbf{J}_d$ is part of the average “diffusion” current density in the fluid (the total diffusion current will have a contribution from $\mathbf{J}_n$ as well),

$$\mathbf{J}_d = - \sum_{\ell=1}^L e z_\ell b_\ell k T \nabla \bar{n}_\ell; \quad (65)$$

$\mathbf{J}_c$ is the average “conduction” current density,

$$\mathbf{J}_c = \frac{1}{V_f} \int_{V_f} \left(\sum_{\ell=1}^L (e z_\ell)^2 b_\ell N_\ell^0 \right) \mathbf{e}_f dV; \quad (66)$$

$\mathbf{J}_s$ is the average “streaming” current density,

$$\mathbf{J}_s = \frac{1}{V_f} \int_{V_f} \left(\sum_{\ell=1}^L e z_\ell N_\ell^0 \right) \mathbf{v} dV; \quad (67)$$

and $\mathbf{J}_n$ is the average current density due to locally induced diffusion currents normal to the pore walls,

$$\mathbf{J}_n = - \frac{1}{V_f} \sum_{\ell=1}^L e z_\ell b_\ell k T \left[\int_{S_w} \mathbf{n} n_\ell dS - \int_{V_f} \frac{\nabla N_\ell^0}{N_\ell^0} n_\ell dV \right]. \quad (68)$$

In this last expression, the equilibrium condition $\mathbf{E}_f^0 = (kT/e z_\ell) \nabla N_\ell^0 / N_\ell^0$ has been employed in the volume integral. In general, a complete description would retain all four of these contributions to the current density. In practice, it will be argued in Sec. IV that $\mathbf{J}_n$ is negligible for the problem at hand of modeling acoustic-electromagnetic wave propagation in porous media. The electromagnetic description will be complete once these current densities have been expressed in terms of the macroscopic fields generating them.

It will now be argued that the macroscopic-diffusion current induced by wave fields, $\mathbf{J}_d$, is negligible in homogeneous-porous media. Taking the divergence of Ampère’s law gives the macroscopic statement of charge conservation:

$$\sum_{\ell=1}^L e z_\ell [i\omega \bar{n}_\ell + b_\ell k T \nabla^2 \bar{n}_\ell] = \nabla \cdot (\mathbf{J}_c + \mathbf{J}_s + \mathbf{J}_n). \quad (69)$$

Comparing the first term on the left-hand side (the net rate of ion accumulation) to the second term (the diffusion flux) gives the dimensionless ratio

$$\frac{|b_\ell k T \nabla^2 \bar{n}_\ell|}{|i\omega \bar{n}_\ell|} \simeq \frac{16 b_\ell k T f}{2\pi c^2}, \quad (70)$$

where c is wave speed and f is frequency in hertz and where it has been assumed that $\bar{n}_\ell$ varies over distances of a quarter wavelength of the macroscopic field inducing $\bar{n}_\ell$ (either acoustic or electromagnetic). Using values of $T = 298$ K, $b_\ell = 3 \times 10^{11}$ N s/m (a typical inorganic ion

such as sodium), and $c = 10^3$ m/s (acoustic waves) gives the condition that if frequencies are much less than 10^{14} Hz, macroscopic-diffusion currents can be neglected in homogeneous media. This condition will always be met.

Alternatively, if the porous material is assumed to possess macroscopic heterogeneities that vary (significantly) over distances h , then the above dimensionless ratio is

$$\frac{|b_\ell k T \nabla^2 \bar{n}_\ell|}{|i\omega \bar{n}_\ell|} \simeq \frac{b_\ell k T}{2\pi f h^2}. \quad (71)$$

If $h = 1$ mm (perhaps the smallest macroscopic length that can be considered in a volume-averaged theory—this, of course, depends on the grain sizes), then diffusion will be negligible if frequencies are much greater than 10^{-3} Hz. This will almost always be the case as well.

It should be noted that the actual macroscopic diffu-

sivity (obtained by allowing for $\mathbf{J}_n$ as well) is always less than $b_\ell k T$ so that the above conditions are overestimating the actual diffusion currents. Because we are only interested in macroscopic deviations $\bar{n}_\ell$ that are induced by wave fields (not those, for example, that are due to injected contaminants), it is safe to assume that diffusion currents associated with these deviations are negligible. This assumption will be made throughout what follows.

It is convenient to rewrite the current-density integrals under the assumption of a thin, uniform, double layer. With the definition of bulk-fluid conductivity,

$$\sigma_f = \sum_{\ell=1}^L (ez_\ell)^2 b_\ell \mathcal{N}_\ell, \quad (72)$$

the conduction-current density becomes

$$\mathbf{J}_c = \frac{\sigma_f}{V_f} \int_{V_f} \mathbf{e}_f dV + \frac{1}{V_f} \int_{S_w} dS \int_0^D \left(\sum_{\ell=1}^L (ez_\ell)^2 b_\ell (N_\ell^0 - \mathcal{N}_\ell) \right) \mathbf{e}_f d\chi. \quad (73)$$

The coordinate χ represents distance measured normal from the wall surface S_w (into the fluid) and D represents the distance over which the charge excess $N_\ell^0 - \mathcal{N}_\ell$ associated with the diffuse double layer is significant (say, a few Debye lengths). In practice, it will be argued that $\mathbf{e}_f$ can be taken as constant across the double layer.

If Coulomb's law for the static field [Eq. (20)] is substituted into the streaming-current integral [Eq. (67)], then

$$\mathbf{J}_s = \frac{\epsilon_0 \kappa_f}{V_f} \int_{V_f} (\nabla \cdot \mathbf{E}_f^0) \mathbf{v} dV \quad (74)$$

$$= \frac{\epsilon_0 \kappa_f}{V_f} \int_{V_f} [\nabla \cdot (\mathbf{E}_f^0 \mathbf{v}) - \mathbf{E}_f^0 \cdot \nabla \mathbf{v}] dV \quad (75)$$

$$= \frac{\epsilon_0 \kappa_f}{V_f} \left[\int_{S_e} \mathbf{n}_e \cdot \mathbf{E}_f^0 \mathbf{v} dS + \int_{S_w} \mathbf{n} \cdot \mathbf{E}_f^0 \mathbf{v} dS - \int_{S_w} dS \int_0^D \mathbf{E}_f^0 \cdot \nabla \mathbf{v} d\chi \right]. \quad (76)$$

In the last equation, S_e is the surface of intersection between the fluid and the averaging-volume surface and $\mathbf{n}_e$ is the outward normal to S_e . We can expect that, on average, the weight $\mathbf{n}_e \cdot \mathbf{E}_f^0$ is as often positive as negative on S_e so that the integral over S_e vanishes (this integral will also vanish if there is no macroscopic variations in the material properties). The second integral also vanishes because $\mathbf{v} = 0$ on S_w . Thus, only the last term is retained:

$$\mathbf{J}_s = \frac{\epsilon_0 \kappa_f}{V_f} \int_{S_w} dS \int_0^D \nabla \Phi^0 \cdot \nabla \mathbf{v} d\chi. \quad (77)$$

Note that, in general, it should not be assumed that $\nabla \mathbf{v}$ has constant components across the thin double layer.

Finally, although $\mathbf{J}_n$ will be neglected later, it is consistent here to also rewrite it under the thin-double-layer assumption. The volume integral within $\mathbf{J}_n$ can be written as

$$\int_{V_f} \frac{\nabla N_\ell^0}{N_\ell^0} n_\ell dV = - \int_{S_w} \mathbf{n} dS \int_0^D \frac{n_\ell}{N_\ell^0} \frac{\partial N_\ell^0}{\partial \chi} d\chi \quad (78)$$

$$= - \int_{S_w} \mathbf{n} dS \int_0^D \left[\frac{\partial}{\partial \chi} \left(N_\ell^0 \frac{n_\ell}{N_\ell^0} \right) - N_\ell^0 \frac{\partial}{\partial \chi} \left(\frac{n_\ell}{N_\ell^0} \right) \right] d\chi \quad (79)$$

$$= \int_{S_w} \mathbf{n} n_\ell dS + \int_{S_w} \mathbf{n} dS \int_0^D N_\ell^0 \frac{\partial}{\partial \chi} \left(\frac{n_\ell}{N_\ell^0} \right) d\chi, \quad (80)$$

so that the final expression for $\mathbf{J}_n$ is

$$\mathbf{J}_n = \frac{1}{V_f} \int_{S_w} \mathbf{n} dS \int_0^D \sum_{\ell=1}^L ez_\ell b_\ell k T N_\ell^0 \frac{\partial}{\partial \chi} \left(\frac{n_\ell}{N_\ell^0} \right) d\chi. \quad (81)$$

Perhaps, now, $\mathbf{J}_n$ is more clearly seen to be the average value of the diffusion currents induced normal to the grain surfaces because of local deviations n_ℓ .

In Sec. IV, the actual induced fields $\mathbf{e}_f$ and $\mathbf{v}$ will be expressed in terms of the macroscopic forces generating them (namely, the macroscopic-electric field and pressure

gradient). Then, in Sec. V, the above integrals will be carried out and the final macroscopic ion-flux-force law will be obtained for $\bar{\mathbf{J}}$.

It remains, here, to average the electromagnetic constitutive laws. Averaging $\mathbf{d}_\xi = \epsilon_0 \kappa_\xi \mathbf{e}_\xi$ gives

$$\bar{\mathbf{D}} = \epsilon_0 [\phi \kappa_f \bar{\mathbf{e}}_f + (1 - \phi) \kappa_s \bar{\mathbf{e}}_s]. \quad (82)$$

From the definition $\bar{\mathbf{E}} = \phi \bar{\mathbf{e}}_f + (1 - \phi) \bar{\mathbf{e}}_s$, we then have

$$\bar{\mathbf{D}} = \epsilon_0 [\phi (\kappa_f - \kappa_s) \bar{\mathbf{e}}_f + \kappa_s \bar{\mathbf{E}}]. \quad (83)$$

Once $\bar{\mathbf{e}}_f$ has been expressed in terms $\bar{\mathbf{E}}$, this law will be complete.

Because the magnetic susceptibilities are assumed negligible in both the fluid and solid phases (iron, nickel, and cobalt are not assumed to be major constituents), we simply have

$$\bar{\mathbf{B}} = \mu_0 \bar{\mathbf{H}}. \quad (84)$$

B. Averaging the mechanical equations

The force balances on the fluid [Eq. (44)] and the solid [Eq. (45)] are averaged and then added to give a macroscopic-force balance on the bulk material (i.e., fluid and solid taken together),

$$-i\omega (\rho_B \dot{\bar{\mathbf{u}}}_s + \rho_f \dot{\bar{\mathbf{w}}}) = \nabla \cdot \bar{\boldsymbol{\tau}}_B + \phi (\mathbf{f}_a + \mathbf{f}_d + \mathbf{f}_n), \quad (85)$$

where the three electrical body forces are defined as

$$\mathbf{f}_a = \frac{Q^0}{V_f} \int_{S_w} \mathbf{e}_s dS, \quad (86)$$

$$\mathbf{f}_d = \frac{1}{V_f} \int_{V_f} \left(\sum_{\ell=1}^L e z_\ell N_\ell^0 \right) \mathbf{e}_f dV, \quad (87)$$

$$\mathbf{f}_n = \frac{1}{V_f} \int_{V_f} \left(\sum_{\ell=1}^L e z_\ell n_\ell \right) \mathbf{E}_f^0 dV. \quad (88)$$

Here, $\mathbf{f}_a$ comes from the boundary conditions and represents the average force acting on the excess charge of the adsorbed layer, $\mathbf{f}_d$ represents the average force acting on the diffuse layer, while $\mathbf{f}_n$ is the average force acting normal to S_w due to any induced-charge excesses in the double layer. In the above force balance, we have introduced the definitions

$$\dot{\bar{\mathbf{w}}} = \phi \bar{\mathbf{v}} = \phi (\dot{\bar{\mathbf{u}}}_f - \dot{\bar{\mathbf{u}}}_s), \quad (89)$$

$$\bar{\boldsymbol{\tau}}_B = \phi \bar{\boldsymbol{\tau}}_f + (1 - \phi) \bar{\boldsymbol{\tau}}_s, \quad (90)$$

$$\rho_B = \phi \rho_f + (1 - \phi) \rho_s, \quad (91)$$

where $\dot{\bar{\mathbf{w}}}$ is called the “filtration velocity” (volume flux of

fluid per unit time and per unit area of porous material) and where a subscript B stands for “bulk.”

In this work, $\mathbf{f}_n$ will be assumed negligible. The conditions of validity for this assumption are essentially the same as ignoring the $\mathbf{J}_n$ contribution to the current density and will be discussed in the next section. Using the charge-balance condition

$$Q^0 = - \int_0^D \sum_{\ell=1}^L e z_\ell N_\ell^0(\chi) d\chi, \quad (92)$$

where χ is the same coordinate used previously to measure distance normal to S_w and where D is several Debye lengths, the body forces $\mathbf{f}_a + \mathbf{f}_d$ within the thin uniform double-layer approximation become

$$\mathbf{f}_a + \mathbf{f}_d = \frac{1}{V_f} \int_{S_w} dS \int_0^D \sum_{\ell=1}^L e z_\ell N_\ell^0 [\mathbf{e}_f - \mathbf{e}_s(0)] d\chi. \quad (93)$$

It will be seen that for thin double layers and when $\kappa_s \ll \kappa_f$ (e.g., quartz and water), $\mathbf{e}_f(\chi) \simeq \mathbf{e}_s(0)$ throughout the double layer so that

$$\mathbf{f}_a + \mathbf{f}_d = 0. \quad (94)$$

This states that the force acting on the adsorbed-layer charge is balanced by that acting on the diffuse-layer counter charge.

Next, the average-force balance on the fluid in relative motion is obtained. The relative-flow velocity $\mathbf{v} = \dot{\bar{\mathbf{u}}}_f - \dot{\bar{\mathbf{u}}}_s$ is substituted into the fluid-force balance [Eq. (44)] along with the fluid-stress tensor and the assumption that the local relative flow is incompressible ($\nabla \cdot \mathbf{v} = 0$) with the result

$$\begin{aligned} -i\omega \rho_f \mathbf{v} = & -\nabla p + i\omega \rho_f \dot{\bar{\mathbf{u}}}_s - \eta \nabla \times \nabla \times \mathbf{v} \\ & + \sum_{\ell=1}^L e z_\ell (N_\ell^0 \mathbf{e}_f + n_\ell \mathbf{E}_f^0). \end{aligned} \quad (95)$$

The incompressibility assumption for oscillatory flow is justified as long as $f^2 r^2 / c_f^2 \ll 1$ where f is frequency in hertz, r is a characteristic grain size, and c_f is the speed of sound in the fluid. This condition is roughly equivalent to the no-grain-scattering condition underlying the entire analysis. Upon averaging over the fluid, we have

$$-i\omega \rho_f \bar{\mathbf{v}} = -\nabla \bar{p} + i\omega \rho_f \dot{\bar{\mathbf{u}}}_s + \mathbf{f}_d + \mathbf{d} - \eta \nabla \times \nabla \times \bar{\mathbf{v}}, \quad (96)$$

where $\mathbf{d}$, the drag force, is defined as

$$\mathbf{d} = \mathbf{d}^{\text{form}} + \mathbf{d}^{\text{fric}}, \quad (97)$$

where

$$\mathbf{d}^{\text{form}} = \frac{1}{V_f} \int_{S_w} -n p dS, \quad (98)$$

$$\mathbf{d}^{\text{fric}} = \frac{\eta}{V_f} \int_{S_w} -\mathbf{n} \times \nabla \times \mathbf{v} dS. \quad (99)$$

Another application of the averaging theorem gives

$$\overline{\nabla \times \mathbf{v}} = \nabla \times \bar{\mathbf{v}} + \frac{1}{V_f} \int_{S_w} \mathbf{n} \times \mathbf{v} dS. \quad (100)$$

Since $\mathbf{v} = 0$ on S_w , we have that

$$\eta \nabla \times \overline{\nabla \times \mathbf{v}} = \eta \nabla \times \nabla \times \bar{\mathbf{v}}. \quad (101)$$

The importance of this macroscopic-shear term is determined by comparing it to the inertial term

$$\frac{|\eta \nabla \times \nabla \times \bar{\mathbf{v}}|}{|i\omega \rho_f \bar{\mathbf{v}}|} \simeq \frac{\eta}{\rho_f c^2} f, \quad (102)$$

where c is the speed of the mechanical wave producing the shear disturbance. For water and, say, $c = 10^3$ m/s, we have that as long as frequencies are much less than 10^{12} Hz, the macroscopic shearing in the fluid is negligible.

In the next section, expressions will be obtained that linearly relate the local-fluid velocity $\mathbf{v}$ and fluid pressure p to the macroscopic forces that are generating them ($-\nabla \bar{p} + i\omega \rho_f \bar{\mathbf{u}}_s$ and $\bar{\mathbf{E}}$). It will be seen that it is more direct to simply volume average the expression for $\mathbf{v}$ to obtain the macroscopic flux-force law than to carry out the integrals contained within $\mathbf{f}_d$ and $\mathbf{d}$ of the above averaged-force balance; however, the two approaches are equivalent.

It remains to average the stress-strain relations so that τ_B and $\bar{p}$ are linearly related to the induced deformation. Because the stress-strain relations utilized here are independent of any explicit electromagnetic coupling (e.g., no piezoelectric deformation is allowed for), the volume-averaged results of Pride, Gangi, and Morgan⁷ are directly applicable. Their results are identical to those of Biot and Willis¹⁰ (who, however, did not employ volume averaging) and are

$$\tau_B = (K_G \nabla \cdot \bar{\mathbf{u}}_s + C \nabla \cdot \bar{\mathbf{w}}) \mathbf{I} + G_{fr} \left(\nabla \bar{\mathbf{u}}_s + \nabla \bar{\mathbf{u}}_s^T - \frac{2}{3} \nabla \cdot \bar{\mathbf{u}}_s \mathbf{I} \right), \quad (103)$$

$$-\bar{p} = C \nabla \cdot \bar{\mathbf{u}}_s + M \nabla \cdot \bar{\mathbf{w}}, \quad (104)$$

where

$$K_G = \frac{K_{fr} + \phi K_f + (1 + \phi) K_s \Delta}{1 + \Delta}, \quad (105)$$

$$C = \frac{K_f + K_s \Delta}{1 + \Delta}, \quad (106)$$

$$M = \frac{1}{\phi} \frac{K_f}{1 + \Delta}, \quad (107)$$

and the parameter Δ is

$$\Delta = \frac{K_f}{\phi K_s^2} [(1 - \phi) K_s - K_{fr}]. \quad (108)$$

The moduli K_{fr} and G_{fr} are the bulk and shear moduli of the "framework of grains" when the fluid is absent—in practice, K_{fr} and G_{fr} can be defined when the saturating fluid is a gas (e.g., air). The boundary-value problem

that defines K_{fr} and G_{fr} will not be discussed in this paper because it is not clearly related to the boundary-value problems for $\mathbf{e}_f$, $\mathbf{v}$, and p . The frame moduli may either be considered experimentally determined (just like K_f , K_s , and G) or may be obtained from one of the many approximate theoretical models available for specific (and very simple) pore-grain geometries (see Refs. 11 and 12 for a review). In conclusion, although the electromagnetic fields do not directly appear in the above stress-strain relations, their influence is present if they induce relative fluid-solid motion $\bar{\mathbf{w}}$.

IV. BOUNDARY-VALUE PROBLEMS

A. Averaging disk

To complete the averaging procedure, the local fields $\mathbf{e}_f$, $\mathbf{v}$, and p involved in the various integrals above must be related to the macroscopic fields. In order to accomplish this, the idea of an averaging disk is introduced. Consider an imaginary volume within the porous material defined by two large plane-parallel faces of area A separated by a distance H . One might imagine a circular disk with radius much greater than thickness. The volume of this disk is AH . We imagine that a macroscopic potential difference exists between the two flat faces. The potential difference may be one of fluid pressure and/or electric potential. This potential difference is defined so that when it is divided by H , one obtains the appropriate macroscopic field (pressure gradient or electric field) in the direction normal to the disk face. The normal to the disk face will be defined as the z direction in what follows. These potential differences represent the boundary values for the local fields in the pores and grains.

The macroscopic-electromagnetic disturbances have wavelengths dramatically larger than H (which may be as small as 10^{-3} m depending on the grain sizes). Thus, it can be assumed that for the purpose of determining transport in the averaging disk, $\nabla \times \mathbf{e}_f = \nabla \times \mathbf{e}_s = 0$ at the pore and grain scale, or, $\mathbf{e}_f = -\nabla \varphi_f$ and $\mathbf{e}_s = -\nabla \varphi_s$. Thus, on the averaging disk, we have that

$$\hat{\mathbf{z}} \cdot \bar{\mathbf{E}} = -\frac{\Delta \phi}{H}, \quad (109)$$

where

$$\Delta \phi = \varphi_f(H) - \varphi_f(0) = \varphi_s(H) - \varphi_s(0) \quad (110)$$

is the potential difference between the two flat faces.

B. Pore-scale electric fields and ion-number-density deviations

In this subsection, the boundary-value problems controlling the electric potentials and deviations n_ℓ will be considered. The conditions required for the neglect of the n_ℓ (at the pore scale) will be defined.

Let us begin by rewriting the current density in the fluid [Eq. (38)] using the static-equilibrium condition $e z_\ell N_\ell^0 \mathbf{E}_f^0 = kT \nabla N_\ell^0$. The result is

$$\mathbf{j}_f = \sum_{\ell}^L e z_{\ell} N_{\ell}^0 \left[\mathbf{v} - b_{\ell} k T \nabla \left(\frac{n_{\ell}}{N_{\ell}^0} + \frac{e z_{\ell}}{k T} \varphi_f \right) \right] \quad (111)$$

and will be used in the conservation of charge,

$$\nabla \cdot \mathbf{j}_f = i \omega \sum_{\ell}^L e z_{\ell} n_{\ell}. \quad (112)$$

Because this conservation law must hold for an electrolyte of arbitrary specifications, we then have

$$\nabla \cdot \left\{ N_{\ell}^0 \left[\mathbf{v} - b_{\ell} k T \nabla \left(\frac{n_{\ell}}{N_{\ell}^0} + \frac{e z_{\ell}}{k T} \varphi_f \right) \right] \right\} = i \omega n_{\ell} \quad (113)$$

for each of the L ionic species. Upon distributing the divergence and noting that $\nabla \cdot \mathbf{v} = 0$, a general statement of ion conservation is obtained (the ion balance):

$$\begin{aligned} \nabla^2 \left(\frac{n_{\ell}}{N_{\ell}^0} + \frac{e z_{\ell}}{k T} \varphi_f \right) + \frac{i \omega}{b_{\ell} k T} \frac{n_{\ell}}{N_{\ell}^0} \\ = \frac{\nabla N_{\ell}^0}{N_{\ell}^0} \cdot \left[\frac{\mathbf{v}}{b_{\ell} k T} - \nabla \left(\frac{n_{\ell}}{N_{\ell}^0} + \frac{e z_{\ell}}{k T} \varphi_f \right) \right]. \end{aligned} \quad (114)$$

The boundary condition of no ion accumulation on S_w ($\mathbf{n} \cdot \mathbf{j}_f = 0$ on S_w) is equivalent to the conditions

$$\mathbf{n} \cdot \nabla \left(\frac{n_{\ell}}{N_{\ell}^0} + \frac{e z_{\ell}}{k T} \varphi_f \right) = 0, \quad (115)$$

$$\mathbf{n} \cdot \mathbf{v} = 0, \quad (116)$$

on S_w .

The right-hand side of the ion balance can be understood as a source term for ion accumulation. The gradient ∇N_{ℓ}^0 is everywhere normal to the grain surfaces and is nonzero only in the double layer. There can only be such a source term if either the flow field $\mathbf{v}$ or the force $\mathbf{f}_{\ell} = -\nabla(n_{\ell}/N_{\ell}^0 + e z_{\ell} \varphi_f/kT)$ driving the ion flux has a component normal to the grain surface within the double layer. From the boundary conditions just given, we expect that when the thickness of the double layer d is much smaller than the radii of curvature r of the grain surfaces (which are responsible for the spatial deviations in $\mathbf{v}$ and $\mathbf{f}_{\ell}$), there effectively will be no normal component to either $\mathbf{v}$ or $\mathbf{f}_{\ell}$ within the double layer. It has previously been discussed that the condition $d \ll r$ is well satisfied for sedimentary media.

Under such thin-double-layer conditions and in the low-frequency limit where $\omega \ll b_{\ell} k T/r^2$ [using values of $b_{\ell} = 3 \times 10^{11}$ Ns/m (typical inorganic ion), $T = 298$ K, and $r = 10^{-4}$ m (e.g., sand grains) gives $f \ll 2 \times 10^{-2}$ Hz; using $r = 10^{-6}$ m (e.g., clay particles) gives $f \ll 200$ Hz], we have the boundary-value problem on the averaging disk,

$$\psi_{\ell} = \frac{n_{\ell}}{N_{\ell}^0} + \frac{e z_{\ell}}{k T} \varphi_f, \quad (117)$$

$$\nabla^2 \psi_{\ell} = 0, \quad (118)$$

$$\mathbf{n} \cdot \nabla \psi_{\ell} = 0 \quad \text{on } S_w, \quad (119)$$

and

$$\psi_{\ell} = \begin{cases} \frac{e z_{\ell}}{k T} \Delta \phi & \text{on } z = H, \\ 0 & \text{on } z = 0. \end{cases} \quad (120)$$

This last condition is a statement that there exists a macroscopic-electric-potential difference of $\Delta \phi$ between the two flat faces of the averaging disk but that no macroscopic gradients in n_{ℓ} exist. A solution for ψ_{ℓ} may be expressed in terms of a related, purely geometric, field Γ possessing units of length and defined to satisfy

$$\nabla^2 \Gamma = 0, \quad (121)$$

$$\mathbf{n} \cdot \nabla \Gamma = 0 \quad \text{on } S_w, \quad (122)$$

$$\Gamma = \begin{cases} H & \text{on } z = H, \\ 0 & \text{on } z = 0. \end{cases} \quad (123)$$

Thus, for thin double layers, we have

$$\frac{n_{\ell}(\mathbf{r})}{N_{\ell}^0} = \frac{e z_{\ell}}{k T} \left[\Gamma(\mathbf{r}) \frac{\Delta \phi}{H} - \varphi_f(\mathbf{r}) \right]. \quad (124)$$

The field $\Gamma(\mathbf{r})$ will be important in all that follows.

We next consider the boundary-value problem for φ_f defined by Coulomb's law

$$\nabla^2 \varphi_f = - \sum_{\ell=1}^L \frac{e z_{\ell}}{\epsilon_0 \kappa_f} n_{\ell} \quad (125)$$

$$= - \sum_{\ell=1}^L \frac{(e z_{\ell})^2}{\epsilon_0 \kappa_f k T} N_{\ell}^0 \left(\Gamma \frac{\Delta \phi}{H} - \varphi_f \right), \quad (126)$$

where on S_w

$$\mathbf{n} \cdot \nabla \varphi_f = \frac{\kappa_s}{\kappa_f} \mathbf{n} \cdot \nabla \varphi_s \quad (127)$$

and on the flat faces

$$\varphi_f = \begin{cases} \Delta \phi & z = H, \\ 0 & z = 0. \end{cases} \quad (128)$$

The presence of φ_s in the boundary condition requires specification of the boundary-value problem for φ_s ,

$$\nabla^2 \varphi_s = 0, \quad (129)$$

$$\varphi_s = \varphi_f \quad \text{on } S_w, \quad (130)$$

$$\varphi_s = \begin{cases} \Delta \phi, & z = H, \\ 0, & z = 0. \end{cases} \quad (131)$$

A particular solution for φ_f (i.e., satisfying the differential equation but not, necessarily, the boundary conditions) is

$$\varphi_f^p = \Gamma \frac{\Delta\phi}{H}. \quad (132)$$

To the extent that $\kappa_s/\kappa_f \ll 1$, so that $\mathbf{n} \cdot \nabla\varphi_f \simeq 0$ on S_w , this particular solution will, in fact, be the actual solution. In this case, we then have $n_\ell \simeq 0$ and the integrals $\mathbf{J}_n$ and $\mathbf{f}_n$ from the volume averaging vanish. For the quartz-electrolyte interface, the contrast is $\kappa_s/\kappa_f \simeq 1/20$, and this approximation is appropriate.

In physical terms, if the dielectric contrast is large, then large polarization-charge densities are induced on those portions of S_w that have $\mathbf{n} \cdot \hat{\mathbf{z}} \neq 0$. These polarization charges produce local fields that keep $\mathbf{n} \cdot \nabla\varphi_s \simeq 0$ everywhere along S_w and, thus, $n_\ell \simeq 0$ throughout the fluid. If the dielectric contrast is small, then electric-field lines can pass through S_w , thus producing an electric field in the fluid that has components normal to S_w near S_w . These normal components will drive an electromigration that must be balanced by an equal and opposite diffusion flux so that ions do not accumulate on S_w [Eq. (115)]; i.e., deviations n_ℓ are necessarily induced near S_w when the dielectric contrast is small.

For the case of a small dielectric contrast, we can express the actual thin-double-layer electric potential as

$$\varphi_f = \Gamma \frac{\Delta\phi}{H} + \delta\varphi, \quad (133)$$

where the field $\delta\varphi$ is defined by the problem

$$\nabla^2\delta\varphi = 0, \quad (134)$$

$$\mathbf{n} \cdot \nabla\delta\varphi = \frac{\kappa_s}{\kappa_f} \mathbf{n} \cdot \nabla\varphi_s \quad \text{on } S_w, \quad (135)$$

$$\delta\varphi = \begin{cases} 0 & z = H, \\ 0 & z = 0; \end{cases} \quad (136)$$

i.e., the source for the $\delta\varphi$ field is the electric field passing normal through S_w . Finally, the ion-number deviations are related to $\delta\varphi$ as

$$\frac{n_\ell}{N_\ell^0} = -\frac{ez_\ell}{kT} \delta\varphi. \quad (137)$$

As frequencies increase, there is progressively less time available for these ion-number deviations to take place and we expect that n_ℓ will decrease. Indeed, when $\omega \gg b_\ell kT/r^2$ we must have $n_\ell \rightarrow 0$ from the ion balance [Eq. (114)].

In order to neglect n_ℓ in our modeling of the fields in the pore space, we therefore must satisfy two conditions: (1) The electric-double layer must be much thinner than the effective grain sizes so that the right-hand side of the ion balance [Eq. (114)] is negligible, and (2) there must be a large contrast in the dielectric constant between the grains and the electrolyte so that $\delta\varphi$ as defined by Eqs. (134)–(136) is negligible. Both of these conditions are

assumed to be met by the materials being considered.

By neglecting n_ℓ , we will be ignoring all grain-scale diffusion currents. Experimentally, it is found that at low frequencies (as defined above) in sedimentary materials containing significant quantities of clay, the electric current has a small phase shift relative to the applied electric field.^{13,14} This phase shift is on the order of a few milliradian. The details of how these small phase shifts arise remains unknown in sedimentary materials; however, the effect is almost certainly due to accumulations n_ℓ induced by the electric field. It can safely be assumed that ignoring this milliradian-phase-shift effect (also sometimes called the “low-frequency dielectric enhancement”) will not produce significant error in the coupled electromagnetic-acoustic wave-propagation problem of interest in this work.

In conclusion, if the n_ℓ are ignored, then we have that for all frequencies the electric-field in the fluid is given by

$$\mathbf{e}_f(\mathbf{r}) = -\nabla\varphi_f(\mathbf{r}) = -\nabla\Gamma(\mathbf{r}) \frac{\Delta\phi}{H}. \quad (138)$$

This field can be assumed to be both constant and tangential to S_w throughout the thin double layer.

C. Pore-scale flow fields

The fluid-flow boundary-value problem is expressed on the averaging disk as [cf. Eq. (95) and note that $\nabla \times \nabla \times \mathbf{v} = -\nabla^2 \mathbf{v}$]

$$\eta \nabla^2 \mathbf{v} + i\omega \rho_f \mathbf{v} = \nabla p + \epsilon_0 \kappa_f (\nabla \cdot \mathbf{E}_f^0) \nabla \Gamma \frac{\Delta\phi}{H}, \quad (139)$$

$$\nabla \cdot \mathbf{v} = 0, \quad (140)$$

$$\mathbf{v} = 0 \quad \text{on } S_w, \quad (141)$$

$$p = \begin{cases} \Delta P \equiv \hat{\mathbf{z}} \cdot (\nabla \bar{p}_f - i\omega \rho_f \hat{\mathbf{u}}_s) H, & z = H, \\ 0, & z = 0. \end{cases} \quad (142)$$

The deviations n_ℓ have been neglected.

The solution for $\mathbf{v}$ and p can be separated into mechanically and electrically induced portions

$$\mathbf{v} = \mathbf{v}_m + \mathbf{v}_e, \quad (143)$$

$$p = p_m + p_e; \quad (144)$$

i.e., the m stands for fields induced by $\Delta P/H$, while the e stands for fields induced by $\Delta\phi/H$. Introducing the viscous skin depth (units of length)

$$\delta = \sqrt{\frac{\eta}{\omega \rho_f}}, \quad (145)$$

the boundary-value problem is then separated as

$$\nabla^2 \mathbf{v}_m + \frac{i}{\delta^2} \mathbf{v}_m = \frac{\nabla p_m}{\eta}, \quad (146)$$

$$p_m = \begin{cases} \Delta P, & z = H, \\ 0, & z = 0, \end{cases} \quad (147)$$

and

$$\nabla^2 \mathbf{v}_e + \frac{i}{\delta^2} \mathbf{v}_e = \frac{\nabla p_e}{\eta} - \frac{\epsilon_0 \kappa_f}{\eta} (\nabla^2 \Phi^0) \nabla \Gamma \frac{\Delta \phi}{H}, \quad (148)$$

$$p_e = \begin{cases} 0, & z = H, \\ 0, & z = 0. \end{cases} \quad (149)$$

We also have that $\mathbf{v}_m = \mathbf{v}_e = 0$ on S_w and $\nabla \cdot \mathbf{v}_m = \nabla \cdot \mathbf{v}_e = 0$ everywhere.

The electrically induced flow problem is treated first. Because the electrical body force inducing flow only acts in the thin double layer and is purely tangential to S_w , we have that $\mathbf{n} \cdot \nabla p_e = 0$ on S_w so that

$$p_e = 0 \quad (150)$$

everywhere and for all frequencies. In the limit of very low frequencies ($\delta \rightarrow \infty$), the flow velocity $\mathbf{v}_{e0}$ is

$$\mathbf{v}_{e0} = -\frac{\epsilon_0 \kappa_f}{\eta} (\Phi^0 - \zeta) \nabla \Gamma \frac{\Delta \phi}{H}. \quad (151)$$

The only vorticity in this flow field is present in the thin double layer. Thus, for general frequency dependence, the flow is governed by the boundary-layer-type equation

$$\frac{\partial^2 \mathbf{v}_e}{\partial \chi^2} + \frac{i}{\delta^2} \mathbf{v}_e = -\frac{\epsilon_0 \kappa_f}{\eta} \frac{\partial^2 \Phi^0}{\partial \chi^2} \nabla \Gamma \frac{\Delta \phi}{H}, \quad (152)$$

where χ is the local coordinate measuring distance normal from the grain surfaces (into the fluid). For water-based electrolytes at room temperature, the Debye length in meters goes as $d = 3 \times 10^{-10} / \sqrt{C}$ where C is salt concentration in moles per liter, while the viscous skin depth goes as $\delta = 4 \times 10^{-4} / \sqrt{f}$ where f is frequency in hertz. Thus, these lengths are related as

$$\frac{d}{\delta} = 10^{-6} \sqrt{\frac{f}{C}}. \quad (153)$$

Taking $f = 10^6$ Hz (the largest possible frequency) it is seen that as long as

$$C \gg 10^{-6} \text{ mol/liter}, \quad (154)$$

we will always be in the regime where

$$\frac{d}{\delta} \ll 1. \quad (155)$$

This condition on concentration will always be met in rocks. Even if a sample is saturated with distilled water, after the water has come to equilibrium with the grains, the condition should again be fulfilled. Thus, the particular solution

$$\mathbf{v}_e^p = -\frac{\epsilon_0 \kappa_f}{\eta} \Phi^0 \nabla \Gamma \frac{\Delta \phi}{H} \quad (156)$$

will hold regardless of frequency; i.e., $|\partial^2 \mathbf{v}_e^p / \partial \chi^2| \gg |i \mathbf{v}_e^p / \delta^2|$ throughout the double layer. To this must be added the homogeneous solution $A \exp(i^{3/2} \chi / \delta)$, which then gives the general frequency-dependent field

$$\mathbf{v}_e = -\frac{\epsilon_0 \kappa_f}{\eta} (\Phi^0 - \zeta e^{i^{3/2} \chi / \delta}) \nabla \Gamma \frac{\Delta \phi}{H}. \quad (157)$$

Note that if the Debye approximation $\Phi^0 \simeq \zeta \exp(-\chi/d)$ is used in Eq. (152), the solution, regardless of the value of d/δ , is

$$\mathbf{v}_e^{\text{Debye}} = -\frac{\epsilon_0 \kappa_f}{\eta} \frac{(\Phi^0 - \zeta e^{i^{3/2} \chi / \delta})}{1 + i d^2 / \delta^2} \nabla \Gamma \frac{\Delta \phi}{H}. \quad (158)$$

The two results are consistent to order d/δ . Because of Eq. (155), we only bother to retain terms to order d/δ throughout what follows.

The mechanically induced flow problem is less tractable because the vorticity is not always confined to a thin boundary layer near S_w . As is explained at the start of the next section, we only have need of the low-frequency and high-frequency solutions defined by whether the inertial term is significant or not. The transition frequency f_t (in hertz) is estimated as

$$f_t = \frac{\eta}{2\pi \rho_f} \frac{|\nabla^2 \mathbf{v}_m|}{|\mathbf{v}_m|} \simeq \frac{\eta}{2\pi \rho_f r^2}, \quad (159)$$

where r can be taken as a typical pore radius. For water and $r = 10^{-5}$ m, this gives $f_t \sim 2000$ Hz.

When $f \ll f_t$ (low frequencies), the vorticity extends throughout the fluid and the inertial term is everywhere negligible. The fluid flow and pressure fields can be represented as

$$\mathbf{v}_{m0}(\mathbf{r}) = \frac{\mathbf{g}(\mathbf{r})}{\eta} \frac{\Delta P}{H} \quad (160)$$

and

$$p_{m0}(\mathbf{r}) = h(\mathbf{r}) \frac{\Delta P}{H}, \quad (161)$$

where $\mathbf{g}(\mathbf{r})$ has the units of length squared and $h(\mathbf{r})$ has the units of length and both are independent of fluid properties. These pore-geometry functions are solutions to the fundamental Stokes problem [obtained from Eqs. (146) and (147) and $\nabla \cdot \mathbf{v}_m = 0$]

$$\nabla^2 \mathbf{g} = \nabla h, \quad (162)$$

$$\nabla^2 h = 0, \quad (163)$$

$$\mathbf{g} = 0 \quad \text{on } S_w, \quad (164)$$

$$h = \begin{cases} H, & z = H, \\ 0, & z = 0. \end{cases} \quad (165)$$

In general, $\mathbf{n} \cdot \nabla h (= \mathbf{n} \cdot \nabla^2 \mathbf{g} = -\mathbf{n} \cdot \nabla \times \nabla \times \mathbf{g}) \neq 0$ on S_w ,

so that $h \neq \Gamma$. The only way for $\mathbf{n} \cdot \nabla h$ to be everywhere zero on S_w is if all the pores are constant-width channels, e.g., $h = z$. Nonetheless, it proves convenient to express the h function as

$$h = \Gamma + \delta h, \quad (166)$$

where δh therefore satisfies

$$\nabla^2 \delta h = 0, \quad (167)$$

$$\mathbf{n} \cdot \nabla \delta h = -\mathbf{n} \cdot \nabla \times \nabla \times \mathbf{g} \quad \text{on } S_w, \quad (168)$$

$$\delta h = \begin{cases} 0, & z = H, \\ 0, & z = 0. \end{cases} \quad (169)$$

The source for the δh field is the rotation of vorticity on the pore walls. As will be seen, the existence of the Stokes-flow geometry field $\mathbf{g}$ is all that is needed to establish relations between the various macroscopic-transport coefficients. Explicit expressions for $\mathbf{g}$ (or h) will not be required.

Continuing now to the case $f \gg f_t$ (high frequencies), the vorticity is confined to a thin boundary layer of thickness δ . The flow in this boundary layer is like flow past a planar wall and, accordingly, has the property that $\mathbf{n} \cdot \nabla p_{m\infty} = 0$ on S_w . Thus, the high-frequency pressure field $p_{m\infty}$ is simply

$$p_{m\infty}(\mathbf{r}) = \Gamma(\mathbf{r}) \frac{\Delta P}{H}. \quad (170)$$

This leads to the boundary-layer equation

$$\frac{\partial^2 \mathbf{v}_{m\infty}}{\partial \chi^2} + \frac{i}{\delta^2} \mathbf{v}_{m\infty} = \frac{\nabla \Gamma}{\eta} \frac{\Delta P}{H}, \quad (171)$$

which has the solution

$$\mathbf{v}_{m\infty} = -\frac{i\delta^2}{\eta} \left(1 - e^{i^{3/2}\chi/\delta}\right) \nabla \Gamma \frac{\Delta P}{H}. \quad (172)$$

All the pore-scale fields required to obtain the macroscopic-transport coefficients have now been defined.

V. TRANSPORT COEFFICIENTS

Using the pore-scale fields just developed on the averaging disk, we must next evaluate the integrals defined by $\mathbf{J}_c$ (conduction current), $\mathbf{J}_s$ (streaming current), and $\bar{\mathbf{v}}$ (relative fluid flow) in order to obtain the macroscopic-transport coefficients. We will obtain flux-force relations in the form

$$\bar{\mathbf{J}} \equiv \phi (\mathbf{J}_c + \mathbf{J}_s) \quad (173)$$

$$= \sigma(\omega) \bar{\mathbf{E}} + L_m(\omega) (-\nabla \bar{p} + i\omega \rho_f \dot{\bar{\mathbf{u}}}_s) \quad (174)$$

and

$$\dot{\bar{\mathbf{w}}} \equiv \phi \bar{\mathbf{v}} \quad (175)$$

$$= \frac{k(\omega)}{\eta} (-\nabla \bar{p} + i\omega \rho_f \dot{\bar{\mathbf{u}}}_s) + L_e(\omega) \bar{\mathbf{E}}. \quad (176)$$

It is demonstrated in Appendix B that

$$L_m(\omega) = L_e(\omega) \quad (177)$$

for all frequencies. The only requirement used to arrive at this statement of Onsager reciprocity is that double layers are thin relative to grain sizes.

The frequency dependence in the coefficients is obtained following the general procedure outlined by Johnson, Koplik, and Dashen¹⁵ as is now described. We have obtained the pore-scale fields in the limits of high and low frequencies. These fields will first be integrated to give the high- and low-frequency definition of the transport coefficients. The two limits are then connected by a simple postulated function. Although there is nonuniqueness in defining such a frequency function, it nonetheless must satisfy certain properties on the complex ω plane in addition to the frequency limits on the real ω axis. For example, causality requires the coefficients to be free of zeros and singularities in the upper-half ω plane.¹⁶ More specifically, in Appendix A of their paper, Johnson, Koplik, and Dashen¹⁵ have shown that the poles and zeros of $k(\omega)$ all lie on the negative imaginary ω axis. Following their argument exactly but using a modified pressure p' given by [cf. Eq. (139)]

$$p' = p + \epsilon_0 \kappa_f (\nabla \cdot \mathbf{E}_f^0) \Gamma \frac{\Delta \phi}{H}, \quad (178)$$

one ends up with the result that both $k(\omega)$ and $L(\omega)$ have all their poles and zeros on the negative imaginary ω axis (for a thin double layer where $\nabla \Gamma$ is perpendicular to the double-layer variation). Thus the frequency functions used to define $k(\omega)$ and $L(\omega)$ should also have this property. Improper location of any singularities in the complex ω plane can lead to spurious contributions from the residue theorem when we (inevitably) return to the time domain via the inverse Fourier transform.

A. Conduction current $\mathbf{J}_c$ and $\bar{\mathbf{e}}_f$

The neglect of the deviations n_t means that both $\mathbf{J}_c$ and $\bar{\mathbf{e}}_f$ are independent of frequency and that they do not depend on the macroscopic-pressure variations. Both are only proportional to $\bar{\mathbf{E}}$. If $\mathbf{e}_f = -\nabla \Gamma \Delta \phi / H$ is averaged over the pore volume of the averaging disk, we have

$$\bar{\mathbf{e}}_f = -\frac{1}{V_f} \int_{V_f} \nabla \phi_f dV \quad (179)$$

$$= -\frac{1}{V_f} \left[\phi H A \hat{\mathbf{z}} + \int_{S_w} \mathbf{n} \Gamma dS \right] \frac{\Delta \phi}{H} \quad (180)$$

$$= -\left[\hat{\mathbf{z}} + \frac{1}{V_f} \int_{S_w} \mathbf{n} \Gamma dS \right] \frac{\Delta \phi}{H}. \quad (181)$$

The macroscopic-electric field is $\bar{\mathbf{E}} = -\hat{\mathbf{z}} \Delta \phi / H$ so that (for isotropic media) we have

$$\bar{\mathbf{e}}_f = \frac{1}{\alpha_\infty} \bar{\mathbf{E}}, \quad (182)$$

where

$$\frac{1}{\alpha_\infty} \equiv 1 + \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \mathbf{n}\Gamma \, dS. \quad (183)$$

The unitless parameter α_∞ is the first of three fundamental porous-material “geometry” terms and is often called the “tortuosity.” It must be identical to the α_∞ defined by Johnson, Koplik, and Dashen¹⁵ which is

$$\mathbf{J}_c = \frac{\sigma_f}{\alpha_\infty} \bar{\mathbf{E}} - \frac{1}{V_f} \int_{S_w} \nabla\Gamma \, dS \int_0^D \sum_{\ell=1}^L (ez_\ell)^2 b_\ell [N_\ell^0(\chi) - \mathcal{N}_\ell] d\chi \frac{\Delta\phi}{H}. \quad (185)$$

If we define

$$\frac{2}{\Lambda} = \frac{\alpha_\infty}{V_f} \int_{S_w} \hat{\mathbf{z}} \cdot \nabla\Gamma \, dS \quad (186)$$

and

$$C_{em} = \int_0^D \sum_{\ell=1}^L (ez_\ell)^2 b_\ell \mathcal{N}_\ell \left[\exp\left(-\frac{ez_\ell \Phi^0}{kT}\right) - 1 \right] d\chi, \quad (187)$$

then (for isotropic media) we have

$$\mathbf{J}_c = \left(\frac{\sigma_f}{\alpha_\infty} + \frac{2C_{em}}{\alpha_\infty \Lambda} \right) \bar{\mathbf{E}}. \quad (188)$$

The parameter Λ is the second fundamental porous-material geometry term. It possesses the units of length and is seen to be a weighted volume-to-surface ratio where the dimensionless weight function $\hat{\mathbf{z}} \cdot \nabla\Gamma$ emphasizes the constricted portions of the pore space (i.e., the “pore throats” where the local field $\nabla\Gamma$ is large in order that $\nabla \cdot \nabla\Gamma = 0$). It will be seen that this “ Λ parameter”

$$\frac{1}{\alpha_\infty} = \frac{1}{V_f} \int_{V_f} \nabla\Gamma \cdot \nabla\Gamma \, dV, \quad (184)$$

thus establishing an identity for the Γ field.

The average conduction current in the fluid $\mathbf{J}_c$ [Eq. (73)] is now

must be identical to that defined by Johnson, Koplik, and Dashen which is

$$\frac{2}{\Lambda} = \frac{\alpha_\infty}{V_f} \int_{S_w} \nabla\Gamma \cdot \nabla\Gamma \, dS, \quad (189)$$

thus establishing another identity for the Γ field. Note that the integrands in Eqs. (186) and (189) cannot be equated because Γ is dependent on S_w .

The parameter C_{em} is the excess conductance associated with the electromigration of double layer ions. For a general multicomponent electrolyte with no restrictions placed on ζ , an exact expression for C_{em} requires a numerical solution of the Poisson-Boltzmann equation followed by a numerical integration. However, it is possible to obtain an extremely accurate estimate by using the Debye approximation $\Phi^0 = \zeta \exp(-\chi/d)$ in the following manner. Define the dimensionless surface potential

$$Z_\ell = \frac{ez_\ell}{kT} \zeta \quad (190)$$

and insert the Debye approximation into C_{em} obtaining

$$C_{em} \simeq \sum_{\ell=1}^L (ez_\ell)^2 b_\ell \mathcal{N}_\ell \int_0^D \left[\exp(-Z_\ell e^{-\chi/d}) - 1 \right] d\chi \quad (191)$$

$$= \sum_{\ell=1}^L (ez_\ell)^2 b_\ell \mathcal{N}_\ell \int_0^D \left[-Z_\ell e^{-\chi/d} + \frac{Z_\ell^2}{2!} e^{-2\chi/d} - \frac{Z_\ell^3}{3!} e^{-3\chi/d} + \dots \right] d\chi \quad (192)$$

$$= \sum_{\ell=1}^L (ez_\ell)^2 b_\ell \mathcal{N}_\ell d \left[-Z_\ell + \frac{Z_\ell^2}{2 \times 2!} - \frac{Z_\ell^3}{3 \times 3!} + \dots \right] \quad (193)$$

$$\simeq 2d \sum_{\ell=1}^L (ez_\ell)^2 b_\ell \mathcal{N}_\ell \left[\exp\left(-\frac{ez_\ell \zeta}{2kT}\right) - 1 \right]. \quad (194)$$

For the special case of a binary symmetric electrolyte ($L = 2$; $z_1 = -z_2$; $\mathcal{N}_1 = \mathcal{N}_2$) this gives

$$C_{em} = 2d(ez_1)^2 \mathcal{N}_1 \left\{ (b_1 + b_2) \left[\cosh\left(\frac{ez_1 \zeta}{2kT}\right) - 1 \right] + (b_1 - b_2) \sinh\left(\frac{ez_1 \zeta}{2kT}\right) \right\}. \quad (195)$$

It turns out that this is the exact result using the exact analytic expression for Φ^0 given in Appendix A for

this special case. Remarkably, the approximation made in the first expression for C_{em} [Eq. (191)] is exactly compensated by the approximate summation used to obtain the final expression. This gives good confidence in the above estimate of C_{em} [Eq. (194)] for a general electrolyte. Under conditions where $|ez_\ell \zeta / kT| \ll 1$, we also have the Debye estimate

$$C_{em}^{\text{Debye}} = -\frac{\zeta d}{kT} \sum_{\ell=1}^L (ez_\ell)^3 b_\ell \mathcal{N}_\ell \left(1 - \frac{ez_\ell \zeta}{4kT} \right). \quad (196)$$

Two terms of the expansion are retained here because if all the b_ℓ are equal, the first term sums to zero.

B. Streaming current $\mathbf{J}_s$

We next consider the average streaming current in the fluid $\mathbf{J}_s$ [Eq. (77)],

$$\mathbf{J}_s = \frac{\epsilon_0 \kappa_f}{V_f} \int_{S_w} dS \int_0^D \nabla \Phi^0 \cdot \nabla \mathbf{v} d\chi, \quad (197)$$

where $\nabla \Phi^0 \cdot \nabla \mathbf{v} = (\partial \Phi^0 / \partial \chi)(\partial \mathbf{v} / \partial \chi)$. Because of the separation into an electrically induced field $\mathbf{v}_e$ and a mechanically induced field $\mathbf{v}_m$, we also have $\mathbf{J}_s = \mathbf{J}_{se} + \mathbf{J}_{sm}$.

Focusing on the electrical portion first with $\mathbf{v}_e$ given by Eq. (157) gives

$$\mathbf{J}_{se} = \frac{2C_{os}}{\alpha_{os}\Lambda} \bar{\mathbf{E}}, \quad (198)$$

where

$$C_{os} = \frac{(\epsilon_0 \kappa_f)^2}{\eta} \int_0^D \frac{\partial \Phi^0}{\partial \chi} \frac{\partial}{\partial \chi} \left(\Phi^0 - \zeta e^{i^{3/2}\chi/\delta} \right) d\chi. \quad (199)$$

C_{os} represents the conductance due to electrically induced streaming (convection) of the excess double-layer ions. This may be called the “electro-osmotic” conductance. Because $\delta \gg d$, the variation of the exponential across the double layer is negligible and we have

$$C_{os} = \frac{(\epsilon_0 \kappa_f)^2}{\eta} \left[\zeta^2 \frac{i^{3/2}}{\delta} + \int_0^D \left(\frac{\partial \Phi^0}{\partial \chi} \right)^2 d\chi \right]. \quad (200)$$

We can obtain a nice approximation for the integral here using exactly the same procedure as for C_{em} . Starting with the Poisson-Boltzmann equation

$$\frac{\partial^2 \Phi^0}{\partial \chi^2} = - \sum_{\ell=1}^L \frac{ez_\ell}{\epsilon_0 \kappa_f} \mathcal{N}_\ell \exp \left(- \frac{ez_\ell}{kT} \Phi^0 \right), \quad (201)$$

both sides are multiplied by $\partial \Phi^0 / \partial \chi$ and the equation integrated to give

$$\left(\frac{\partial \Phi^0}{\partial \chi} \right)^2 = \frac{2kT}{\epsilon_0 \kappa_f} \sum_{\ell=1}^L \mathcal{N}_\ell \left[\exp \left(- \frac{ez_\ell}{kT} \Phi^0 \right) - 1 \right]. \quad (202)$$

Integrating this over the double layer using the sequence of steps defined by Eqs. (191)–(194) then gives the estimate

$$C_{os} = \frac{(\epsilon_0 \kappa_f)^2}{\eta} \left\{ \frac{4kTd}{\epsilon_0 \kappa_f} \sum_{\ell=1}^L \mathcal{N}_\ell \left[\exp \left(- \frac{ez_\ell \zeta}{2kT} \right) - 1 \right] + \frac{i^{3/2}}{\delta} \zeta^2 \right\}. \quad (203)$$

There is a weak frequency dependence through the presence of the skin depth $\delta = \sqrt{\eta/\rho_f \omega}$ in the last term. It provides insight to expand the exponent to second order

in $ez_\ell \zeta / 2kT$ (note that the first-order term vanishes from bulk-charge neutrality) and obtain

$$C_{os} = \frac{(\epsilon_0 \kappa_f)^2 \zeta^2}{2d\eta} \left(1 + 2i^{3/2} \frac{d}{\delta} \right), \quad (204)$$

where the definition of the Debye length has been used. This is valid whenever $|ez_\ell \zeta / 2kT| \ll 1$. Alternatively, if the Debye-approximation flow field [Eq. (158)] is used in $\mathbf{J}_s$ along with $\Phi^0 = \zeta \exp(-\chi/d)$, then we obtain the Debye approximation estimate C_{os}^{Debye} that is valid (if $|ez_\ell \zeta / 2kT| \ll 1$) regardless of the d/δ magnitude:

$$C_{os}^{\text{Debye}} = \frac{(\epsilon_0 \kappa_f)^2 \zeta^2}{2d\eta} \left(1 - i^{3/2} \frac{d}{\delta} \right)^{-2}. \quad (205)$$

Thus, when terms only to first order in d/δ are kept, Eqs. (204) and (205) are seen to be consistent.

It is important to note, however, that the frequency dependence defined by C_{os}^{Debye} has no singularities in the complex ω plane; nor can it lead to there being zeros of $\sigma(\omega)$ as long as analysis is carried out on a Riemann surface (there is a branch point at the origin due to $\delta^{-1} = \sqrt{\omega \rho_f / \eta}$ that must be respected). Contrarily, the frequency dependence of Eq. (204) can lead to $\sigma(\omega)$ having zeros in the upper-half ω plane ($\omega_I > 0$); i.e., C_{os} can be a real negative number when $\omega_I > 0$. Causality requires that $\sigma(\omega)$ be free from zeros or singularities when $\omega_I > 0$. Thus, because we are only developing a theory to first order in d/δ , we can rewrite Eq. (203) as

$$C_{os} = \frac{(\epsilon_0 \kappa_f)^2 \zeta^2}{2d\eta} P \left(1 - \frac{2i^{3/2}}{P} \frac{d}{\delta} \right)^{-1}, \quad (206)$$

where the dimensionless parameter P is greater than or equal to 1 and is defined as

$$P = \frac{8kTd^2}{\epsilon_0 \kappa_f \zeta^2} \sum_{\ell=1}^L \mathcal{N}_\ell \left[\exp \left(- \frac{ez_\ell \zeta}{2kT} \right) - 1 \right]. \quad (207)$$

P goes to one when $|ez_\ell \zeta / 2kT| \ll 1$. This estimate leads to no problems when $\omega_I > 0$.

Finally, we compare the electroosmotic conductance C_{os} (at frequencies where d/δ can be completely neglected) to the double-layer electromigration conductance C_{em} for the special case of a binary-symmetric electrolyte with $b_1 = b_2 = b$ (e.g., KCl). At room temperature we have

$$\frac{C_{os}}{C_{em}} = \frac{2\epsilon_0 \kappa_f kT}{(ez)^2 \eta b} \simeq \frac{1}{2}. \quad (208)$$

Thus, we have the important conclusion that C_{os} contributes nearly as much as C_{em} to the total conduction and should, therefore, not be neglected.

The mechanically induced streaming current is now determined. If the identity $\nabla \Phi^0 \cdot \nabla \mathbf{v}_m = \nabla \cdot (\Phi^0 \nabla \mathbf{v}_m) - \Phi^0 \nabla^2 \mathbf{v}_m$ is introduced into Eq. (197), then, because $\nabla^2 \mathbf{v}_m$ (unlike $\nabla^2 \mathbf{v}_e$) varies negligibly across the thin double layer compared to Φ^0 , we have

$$\mathbf{J}_{sm} = \frac{\epsilon_0 \kappa_f \zeta}{V_f} \int_{S_w} (\mathbf{n} \cdot \nabla \mathbf{v}_m - \tilde{d} \nabla^2 \mathbf{v}_m) dS, \quad (209)$$

where the length $\tilde{d}$ is defined as

$$\tilde{d} = \int_0^D \frac{\Phi^0(\chi)}{\zeta} d\chi. \quad (210)$$

For the Debye approximation, we have $\tilde{d} = d$ while, in general, $\tilde{d} \leq d$.

As previously discussed, the mechanically induced flow separates into low-frequency and high-frequency regimes. Starting at low frequencies, we insert Eq. (160) into Eq. (209) and obtain

$$\phi \mathbf{J}_{sm0} = L_{m0} (-\nabla \bar{p} + i\omega \rho_f \dot{\mathbf{u}}_s), \quad (211)$$

where the subscripts $sm0$ stand for “streaming, mechanical, and low frequencies” and where the low-frequency coupling coefficient L_{m0} is

$$L_{m0} = -\phi \frac{\epsilon_0 \kappa_f \zeta}{\eta} \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} (\mathbf{n} \cdot \nabla \mathbf{g} - \tilde{d} \nabla h) dS, \quad (212)$$

where $\nabla^2 \mathbf{g} = \nabla h$ [Eq. (162)] was used. If $\nabla^2 \mathbf{g} = \nabla h$ is integrated over the pore volume and $h = \Gamma + \delta h$ introduced, one obtains

$$\frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \mathbf{n} \cdot \nabla \mathbf{g} dS = \frac{1}{\alpha_\infty} + \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \mathbf{n} \delta h dS. \quad (213)$$

Using this and the definition of Λ [Eq. (186)] in L_{m0} then gives

$$L_{m0} = -\frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \left[1 - 2 \frac{\tilde{d}}{\Lambda} + \alpha_\infty \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} (\mathbf{n} \delta h - \tilde{d} \nabla \delta h) dS \right]. \quad (214)$$

We leave this low-frequency definition for now and will consider it again after L_{e0} has been defined.

At high frequencies, the coupling coefficient $L_{m\infty}$ is obtained by inserting Eq. (172) into Eq. (209) which gives

$$\phi \mathbf{J}_{sm\infty} = L_{m\infty} (-\nabla \bar{p} + i\omega \rho_f \dot{\mathbf{u}}_s), \quad (215)$$

where

$$L_{m\infty} = -2 \frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \frac{i^{1/2} \delta}{\Lambda} \left(1 + i^{3/2} \frac{\tilde{d}}{\delta} \right) \quad (216)$$

$$= -2 \frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \frac{i^{1/2} \delta}{\Lambda} \left(1 - i^{3/2} \frac{\tilde{d}}{\delta} \right)^{-1}. \quad (217)$$

Equation (217) leads to no problems when $\omega_I > 0$ (again, the branch point at the origin must be allowed for) and is the estimate to be employed. Note that if the Debye approximation is used, then, regardless of the d/δ ratio, we have

$$L_{m\infty}^{\text{Debye}} = -2 \frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \frac{i^{1/2} \delta}{\Lambda} \left(1 - i^{3/2} \frac{d}{\delta} \right)^{-1}, \quad (218)$$

thus confirming the form of Eq. (217). Note that the general estimate [Eq. (217)] involves $\tilde{d}$ while the Debye estimate involves d .

C. Relative flow $\bar{\mathbf{v}}$

The average value of the relative fluid flow $\bar{\mathbf{v}}$ can be separated into electrically- and mechanically-induced portions $\bar{\mathbf{v}} = \bar{\mathbf{v}}_e + \bar{\mathbf{v}}_m$, as well as analyzed in the limits of low and high frequencies. Furthermore, it can be evaluated either by directly averaging the pore-scale field or by utilizing the averaged-force balance [Eq. (96)] and evaluating the drag integral $\mathbf{d}$ and the electrical body force $\mathbf{f}_d$. It is more efficient to simply volume average the pore-scale field.

Consider first the mechanically induced flow. In the limit of low frequencies, a direct averaging of Eq. (160) gives

$$\phi \bar{\mathbf{v}}_{m0} = \frac{k_0}{\eta} (-\nabla \bar{p} + i\omega \rho_f \dot{\mathbf{u}}_s), \quad (219)$$

where ϕ is porosity and

$$k_0 = -\frac{\phi}{V_f} \int_{V_f} \hat{\mathbf{z}} \cdot \mathbf{g} dV. \quad (220)$$

The negative sign arises because $\mathbf{g}$, by definition of its boundary-value problem on the averaging disk, is directed on average in the $-\hat{\mathbf{z}}$ direction. The parameter k_0 is the dc permeability and is our third and final pore-geometry parameter.

In the limit of high frequencies, Eq. (172) is directly averaged to give

$$\phi \bar{\mathbf{v}}_{m\infty} = \frac{k_\infty}{\eta} (-\nabla \bar{p} + i\omega \rho_f \dot{\mathbf{u}}_s), \quad (221)$$

where

$$k_\infty = i\delta^2 \phi \frac{\hat{\mathbf{z}}}{V_f} \cdot \left[\int_{V_f} \nabla \Gamma dV - \int_{S_w} \nabla \Gamma dS \times \int_0^{D_v} e^{i^{3/2} \chi / \delta} d\chi \right] \quad (222)$$

$$= i\delta^2 \frac{\phi}{\alpha_\infty} \left(1 - 2i^{1/2} \frac{\delta}{\Lambda} \right) \quad (223)$$

$$= i\delta^2 \frac{\phi}{\alpha_\infty} \left(1 + 2i^{1/2} \frac{\delta}{\Lambda} \right)^{-1}. \quad (224)$$

In the high-frequency limit ($\delta \rightarrow 0$), Eqs. (224) and (223) are equivalent; however, Eq. (224) leads to no problems when $\omega_I > 0$ (as long as the branch point at the origin is respected). The distance D_v in Eq. (222) can be taken as a few viscous-skin depths.

Next, Eq. (151) for the low-frequency electrically induced flow is integrated to give

$$\phi \bar{\mathbf{v}}_{e0} = L_{e0} \bar{\mathbf{E}}, \quad (225)$$

where the low-frequency coupling coefficient L_{e0} is

$$L_{e0} = \phi \frac{\epsilon_0 \kappa_f}{\eta} \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{V_f} (\Phi^0 - \zeta) \nabla \Gamma dV \quad (226)$$

$$= -\frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \left(1 - 2 \frac{\tilde{d}}{\Lambda}\right). \quad (227)$$

Comparing this expression to Eq. (214) for L_{m0} and using Onsager reciprocity ($L_e = L_m$) as independently derived in Appendix B, then gives the identities

$$\frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \mathbf{n} \delta h dS = 0, \quad (228)$$

$$\frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \nabla \delta h dS = 0. \quad (229)$$

When these are introduced into Eqs. (213) and (186), we have the apparently new results involving the Stokes flow:

$$\frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \mathbf{n} \cdot \nabla \mathbf{g} dS = \frac{1}{\alpha_\infty}, \quad (230)$$

$$\frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \nabla^2 \mathbf{g} dS = \frac{2}{\alpha_\infty \Lambda}. \quad (231)$$

Heretofore, α_∞ and Λ were defined solely in terms of the potential field Γ (e.g., as in Ref. 15).

In the high-frequency limit, Eq. (157) is volume integrated to give

$$\phi \bar{\mathbf{v}}_{e\infty} = L_{e\infty} \bar{\mathbf{E}}, \quad (232)$$

where the high-frequency coupling coefficient is

$$L_{e\infty} = -\phi \frac{\epsilon_0 \kappa_f \zeta}{\eta} \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \nabla \Gamma dS \left[\int_0^{D_v} e^{i^{3/2} \chi / \delta} d\chi - \int_0^D \frac{\Phi}{\zeta} d\chi \right] \quad (233)$$

$$= -2 \frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \frac{i^{1/2} \delta}{\Lambda} \left(1 + i^{3/2} \frac{\tilde{d}}{\delta}\right) \quad (234)$$

$$= -2 \frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \frac{i^{1/2} \delta}{\Lambda} \left(1 - i^{3/2} \frac{\tilde{d}}{\delta}\right)^{-1}. \quad (235)$$

It is seen that $L_{e\infty} = L_{m\infty}$, thus providing a check on the validity of this expression.

D. Final coefficients

We have obtained above the low-frequency and high-frequency limits of the coupling coefficient $L (= L_e = L_m)$ and the permeability k . It is easy to verify that (note $\delta = \sqrt{\eta/\rho_f \omega}$)

$$\frac{k(\omega)}{k_0} = \left[\left(1 - i \frac{\omega}{\omega_t} \frac{4}{m}\right)^{\frac{1}{2}} - i \frac{\omega}{\omega_t} \right]^{-1} \quad (236)$$

and

$$\frac{L(\omega)}{L_0} = \left[1 - i \frac{\omega}{\omega_t} \frac{m}{4} \left(1 - 2 \frac{\tilde{d}}{\Lambda}\right)^2 \left(1 - i^{3/2} \tilde{d} \sqrt{\frac{\omega \rho_f}{\eta}}\right)^2 \right]^{-\frac{1}{2}} \quad (237)$$

are two possible functions that simply and smoothly connect these frequency limits. These functions have the correct behavior on the complex ω plane. The transition frequency ω_t separating low-frequency viscous flow and high-frequency inertial flow is defined as

$$\omega_t \equiv \frac{\phi}{\alpha_\infty k_0} \frac{\eta}{\rho_f}. \quad (238)$$

The dimensionless number m is defined as

$$m \equiv \frac{\phi}{\alpha_\infty k_0} \Lambda^2 \quad (239)$$

and consists only of the pore-space geometry terms. The low-frequency coupling coefficient L_0 is defined

$$L_0 = -\frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \left(1 - 2 \frac{\tilde{d}}{\Lambda}\right). \quad (240)$$

All the remaining parameters have been defined in the previous sections.

Despite the negative sign in its definition, L_0 will lead to positive streaming-current densities for positive values of $-\nabla \bar{p}$. This is because the ζ potential is negative when the diffuse layer contains excess positive charge and positive when the diffuse layer contains excess negative charge. Note also that the correction term $1 - 2\tilde{d}/\Lambda$ is only valid under the thin-double-layer assumption and at most represents a correction of a few percent. We do not have that $L_0 \rightarrow 0$ as $\tilde{d} \rightarrow \Lambda/2$. For the special case of a single cylindrical tube of radius R , it easy to show (using the Debye approximation) that the coupling coefficient is given by

$$L_0^{\text{tube}} = -\frac{\epsilon_0 \kappa_f \zeta}{\eta} \frac{I_2(R/d)}{I_0(R/d)}, \quad (241)$$

where I_2 and I_0 are modified Bessel functions. In the limit as $R/d \rightarrow \infty$ (thin double layers), we have that $I_2(R/d)/I_0(R/d) \rightarrow 1 - 2d/R$. It will be seen that for a cylindrical tube, $\Lambda = R$. Thus the form of Eq. (240) is confirmed for this special case. However, when $R/d = 2$ (double layers half the radius of the tube), we have that $I_2(2)/I_0(2) = 0.3$, while the correction factor of Eq. (240) would predict zero. Again, the thin-double-layer assumption must be satisfied for the analysis of this paper to be valid.

The above expression for $k(\omega)$ is identical to the one defined by Johnson, Koplik, and Dashen¹⁵ using energy arguments thus giving that the Λ defined by Eq. (186) is identical to the Λ defined by Eq. (189).

The conductivity $\sigma(\omega)$ of the material is defined by Eqs. (173) and (174), which then gives [from Eqs. (188) and (198)]

$$\sigma(\omega) = \frac{\phi\sigma_f}{\alpha_\infty} \left[1 + \frac{2[C_{em} + C_{os}(\omega)]}{\sigma_f\Lambda} \right]. \quad (242)$$

The weak dependence on frequency comes only from the electro-osmotic conductance and is due to the electrically driven viscous-flow to inertial-flow transition. The definitions of σ_f , C_{em} , and $C_{os}(\omega)$ are given by Eqs. (72), (194), and (206), respectively. Again, it is emphasized that C_{os} is of the same order of magnitude as C_{em} and should not be neglected [Eq. (208)].

A central question is whether the pore-geometry number m is a universal constant for all porous media. Rearranging the definition of m gives an equation for the dc permeability:

$$k_0 = \frac{1}{m} \frac{\phi}{\alpha_\infty} \Lambda^2. \quad (243)$$

There is both laboratory^{17,18} and numerical^{15,19,20} evidence suggesting that when k_0 , ϕ , α_∞ , and Λ are independently measured, m lies in the range

$$4 \leq m \leq 8 \quad (244)$$

for a variety of porous media ranging from grain packings to networks of variable-radii tubes. However, it is difficult to independently measure Λ and α_∞ in rocks that contain significant amounts of clay so that the value of m in, for example, shaley sandstones is uncertain.

In this work, we have developed expressions for k_0 , α_∞ , and Λ . These can be expressed as

$$\frac{k_0}{\phi} = -\frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{V_f} \mathbf{g} dV, \quad (245)$$

$$\frac{1}{\alpha_\infty} = \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \mathbf{n} \cdot \nabla \mathbf{g} dS \left(= 1 + \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \mathbf{n} \Gamma dS \right), \quad (246)$$

$$\frac{2}{\alpha_\infty \Lambda} = \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \nabla^2 \mathbf{g} dS \left(= \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{S_w} \nabla \Gamma dS \right). \quad (247)$$

Note that α_∞ and Λ can also be defined by Eqs. (184) and (189) as derived in Ref. 15. Although all three pore-geometry terms are expressed as integrals of the Stokes flow $\mathbf{g}$ [defined by Eqs. (162)–(165)], it has proven difficult to use these relations to make general correlations between m and material types. It is trivial to demonstrate that for a cylindrical tube of radius R with axis directed in the $\hat{\mathbf{z}}_1$ direction (perhaps different than $\hat{\mathbf{z}}$), the Stokes-flow vector is (in cylindrical coordinates) $\mathbf{g} = \hat{\mathbf{z}}_1(r^2 - R^2)/4$, and Eqs. (245)–(247) satisfy Eq. (243) with $m = 8$. In this particular case, we have $\Lambda = R$ as pointed out in Ref. 15.

VI. SUMMARY

We now summarize the final form of the macroscopic governing equations. All overbars denoting averaging are dropped. For the problem of modeling the propagation of coupled electromagnetic and mechanical disturbances in an isotropic-porous material, the following equations represent a complete set:

$$\nabla \times \mathbf{E} = i\omega \mathbf{B}, \quad (248)$$

$$\nabla \times \mathbf{H} = -i\omega \mathbf{D} + \mathbf{J}, \quad (249)$$

$$\nabla \cdot \boldsymbol{\tau}_B = -\omega^2 (\rho_B \mathbf{u}_s + \rho_f \mathbf{w}), \quad (250)$$

$$\mathbf{J} = \sigma(\omega) \mathbf{E} + L(\omega) (-\nabla p + \omega^2 \rho_f \mathbf{u}_s), \quad (251)$$

$$-i\omega \mathbf{w} = L(\omega) \mathbf{E} + \frac{k(\omega)}{\eta} (-\nabla p + \omega^2 \rho_f \mathbf{u}_s), \quad (252)$$

$$\mathbf{D} = \epsilon_0 \left[\frac{\phi}{\alpha_\infty} (\kappa_f - \kappa_s) + \kappa_s \right] \mathbf{E}, \quad (253)$$

$$\mathbf{B} = \mu_0 \mathbf{H}, \quad (254)$$

$$\boldsymbol{\tau}_B = (K_G \nabla \cdot \mathbf{u}_s + C \nabla \cdot \mathbf{w}) \mathbf{I} + G_{fr} \left(\nabla \mathbf{u}_s + \nabla \mathbf{u}_s^T - \frac{2}{3} \nabla \cdot \mathbf{u}_s \mathbf{I} \right), \quad (255)$$

$$-p = C \nabla \cdot \mathbf{u}_s + M \nabla \cdot \mathbf{w}. \quad (256)$$

All coupling is present in the transport equations [Eqs. (251) and (252)] and comes from the coefficient $L(\omega)$. If, for some reason, we want to calculate the macroscopic-charge density $\rho \equiv \phi \sum_{\ell} e z_{\ell} \bar{n}_{\ell}$ induced in the porous material by the above disturbances, we utilize the macroscopic statement of Coulomb's law $\rho = \nabla \cdot \mathbf{D}$, once $\mathbf{D}$ has been determined; however, it is unclear why such a calculation would be of interest. The definitions of $\sigma(\omega)$, $k(\omega)$, and $L(\omega)$ have been given in the previous section [Eqs. (242), (236), and (237), respectively]. There are two distinct frequency relaxations present in the above equations. One is in the transport equations and is due to the transition from viscous flow to inertial flow while the other is in Ampère's law [Eq. (249)] and is due to the transition from ion transport $\mathbf{J}$ to dielectric-displacement current $-i\omega \mathbf{D}$.

Two interesting observations can be made concerning the relaxation in the transport equations. First, $k(\omega)$ relaxes at the frequency ω_t [Eq. (238)] while $L(\omega)$ relaxes at the frequency $4\omega_t/m$. Observation of the two relaxation frequencies would provide an independent means for determining m . Second, $k(\omega)$ falls off as ω when $\omega > \omega_t$ while $L(\omega)$ falls off as $\omega^{1/2}$ when $\omega > 4\omega_t/m$. If it is possible to stress the material with equal amplitude over a wide range of frequencies, then $|\nabla p + \omega^2 \rho_f \mathbf{u}_s|$ increases directly as ω so that $|\mathbf{w}|$ remains constant above ω_t while $|\mathbf{J}|$ continues to increase (as $\omega^{1/2}$) above $4\omega_t/m$; this states that there continues to be enhancement in the wave-induced streaming current even above the relaxation frequency.

Finally, if flow is being induced by applied pressure gradients and electric fields that are steady in time, we may set

$$\mathbf{E} = -\nabla \Phi \quad (257)$$

and obtain the following complete set of static equations:

$$\nabla \cdot \mathbf{J} = 0, \quad (258)$$

$$\nabla \cdot \mathbf{w} = 0, \quad (259)$$

$$\mathbf{J} = \frac{\phi}{\alpha_\infty} \sigma_f \left[1 + \frac{2(C_{em} + C_{os})}{\sigma_f \Lambda} \right] (-\nabla \Phi) - \frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \left[1 - 2 \frac{\tilde{d}}{\Lambda} \right] (-\nabla p) \quad (260)$$

$$\dot{\mathbf{w}} = -\frac{\phi}{\alpha_\infty} \frac{\epsilon_0 \kappa_f \zeta}{\eta} \left[1 - 2 \frac{\tilde{d}}{\Lambda} \right] (-\nabla \Phi) + \frac{k_0}{\eta} (-\nabla p), \quad (261)$$

where the dc transport coefficients have been explicitly written out [σ_f , C_{em} , C_{os} , and $\tilde{d}$ are defined by Eqs. (72), (194), (206), and (210), respectively, with the skin-depth term in Eq. (206) neglected]. When gravity effects are important, the pressure gradient in Eqs. (260) and (261) must be generalized to $\nabla p + \rho_f \mathbf{g}$, where $\mathbf{g}$ is the acceleration of gravity. If it is known that the applied frequencies are greater than zero but less than the above mentioned transition frequencies (i.e., quasistatic disturbances), then Eq. (259) should be replaced by Eqs. (250), (255), and (256). For hydrology problems in which flow occurs due to externally applied quasistatic pressure gradients or electric fields, if the material is highly consolidated so that $K_{fr} \gg K_f$, it is further possible to ignore the $\nabla \cdot \mathbf{u}_s$ term in Eq. (256) and replace Eq. (259) simply by $i\omega p = M \nabla \cdot \dot{\mathbf{w}}$. Note, however, that this simplification cannot be made (even if $K_{fr} \gg K_f$) if flow is being induced by mechanical (seismic) waves or by deformation of the bulk-porous material in general.

In conclusion, it is emphasized that the above results have been obtained by volume averaging and not simply postulated. Boundary conditions and wave properties for the governing equations are discussed in Ref. 9.

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APPENDIX A: DOUBLE-LAYER DESCRIPTION

As discussed in the text, we need to have a solution for the Poisson-Boltzmann [Eq. (27)] next to a plane wall. Writing the equation as

$$\frac{\partial^2 \Phi^0}{\partial \chi^2} - \sum_{\ell=1}^L \frac{(ez_\ell)^2 \mathcal{N}_\ell}{\epsilon_0 \kappa_f kT} \left[1 - \frac{1}{2!} \frac{ez_\ell}{kT} \Phi^0 + \dots \right] \Phi^0 = 0, \quad (A1)$$

it is seen that if $|ez_\ell \zeta / kT| \ll 2$, where $\zeta = \Phi^0(0)$ (i.e., the potential at the shear plane), then the equation linearizes, and the solution is the Debye approximation

$$\Phi^0 = \zeta e^{-\chi/d}, \quad (A2)$$

where the Debye length d is given by Eq. (29). Using this estimate, the ion-number densities are estimated as

$$N_\ell^0 = \mathcal{N}_\ell \exp(-Z_\ell e^{-\chi/d}), \quad (A3)$$

where

$$Z_\ell = \frac{ez_\ell \zeta}{kT}. \quad (A4)$$

This estimate of N_ℓ^0 is quite accurate even when $|Z_\ell| > 2$, i.e., even when the linearization of Eq. (A1) is not valid. However, the linearized estimate [obtained by expanding Eq. (A3)]

$$N_\ell^0 \simeq \mathcal{N}_\ell \left(1 - Z_\ell e^{-\chi/d} \right) \quad (A5)$$

should only be used when $|Z_\ell| \ll 1$.

One way that Eq. (A2) can lead to considerable error when $|Z_\ell| > 2$ is if it is used to relate ζ to the surface-charge density Q^0 . A blind application of the boundary condition $\epsilon_0 \kappa_f \mathbf{n} \cdot \mathbf{E}_f^0 = Q^0$ would give $\zeta = Q^0 d / (\epsilon_0 \kappa_f)$, which is a very poor approximation when $|Z_\ell| > 2$. It is far better to obtain the relation between ζ and Q^0 using Eq. (A3) and the assumption that there is bulk-charge neutrality,

$$Q^0 \simeq - \int_0^\infty \sum_{\ell=1}^L ez_\ell \mathcal{N}_\ell \exp(-Z_\ell e^{-\chi/d}) \quad (A6)$$

$$\simeq -2d \sum_{\ell=1}^L ez_\ell \mathcal{N}_\ell \exp\left(-\frac{ez_\ell \zeta}{2kT}\right), \quad (A7)$$

where the integral is evaluated using the steps defined in Eqs. (191)–(194). This is a generally valid result even when $|Z_\ell| > 2$.

Indeed, for the special case of a binary symmetric electrolyte ($L = 2$, $z_1 = -z_2$, and $\mathcal{N}_1 = \mathcal{N}_2$) we have the exact solution of the Poisson-Boltzmann equation

$$\frac{ez_1 \Phi^0}{2kT} = \ln \left[\frac{1 + \exp(-\chi/d) \tanh[ez_1 \zeta / (4kT)]}{1 - \exp(-\chi/d) \tanh[ez_1 \zeta / (4kT)]} \right], \quad (A8)$$

which gives

$$\zeta = \frac{2kT}{ez_1} \sinh^{-1} \left(\frac{Q^0}{2ez_1 \mathcal{N}_1 d} \right). \quad (A9)$$

But this exact result is identical to our result [Eq. (A7)] for this special case. This gives good confidence in both Eq. (A7) and Eq. (A3) for more general electrolytes even when $|Z_\ell| > 2$.

APPENDIX B: ONSAGER RECIPROCITY

In this appendix, Onsager reciprocity is derived for frequency-dependent electrokinetic phenomena in porous media satisfying the thin-double-layer assumption.

The general boundary-value problem on the averaging disk for mechanically induced flow $\mathbf{v}_m$ is given by Eqs. (146) and (147). Note that the problem can also be stated as

$$\eta \nabla^2 \mathbf{v}_m + i\omega \rho_f \mathbf{v}_m = \nabla \left(\delta p_m + \Gamma \frac{\Delta P}{H} \right), \quad (B1)$$

where $\delta p_m = 0$ on the disk faces. The problem for the electrically induced flow $\mathbf{v}_e$ is

$$\eta \nabla^2 \mathbf{v}_e + i\omega \rho_f \mathbf{v}_e = \nabla p_e + \rho^0 \nabla \Gamma \frac{\Delta \phi}{H}, \quad (\text{B2})$$

where $p_e = 0$ on the disk faces and ρ^0 is the excess-charge density in the diffuse double layer ($\rho^0 = \sum_{l=1}^L e z_l N_l^0$). We also have that $\nabla \cdot \mathbf{v}_m = \nabla \cdot \mathbf{v}_e = 0$ everywhere and $\mathbf{v}_m = \mathbf{v}_e = 0$ on S_w .

If $\mathbf{v}_e$ is dotted into Eq. (B1) and $\mathbf{v}_m$ is dotted into Eq. (B2) and the result subtracted and integrated over the pore volume V_f , one obtains

$$\begin{aligned} & \int_{V_f} \nabla \cdot [\eta (\nabla \mathbf{v}_m \cdot \mathbf{v}_e - \nabla \mathbf{v}_e \cdot \mathbf{v}_m) + \delta p_m \mathbf{v}_e - p_e \mathbf{v}_m] dV \\ &= \int_{V_f} \left\{ \nabla \cdot \left[\Gamma \left(\mathbf{v}_e \frac{\Delta P}{H} - \rho^0 \mathbf{v}_m \frac{\Delta \phi}{H} \right) \right] \right. \\ & \quad \left. + \Gamma \nabla \rho^0 \cdot \mathbf{v}_m \right\} dV. \end{aligned}$$

We have used the identities $\nabla \cdot (\nabla \mathbf{v}_m \cdot \mathbf{v}_e) = (\nabla^2 \mathbf{v}_m) \cdot \mathbf{v}_e + \nabla \mathbf{v}_e : (\nabla \mathbf{v}_m)^T$ and $\nabla \mathbf{v}_e : (\nabla \mathbf{v}_m)^T = (\nabla \mathbf{v}_e)^T : \nabla \mathbf{v}_m$ plus the fact that $\nabla \cdot \mathbf{v}_m = \nabla \cdot \mathbf{v}_e = 0$. Because Eq. (B2) is only valid in the thin-double-layer approximation, we are justified in setting $\nabla \rho^0 \cdot \mathbf{v}_m = 0$ (the two vectors are orthogonal when double layers are thin). The divergence theorem plus the boundary conditions then gives

$$\begin{aligned} V_f \frac{\Delta \phi}{H} \frac{\Delta P}{H} (L_e - L_m) &= \hat{\mathbf{z}} \cdot \int_{V_f} \left(\mathbf{v}_e \frac{\Delta P}{H} - \rho^0 \mathbf{v}_m \frac{\Delta \phi}{H} \right) dV \\ &= \int_{V_f} \nabla \cdot \left[z \left(\mathbf{v}_e \frac{\Delta P}{H} - \rho^0 \mathbf{v}_m \frac{\Delta \phi}{H} \right) \right] dV \\ &= H \int_{z=H} \hat{\mathbf{z}} \cdot \left(\mathbf{v}_e \frac{\Delta P}{H} - \rho^0 \mathbf{v}_m \frac{\Delta \phi}{H} \right) dS, \end{aligned}$$

where the thin-double-layer assumption $\nabla \rho^0 \cdot \mathbf{v}_m = 0$ was used again. Thus, from Eq. (B3), the desired result $L_e = L_m$ is arrived at.

$$\begin{aligned} & \eta \int_{z=H} \hat{\mathbf{z}} \cdot (\nabla \mathbf{v}_m \cdot \mathbf{v}_e - \nabla \mathbf{v}_e \cdot \mathbf{v}_m) dS \\ & - \eta \int_{z=0} \hat{\mathbf{z}} \cdot (\nabla \mathbf{v}_m \cdot \mathbf{v}_e - \nabla \mathbf{v}_e \cdot \mathbf{v}_m) dS \\ &= H \int_{z=H} \hat{\mathbf{z}} \cdot \left(\mathbf{v}_e \frac{\Delta P}{H} - \rho^0 \mathbf{v}_m \frac{\Delta \phi}{H} \right) dS. \end{aligned}$$

If the material has no significant macroscopic variation over distances H so that the distribution of grains (and, therefore, flow) near $z = 0$ is similar to that near $z = H$, then the two integrals on the left-hand side will cancel (one may additionally establish that each of the integrals on the left-hand side individually goes to zero). Thus, we have

$$H \int_{z=H} \hat{\mathbf{z}} \cdot \left(\mathbf{v}_e \frac{\Delta P}{H} - \rho^0 \mathbf{v}_m \frac{\Delta \phi}{H} \right) dS = 0. \quad (\text{B3})$$

The definitions of the cross-coupling coefficients L_e and L_m are now introduced:

$$\begin{aligned} \hat{\mathbf{z}} \cdot \bar{\mathbf{v}}_e &= L_e \frac{\Delta \phi}{H} \\ &= \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{V_f} \mathbf{v}_e dV \end{aligned}$$

and

$$\begin{aligned} \hat{\mathbf{z}} \cdot \bar{\mathbf{j}}_m &= L_m \frac{\Delta P}{H} \\ &= \frac{\hat{\mathbf{z}}}{V_f} \cdot \int_{V_f} \rho^0 \mathbf{v}_m dV. \end{aligned}$$

We can thus form

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Electroseismic Wave Theory of Frenkel and More Recent Developments

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Abstract: Frenkel's 1944 theory of porous media acoustics is carefully scrutinized. After some manipulation, Frenkel's equations are seen to have nearly identical form to Biot's 1962 equations. The only difference is that Frenkel includes an extraneous fluid-pressure gradient in his bulk force balance. Frenkel also makes a slight error in the development of his effective poroelastic moduli that prevents him from being the first to obtain the so-called "fluid-substitution" relations. Outside of these two small problems, Frenkel's analysis can be considered correct. He limits his treatment of electroseismic phenomena to explain the electric field that accompanies a compressional seismic wave in a homogeneous material. He predicts that the electric field in a compressional wave is directly proportional to the particle acceleration and this has been verified by the recent field measurements of Garambois and Dietrich. However, such electric fields are only a small part of the total electroseismic response of the earth. Accordingly, some of the additional phenomena not discussed by Frenkel are also presented and discussed.

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Introduction

Frenkel (1944) is the first writer to have developed a complete set of equations governing the acoustics of isotropic porous media. Kosten and Zwikker (1941) proposed two coupled force-balance equations for the average fluid and solid response of a porous material, and predicted the existence of two compressional-response modes (a "fast" and "slow" wave). However, Kosten and Zwikker (1941), unlike Frenkel (1944), proposed a purely scalar theory that failed to allow for shear and that did not define the effective compressibility moduli in terms of drained and undrained experiments. The coefficients in the Kosten and Zwikker (1941) theory are entirely phenomenological.

Frenkel only wrote one paper on poroelasticity. His principal motivation for developing a theory of porous-media acoustics is to quantitatively explain the so-called "E effect" (a nomenclature apparently no longer used in the literature) which is the phenomenon by which a pair of electrodes attached to the earth registers a voltage difference as a seismic wave traverses the electrode pair. Ivanov (1939) measured this phenomenon in the field and suggested that the explanation of the recorded electric field was electrokinetic in nature. The stated purpose of Frenkel's (1944) paper is to develop Ivanov's idea into a quantitative theory.

The purpose of the present paper is to: (1) carefully compare

Frenkel's porous-media acoustics equations to those of Biot (1956a and 1956b) and determine the reason for any discrepancies; (2) determine whether Frenkel's analysis of the electric field contained within a compressional wave is correct; and (3) present results for those effects not considered by Frenkel which include relaxation in the transport coefficients and an assortment of electrokinetic couplings between seismic waves (both compressional and shear) and electric and magnetic fields.

Frenkel's Theory of Porous-Media Acoustics

With some effort, the various macroscopic fields in Frenkel's theory can be identified and his governing equations rewritten in a form that is perhaps more familiar.

To achieve this, the response fields used in the present paper are slightly different from those in Frenkel's paper. The poroelastic displacements are taken to be $\mathbf{u}$ and $\mathbf{w}$ where $\mathbf{u}$ is the average displacement in the solid phase of a porous sample, and $\mathbf{w}$ is defined $\mathbf{w} = \phi(\mathbf{u}_f - \mathbf{u})$ where ϕ is porosity and $\mathbf{u}_f$ is the average displacement of the fluid in the pores. The time derivative $\partial \mathbf{w} / \partial t$ thus represents the Darcy filtration velocity induced by the wave. The stress variables are the bulk stress tensor $\boldsymbol{\tau}$ and the average fluid pressure in the pores p_f . The bulk stress tensor represents the average stress throughout both the solid and fluid phases of a porous sample and can be identified as $\boldsymbol{\tau} = (1 - \phi)\boldsymbol{\tau}_s - \phi p_f \mathbf{I}$, where $\boldsymbol{\tau}_s$ is the average stress tensor of the solid phase. The confining pressure is defined as $P_c = -\text{tr}\{\boldsymbol{\tau}\}/3 = (1 - \phi)p_s + \phi p_f$ and both P_c and p_f are used as the two independent pressure variables in the compressibility laws. In Table 1, Frenkel's (1944) variables are given in terms of these porous-continuum variables.

Frenkel is only concerned with developing a theory of linear acoustics. As such, we forego placing a small d or δ in front of the stress and displacement variables, it being understood that all stresses and displacements are increments induced by a passing wave.

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Table 1. Identification of Frenkel Variables

Frenkel variable ^a	Equivalent definition ^b
T (partial stress tensor) (associated with solid)	$(1-\phi)\tau_s = \tau + \phi p_f \mathbf{I}$
f (porosity)	ϕ
p_2 (fluid pressure)	p_f
$p_1 = -\text{tr}\{\mathbf{T}\}/3$ (partial pressure) (associated with solid)	$(1-\phi)p_s = P_c - \phi p_f$
u (average displacement) (of solid phase)	u
$\theta = \Delta V/V = \nabla \cdot \mathbf{u}$ (dilatation) (of porous sample)	$\nabla \cdot \mathbf{u}$
$\varphi = -\Delta p_2/p_2$ (dilatation) (of fluid volume)	$-p_f/K_f$
v ₁ (average velocity) (of solid phase)	$\partial \mathbf{u} / \partial t$
v ₂ (average velocity) (of fluid phase)	$\phi^{-1} \partial \mathbf{w} / \partial t + \partial \mathbf{u} / \partial t$
Δf (porosity change)	$\Delta \phi = -\phi \nabla \cdot \mathbf{u} - \nabla \cdot \mathbf{w} - \phi p_f / K_f$

^aFrenkel (1944).^bPresent paper.

Frenkel's Stress/Strain Relations

In short, Frenkel obtains the proper form of the stress/strain relations but, because of a small error to be described below, does not obtain the so-called Gassmann (1951) "fluid substitution relations" that give the poroelastic compressibility moduli in terms of the underlying fluid bulk modulus, solid bulk modulus, and drained bulk modulus.

Frenkel obtains his macroscopic compressibility laws by focusing on two thought experiments. Adding the results of the two thought experiments yields the functions $\nabla \cdot \mathbf{u}(P_c, p_f)$ and $\Delta \phi(P_c, p_f)$ that describe how sample volume and porosity are changed due to applied increments in confining pressure P_c and fluid pressure p_f . Frenkel makes the not so obvious identification that in a porous material

$$\nabla \cdot \mathbf{u} = \Delta V/V \quad (1)$$

where $\mathbf{u}$ =average displacement in the solid phase and V =sample volume. Pride and Berryman (1998) demonstrate that only when the geometrical center of the grain space coincides with the geometrical center of the pore space in a porous sample does $\nabla \cdot \mathbf{u}$ become exactly the volumetric dilatation of the sample. However, most writers starting from Biot and Willis (1957) also invoke Frenkel's Eq. (1) and it can be considered a reasonable approximation.

In his first "drained" thought experiment, Frenkel applies a confining pressure to a sample with no change in the fluid pressure. Under such drained conditions, one has $P_c = (1-\phi)p_s = p_1$ (where p_1 is Frenkel's variable for the partial pressure in the solid). Frenkel introduces two poroelastic moduli K and α_F to describe the drained response [α_F is used here instead of Frenkel's original α so as to avoid later confusion with the Biot and Willis (1957) parameter]

$$\nabla \cdot \mathbf{u}(P_c, 0) = -\frac{P_c}{K} \quad (2)$$

$$\Delta \phi(P_c, 0) = -\left(\frac{1}{1+\alpha_F} - \phi\right) \frac{P_c}{K} \quad (3)$$

Here, K =so-called "drained" bulk modulus and α_F has no standard name ("Frenkel's alpha") but can either be defined from Eq. (3) or can be identified, as Frenkel properly does, as the ratio of the increment in solid volume ΔV_s to the increment of pore volume ΔV_ϕ under drained conditions; i.e., $\alpha_F = \Delta V_s / \Delta V_\phi|_{p_f=0}$.

In his second thought experiment, Frenkel applies a fluid pressure p_f everywhere throughout the pore space and, simultaneously, a confining pressure $P_c = p_f$ to the external surface of the sample. One way to realize this in practice is to immerse an unjacketed sample into a fluid reservoir and change the pressure of the reservoir by p_f . In this case, Frenkel notices that the effect of the applied-pressure increment is to simply scale down (or up) the grain pack without changing its relative geometry. Accordingly, the porosity remains unchanged in this experiment. This is only exactly correct when the grains are made of an isotropic material having a bulk modulus K_s that is spatially uniform. Under this isotropic monomineral assumption (that Frenkel does not discuss), he properly obtains

$$\nabla \cdot \mathbf{u}(p_f, p_f) = -\frac{p_f}{K_s} \quad (4)$$

$$\Delta \phi(p_f, p_f) = 0 \quad (5)$$

Up to this point, Frenkel has done everything correctly.

His error comes when he adds together the results of the two thought experiments. Because of the linear nature of the incremental response, the proper additions yield

$$\begin{aligned} \nabla \cdot \mathbf{u}(P_c, p_f) &= \nabla \cdot \mathbf{u}(P_c - p_f, 0) + \nabla \cdot \mathbf{u}(p_f, p_f) \\ &= -\frac{P_c}{K} + \left(1 - \frac{K}{K_s}\right) \frac{p_f}{K} \end{aligned} \quad (6)$$

$$\begin{aligned} \Delta \phi(P_c, p_f) &= \Delta \phi(P_c - p_f, 0) + \Delta \phi(p_f, p_f) \\ &= -\left(\frac{1}{1+\alpha_F} - \phi\right) \frac{(P_c - p_f)}{K} \end{aligned} \quad (7)$$

However, Frenkel performs the meaningless addition

$$\begin{aligned} \nabla \cdot \mathbf{u}(P_c - \phi p_f, 0) + \nabla \cdot \mathbf{u}(p_f, p_f) \\ = \nabla \cdot \mathbf{u}[P_c + (1-\phi)p_f, p_f] \neq \nabla \cdot \mathbf{u}(P_c, p_f) \end{aligned}$$

and similarly for $\Delta \phi$, which results in his Eqs. (13) and (24a) being incorrect. He thus makes the misidentifications

$$\nabla \cdot \mathbf{u}(P_c, p_f) = -\frac{P_c}{K} + \left(\phi - \frac{K}{K_s}\right) \frac{p_f}{K} \quad (8)$$

$$\Delta \phi(P_c, p_f) = -\left(\frac{1}{1+\alpha_F} - \phi\right) \frac{(P_c - \phi p_f)}{K} \quad (9)$$

It is this error that prevents Frenkel from properly identifying the poroelastic compressibility moduli.

The constitutive equation for $\Delta \phi$ can be translated into a constitutive equation for the increment in fluid content $\nabla \cdot \mathbf{w}$ by rewriting Frenkel's Eq. (20) for the conservation of fluid mass in terms of the variables of the present paper as

$$\Delta\phi + \frac{\phi}{K_f} p_f + \nabla \cdot \mathbf{w} + \phi \nabla \cdot \mathbf{u} = 0 \quad (10)$$

which is exact for linear response. If the proper results of Eqs. (6) and (7) are then employed, Frenkel's "corrected" compressibility laws can be written

$$\begin{bmatrix} \nabla \cdot \mathbf{u} \\ \nabla \cdot \mathbf{w} \end{bmatrix} = -\frac{1}{K} \begin{bmatrix} 1 & -1(1 - K/K_s) \\ -(1 + \alpha_F)^{-1} & \phi(K/K_f - K/K_s) + (1 + \alpha_F)^{-1} \end{bmatrix} \times \begin{bmatrix} P_c \\ P_f \end{bmatrix} \quad (11)$$

These equations can further be used to identify the nature of the parameter α_F , an issue not addressed by Frenkel. Since the compressibility laws can also be derived by taking derivatives of a strain-energy function, one has the Maxwell relation that

$$\frac{1}{1 + \alpha_F} = 1 - \frac{K}{K_s} \quad (12)$$

Frenkel does not state this result for α_F which can also be given as $1/(1 + \alpha_F) = \alpha$, where α is known as the Biot and Willis (1957) constant. As has been shown by Berge et al. (1993), α is always independent of the fluid's bulk modulus for any porous material and is only given exactly by $\alpha = 1 - K/K_s$ in the special case of isotropic monomineral grains.

In passing, we give the compressibility laws in perhaps their most useful form (applicable to porous media with anisotropy at either the grain or macroscales and with arbitrary heterogeneity in the mineral properties)

$$-\begin{bmatrix} P_c \\ P_f \end{bmatrix} = K_U \begin{bmatrix} 1 & B \\ B & B/\alpha \end{bmatrix} \begin{bmatrix} \nabla \cdot \mathbf{u} \\ \nabla \cdot \mathbf{w} \end{bmatrix} \quad (13)$$

where the coefficients K_U (undrained bulk modulus) and B [Skempton's (1954) coefficient] are defined as

$$K_U = -\left. \frac{P_c}{\nabla \cdot \mathbf{u}} \right|_{\nabla \cdot \mathbf{w}=0} \quad (14)$$

$$B = \left. \frac{P_f}{P_c} \right|_{\nabla \cdot \mathbf{w}=0} \quad (15)$$

Both K_U and B are measured on samples that have their exterior surface sealed ($\nabla \cdot \mathbf{w}=0$). A generally valid definition of the Biot and Willis (1957) coefficient is

$$\alpha = (1 - K/K_U)/B \quad (16)$$

so that the three fundamental compressibility constants of poroelasticity can be taken to be K_U , K , and B . Under the restriction to isotropic monomineral grains, one obtains

$$\alpha = 1 - \frac{K}{K_s} \quad (17)$$

$$B = \frac{\alpha}{\alpha + \phi(K/K_f - K/K_s)} \quad (18)$$

$$K_U = \frac{K}{1 - B\alpha} \quad (19)$$

Gassmann (1951) only considers the undrained response ($\nabla \cdot \mathbf{w}=0$) and, accordingly, only obtains results for K_U and B . It is Biot and Willis (1957) who first properly obtain the entirety of Eqs. (13)–(19). Frenkel also obtains poroelastic compressibility

laws of the form of Eq. (13). However, his error discussed earlier and his lack of use of the symmetry condition prevent him from obtaining the fluid-substitution relations of Eqs. (17)–(19).

Finally, like Gassmann (1951) and Biot and Willis (1957) who follow him, Frenkel makes the assumption that the fluid in the pores has no influence on the shear properties of an isotropic porous material. Frenkel thus assumes that the deviatoric stress tensor $\boldsymbol{\tau}^D = \boldsymbol{\tau} + P_c \mathbf{I}$ acting on the bulk material is related to the deviatoric shear by a fluid-independent constant G

$$\boldsymbol{\tau}^D = G(\nabla \mathbf{u} + \nabla \mathbf{u}^T - \frac{2}{3} \nabla \cdot \mathbf{u} \mathbf{I}) \quad (20)$$

At the low frequencies used in seismic exploration, Frenkel's assumption of fluid independent shear is usually appropriate.

Frenkel's Dynamics of Porous Media

Frenkel proposes force-balance equations for the fluid in relative motion to the solid [his Eq. (22)] and for the solid skeleton [his Eq. (23)] that, except for one slight problem discussed below, can be considered equivalent to those of Biot (1956a,b).

Frenkel states Darcy's law [his Eq. (18)] in a nonstandard form. The relative velocity $\mathbf{v}_2 - \mathbf{v}_1$ (his notation) that Frenkel uses in Darcy's law represents the average speed at which the fluid is moving relative to the solid and is not a filtration velocity like it should be. A filtration velocity is the volume of fluid traversing unit area of porous material in unit time and is the porosity ϕ times the relative fluid speed. In the notation of the present paper, the filtration velocity is denoted $\partial \mathbf{w} / \partial t$ where $\mathbf{w} = \phi(\mathbf{u}_f - \mathbf{u})$ is called the filtration displacement. Because of this, the steady-flow permeability k_0 used in the present paper (standard definition) is ϕ times the permeability used by Frenkel.

With this in mind, Frenkel's relative force balance [his Eq. (22a)] can be directly rewritten using the variables of the present paper as

$$\frac{\rho_f}{\phi} \frac{\partial^2 \mathbf{w}}{\partial t^2} + \frac{\eta_f}{k_0} \frac{\partial \mathbf{w}}{\partial t} = -\nabla p_f - p_f \frac{\partial^2 \mathbf{u}}{\partial t^2} \quad (21)$$

where η_f =fluid's shear viscosity. This force-balance can be considered equivalent to the one later developed by Biot. In modern retrospect, we now know (Brown 1980) that the effective fluid inertia in this force balance is more generally $\rho_f F$, where F is the electrical formation factor in the porous material when surface electrical conduction is not important. In models of the pore space where the current lines are straight, one has that $F=1/\phi$, which then reduces to the Frenkel statement. Biot (1956b), and later Johnson et al. (1987) and others, have provided models that allow for the development of viscous-boundary layers in the pores at sufficiently high frequencies. However, for most earth materials of interest and across the entire frequency band of interest in seismic exploration, the flow in the pores is well modeled as being parabolic. As such, the inertial effect captured by the term $\rho F \partial^2 \mathbf{w} / \partial t^2$, as well as all viscous boundary layer effects, can normally be neglected in Eq. (21). This is particularly true for the seismic (as opposed to ultrasonic) applications of the theory that Frenkel has in mind.

The total force balance for all the material in an averaging volume is obtained by adding Frenkel's Eqs. (22) and (23) to directly obtain

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} + \rho_f \frac{\partial^2 \mathbf{w}}{\partial t^2} = \Phi^{(1)} - \phi \nabla p_f \quad (22)$$

where $\rho = (1 - \phi)\rho_s + \phi\rho_f$ = bulk density and where Frenkel defines $\Phi^{(1)}$ as the “elastic force acting on the solid skeleton.” Frenkel should therefore have identified this as $\Phi^{(1)} = (1 - \phi)\nabla \cdot \boldsymbol{\tau}_s = \nabla \cdot \boldsymbol{\tau} + \phi \nabla p_f$ so that the right-hand side of Eq. (22) becomes simply the total elastic force $\nabla \cdot \boldsymbol{\tau}$. Written in this way, Frenkel’s Eq. (22) is identical to the results of Biot (1956a,b).

However, Frenkel for some reason proposes that $\Phi^{(1)} = (1 - \phi)\nabla \cdot \boldsymbol{\tau}_s - (1 - \phi)\nabla p_f$ [his Eq. (16)]. He somehow pictures an additional fluid-pressure gradient acting directly on the solid frame which is erroneous. Because of this, his equation governing the total force balance has an extra fluid-pressure gradient term present that should not be there.

Frenkel’s Wave Properties

As seen above, the form of Frenkel’s equations, after some manipulation, are nearly identical to those of Biot (1956a,b). However, because his analysis misidentifies two of the three compressibility constants and adds an erroneous fluid-pressure gradient to the total force balance, his detailed results for the nature of compressional (longitudinal) slownesses are not correct. Since he correctly writes down the force balance governing the relative flow, he properly predicts the existence of two distinct longitudinal modes. Unfortunately, he gives little importance to the fluid-pressure-diffusion (or “slow-wave”) mode and even goes so far as to say it is “nonexistent.”

Something Frenkel gets entirely correct is the complex shear-wave slowness s_s

$$s_s = \sqrt{\frac{\rho + i\omega k_a p_f^2 / \eta}{G}} \quad (23)$$

This is because the shear-wave slowness is independent of the presence of fluid-pressure gradients and the compressibility moduli.

In the balance, Frenkel certainly deserves more recognition than he has received for his pioneering contribution to porous-media acoustics. He did many nonobvious things correctly, not the least of which is defining appropriate dependent and independent variables for a theory of porous-media acoustics.

Frenkel’s Electro seismic Theory

Ivanov (1939) measured an electric field as seismic waves passed electrode pairs on the earth’s surface. He proposes that such coupling is electrokinetic in nature. In soils, rocks, and other porous materials, the solid grains have an excess charge fixed to their surfaces that is balanced by free counter ions diffusely distributed in the fluid layers immediately adjacent to the grain surfaces. The bound charge on the grain surfaces is immobile while the free counter ions in the fluid are able to move in response to applied forces. The bound charge and diffuse counter charge together is called the “electric double layer.” Electrokinetic phenomena are, by definition, processes associated with the transport of the diffuse counter charge.

Frenkel states that his principal motivation for developing a theory of porous-media acoustics is to explain the electro seismic measurements of Ivanov. Both Ivanov and Frenkel envision that a compressional wave creates fluid-pressure gradients at the scale of the wavelength and that the associated flow from the compressions to dilatations transports the counter charge of the double

layer relative to the bound charge. In this way, counter charge accumulates in the regions of dilation and the bound charge becomes “exposed” in the regions of compression so that an electric field is created perpendicular to the wavefront at the scale of the wavelength. This electric field moves along with the seismic wave as part of the material response and can be recorded as the seismic wave traverses an electrode pair.

We have seen above that Frenkel produces a theory capable of predicting (properly so if he hadn’t made his small mistakes) the macroscopic fluid-pressure gradients ∇p_f associated with a compressional wave. It only remains to relate this seismic response to the associated electric field.

To do so, Frenkel simply quotes the famous result of von Smoluchowski (1903) for the electric field created by a static fluid-pressure gradient. In the present context in which a compressional wave is driving a relative fluid flow, the right-hand side of Eq. (21) identifies the applied force as $-\nabla p_f - \rho_f \partial^2 \mathbf{u} / \partial t^2$. von Smoluchowski’s result for the electric field $\mathbf{E}$ created by this applied force is

$$\mathbf{E} = -\frac{\varepsilon_f \zeta}{\eta_f \sigma_f} \left(\nabla p_f + \rho_f \frac{\partial^2 \mathbf{u}}{\partial t^2} \right) \quad (24)$$

where ε_f = fluid’s dielectric permittivity; σ_f = fluid’s electrical conductivity; η_f = again the fluid’s viscosity; and ζ = property of the grain–fluid interface that quantifies the electric potential at the interface separating the bound charge from the diffuse-layer charge. The “zeta potential” ζ is thus a measure of the amount of charge separation present in the electric double layer.

For a plane P -wave having $e^{-i\omega t}$ time dependence and propagating in the x direction in a uniform porous material, Frenkel first calculates the $\partial p_f / \partial x$ associated with the wave and then determines E_x using Eq. (24). If the corrected variant of Frenkel’s equations are used, one obtains

$$\frac{E_x}{-\omega^2 u_x} = \frac{\rho_f \varepsilon_f \zeta}{\eta_f \sigma_f} \left(1 - \frac{\rho C}{\rho_f H} \right) \quad (25)$$

where u_x = amplitude of the particle displacement associated with the wave in the x direction; and where C and H = elastic moduli (as defined by Biot) related to the undrained bulk modulus K_U and Skempton’s coefficient B as

$$C = BK_U \quad (26)$$

$$H = K_U + 4G/3. \quad (27)$$

Eq. (25) for E_x , which corresponds to the low-frequency limit of an expression given by Pride and Haartsen (1996) and to an expression given by Garambois and Dietrich (2001), is the same as the culminating Eq. (46) of Frenkel’s paper except for a slightly different definition of the elastic moduli ratio C/H due to Frenkel’s errors associated with the longitudinal response as previously discussed.

The message of Frenkel’s result [Eq. (25)] is that the electric field E_x moving along with a P -wave as part of the material response is directly proportional (without phase adjustments) to the particle acceleration. This has been experimentally verified by Garambois and Dietrich (2001). In Fig. 1, the recorded particle accelerations (time derivatives of the geophone response which is proportional to the particle velocities) and associated electric fields are plotted together on the same time axis for receivers on the earth’s surface at different distances from the seismic source (a hammer). A horizontal geophone was at the center of each 50 cm electric-dipole antenna (two metal rods driven vertically

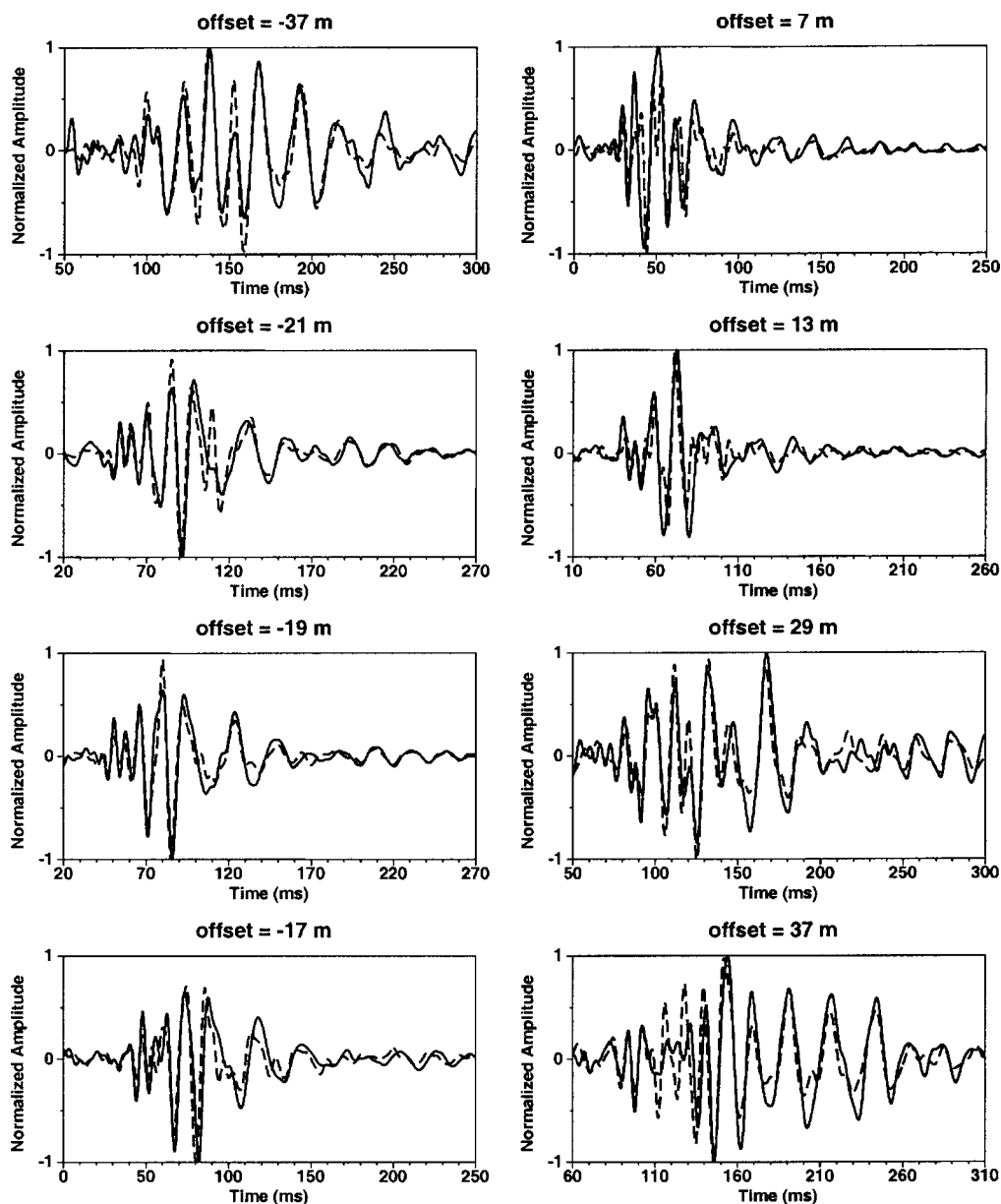


Fig. 1. Field data of Garambois and Dietrich (2001). Horizontal component of electric field (solid lines) along with horizontal component of particle acceleration (dashed lines) as recorded at earth's surface at different offsets from seismic source which is hammer impacting a plate on earth's surface.

into the earth, 50 cm apart) and the time difference between when the seismic wave arrives at the first electrode and when it arrives at a geophone have been allowed for (roughly 1 ms). The experimental field data of Fig. 1 provide the ultimate testimony to the lucidity and correctness of Frenkel's pioneering work.

Modern Theory of Electrostatic Phenomena

Frenkel limited his electrostatic analysis to explaining the electric field that accompanies a propagating longitudinal wave. However, there are many other aspects to the electric and magnetic fields generated by both compressional and shear waves that Frenkel did not attempt to address as we now go on to discuss.

Governing Equations

In order to model these additional effects, equations of a more general character are required. Pride (1994) obtains the equations governing the coupled seismic and electromagnetic response of a porous material when electrokinetics is responsible for the coupling. Assuming an $e^{-i\omega t}$ time dependence, these equations are

$$-\omega^2 \rho \mathbf{u} = -\nabla P_c + \nabla \cdot \boldsymbol{\tau}^D + \omega^2 \rho_f \mathbf{w} \quad (28)$$

$$\boldsymbol{\tau}^D = G \left(\nabla \mathbf{u} + \nabla \mathbf{u}^T - \frac{2}{3} \nabla \cdot \mathbf{u} \mathbf{I} \right) \quad (29)$$

$$-\begin{bmatrix} P_c \\ p_f \end{bmatrix} = K_U \begin{bmatrix} 1 & B \\ B & B/\alpha \end{bmatrix} \begin{bmatrix} \nabla \cdot \mathbf{u} \\ \nabla \cdot \mathbf{w} \end{bmatrix} \quad (30)$$

$$\begin{bmatrix} -i\omega \mathbf{w} \\ \mathbf{J} \end{bmatrix} = \begin{bmatrix} k(\omega)/\eta_f & L(\omega) \\ L(\omega) & \sigma(\omega) \end{bmatrix} \begin{bmatrix} -\nabla p_f + \omega^2 \rho_f \mathbf{u} \\ \mathbf{E} \end{bmatrix} \quad (31)$$

$$\nabla \times \mathbf{H} = -i\omega \epsilon \mathbf{E} + \mathbf{J} \quad (32)$$

$$\nabla \times \mathbf{E} = i\omega \mu \mathbf{H} \quad (33)$$

and are simply the Frenkel/Biot equations for porous media acoustics along with the Maxwell equations for the electric and magnetic fields $\mathbf{E}$ and $\mathbf{H}$. The electroseismic coupling occurs in the transport laws of Eq. (31) where $\mathbf{J}$ is the electric-current density and $-i\omega \mathbf{w}$ the Darcy filtration velocity. If the coupling coefficient L were set to zero then there would be complete decoupling between the poroelastic and electromagnetic response fields.

Two types of electroseismic coupling are present in Eq. (31). Seismic waves generate a force $-\nabla p_f + \omega^2 \rho_f \mathbf{u}$ that in addition to driving a Darcy fluid filtration $(k/\eta_f)(-\nabla p_f + \omega^2 \rho_f \mathbf{u})$, also transports the diffuse charge of the double layer relative to the bound charge on the grain surfaces resulting in a "streaming" electric current $L(-\nabla p_f + \omega^2 \rho_f \mathbf{u})$. Such generation of an electric current from an applied fluid-pressure gradient is known as "electro-filtration." Conversely, when an applied electric field $\mathbf{E}$ acts on a porous material, in addition to driving a conduction current given by $\sigma \mathbf{E}$, it also acts as a body force on the excess charge of the diffuse double layer resulting in a net fluid filtration given by $L\mathbf{E}$. Such generation of a fluid filtration from an electric field is known as "electro-osmosis."

As stated, Frenkel (1944) only concerns himself with the electrofiltration (or "E") effect. He makes the implicit assumption (correctly), that within a propagating compressional wave in a homogeneous material, there is no net electric current. The streaming electric current induced by the wave causes counter charge to accumulate in the troughs of the wave and the bound charge to become exposed in the peaks. The electric field associated with this wavelength-scale charge separation drives a conduction current that exactly balances the streaming current. Thus, for a P -wave propagating in a homogeneous material, $\mathbf{J}=0$ in Eq. (31) yielding $\mathbf{E}=-(L/\sigma)(-\nabla p_f + \omega^2 \rho_f \mathbf{u})$, which, as will be seen next, is equivalent to Eq. (24).

Porous Media Transport Coefficients

The main purpose of this subsection is to address what is known about the frequency relaxation in the transport equations; an issue not addressed by Frenkel (1944).

Pride (1994) obtains analytic expression for the three porous-media transport coefficients: permeability $k(\omega)$, electrokinetic-coupling coefficient $L(\omega)$, and electric conductivity $\sigma(\omega)$. The results are valid to leading order in the dimensionless ratio d/Λ , where Λ is a characteristic pore-throat radius defined by Johnson et al. (1987) and d is a skindepth measure of the diffuse-charge-layer thickness known as the Debye length given by

$$d = \sqrt{\frac{\epsilon_f kT}{e^2 z^2 N}} \quad (34)$$

where kT =thermal energy; and ez =electric charge of each ion contributing to a symmetric electrolyte having ionic-number density N (number of ions having valence z per cubic meter). For most salinity conditions in the earth, d is on the order of nanometers and so the small d/Λ limit can be considered appropriate.

The results are

$$\frac{k(\omega)}{k_o} = \left[\left(1 - i \frac{4}{m} \frac{\omega}{\omega_t} \right)^{1/2} - i \frac{\omega}{\omega_t} \right]^{-1} \quad (35)$$

$$\frac{L(\omega)}{L_o} = \left[1 - i \frac{\omega}{\omega_t} \right]^{-1/2} \quad (36)$$

$$\frac{\sigma(\omega)}{\sigma_o} = 1 \quad (37)$$

where the following expressions for the viscous-to-inertial relative-flow transition frequency ω_t and the dimensionless number m have been used

$$\omega_t = \frac{\eta_f}{F k_o \rho_f}$$

and

$$m = \frac{\Lambda^2}{F k_o} \quad (38)$$

The effective pore-throat radius Λ and the electrical formation factor F both have precise mathematical definition given by Johnson et al. (1987) in terms of the dimensionless electric field $-\nabla \Phi$ in the pore space of an averaging volume of porous material of volume V

$$\frac{1}{F} = \frac{1}{V} \int_{\Omega_p} \nabla \Phi \cdot \nabla \Phi dV \quad (39)$$

$$\frac{2}{\Lambda F} = \frac{1}{V} \int_{\partial \Omega_{gp}} \nabla \Phi \cdot \nabla \Phi dS \quad (40)$$

Here, Ω_p =pore space and $\partial \Omega_{gp}$ =surface separating the grains from the pores. The electric potential Φ has units of length. Within a cylindrical disk of porous material having a thickness H , Φ satisfies the Laplace equation $\nabla^2 \Phi = 0$ throughout the pore space, and the boundary conditions that: (1) $\Phi = H$ on the flat surface $z = H$ of the disk; (2) $\Phi = 0$ on $z = 0$, and (3) $\mathbf{n} \cdot \nabla \Phi = 0$ on both the circumferential surface of the disk as well as on the grain surfaces. In clean porous rocks where secondary clay on the grain surfaces has not created a large grain-surface area, Johnson et al. (1987) suggest that a good model for Λ is to take $m=8$ so that $\Lambda = \sqrt{8 F k_o}$.

The dynamic permeability of Eq. (35) was first obtained by Johnson et al. (1987) and allows for three distinct phenomena associated with the relative flow. At low enough frequencies $\omega \ll \omega_t m/4$, there is parabolic flow in the pores resulting in a permeability k_o that depends only on the topology and size of the pores. As frequencies increase and $\omega \approx \omega_t m/4$, viscous boundary layers develop in the pores. Finally, in the high-frequency limit where $\omega \gg \omega_t$, the viscous boundary layers have a thickness that diminishes with the square root of frequency, resulting in a relative motion dominated by inertial plug (ideal) flow. The real and imaginary parts of $k(\omega)$ in the high-frequency limit are

$$k(\omega) \sim \frac{\sqrt{2}}{F \Lambda} \left(\frac{\eta_f}{\rho_f \omega} \right)^{3/2} + \frac{i}{F} \frac{\eta_f}{\omega \rho_f} \text{ as } \omega \rightarrow \infty \quad (41)$$

So above the transition frequency ω_t , inertial forces dominate viscous forces, and the amplitude of the relative flow is reduced.

The relaxation in the dynamic coupling coefficient $L(\omega)$ of Eq. (36) is also due to the development of viscous boundary

layers in the pores. Pride (1994) allows for an additional relaxation in $L(\omega)$ as the viscous skin depth $\sqrt{\eta_f/(\rho_f\omega)}$ associated with the viscous boundary layers becomes smaller than the Debye length; however, over the range of frequencies and salinities normally encountered in the earth, this relaxation never takes place and as such has not been included in Eq. (36). The steady-state coupling coefficient L_0 is given by

$$L_0 = -\frac{\varepsilon_f \zeta}{\eta_f F} \quad (42)$$

to leading order in d/Λ where F is the same formation factor defined by Eq. (39).

There is no important relaxation in $\sigma(\omega)$ over frequencies where the viscous skin depth is greater than the Debye length; i.e., when $\sqrt{\eta_f/(\rho_f\omega)} > d$ which corresponds to the frequency range of all seismic applications (c.f., Pride 1994). To leading order in d/Λ , one simply has

$$\sigma(\omega) = \sigma_0 = \frac{\sigma_f}{F} \quad (43)$$

with the formation factor again defined by Eq. (39). Induced polarization effects associated with ion accumulations in the electric double layers surrounding the grains exist across the seismic band of frequencies, but manifest themselves as milliradian phase differences between the applied electric field and induced current that result in negligible changes in $\sigma(\omega)/\sigma_0$. Other polarization effects resulting in more significant variations in $\sigma(\omega)/\sigma_0$ due to interfaces between ionic and electronic conductors or due to clay membranes are not modeled here.

An immediate consequence of these results is that at frequencies where viscous boundary layers have not yet developed, one has that when there is no net current (e.g., in a longitudinal wave)

$$\left. \frac{E_x}{-\partial p_f / \partial x + \omega^2 \rho_f u_x} \right|_{J=0} = \frac{\varepsilon_f \zeta}{\eta_f \sigma_f} \quad (44)$$

to leading order in d/Λ which is the famous proposition of von Smoluchowski (1903). It is through Eq. (44) that the zeta potential is often measured. If the left-hand side is directly measured after applying a pressure drop across a porous sample, and if the fluid properties ε_f , η_f , and σ_f are known, ζ is determined. For quartz surfaces in contact with a saline solution having a given salinity C (moles/liter) and pH, ζ is observed to obey the empirical formula

$$\zeta(\text{Volts}) = (0.01 + 0.025 \log C) \frac{(\text{pH} - 2)}{5} \quad (45)$$

which gives the measured dependence on C when pH=7, and gives the appropriate general trend for the pH dependence.

In conclusion, the important relaxation in the porous-media transport coefficients is associated with the onset of viscous boundary layers in the pores. However, since the transition frequency $\omega_t = \eta_f/(\rho_f F k_o)$ normally lies above the frequency band of 10–10³ Hz used in seismic exploration, it is usually appropriate to ignore the frequency dependence in these coefficients (as did Frenkel 1944) and simply take $k(\omega) = k_o$, $L(\omega) = \varepsilon_f \zeta/(\eta_f F)$, and $\sigma(\omega) = \sigma_f/F$. In this case, only two porospace topology parameters k_o and F must be specified to model the electroseismic transport.

Diverse Electrostatic Wave Phenomena

Eqs. (28)–(33) allow for a wide range of electroseismic phenomena beyond the electric field contained in a compressional wave

that was the sole focus of Frenkel (1944). A few of the many possible phenomena/applications are now discussed.

Magnetic Field in Shear Wave

A shear wave in a homogeneous material does not cause the counter charge in the double layer to accumulate. Because of this, the grain accelerations produce a nonzero net current $\mathbf{J}$ in the plane of the wavefront and such current sheets produce magnetic fields as part of the material response. There is a small electric field produced via induction but this is extremely small compared to the electric field in a compressional wave of similar amplitude and does not produce a conduction current that significantly opposes the streaming current. For a shear wave propagating in the x direction with a displacement amplitude u_y , Pride and Haartsen (1996) have determined the exact expression for the magnetic field which at low frequencies and to leading order in the dimensionless number $\omega k_o \rho_f^2/(\rho \eta_f)$ reduces to

$$\frac{H_z}{-i\omega u_y} = \sqrt{\frac{G}{\rho}} \frac{\rho_f \varepsilon_f \zeta}{\eta_f F} \quad (46)$$

This result was also given by Garambois and Dietrich (2001) and states that the magnetic field in a seismic shear wave is directly proportional to the particle velocity which is the material response recorded by a geophone. The magnetic field measured by a magnetometer is $B_z = \mu H_z$, where μ is the magnetic susceptibility of the material (only different from that of vacuum when iron is significantly present in the soil). If the particle velocities of a shear wave are on the order of 10⁻³ m/s (a large amplitude but linear seismic response) and if characteristic values appropriate to water are used, one obtains that B_z is on the order of 10⁻² nT which is at the extreme limit of being measurable. Using magnetometers as shear-selective recording devices at seismic frequencies is, therefore, probably not a commercially interesting possibility.

Converted Seismic-to-Electromagnetic Fields at Interfaces

When a compressional (or shear) wave traverses an interface in which any of the transport properties or elastic moduli change, there is a dynamic imbalance of the streaming current that results in additional charge separation across the interface. Because this charge separation is concentrated in space and has a strong dipolar component, particularly so as the first Fresnel zone traverses the interface, there are electric (and magnetic) fields created that have extent outside the support of the seismic waves and that can be recorded at the earth's surface. Over the seismic band width of 10 Hz < f < 10³ Hz, the electromagnetic skin depth $\delta = 1/\sqrt{2\pi\mu_o\sigma f}$ varies as 10³ m > δ > 10² m. So if the interface of interest is less than 100 m from the electrical antennas, the recorded response is in the electrostatic and magnetostatic near field of the charge separation at the interface and induction effects may be neglected.

A numerical example of the electric and magnetic fields generated at an interface is now given for the situation depicted in Fig. 2. An isotropic explosion is located at $z=5$ m, a material-property interface is at $z=50$ m, and a line of geophones and electric-dipole antennas is at $z=0$ m. A numerical reflectivity algorithm (Haartsen and Pride 1997; Garambois and Dietrich 2002) is used to solve the complete set of governing equations [Eqs. (28)–(33) with no terms neglected] for this geometry. The interface separates a higher-porosity more compressible sandstone upper layer, from a lower-porosity stiffer lower layer. The results of the numerical simulation are displayed in Fig. 3.

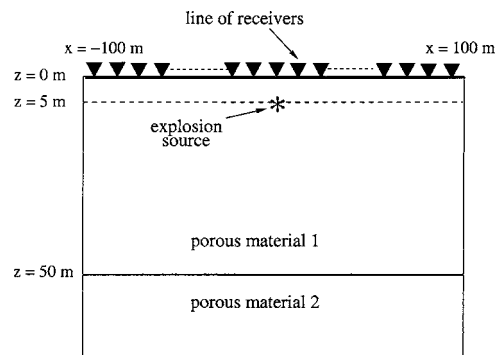


Fig. 2. Receiver, source, and interface geometry for numerical example given in Fig. 3

Of interest from an exploration perspective are the “flat” events shown on the electric and magnetic sections that are generated at $t=21$ ms as the P -wave traverses the interface. Such converted fields provide information about the interface that is distinct from the information in the seismic reflection (e.g., they are sensitive to fluid chemistry changes while seismic waves are not). Field data that have actually recorded such interface conversions (from P to EM) have been reported by Martner and Sparks (1959), Thompson and Gist (1993), and Garambois and Dietrich (2001). However, much further work is required before electrical

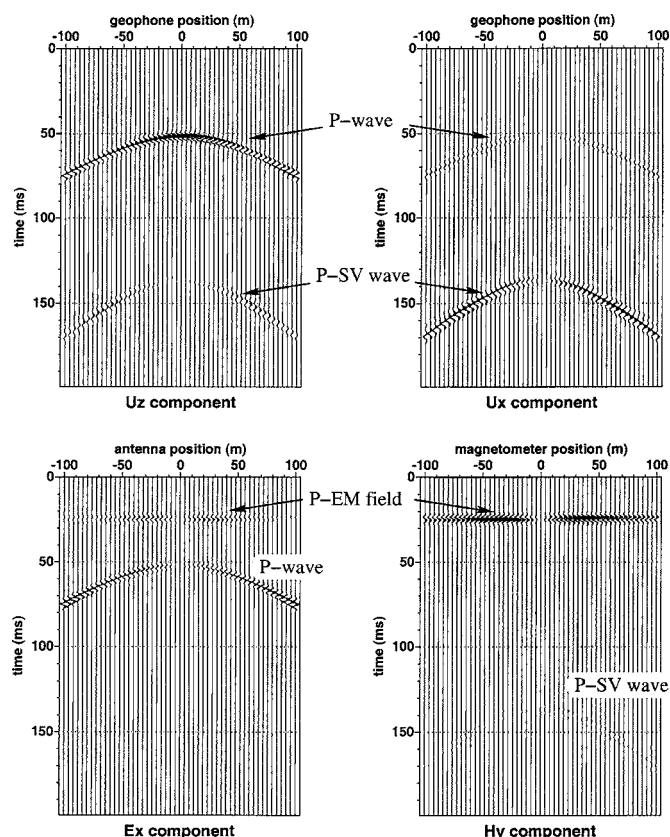


Fig. 3. Electro seismic example for geometry depicted in Fig. 2. Interface is located at P -wave travel time of 21 ms. The no-moveout P -EM converted fields have been multiplied by 500 in both electric and magnetic sections. P -SV conversion is just barely visible on magnetic section (the hyperbolic arrival at 130 ms). Direct P , wave from source to receivers is not shown.

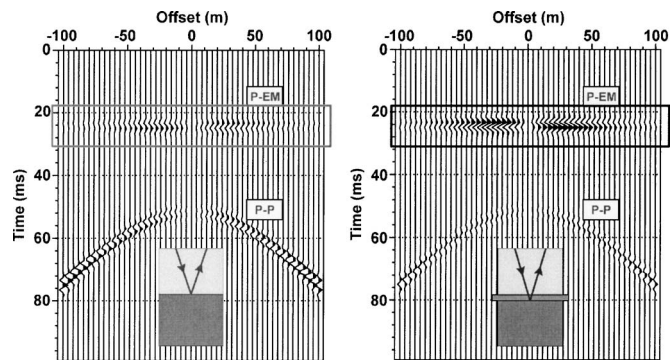


Fig. 4. Electrical section to left corresponds to same situation shown in Fig. 3 while electrical section to right corresponds to when a thin (1 cm) impermeable layer has been sandwiched between two half spaces. Converted electric field is roughly ten times larger in situation in which thin layer is present.

antenna data can be routinely processed to obtain the no-moveout interface fields. A principal challenge is that the electric fields created at an interface are far smaller than the fields contained within the seismic waves and so the electrical section has to be filtered to remove all events that move out as a function of time (since these are due to seismic waves traversing the antennae).

A particularly interesting aspect of the electro seismic conversion at an interface is that the amplitude of the converted electric field can be drastically increased if there is a thin layer of third material present at the interface. In Fig. 4, the same example as in Fig. 3 is considered but with the addition of a thin 1 cm layer of very low permeability material (a shale aquitard) sandwiched between the two half spaces. The amplitude of the converted electric field is increased by roughly a factor of 10 in the case where the thin shale layer is present, while the reflected seismic wave is essentially unaffected by the presence of the thin layer. This has the potential for being exploited as an aquitard detector in hydrological applications.

Applications to Electric Fields Created by Earthquakes

A fault that has undergone a shear dislocation creates lobes of compression and dilation in the crust surrounding the limits of the fault segment that slipped. Immediately after the earthquake, the poroelastic response is undrained (no fluid mass enters or leaves each mass element of the crust). However, the fluid pressure created by such undrained compression will subsequently equilibrate by diffusion (the slow wave) producing electric fields in the process by electrofiltration. Modeling the complete poroelastic and electric response as the fluid pressure equilibrates through time in a uniform crust after an earthquake has been the subject of a recent study (Pride et al. 2004).

An example is given in Fig. 5. The earthquake is modeled as 50 cm of uniform slip on a normal fault dipping at 45° . The slip surface is 5 km in the direction of slip and 20 km wide and is denoted in the top panel of Fig. 5 by the dashed-white line. This corresponds to a Magnitude 6 earthquake and the stress changes created immediately after the slip are numerically determined using the algorithm of Okada (1992). The subsequent fluid-pressure variations through time and space were numerically calculated using finite differences. The displayed vertical component of the electric field has an amplitude on the order of milliVolt/meter which is 100–1000 times larger than the telluric fields that are routinely measured (the telluric fields are electric

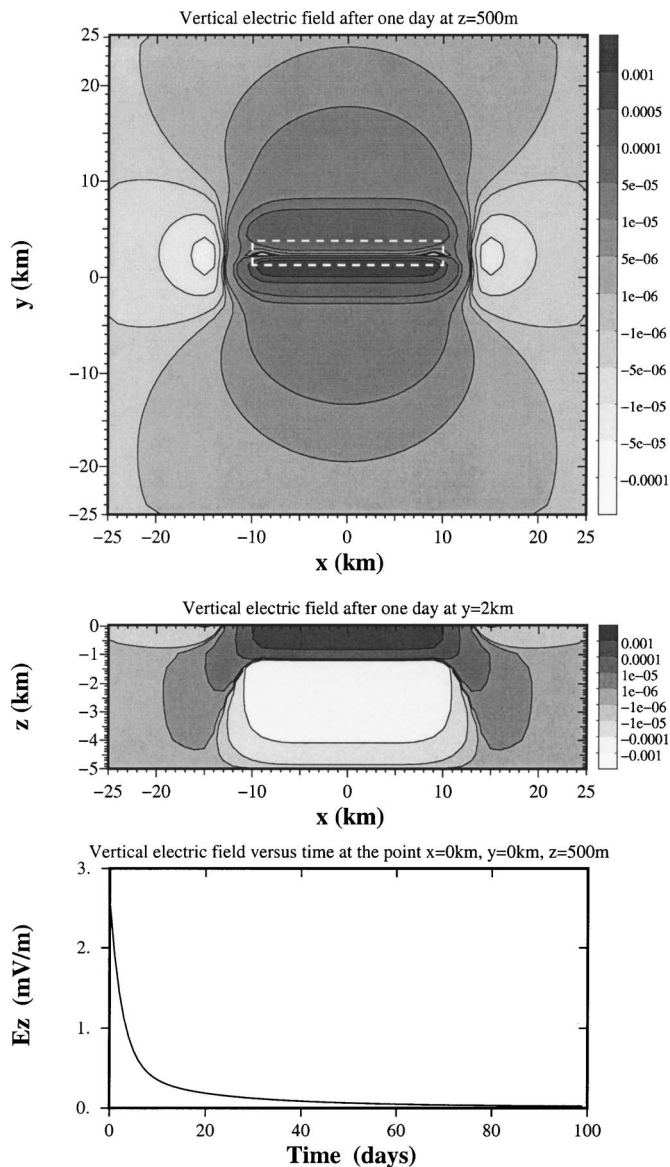


Fig. 5. Electrical field (V/m) generated in uniform crust after magnitude 6 earthquake on reverse fault dipping at 45° and positioned as shown in top panel with dashed-white line. Vertical-depth axis is z .

fields that enter the earth from the atmosphere each day as the ionosphere is heated by the sun). Direct measurement of the decay of the crust's electric field through time after an earthquake has not yet been attempted.

Electro-Osmotic Coupling

One of the greatest potential applications of Eqs. (28)–(33), concerns injecting time-varying current into the earth with the goal of generating seismic waves at interfaces at depth. The idea is that an applied electric field will drive an electro-osmotic flow of fluid $\mathbf{LE}$. At interfaces where either L or $\mathbf{n} \cdot \mathbf{E}$ change ($\mathbf{n}$ being the normal to the interface), there is an accumulation of fluid and an associated volumetric dilatation of the porous material that generates a compressional wave having the same time signature as the applied current.

Although this possibility has been discussed by Thompson and Gist (1993), no studies have ever been published in which such

electrically generated seismic waves have been measured. Numerical simulations indicate that for safe levels of injected current (say less than 10 A at the injection electrode), the generated seismic waves will have a very small amplitude (displacements less than angstroms). However, because the source is a nondestructive electric current that can be applied for long durations with a controlled time signature, it should be possible to extract even small-amplitude seismic waves from the ambient noise by cross-correlating the seismic data with the known time signature of the electric current. This is the subject of ongoing research.

Conclusions

Frenkel (1944) produced equations having nearly identical form to those of Biot (1956a,b). The only difference in the form of Frenkel's equations is an extra fluid pressure gradient term in the total conservation of momentum equation that should not be present. As was seen here, Frenkel also made a slight error in developing his effective poroelastic compressibility coefficients with the result that he did not obtain the fluid-substitution relations of Gassmann (1951) and Biot and Willis (1957). Frenkel's estimate of the electric field contained in a compressional wave (the focus of his electroseismic investigation) is correct if the proper compressibility constants are used and if the extra fluid-pressure gradient in the bulk-force balance is neglected. He did not attempt to analyze any of the other electroseismic phenomena that were discussed above.

Frenkel covered a lot of ground in his only paper on poroelastic phenomena, and the vast majority of his intuitive approach to obtaining the governing equations is correct. He certainly deserves a greater recognition for his pioneering contribution than he has so far been accorded. From the present writer's perspective, it seems entirely appropriate to refer to the equations governing the acoustics of isotropic porous materials as the "Frenkel/Biot equations" instead of just the "Biot equations" as is the current fashion.

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